Methods in Molecular Biology 2929

Springer Protocols

Cristina Ruiz-Romero
Lucía Lourido
Valentina Calamia *Editors*

Protein Arrays

Methods and Protocols

Humana Press

Methods in Molecular Biology

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School of Life and Medical Sciences
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Edited by

Cristina Ruiz-Romero, Lucía Lourido and Valentina Calamia

Grupo de Investigación de Reumatología, INIBIC, A Coruña, Spain

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Editors
Cristina Ruiz-Romero
Grupo de Investigación de Reumatología
INIBIC
A Coruña, Spain

Lucía Lourido
Grupo de Investigación de Reumatología
INIBIC
A Coruña, Spain

Valentina Calamia
Grupo de Investigación de Reumatología
INIBIC
A Coruña, Spain

ISSN 1064-3745 ISSN 1940-6029 (electronic)
Methods in Molecular Biology
ISBN 978-1-0716-4594-9 ISBN 978-1-0716-4595-6 (eBook)
https://doi.org/10.1007/978-1-0716-4595-6

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The registered company address is: 1 New York Plaza, New York, NY 10004, U.S.A.

Preface

In the era of high-throughput biology, the ability to analyze proteins on a large scale has transformed our understanding of complex biological systems and disease mechanisms. Protein microarrays stand at the forefront of this revolution, providing a robust platform for studying protein expression, interactions, and modifications with an unprecedented level of detail. These powerful tools enable scientists to explore vast proteomic landscapes efficiently, identifying novel biomarkers, deciphering disease pathways, and advancing new therapeutic interventions. The aim of this book is to describe novel methods and applications of protein microarrays, guiding readers through their development and fabrication, strategies for their data analysis, and wide-reaching applications in biology and disease research.

The intended audience includes graduate students, postdoctoral researchers, laboratory scientists, and industry professionals who wish to deepen their understanding of protein microarray technology. Whether embarking on their first protein array experiment or seeking advanced insights to refine their methodologies, this book offers valuable strategies and practical guidance, as it has been designed to help researchers of all levels harness the power of protein microarrays in their own work.

We are deeply grateful to the authors whose dedication has made this book a reality, and whose work continues to inspire new applications and methodologies. Special thanks also go to the readers, whose curiosity and commitment to advancing scientific knowledge are the driving forces behind this book. It is our hope that it will serve as a valuable resource in their exploration of protein microarrays, sparking new ideas and advancing research in ways that will contribute to the broadest goals of biology and a future of precision medicine.

A Coruña, Spain

Cristina Ruiz-Romero
Lucía Lourido
Valentina Calamia

Contents

Contributors

VITALY ABLAMUNITS • *Pharmacokinetics, Dynamics and Metabolism, Pfizer, Inc., Andover, MA, USA*
RENÉ P. ACHTERBERG • *Wageningen Bioveterinary Research, Lelystad, The Netherlands*
MARCOS J. C. ALCOCER • *Division of Food Sciences, School of Biosciences, University of Nottingham, Loughborough, UK*
ELENA ANAYA-CUBERO • *Clinical Neuroproteomics Unit, Proteomics Platform, Navarrabiomed, Complejo Hospitalario de Navarra (CHN), Universidad Pública de Navarra (UPNA), IdiSNA, Proteored-ISCIII, Pamplona, Spain*
ENI ANDERSSON • *SciLifeLab Autoimmunity and Serology Profiling Unit, Department of Protein Science, KTH Royal Institute of Technology, Stockholm, Sweden*
CARLOTA ARIAS-HIDALGO • *Department of Medicine and General Cytometry Service-Nucleus, CIBERONC CB16/12/00400, Cancer Research Centre (IBMCC/CSIC/USAL/IBSAL), IBSAL, Universidad de Salamanca-CSIC, Salamanca, Spain; Proteomics Unit, Instituto de Investigación Biomédica de Salamanca (IBSAL), Salamanca, Spain*
NURIA BARBARROJA • *Cobiomic Bioscience SL, EBT UCO/IMIBIC, Cordoba, Spain; Maimonides Institute for Research in Biomedicine of Cordoba (IMIBIC)/Reina Sofia University Hospital/University of Cordoba (UCO), Córdoba, Spain*
RODRIGO BARDERAS • *Functional Proteomics Unit, Chronic Disease Programme (UFIEC), Instituto de Salud Carlos III, Madrid, Spain; CIBER of Frailty and Healthy Aging (CIBERFES), Madrid, Spain*
SOMA BASAK • *Pharmacokinetics, Dynamics and Metabolism, Pfizer, Inc., Andover, MA, USA*
FRANCISCO J. BLANCO GARCÍA • *Grupo de Investigación de Reumatología (GIR), Instituto de Investigación Biomédica de A Coruña (INIBIC), Hospital Universitario A Coruña (HUAC), Universidade da Coruña (UDC), A Coruña, Spain*
TERESA M. CAIAZZO • *Pharmacokinetics, Dynamics and Metabolism, Pfizer, Inc., Andover, MA, USA*
VALENTINA CALAMIA • *Grupo de Investigación de Reumatología, (INIBIC), A Coruña, Spain*
JULIÁN CANDIA • *Longitudinal Studies Section, Translational Gerontology Branch, National Institute on Aging, National Institutes of Health, Baltimore, MD, USA*
PAZ CARTAS-CEJUDO • *Clinical Neuroproteomics Unit, Proteomics Platform, Navarrabiomed, Complejo Hospitalario de Navarra (CHN), Universidad Pública de Navarra (UPNA), IdiSNA, Proteored-ISCIII, Pamplona, Spain*
PABLO DOMÍNGUEZ-GUERRERO • *Unidad de Proteómica, Grupo de Investigación de Reumatología (GIR), Instituto de Investigación Biomédica de A Coruña (INIBIC), Hospital Universitario A Coruña (HUAC), Universidade da Coruña (UDC), A Coruña, Spain*
LEIRE EXTRAMIANA • *Clinical Neuroproteomics Unit, Proteomics Platform, Navarrabiomed, Complejo Hospitalario de Navarra (CHN), Universidad Pública de Navarra (UPNA), IdiSNA, Proteored-ISCIII, Pamplona, Spain*
FRANCO H. FALCONE • *Institute of Parasitology, Justus Liebig University Giessen, Giessen, Germany*

Joaquín Fernández-Irigoyen • *Clinical Neuroproteomics Unit, Proteomics Platform, Navarrabiomed, Complejo Hospitalario de Navarra (CHN), Universidad Pública de Navarra (UPNA), IdiSNA, Proteored-ISCIII, Pamplona, Spain*

Manuel Fuentes • *Department of Medicine and General Cytometry Service-Nucleus, CIBERONC CB16/12/00400, Cancer Research Centre (IBMCC/CSIC/USAL/IBSAL), IBSAL, Universidad de Salamanca-CSIC, Salamanca, Spain; Proteomics Unit, Instituto de Investigación Biomédica de Salamanca (IBSAL), Salamanca, Spain*

Rafael Gongora • *Department of Medicine and General Cytometry Service-Nucleus, CIBERONC CB16/12/00400, Cancer Research Centre (IBMCC/CSIC/USAL/IBSAL), IBSAL, Universidad de Salamanca-CSIC, Salamanca, Spain*

Marloes Heijne • *Wageningen Bioveterinary Research, Lelystad, The Netherlands*

Ceke Hellström • *SciLifeLab Autoimmunity and Serology Profiling Unit, Department of Protein Science, KTH Royal Institute of Technology, Stockholm, Sweden*

Angela-Patricia Hernández • *Department of Medicine and General Cytometry Service-Nucleus, CIBERONC CB16/12/00400, Cancer Research Centre (IBMCC/CSIC/USAL/IBSAL), IBSAL, Universidad de Salamanca-CSIC, Salamanca, Spain; Department of Pharmaceutical Sciences, Organic Chemistry, Faculty of Pharmacy, University of Salamanca, CIETUS, IBSAL, Salamanca, Spain*

Shixia Huang • *Advanced Technology Cores, Baylor College of Medicine, Houston, TX, USA; Department of Molecular and Cellular Biology, Baylor College of Medicine, Houston, TX, USA; Department of Education, Innovation and Technology, Baylor College of Medicine, Houston, TX, USA*

An Jacobs • *Flemish Institute for Technological Research (VITO), Environmental Intelligence, Mol, Belgium*

Pablo Juanes-Velasco • *Department of Medicine and General Cytometry Service-Nucleus, CIBERONC CB16/12/00400, Cancer Research Centre (IBMCC/CSIC/USAL/IBSAL), IBSAL, Universidad de Salamanca-CSIC, Salamanca, Spain; Proteomics Unit, Instituto de Investigación Biomédica de Salamanca (IBSAL), Salamanca, Spain*

John Kamerud • *Pharmacokinetics, Dynamics and Metabolism, Pfizer, Inc., Andover, MA, USA*

Mercedes Lachén-Montes • *Clinical Neuroproteomics Unit, Proteomics Platform, Navarrabiomed, Complejo Hospitalario de Navarra (CHN), Universidad Pública de Navarra (UPNA), IdiSNA, Proteored-ISCIII, Pamplona, Spain*

Rosemary Lawrence-Henderson • *Pharmacokinetics, Dynamics and Metabolism, Pfizer, Inc., Andover, MA, USA*

Adrián Llamas-Urbano • *Cobiomic Bioscience SL. EBT UCO/IMIBIC, Cordoba, Spain*

Lucía Lourido • *Grupo de Investigación de Reumatología, (INIBIC), A Coruña, Spain*

Julio Manuel Martínez-Moreno • *Cobiomic Bioscience SL. EBT UCO/IMIBIC, Cordoba, Spain*

Ana Montero-Calle • *Functional Proteomics Unit, Chronic Disease Programme (UFIEC), Instituto de Salud Carlos III, Madrid, Spain*

Inge Nelissen • *Flemish Institute for Technological Research (VITO), Environmental Intelligence, Mol, Belgium*

Michael Hoang Nguyen • *Advanced Technology Cores, Baylor College of Medicine, Houston, TX, USA*

Ana Nuño-Soriano • *Department of Medicine and General Cytometry Service-Nucleus, CIBERONC CB16/12/00400, Cancer Research Centre (IBMCC/CSIC/USAL/IBSAL), IBSAL, Universidad de Salamanca-CSIC, Salamanca, Spain; Proteomics Unit, Instituto de Investigación Biomédica de Salamanca (IBSAL), Salamanca, Spain*

JENNIE OLOFSSON • *SciLifeLab Autoimmunity and Serology Profiling Unit, Department of Protein Science, KTH Royal Institute of Technology, Stockholm, Sweden*
ROCÍO PAZ-GONZÁLEZ • *Unidad de Proteómica Grupo de Investigación de Reumatología (GIR), Instituto de Investigación Biomédica de A Coruña (INIBIC), Hospital Universitario A Coruña (HUAC), Universidade de Coruña (UDC), A Coruña, Spain*
CARLOS PÉREZ-SÁNCHEZ • *Cobiomic Bioscience SL. EBT UCO/IMIBIC, Cordoba, Spain; Maimonides Institute for Research in Biomedicine of Cordoba (IMIBIC)/Reina Sofia University Hospital/University of Cordoba (UCO), Córdoba, Spain; Department of Cell Biology, Immunology and Physiology, Agrifood Campus of International Excellence, University of Córdoba (UCO), ceiA3, Córdoba, Spain*
THOMAS B. G. POULSEN • *Department of Health Science and Technology, Aalborg University, Aalborg, Denmark*
PATRICIA QUARANTA • *Unidad de Proteómica Grupo de Investigación de Reumatología (GIR), Instituto de Investigación Biomédica de A Coruña (INIBIC), Hospital Universitario A Coruña (HUAC), Universidade de Coruña (UDC), A Coruña, Spain*
SELVA RIVA-MENDOZA • *Unidad de Proteómica Grupo de Investigación de Reumatología (GIR), Instituto de Investigación Biomédica de A Coruña (INIBIC), Hospital Universitario A Coruña (HUAC), Universidade de Coruña (UDC), A Coruña, Spain*
CRISTINA RUIZ-ROMERO • *Grupo de Investigación de Reumatología, (INIBIC), A Coruña, Spain*
PABLO SAN SEGUNDO-ACOSTA • *Functional Proteomics Unit, Chronic Disease Programme (UFIEC), Instituto de Salud Carlos III, Madrid, Spain*
ENRIQUE SANTAMARÍA • *Clinical Neuroproteomics Unit, Proteomics Platform, Navarrabiomed, Complejo Hospitalario de Navarra (CHN), Universidad Pública de Navarra (UPNA), IdiSNA, Proteored-ISCIII, Pamplona, Spain*
ZHONGCHENG SHI • *Advanced Technology Cores, Baylor College of Medicine, Houston, TX, USA*
RONALD SJÖBERG • *SciLifeLab Autoimmunity and Serology Profiling Unit, Department of Protein Science, KTH Royal Institute of Technology, Stockholm, Sweden*
JACOB SKALLERUP • *Department of Health Science and Technology, Aalborg University, Aalborg, Denmark*
LOVISA SKOGLUND • *SciLifeLab Autoimmunity and Serology Profiling Unit, Department of Protein Science, KTH Royal Institute of Technology, Stockholm, Sweden*
ALLAN STENSBALLE • *Department of Health Science and Technology, Aalborg University, Aalborg, Denmark; Clinical Cancer Research Center, Aalborg University Hospital, Aalborg, Denmark*
LUIS ANTONIO TORTAJADA-GENARO • *Instituto Interuniversitario de Investigación de Reconocimiento Molecular y Desarrollo Tecnológico (IDM), Universitat Politècnica de València, Universitat de València, Camino de Vera s/n, Valencia, Spain*
FIMME J. VAN DER WAL • *Wageningen Bioveterinary Research, Lelystad, The Netherlands*
DANIEL WAN • *School of Pharmacy, University of Nottingham, Nottingham, UK*
YUAN YAO • *Advanced Technology Cores, Baylor College of Medicine, Houston, TX, USA*

Part I

Strategies in Protein Microarrays Development

Chapter 1

Printing High-Density Human Protein Fragments on Epoxy-Treated Slides

Eni Andersson, Jennie Olofsson, Lovisa Skoglund, Ceke Hellström, and Ronald Sjöberg

Abstract

The deposition of affinity reagents on to spatially discrete solid substrates is utilized for constructing protein microarrays. Protein microarrays have found their way into various applications within characterisation of proteins, affinity binders, and diseases, for diagnostics and biomarker discovery purposes. The most common methods of producing protein microarrays are by immobilizing the proteins on a solid support consisting of either microspheres or functionalized planar microscope slides, both of which have different pros and cons as pertaining to throughput, storage stability, accessibility, etc. For extensive collections of proteins, the planar format is often preferred as planar substrates, such as functionalized microscope slides, can hold multiple tens of thousands of different proteins if arrayed at a high enough density. In this chapter, we detail our production of high-density planar protein microarrays comprising of 42,100 protein fragments derived from 18,955 Ensembl Gene ID (Ensembl Release 112 (May 2024)).

Key words Affinity proteomics, Protein microarrays, Antigen microarrays, High-density arrays

1 Introduction

Planar protein microarrays consist of minute volumes of protein solutions that have been deposited on a solid support where they form small spots of known content in known spatial positions. The first result of the term "microspot" in the PubMed database is from 1964 and an article published by Joseph G. Feinberg in Nature. The article details how a microscope cover-glass was coated with agar and small volumes of antigens were deposited using capillaries, forming small spots that were then used for detecting antigen-antibody interaction [1]. Later on, in 1989, Roger P. Ekins formulated the "ambient analyte immunoassay" concept, which describes the use of small spots for achieving highest possible local signal density without affecting the local analyte concentration [2]. And then, in 1995, what is usually considered the first true

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_1,

microarray was published by Patrick O. Brown [3]. Today, protein arrays enable large scale screening of patient samples for biomarker detection and drug target discovery [4–6], affinity binder validation, studying protein-protein interactions, post-translational modification etc. [7, 8].

Arrays can be built using other substrates than functionalized microscope slides, such as microspheres. This approach allows for a higher sample throughput over a limited number of analytes per array, which requires the analytes to be divided into subarrays with different analytes in each array if large numbers of analytes are to be tested [9]. This can complicate data analysis by requiring additional normalization steps to account for the batch effects, and therefore the planar format can still be the preferred configuration for arraying large sets of proteins. However, powerful synergistic effects can be attained in the combination of planar and microsphere-based arrays. Initial screening of smaller subset of samples on large high-density planar arrays can be leveraged into more high-powered screening of large sample cohorts on smaller subsets of proteins using a microsphere-based platform [10].

The planar microarrays are manufactured by depositing minute volumes, usually in the pico to nanoliter range, of protein solution within a defined spatial area on the substrate. Each volume then forms a feature (or "spot") with diameters in the micrometer range. These comparably small features means that a large number of features, up to multiple tens of thousands, can be arrayed on to each slide.

Arraying such small volumes of solution presents a number of challenges as variation in humidity, temperature, pressure, viscosity, and buffer and substrate composition, will affect the precision by which the volumes can be deposited, and thereby how closely they can be packed before they stop being spatially discreet and morphologically homogenous. Higher density arrays are generally more desirable as they allow for a larger number of tests to be performed using the same sample volume, with less variation introduced by spatial effects. However, achieving the required density involves extensive development and testing of suitable combinations of buffers and substrates.

Over the years different instruments using different techniques for producing planar arrays have been available on the market or presented in research articles [11]. Today one of the more dominant technologies within production of planar protein microarrays are non-contact arraying using inkjet technology to eject droplets from a reservoir some distance above the substrate, which then fly through the air onto the substrate [12]. This "on-the-fly" arraying greatly speeds up the process and eliminates the risk of damaging the substrate due to physical impact. The comparably large reservoirs used also ensures that more droplets can be deposited before reloading, which improves reproducibility. However, adsorption of proteins and deposition of salt from buffers can still affect the ability to dispense droplets, which can result in missing or out of alignment features, carry-over between loadings, and small satellite spots [13].

While alignment is important for high-density packing of protein microarrays, the morphology of the features is also important. A typical issue with microarrays produced by depositing droplets on planar surfaces are nonuniformity of the features, often in the form of rings. Thin liquid films of particle solutions that are spilled and allowed to evaporate often takes the shape of rings due to transport of material from the center of the droplet to the gas/liquid interface by capillary flow. For protein solutions, this is additionally driven by accumulation of proteins in the gas/liquid interface as different parts of the protein will have different preference for either side of the interface depending on its hydrophobic and hydrophilic parts. This effect often becomes more pronounced the faster the droplet dries and therefore the inclusing of low vapor pressure components, such as glycerol, to the buffer that is used for dilution of the proteins, can greatly improve the uniformity of the feature morphology [14–17].

The substrate that is chosen for the array is additionally of great importance. Immobilization chemistries can be divided into physical adsorption, affinity binding, and covalent binding, while the immobilization geometries can be thought of as two-dimensional or three-dimensional. The choice of chemistry is often dependent on how important correct folding and function of the arrayed proteins is. If enzymatic activity is of great importance, a directed immobilization chemistry might be preferred over random immobilization, and dynamic range can be improved by utilizing more three-dimensional immobilization strategies such as hydrogels and nitrocellulose films [18–20]. However, utilizing more complex immobilization strategies tend to require both more development efforts as well as financial investments, so less complex two-dimensional strategies still full-fill a role.

In the following protocol, we will detail how we generate our in-house high-density arrays with up to 40,000 recombinant protein fragments on functionalized microscope slides. These protein fragments have been produced by the Human Protein Atlas for the use as antigens for production of antibodies [21, 22]. The fragments are similar in size (median 80 amino acids, range 20–200 amino acids) and are all stored in the same buffer, at approximately the same concentration, and have been designed in silico to be unique representations of the protein they are derived from, with lowest possible homology to the rest of the proteome [23].

The fact that the fragments are rather similar in their composition helps in improving the density by which the features can be packed, but the use of a high-viscosity buffer also helps greatly in ensuring that the effect of any variation in concentration etc. is reduced. Additionally they all contain an N-terminal hexa-histidine albumin binding protein tag (His_6ABP) for purification and solubility purposes, further contributing to the similarity between fragments [24].

The protein fragments were originally designed for producing polyclonal antibodies against linear epitopes and while they are long enough to start folding, there is no guarantee that they fold correctly, or even fold at all. We have previously extensively used the protein fragments for validation of binding for the antibodies produced by the protein atlas [25]. However, we have also found that they are useful for screening for autoantibody reactivity in body fluids in the context of autoimmune conditions as well [26]. Due to the lack of correct folding of the fragments, and the characteristics of autoantibody interaction screening we have found that two-dimensional substrates that offer covalent attachment of the arrayed proteins, such as epoxy functionalized slides, are an effective alternative offering low cost and good storage characteristics.

2 Materials

1. Microarray printer: ArrayJet Super Marathon with a Jetmosphere Max cold room (ArrayJet Ltd.).
2. 384-well microplates: JetStar Microarray-Specific 384 Microplates (ArrayJet Ltd.).
3. Plate lids: JetGuard Probe Protector (ArrayJet Ltd.).
4. Microarray substrate: Schott Nexterion Epoxysilane (E) coated slides (Schott).
5. Antigens: His_6ABP-tagged human protein fragments stored at 0.8 mg/mL in 0.1 M urea in −20 °C.
6. Printing buffer: 0.05 M carbonate-bicarbonate buffer, pH = 9.6, supplemented with 49.3% glycerol. Prepare the printing buffer by dissolving one carbonate-bicarbonate tablet in 42 mL ultrapure water before adding 58 mL of 85% glycerol (by first dissolving the carbonate-bicarbonate tablet in water, the dissolving process is sped up). Once the buffer is prepared it should be stored at 8 °C and preferably used within 1 week (*see* **Note 1**).
7. Blocking buffer: Phosphate-buffered saline buffer (1xPBS), pH = 7.4, supplemented with 0.1% (v/v) Tween-20 (1xPBS-T) and 3% bovine serum albumin (BSA).
8. Wash buffers: 1xPBS and 1xPBS-T.

3 Methods

Production of high-density planar protein microarrays can be performed using arraying robots based on different principles for deposition of the antigens on the slides. However, for the required precision of deposition the use of an inkjet arrayer is strongly recommended. In the following, we describe the arraying procedure using an ArrayJet Super Marathon printer that utilizes inkjet technology for on-the-fly deposition of proteins, and which is suitable to use with glycerol-based buffers with a higher viscosity than water. Please note that other arrayers might not be compatible with glycerol-based buffers.

1. Dilute the antigens to 0.08 mg/mL in printing buffer into 384-well microplates and store the prepared plates at −20 °C until arraying.
2. Allow the plates to reach 20 °C prior to arraying procedure. Ensure that the diluted antigens in the plate wells are free from air bubbles by vortexing the plates and spinning them down in a plate centrifuge before placing the plates together with the glass slides in the arrayer (*see* **Note 2**).
3. Ensure that the temperature and humidity in the arrayer stabilizes at 20 °C and 50%, respectively, before starting the printing process (*see* **Note 3**).
4. Array the antigens in a hexagonal lattice with a horizontal gap and pitch of approximately 0.150 mm. The arrayed antigens will produce features up to 0.08 mm in diameter. Using a height of the array of 59 mm and a width of 20.2 mm up to 150 plates, up to 57,600 features can be arrayed on each slide (*see* **Note 4**).
5. Remove the slides from the printer after arraying and allow the deposited drops to dry by quickly placing the arrayed slides into a heat cabinet at 37 °C until dry. If the slides are not dried properly, features can merge due to the hygroscopic effect of the glycerol or tailing and bleed-off of the features can occur during the following blocking step (*see* **Note 5**).
6. Once they are dry, block the arrayed slides by gently submerging them in the blocking buffer for 1 h on an orbital shaker at 85 rpm (*see* **Note 6**).
7. Wash the blocked slides by submerging them in 1xPBS-T for 5 min on an orbital shaker at 85 rpm, repeat this once. Wash with 1xPBS to wash away any residual Tween-20 and finish the washing process by a quick rinse in ultrapure water to remove any residual salts and particles. Spin-dry the slides before storing them in air-tight containers at 8 °C.

4 Notes

1. It is strongly recommended to ensure that a large enough volume of the printing buffer is prepared as the precision of alignment is strongly dependent on an even viscosity in the different wells in each print plate. The inkjet printhead aspirates from multiple wells simultaneously into multiple nozzles from which the protein solution is then ejected on to the substrate. Varying viscosities will result in different volumes being ejected, and thus result in different "flight-characteristics," affecting the speed and distance travelled by the droplets such that the precision in alignment can be affected. The more similar the viscosity of the solutions, the higher the precision, which results in more tightly packed features and a higher density can be achieved.
2. Always take care to only handle the slides using gloves and/or tweezers, and only by touching them by their edges or their barcode area if the slides are barcoded. Additionally, take care not to damage the barcode if tweezers are used as this might prevent the barcode from being read properly by the microarray scanner. Take good care to ensure that the slides are free from dust and other blemishes, even minor damages or dust can lead to noticeable data loss if the features are packed together tightly. It is strongly recommended to inspect the print plates after centrifugation to check for any air bubbles. If necessary, continue centrifuging the plates until the bubbles are no longer present. Due to the high-viscosity nature of glycerol, any bubbles introduced by preparing the plates will be comparably difficult to centrifuge away and in case they are not removed they can enter the capillaries of the printhead and cause disturbances in alignment. Thorough inspection with a magnifying glass can detect issues with particles, fibers from clothes, etc. that might have been introduced into the wells during plate-preparation, all which can cause printing irregularities.
3. Arraying at lower temperatures is possible, but as glycerol is somewhat hygroscopic it is important to ensure that the entire instrument is allowed to slowly reach ambient temperature before opening the doors to the instrument. If the slides are too cold compared to the humidity, merging of features in the array can appear even if there is no condensation visible to the naked eye. While waiting for the instrument to equilibrate with the outside temperature and humidity might not be an issue when producing smaller arrays, it can become problematic when producing larger arrays. Large arrays require pausing the instrument for changing print plates and buffers at different

intervals, and waiting for the instrument to equilibrate with ambient conditions each time can become time consuming.

4. Fitting this many features onto each slide assumes that the viscosity of the protein solutions is similar. Evaporation from the plates is inevitable and if the plates have been left out for longer periods, such as through multiple long arraying sessions, the outer wells will evaporate more and the imbalance in the viscosity will affect the alignment. It is then advised to remake the print plates if the maximum density is necessary. Optionally, the set of proteins can be spread out on to multiple slides. It is then possible to increase the distance between the features and continue using the same plates for making lower density arrays. The increased space between the features minimizes merging and ensures a good separation between the features, but at the cost of introducing batch effects due to the use of multiple slides for analysis of the full set of proteins. The minimum distance between features suggested in this protocol are only achievable using freshly produced print plates on the specified substrate.
5. Drying large arrays can take considerable time. When using a heat cabinet for this purpose, make sure that there is plenty of space between the slides and sufficient airflow to prevent uneven drying. Uneven drying is not necessarily problematic if all features are properly dried in the end. However, if parts of the arrays are still wet when introduced to any blocking or washing steps, there is a risk of smearing or spatially uneven signal strength to appear. Smaller arrays can be left to dry in the arrayer as this will ensure minimal disturbances in a controlled environment with filtered air. However, for arrays with tens of thousands of features the drying can take multiple days at room temperature and the use of a heat-cabinet might be a suitable option to free up the arrayer for continued operation.
6. The slides do not necessarily need to be blocked directly after drying, instead blocking can wait until just before use. In our experience, blocking all slides simultaneously result in a more similar blocking of all slides and can be a timesaver compared to performing a blocking procedure each time a slide is to be used. However, if the slides are to be used for different protocols blocking each time a slide is used will be more flexible.

Acknowledgments

We thank the Human Protein Atlas for suppling us with all the antigens and SciLifeLab for financing our work. The authors declare no conflicts of interest.

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Chapter 2

Building Reversed-Phase Microarrays on Low-Cost Biosensing Chips: Application to Multiplexed Detection of Food Allergies

Luis Antonio Tortajada-Genaro

Abstract

Microanalytical methods that provide multianalyte profiling information are in high demand. However, the selection of selective probes is sometimes challenging for protein biomarkers. Thus, a reversed-phase array is a reliable analytical system because these probes can better recognize the intrinsic variety of responses compared to single-protein approaches. This chapter also describes the combination of a biorecognition assay and a cheap portable optical biosensing as a powerful solution when an accurate multiple diagnosis is required.

As a proof-of-concept, an array-based method was developed as an up-and-coming in vitro diagnostic tool for effectively identifying patients with food allergies. The microanalytical multiplexed assay was supported by compact disc technology (disc, reader, software) as a portable, simple, cost-effective sensing device. Biochip arrays containing a set of immobilized allergens were a feasible solution for detecting 12 food allergies in a single experiment and with a small serum volume. The results were excellent compared to the reference method and other technologies for detecting of specific immunoglobulin E.

Key words Protein array, Micro-immunoassay, Optical biosensor, Serum biomarker, Food allergy

1 Introduction

Advanced diagnostic methods require a highly sensitive, parallel analysis enabling the simultaneous detection of diverse biomarkers within a single assay facilitated by cost-effective portable technology [1]. Protein microarrays are pivotal in conducting multiplexed analyses across various diagnostic areas [2]. This approach is compact since each array of capture molecules immobilized occupies a small area (μm^2) and demands minimal sample and reagent volumes (nL-μL) for operation. Numerous researchers have devised chip assays in a miniaturized array layout, aiming to pinpoint specific biomarkers related to human diseases concurrently. However, a

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_2,

significant limitation of forward-phase arrays is the restricted availability of proteins in their recombinant or natural form.

Conversely, reverse-phase arrays represent a vital category within protein abundance microarrays [3], employing immobilized cellular or protein-based lysates [4]. Nitrocellulose is a prevalent substrate due to its high capacity for protein binding (e.g., 0.25 mg/mL total protein). These assays can adopt multipad or sector formats, encompassing multiple nitrocellulose pads on a single glass slide or 96-well plates [3]. In the realm of allergies, a proposed microarray relies on protein extracts derived from food components [5, 6]. Glass slides coated with a nitrocellulose polymer have been utilized to capture proteins through noncovalent means, using a four-color fluorescence scanner for detection.

The traditional approach to food allergies involves utilizing a forward-phase capturing array, wherein each allergen (isolated or recombinant protein) is immobilized on a glass slide [7]. Subsequently, the sample undergoes incubation on the chip, facilitating the recognition of specific biomarkers. Finally, a fluorescence scanner is commonly employed to detect the intensities of spots generated by antibodies stained with fluorophores. Alternative assay substrates, such as silicon chips coated with a functional polymer and three-dimensional gel-based microchips, have also been proposed [8, 9]. The ImmunoCAP ISAC (Immuno Solid-phase Allergen Chip, ThermoFisher Scientific) is a commercially available microarray designed for IgE antibody testing. This method is considered the gold standard in food allergy diagnoses from serum.

This study introduces a reliable, cost-effective assay to measure serum IgE levels using protein extracts and versatile digital technology (DVD). This innovative consumer electronics device holds the potential for creating compact biosensing systems tailored for point-of-care testing of various biomarkers [10, 11]. The methodology mirrors the steps of slide microarrays, employing polymeric discs as supports for conducting assays with minimal reagent volumes, with the disc drive serving as the imaging system. Optoelectronic technology has shown promise in clinical diagnostics, particularly in the immunorecognition of allergenic proteins present in food samples, for ensuring safety and quality control [12]. What distinguishes our present approach is the development of a method that quantifies sIgE in serum samples by combining the immobilization of allergenic proteins (forward phase) and food lysates (reverse phase) on the same disc [13]. The objective is to showcase the viability of versatile allergen microarrays capable of optically detecting a panel of biomarkers in minimally specialized clinical labs. To achieve a fully integrated assay on a disc accommodating multiple samples, various stages of this microimmunoassay have been devised, encompassing probe immobilization, sample incubation, washing, staining, and analysis. Consequently, the study seeks to assess whether utilizing protein extracts to identify and measure specific biomarkers in an array format offers advantages over existing methods reliant on highly purified proteins in a singular format.

2 Materials

2.1 Probes of Forward Phase Array

The pure proteins were provided by ALK-Abelló (Spain), Leti Pharma (Spain), and Roxall (Germany). The proteins were profilin, ovalbumin, beta-lactoglobulin, lipid transporter proteins from peach (LTP), and casein.

2.2 Probes of Reverse Phase Array

The proteins from food sources were provided by ALK-Abelló (Spain), Leti Pharma (Spain), Roxall (Germany), and Allergy Therapeutics (UK).

1. Protein extract of wheat (*Triticum aestivum*).
2. Protein extract of barley (*Hordeum vulgare*).
3. Protein extract of kiwi (*Actinidia deliciosa*).
4. Protein extract of oily fish.
5. Protein extract of prawn (*Parapenaeus longirostris*).
6. Protein extract of peach (*Prunus persica*).
7. Protein extract of sesame (*Sesamum indicum*).
8. Protein extract of cow's milk (*Bos taurus*).
9. Protein extract of chicken egg (*Gallus gallus*).
10. Protein extract of walnut (*Juglans regia*).
11. Protein extract of peanut (*Arachis hypogaea*).
12. Protein extract of pistachio (*Pistacia vera*).

2.3 Protein Array

1. Digital versatile discs (DVDs) (MPO Iberica, Madrid, Spain).
2. Spotting buffer: 50 mM carbonate buffer at pH 9.6 and 1% glycerol (v/v).
3. Incubation buffer: Phosphate-buffered saline (PBS), 0.8% NaCl, 0.02% KCl, 0.02% KH_2PO4, 0.3% Na_2HPO_4, pH 7.5.
4. Washing buffer: PBS with 0.05% Tween-20 (PBST).
5. Primary antibody: Monoclonal anti-human IgE antibody and mouse anti-human IgE monoclonal antibody (Ingenasa-Eurofins, Spain).
6. Secondary antibody: Polyclonal anti-human IgE antibody (pAb) conjugated to horseradish peroxidase (HRP) was employed (Dr. Fooke, Germany).
7. Colorimetric substrate: tetramethylbenzidine (TMB) substrate (ep(HS)TMB-mA, SDT GmbH, Germany).

8. The positive controls, human IgE (Abcam) at 10 ng/μL.
9. The negative control, human serum albumin at 10 ng/μL (HSA, Sigma-Aldrich).
10. Calibrators of IgEs: The third international standard for serum IgE, 11/234 (NIBSC, Hertfordshire, UK).

2.4 Instruments and Software

1. Liquid dispenser robot (Biodot AD1500, Irvine, CA).
2. Compact disc drive. The assays performed on digital versatile discs (DVD) were directly read by a DVD drive (LG DVD GSA-H42N, LG Electronics Inc., USA), which incorporated a data acquisition board model (DT9832A- 02-OEM; Data Translation, Marlboro, MA, USA).
3. Custom software for drive control and data acquisition [14].
4. ImageJ free-access software (National Institutes of Health, USA).

3 Methods

3.1 Preparation of DVD-Based Chip

1. Prepare discs by gentle ethanol washing, water rinsing, and drying by spinning for 1 min at 800 rpm.
2. Prepare dilutions of pure proteins or protein extracts in the spotting buffer.
3. Transfer the solutions to discs using the liquid dispenser robot at 25 °C and 80% relative humidity. Set the distance between flanking spots and ensure a reproducible spot diameter, for instance, 1.5 mm and 500 ± 10 μm, respectively. Figure 1 describes the array layout applied to the food allergy diagnosis.
4. Incubate discs overnight at room temperature, promoting the surface coating by passive adsorption on the DVD substrate (*see* **Note 1**). A blocking step is not required.

Probe	Target	Probe	Target
1	Wheat	8	Cow's milk
2	Barley	9	Egg
3	Kiwi	10	Walnut
4	Fish	11	Peanut
5	Prawn	12	Pistachio
6	Peach	T	Total IgE
7	Sesame	+	Positive control

Fig. 1 Array layout for food allergy diagnosis. The probes were spotted in a 6×7 format per array and 20 arrays per DVD. Each array has 3 replicate spots corresponding to 12 food extracts (three replicates per extract), anti-total IgE antibodies, and positive controls

5. Store the discs in a sealed box at 4 °C until use. Control experiments can confirm that the probes are active for at least 30 days.

3.2 Immunoassay Protocol for Biomarker Determination

1. Mix standards or serum (25 μL) with 0.5% Tween-20 (25 μL) and dispense on each disc array (14 samples per disc).
2. After incubation for 30 min at room temperature, wash the disc with PBST and water. Dry the discs by exposure to air.3. Dispense the HRP-labeled primary antibody in PBS (1/800 dilution) for 30 min, and wash the disc as before.
3. Perform colorimetric staining by directly dispensing TMB substrate dispensation, followed by a 10-min incubation period, and stop the reaction by washing with deionized water.

3.3 Reading and Data Processing

1. Acquire surface images by the modified DVD drive. Use custom software to control the reader and select scanning parameters: speed 8×, gain 25 dB, and processing diameter 250 mm (460 pixels per spot). The detection principle is described in Fig. 2 to generate an image for each microarray in jpeg format (*see* **Note 2**).
2. Quantify spot signals by an image analysis. Divide the image into a set of objects that identify each one inside a position of the array. Obtain the intensity of each spot by subtracting the local background from the mean signal value of the spot pixels.
3. Estimate biomarker concentrations from the calibration curve obtained from signals registered corresponding to calibrators. For allergy diagnosis, the calibrators are WHO standards (immunoglobulin E human serum) in a human control serum at different concentrations (0.3–100 kIU/L).

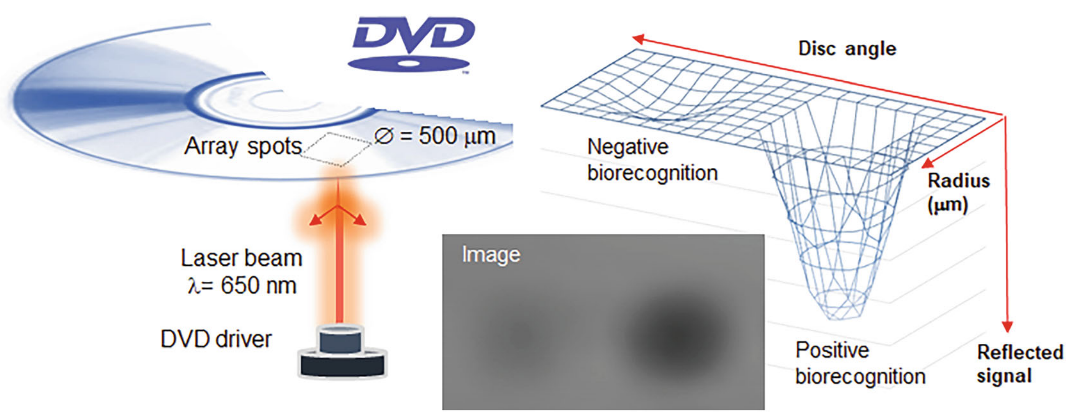


Fig. 2 Reading principle on the disc surface by the DVD reader. In the reflection mode, the intensity of the reflected beam changes in the presence of biorecognition elements on the disc surface. The graph represents the optical intensity depending on disc coordinates, generating a gray-scale image

3.4 Comparison of Pure Protein and Protein Extracts

1. Prepare the chips containing pure proteins or protein extracts as probes for recognizing the same biomarkers.
2. Obtain the optimal concentration of each coated probe selected by checkerboard titration assays (Fig. 3). Prepare serial dilutions to create a range of concentrations. Dispense the diluted solutions into the array rows and replicate them into the columns, creating a checkerboard pattern. Perform the assay using a reference positive sample.
3. Interpret the results to assess the effect of coating concentration on the spot signals. Select specific concentrations for each protein or extract because differences can be observed depending on the food ingredient. Reference values coating

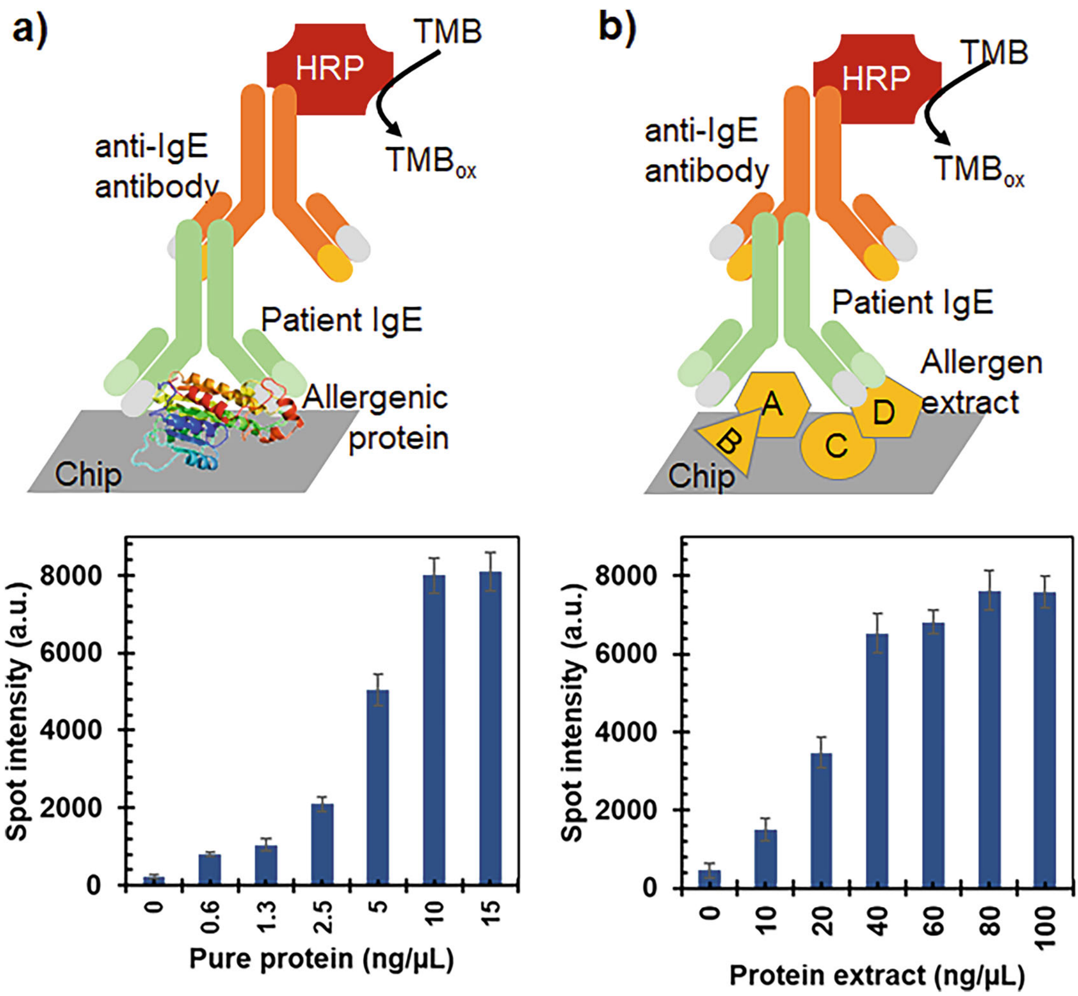


Fig. 3 Immobilization yield: (**a**) forward-phase array based on specific antibody or allergen protein (lipid transporter protein) and (**b**) reverse-phase array based on protein extract (wheat). Serum samples from allergic patients: peach (**a**) and wheat (**b**). Replicates: 5 (mean ± standard deviation)

concentrations are 10 ng/μL for pure proteins and 80 ng/μL (total protein) in the carbonate buffer.

4. Revise the results to confirm that optical disc-based technology is a high-throughput proteomics platform for reverse-phase arrays.

3.5 Estimation of Analytical Performances

1. Apply the following protocol to assess the method capability of detecting several specific biomarkers simultaneously (*see* **Note 3**).
2. Prepare discs with an array of immobilized probes, including replicates and controls. Obtain a series of standard samples with known concentrations of the target analyte, covering the expected range of concentrations. Also, generate pooled sera containing the target biomarkers from the mixtures of samples from the patients.
3. Reproducibility: Obtain assay precision from replicated assays on the same day and between different days. According to our experiments, suitable relative standard deviations are 2–7% (intra-day) and 10–25% (inter-day).
4. Qualitative analysis: Study the recognition profiles to find that spots presented a significantly different response from the background. The assay selectivity is satisfactory if the biomarker is specifically recognized only by their corresponding immobilized probes.
5. Quantitative analysis: Measure the signals that the assay generates for each standard. Confirm that the assay fits a linear relation between the mean spot intensities and the biomarker concentration for all probes. Evaluate the goodness-of-fit of the curve (regression coefficient > 99).
6. Sensitivity evaluation: Analyze biomarker dilutions in a control serum. Calculate the assay sensitivity, typically defined as the slope of the calibration curve. For the detection limit, obtain a set of blank samples that do not contain the target analyte. Perform the assay following the assay protocol and calculate the detection limit from the signal mean and standard deviation. Based on our experiments, the suitable detection limits are 0.30 ± 0.04 kIU/L (0.72 ± 0.10 ng/mL).
7. Multiplexing capability: Develop a microsystem capable of simultaneously determining multiple targets in a single run using a small serum volume. For this purpose, study the method initially with a low number of targets and then expand it to accommodate a higher number of targets by increasing the number of probes spotted on the same array. Apply statistical analysis to ensure that spot signals are comparable with both the t-test and F-test, as demonstrated in Fig. 4. The maximum

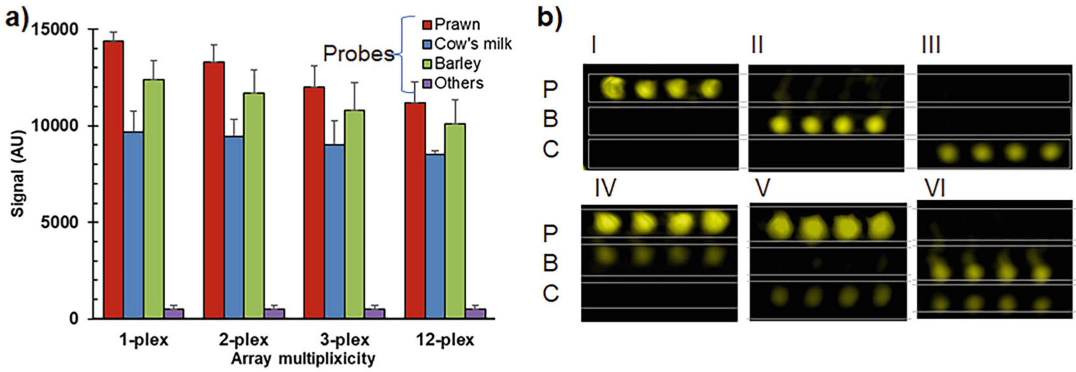


Fig. 4 (**a**) Evaluation of multiplexing capability based on spot intensities (mean ± standard deviation). (**b**) Array images for the 3-plex format. Probes: extracts from prawn (P), barley (B), and cow's milk (C), replicates = 4 spots. Samples classified depending on patient allergy: I, P; II, B; III, C; IV, P and B; V, P and C; VI, B and C

number of simultaneous assays until p-values greater than 0.05 are obtained.

3.6 Application to Patient Diagnostics

1. Examine the capability of the developed system to identify patients suffering from multiple food allergies.
2. Perform the DVD-based assay to detect which specific IgEs are present, incubating serum samples against the panel of allergens immobilized in a microarray. Figure 5 illustrates a design format example for 12 food allergies.
3. Evaluate the method performance, evaluating the response patterns. Confirm that the positive control spots are observed for all the patients. Review and identify the culprit food allergy by examining the protein extracts that yield signal intensities higher than the detection threshold.
4. Figure 6 depicts the identification of individuals showing positive responses for multiple extracts. In our pilot study (55 samples), 30 individuals were classified as positive for at least one allergen. Eleven food allergy types were found, with peach being the most detected food allergy (11 patients).
5. Validate the results by comparing the results with those obtained by a reference method. Confirm the strong agreement in classifying patient groups with ImmunoCAP, recognized as the gold standard in vitro diagnostic test for food allergies (*see* **Note 4**).
6. Confirm that this tool might help to prevent misdiagnosis of allergies and guide food avoidance, overcoming the performance of current technologies (*see* **Note 5**).

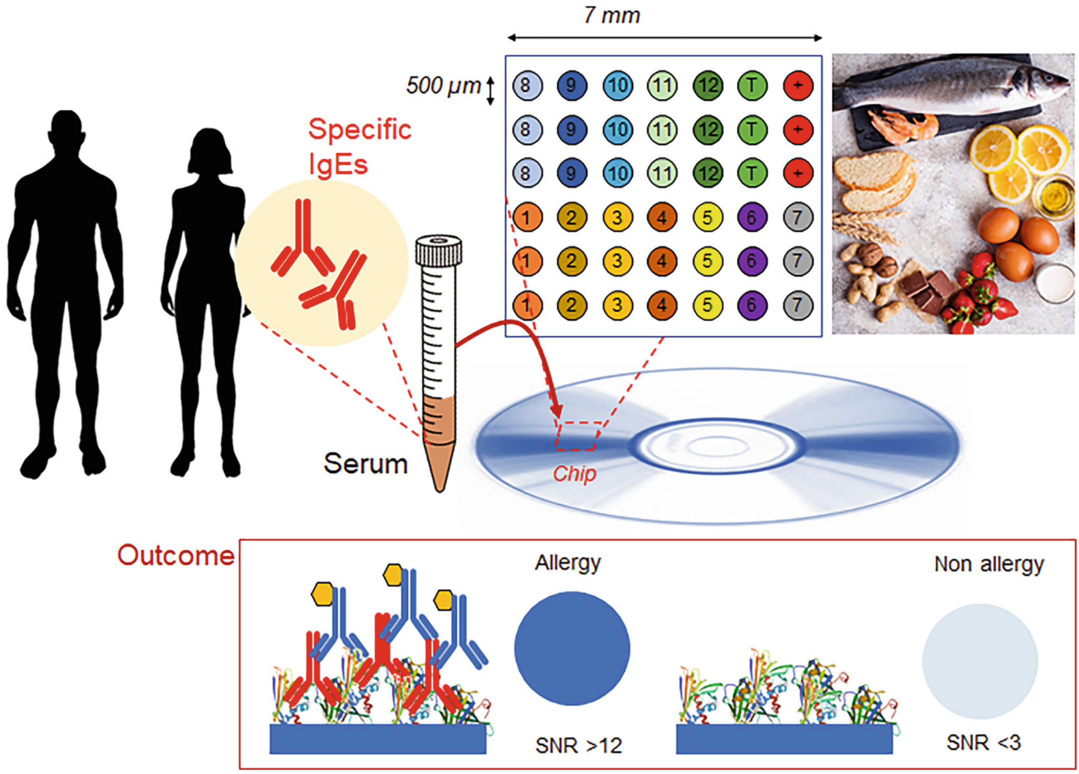
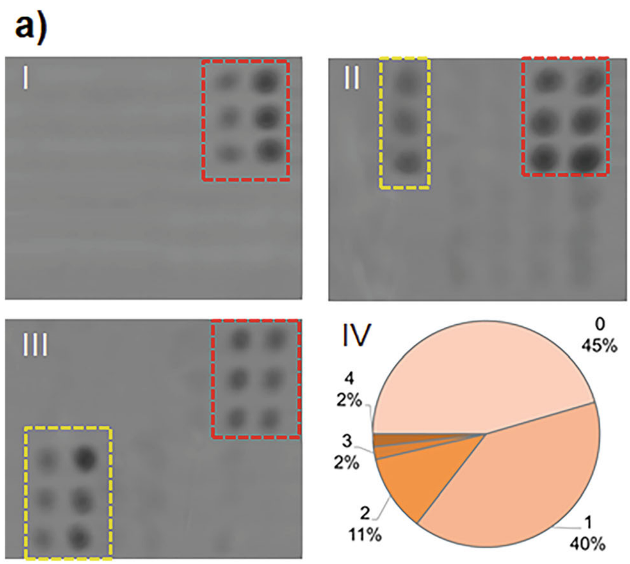


Fig. 5 Scheme of micro-immunoassay for the analysis of patient samples. Protein extracts (probes) are immobilized on the DVD surface in a microarray format. Insert indicates the optical transduction of a biorecognition event depending on the analyte concentration. SNR: signal-to-noise ratio

4 Notes

1. The surface chemistry that leads to probe immobilization was one of the most crucial aspects of the protein array architecture. In this study, the direct attachment of allergens by passive adsorption was approached by the native functional groups of proteins. Thus, the primary immobilization approach was based on a hydrophobic bonding between the polycarbonate surface and non-polar amino acid residues in proteins, such as alanine, valine, leucine, isoleucine, phenylalanine, tyrosine, tryptophan, proline, and methionine.
2. The selected detection principle is based on compact disc biosensing. This technology primarily relates to storing and reading digital data. A laser beam reads the stored data. This process can be adapted for bioanalytical purposes. The assay is performed on the disc surface, generating a solid product. During disc scanning, the laser hits the immunoreaction product, which modifies the reflection properties of the disc surface. The optical density of the immunoreaction product is



b)

	Allergy	Patient cases	%
1	Wheat	3	8.8
2	Barley	2	5.9
3	Kiwi	3	8.8
4	Fish	4	11.8
5	Prawn	8	23.5
6	Peach	11	32.4
7	Sesame	2	5.9
8	Cow-milk	4	11.8
9	Egg	5	14.7
10	Walnut	0	0.0
11	Peanut	3	8.8
12	Pistachio	1	2.9

Fig. 6 Results obtained analyzing 55 patients sera samples. (**a**) Images obtained for assays in the 12-plex format. Red rectangle: Control spots. Yellow rectangle: detected allergens. Panel I, nonallergic patient; Panel II, allergic patient to egg allergy; Panel III, allergic patient to wheat and barley; Panel IV, percentage of detected allergens per serum sample. (**b**) Number and percentage of positive cases for the studied food allergies

proportional to the analyte concentration. The result is a competitive technology, given its low cost and portability compared to other imaging technologies.

3. Multitarget assays can present cross-reactivity; the analytical performances associated with a particular biomarker could depend on the nature and number of targets that had been simultaneously assayed. Thus, assay features must be confirmed under different configurations before being applied. In the proof-of-concept test, we prepared tests capable of determining specific IgE associated with allergies to several foods. The method was extended from a panel of 3 allergens to 12 by following the same strategy of immobilizing protein extracts in a microarray assay.

4. The historical threshold (0.35 kIU/L) was used as a cut-off for positive and negative results in an in vitro diagnostic test called ImmunoCAP. This test is based on the principle of the enzyme-linked immunosorbent assay (ELISA), specifically designed for the quantification of IgE antibodies. Allergens are purified and immobilized onto a solid-phase carrier, usually small polymer beads. A small amount of the patients' blood serum, containing IgE antibodies, is mixed with these allergen-coated beads. A secondary antibody that recognizes human IgE antibodies is added. This secondary antibody is conjugated with a detectable enzyme or fluorophore. The amount of IgE antibodies present

Table 1
Comparison of biosensing methods employed for food-specific IgE determination

Biosensor type	LOD[a] (ng/mL)	Advantages	Drawbacks
Immunochemical	0.6–2.5	Conventional instruments Low cost Cost-effective High throughput	Low multiplexing capability Limited sensitivity Large sample amount Time-consuming
Electrochemical	0.005–5	Simple transducers Easy to miniaturized Possibility of mass production	Low number of studies Low multiplexing capability
Arrays	0.4–4	Multiplexing capability Low consumption of reagents Small sample amount High throughput	High-cost instrumentation
Nanomaterial based	20–80	Modulable properties Easier miniaturization Versatility	Limited sensitivity Partially explored field
Label-free	190	Low consumption of reagents Lower cross-reactivity Fast response	Limited sensitivity Sophisticated instrumentation Low multiplexing capability Few studies on serum
Microfluidics	1–2.5	High automatization Miniaturization Low consumption of reagents Small sample amount Portability Simple for end-users	Complicated design Difficult selection of materials Difficulties for reagent storage
Developed method	*0.7*	*Multiplexing capability* *Low-cost instrumentation* *Small sample amount* *Portability* *High throughput*	*Partial automatization*

[a]LOD: limit of detection (1 ng/mL ≡ 0.417 kIU/L)

is directly proportional to the intensity of the generated signal. This signal is quantified using a specialized instrument.

5. The developed technology can support the diagnosis of allergies that result from hidden allergens, i.e., ingredients not declared on product labels, and might unexpectedly reach end food products through several routes, such as accidental contamination. The advanced formats of analytical devices can be incorporated to improve method feasibility and working capacity; i.e., the number of samples per assay or assay automation. Table 1 summarizes relevant features of commercialized technologies for detecting specific IgEs. The novel in vitro method is a low-cost, ubiquitous, and portable device with an accurate,

sensitive, and simple assay (12 allergens, 25 μL of sample, 90 min). The results demonstrate that microanalytical systems based on allergen arrays can potentially diagnose multiple food allergies and can be easily implemented in primary care laboratory settings.

Acknowledgments

This work was supported by project WEAROPSENS PID2022-140653OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by "ERDF/EU". Financial support received from H2020 program (project COBIOPHAD, UE grant agreement No. 688448).

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Chapter 3

Bead-Based Suspension Array Using Peptides as Antigens to Capture and Identify Serum Antibodies

Fimme J. van der Wal, Marloes Heijne, and René P. Achterberg

Abstract

Bead-based suspension arrays built with multiple antigens enable detection and identification of specific antibodies in human and animal serum. Here we describe how to use synthetic peptides as antigens in a serological Luminex assay. Biotinylated peptides immobilized on avidin beads are used to capture serum antibodies, which in turn are detected using a generic immunoglobulin-binding protein or broad anti-bird antibodies in resp. mammalian and avian sera. Background is suppressed by pre-blocking sera and by using a blocking agent during antibody capture. By employing an internal negative control, serum-specific cutoffs can be determined and/or signals can be normalized. The provided protocols allow creating multiplex bead-based suspension arrays using peptides as antigens to capture and detect specific serum antibodies from mammalians, including humans, and avian species.

Key words Serum, Antibody detection, Multiplex, Bead-based suspension array, Luminex, Synthetic peptides, Serum-specific background, Host-independent

1 Introduction

Detection of antibodies (serology) against pathogens is used to establish if individuals have encountered a particular pathogen, or if certain pathogens are circulating in a population. In the context of animal production systems serology is performed as part of monitoring programs that aim to prevent spreading of contagious animal diseases or zoonotic diseases (e.g. [1, 2]).

For antibody detection, a number of methods can be used, which all involve antigens to which serum antibodies are generated during infection. For animal sera, next to hemagglutination inhibition and virus neutralization, ELISAs are universally being employed. Various formats exist [3], e.g. indirect ELISA, where immobilized antigens capture serum antibodies that are then detected by labelled secondary antibodies, or, among others, competitive ELISA in which serum antibodies and labelled antibodies

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_3,

compete for antigens that are immobilized in wells of a microplate through passive coating or by antibody-mediated specific capture.

ELISA is well-established technique, and for detection of antibodies against veterinary pathogens many ELISA kits are commercially available. In general, these are designed for detecting antibodies against one pathogen, and often are for use with sera from one host species. When multiple pathogens play a role in a disease syndrome [4], or a distinction between related pathogens is desirable [5], a technology that enables serological assays with multiple antigens in a multiplex format is ideal. Various multiplexing technologies are available (e.g. Ella, Meso Scale Discovery, Luminex). Focus of this chapter is on bead-based suspension arrays on the Luminex platform. Assays on this platform are relatively easy to setup, and the workflows that are required are very similar to those used for indirect ELISA.

The Luminex technology employs colour-coded bead sets where each bead set serves as the solid support carrying antigens, comparable to a well in indirect ELISA. By combining differently coloured bead sets, each carrying specific antigens, multiplex serology in one well becomes possible. For readout of assays, a Luminex instrument is used that examines each bead: beads (and hence the antigen) are identified by their colour, while presence of captured serum antibodies is investigated by detecting presence (or absence) of fluorescence from a labelled secondary antibody. By using paramagnetic beads, workflows similar to ELISA are possible. For Luminex assays (and ELISA) limitations concerning the host range can be alleviated by using broad secondary antibodies for avian hosts [6] or broad Ig-binding proteins for mammals [7]. Bead-based assays on this platform are possible in many different formats and are not limited to indirect antibody detection [8, 9].

The types of antigens used for antibody detection range from crude preparations of unpurified inactivated pathogens to purified components such as proteins or carbohydrates, recombinant proteins expressed in another host, or completely synthetic compounds like peptides. For common serology, the antigens used ideally should not cross-react because cross-reactivity can complicate diagnosis, as observed for Pestivirus in swine [10] and Flavivirus in horses [11] and humans [12]. This is especially true for indirect assays (ELISAs and Luminex) where the assay background can be higher than in competition assays [13]. In indirect antibody detection assays, often proteins are used as antigens. To diminish or avoid cross-reactivity with proteinaceous antigens, proteins can be trimmed down to small units that contain fewer epitopes, or synthetic peptides containing one linear epitope can be used [14], as shown for detection of serum antibodies against Classical Swine Fever Virus [15] and Zika Virus [16]. Once seroreactive peptides have been identified—which can be challenging but is not the subject of this chapter—they can be used to setup assays in any format, such as ELISA, planar microarrays, or bead-based suspension arrays (Luminex).

This chapter describes how to construct and perform bead-based suspension arrays for serological assays employing the xMAP technology from Luminex, using paramagnetic beads carrying immobilized seroreactive peptides. The protocols given are based on our published [5, 17, 18] and unpublished experiences using peptides as antigens in Luminex assays. Included are protocols that allow antibody detection independent of the host from which the serum originates, procedures to block assay background, and the use of an internal negative control to determine the serum-specific background. The latter can be used to calculate serum-specific cutoffs for all peptides, or for normalization of signals, without having to establish separate cutoffs for each peptide.

2 Materials

Use low-binding plastics, preferably as advised by the manufacturer of Luminex instruments and reagents [19].

2.1 Equipment

1. Magnet for microcentrifuge tubes, for example the Dynabeads MPC-S Magnetic Particle Concentrator.
2. Magnet for microplates, for example the LifeSep 96F magnetic separator unit.
3. End-over-end mixer, for example the HulaMixer Sample Mixer.
4. Luminex xMAP instrument (*see* **Note 1**).

2.2 Beads, Peptides, and Specialty Reagents

1. MagPlex-avidin microspheres (*see* **Note 2**).
2. Synthetic peptides with the following structure: a biotin separated from the N-terminal residue of the peptide by a spacer (five units of [2-(2-aminoethoxy) ethoxy] acetic acid), followed by the peptide sequence, and an amide at the C-terminal residue (*see* **Notes 3–6**).
3. Luminex xMAP blocking buffer (Prime Diagnostics, Wageningen, the Netherlands).
4. Goat Anti-Chicken IgG(H + L)-PE (Southern Biotech) (*see* **Note 7**).
5. Protein A/G (*see* **Note 8**).
6. Lightning-Link R-Phycoerythrin Conjugation Kit (Abcam) (*see* **Note 9**).

2.3 Buffers and Solutions

1. DMSO.
2. PBS: 0.01 M Phosphate Buffered Saline, pH 7.2.
3. PBS-BSA: 1% BSA in PBS. Add 1 g BSA gently to the meniscus of 100 mL PBS. Let the BSA dissolve slowly and without vortexing vigorously to avoid clumping. Stir gently by using a stirring bar at 5–10 rpm. Filter sterilize and store at 4 °C.
4. PBS-TBN: 0.1% BSA, 0.02% Tween-20, 0.05% sodium azide in PBS. Add 0.1 g BSA, 20 μL Tween-20 and 50 ug sodium azide to PBS. After dissolving filter sterilize and store at 4 °C.
5. PBS-T: 0.05% Tween-20 in PBS. Add 50 μL to 100 mL PBS, after dissolving completely filter sterilize and store at 4 °C.
6. Biotin stock: 1 mg/mL biotin in water.
7. Dilution Buffer: PBS + 0.05% Tween-20 and 10% Luminex xMAP blocking buffer. Add 50 μL Tween-20 and 10 mL Luminex xMAP blocking buffer to 90 mL PBS. Store at 4 °C.
8. Neutravidin stock: 10 mg/mL neutravidin in water. Dilute to 10 μg/mL neutravidin just before use.
9. PE-conjugated Protein A/G: conjugate Protein A/G with PE according to the manual from the Lightning-Link R-Phycoerythrin Conjugation Kit (*see* **Note 9**).
10. Detection Buffer: dilute the PE-conjugated secondary antibody or PE-conjugated Protein A/G in PBS-T in the desired concentration (*see* **Note 10**).

3 Methods

3.1 Preparing Peptides

Design a peptide panel that suits the research goal (*see* **Notes 11** and **12**) and obtain these with an N-terminal biotin separated from the peptide sequence by a spacer (*see* **Notes 3–6**). Level of purity of the peptides is not crucial but does affect seroreactivity (*see* **Note 13**). In addition, a similar compound that has the same composition as the peptides (i.e. biotin, spacer, peptide, C-terminal group e.g. amide) but without a real peptide moiety ('empty peptide'), can be used as internal negative control (INC; *see* Subheading 3.6).

1. Bring the lyophilized peptides to room temperature.
2. Dissolve the peptides according to the manufacturer's advice in the highest concentration possible.
3. Store the dissolved peptides at −20 °C (shelf life ~3 months, for longer storage (up to 5 years) store lyophilized at −20 °C).

3.2 Coupling Peptides to Beads

Biotinylated peptides are immobilized on paramagnetic beads carrying avidin. The protocol described here is in essence as described by the manufacturer of Luminex instruments and consumables

[19], with modifications [5]. Do this for each desired combination of a particular bead set and a specific peptide (and for the internal negative control) so that all peptides get their own dedicated bead set.

1. Resuspend the MagPlex-Avidin beads by vortexing approximately 20 s (*see* **Note 14**).
2. For each bead set transfer 2.0×10^5 (80 μL) beads to a microcentrifuge tube (*see* **Note 15**).
3. Pellet the beads from the liquid by placing the tube in a magnetic separator, for example a Dynabeads MPC-S. Wait for at least 1 min to allow the beads to settle against the wall of the tube. Carefully remove the liquid using a pipette.
4. Resuspend the pelleted beads in 500 μL PBS–BSA by vortexing for 20 s.
5. Dilute the dissolved biotinylated peptides to 1 mM in PBS–BSA. Add 2 μL of this dilution to the 500 μL beads (end concentration: 4×10^{-6} M peptide).
6. Couple the peptides to the beads by an incubating of 30 min at room temperature, whilst mixing end-over-end at 20 rpm using for example a HulaMixer (*see* **Note 16**).
7. Pellet the beads as described in **step 3**, and resuspend the beads in 500 μL PBS–TBN by vortexing for 20 s.
8. Repeat pelleting and resuspension from **step 7**.
9. Pellet the coupled beads (*see* **step 3**) and resuspend in 500 μL PBS-T.
10. Dilute the biotin stock (1 mg/mL in water) 10× in PBS-T and add 5 μL to the coupled beads (end concentration: 4×10^{-6} M biotin) (*see* **Note 17**).
11. Incubate 30 min at room temperature, whilst mixing end-over-end (*see* **step 6**).
12. Pellet the beads (*see* **step 3**), remove the liquid, and resuspend the beads in 100 μL PBS–TBN by vortexing for 20 s.
13. Store the beads in a refrigerator, protected against light (*see* **Note 14**).

3.3 Preparing Bead Mixes for Multiplexing

1. For each coupled bead set determine the number of beads per volume by diluting 1 μL of the coupled beads in 200 μL PBS-T. Divide this volume over two wells in a microplate (*see* **Note 14**) and use the Luminex instrument to count the number of beads present in 50 μL. Average the value from these two wells to calculate the bead concentration in 1 μL.
2. Calculate the number of beads needed for the number of assays to be performed, aim for 750 beads per bead set for each

sample. For example, for 200 samples, add 150,000 beads per bead set (using the bead concentration determined in **step 1**) to a microcentrifuge tube.

3. To the same tube, add the required number of beads of each bead set that will be part of the bead mix (*see* **Note 18**).
4. Pellet the beads of the bead mix using the magnetic separator and remove the liquid (*see* **step 3** from Subheading 3.2). Resuspend the pelleted bead mix in 200 μL PBS-TBN (i.e. 1 μL bead mix for each sample to be tested).
5. When preparing a run, for each sample to be tested add 1 μL bead mix to 50 μL Dilution Buffer. For 20 samples this will result in 1000 μL bead mix. Of this bead mix, 50 μL is used for each sample (*see* Subheading 3.5). Prepare a slightly larger volume as to ensure there is sufficient bead mix for each sample.

3.4 Preparing Sera

1. Dilute each serum 200× in Dilution Buffer (*see* **Note 19**). End volume is 200 μL.
2. Add 2 μL of a 10 μg/mL neutravidin solution (a 1000× diluted stock solution) to the 200 μL diluted serum to pre-block the sample (*see* **Note 19**).
3. Incubate 30 min at room temperature while shaking at ~600 rpm on a microplate shaker.
4. Proceed to the actual assay.

3.5 Assay Protocol

1. Mix 50 μL of the bead mix prepared in 3.3 with 50 μL of the pre-blocked diluted serum in a microplate (*see* **Note 20**). To prevent long exposure of beads to light (*see* **Note 14**) and to assure similar incubation times, first administer the diluted pre-blocked sera to each well, then add the bead mix to all wells.
2. Incubate 60 min at room temperature, while shaking at ~600 rpm on a microplate shaker.
3. Pellet the beads by placing the microplate at least 1 min on an appropriate magnet for microplates.
4. Use a (multichannel) pipet to remove the liquid from the wells, or keep the plate fixed on the magnet and manually throw away the liquid in a waste container, similar to removing liquid from a plate during washing steps when performing ELISA by hand.
5. Add 100 μL PBS-T to each well.
6. Repeat **steps 3–5** two more times for a total of three washes.
7. Add 100 μL of the Detection Buffer containing the optimal concentration of PE-labelled secondary antibody (Goat Anti-Chicken IgG(H + L)-PE (for bird sera) or PE-labelled Protein A/G (for mammalian sera) to each well (*see* **Note 10**).

8. Incubate 60 min at room temperature while shaking at ~600 rpm.
9. Wash 3× with 100 μL PBS, *see* **steps 3–5**.
10. Resuspend in 100 μL PBS-T.
11. Analyse on a Luminex instrument by counting a minimum of 50 beads per bead set (*see* **Note 21**). For the Luminex LX200 we use standard settings, i.e. analysis at room temperature, low PMT, minimal count of 50 or 100 beads per bead set. See the manual of your system for detailed programming information.

3.6 Data Analysis Including Using the Internal Negative Control

Luminex xPONENT software creates an csv (comma-separated values) file for each so-called 'batch', i.e. a set of samples analysed with a detection protocol in xPONENT. This csv file contains raw data produced during the analysis, such as bead counts (the number of beads of each bead set that have been investigated), MFI values (median fluorescence intensity, a measure for the amount of signal from the fluorescent label), and annotation of the samples. Similar software from other suppliers will have similar output capabilities. Differences in MFI values between bead sets indicate which antigens have captured serum antibodies. Data can be processed in Microsoft Excel to help interpretation of results for example by setting a rule-based cutoff (*see* **Note 22**). For bead-based serology using peptides, we assume all peptides give the same assay background. To make the interpretation of data from a peptide suspension array easy, an internal negative control can be used (*see* **Note 23**).

1. Determine for each serum the specific assay background. This is the signal on the bead set carrying the internal negative control. This value will differ for each serum (*see* **Note 23**).
2. Establish the serum-specific cutoff, i.e. the cutoff for all bead sets in a bead mix used to test a particular serum sample. This has to be done for each serum by taking a multiple of the serum-specific assay background. The multiplication of the observed background should be based on observations with known seropositive and seronegative sera (*see* **Note 24**).
3. Evaluate the serostatus of all tested sera by applying the serum-specific cutoffs (*see* **Note 25**).

4 Notes

1. Different Luminex instruments with different multiplexing capabilities are available. The currently available systems are flow cytometers that investigate each bead: a red laser identifies the colour of the beads and therewith the antigen, a green laser

investigates the amount of labelled (phycoerythrin) secondary antibody present on that particular bead. The LX200 system can distinguish up to 100 bead sets, while the Flexmap 3D system can distinguish 500 bead sets. The new Intelliflex system is able to recognize two different fluorescent labels, enabling for example the distinction between IgG and IgM in the same assay.

2. When selecting bead sets, compatibility with the various available instruments needs to be taken into account, especially when a universally applicable assay is desired.

3. There is no rule for the length of peptides. In bead-based serology for *Chlamydia* we have used known seroreactive peptides varying in length from 16 to 40 amino acids [5]. Seroreactivity of these peptides was previously shown by a research group that tested large numbers of peptides in ELISA, and subsequently a subset in a peptide microarray [20–22]. In unpublished work on CSFV, we found that in ELISA 20-mers (with predicted linear epitopes) consistently performed better than 15-mers or 10-mers. In our unpublished work on influenza virus hardly any of the 20-mers with predicted linear epitopes were seroreactive. In fact, it is estimated that only a small percentage of antibodies is aimed against linear epitopes [23]. In other work, we observed that (known seroreactive) peptides of 9, 13, and 14 amino acids, immobilized on beads, were recognized perfectly well by the corresponding antibodies [18]. So, prior knowledge of seroreactivity is more important than length.

4. Biotinylated peptides have the advantage of having the group used for bead-immobilization at one of the termini. With more traditional carbodiimide coupling from free amines, i.e. the N-terminus and lysines, epitopes that contain lysines may be destroyed.

5. Biotinylated molecules are easy to immobilize on beads carrying avidin. Routinely N-terminal biotinylation is used, but C-terminal biotinylation is also possible, e.g. by addition of a biotinylated lysine residue at the C-terminus. In our experience, if a known seroreactive peptide is used, immobilization on beads through the N-terminus results in beads that are capable to capture antibodies. However, it may depend on the position of the epitope in a peptide if accessibility and hence seroreactivity is affected by immobilization. If seroreactive peptides are identified by ELISA using peptides that are immobilized from the N-terminus, it is likely the same approach will work with beads.

6. Indications are that longer spacers can result in higher signals [17, 18]. Since these findings, we always use a long spacer, i.e. five units of [2-(2-aminoethoxy) ethoxy] acetic acid (AEEA), in which one AEEA unit is maximal 9.6 Å long.
7. For indirect ELISAs and analogous Luminex assays, a secondary antibody is required. Often a host-specific secondary antibody is used. For birds, we use broad secondary antibodies that are reactive against antibodies of at least chickens, turkeys, pheasants, guinea fowl, ducks, and swans [6].
8. For mammalian species, a generic Ig-binding protein can be used instead of a secondary antibody, such as Protein A/G that recognizes IgG of many species [7, 24, 25] (and IgM of some species). For detection of IgM and IgA specific secondary antibodies would be required, or if no distinction is needed, an assay could be developed that does not require secondary antibodies, such as an analogue of a double-antigen sandwich ELISA [26].
9. Phycoerythrin (PE) is a fluorophore that is suitable for use in Luminex instruments. When antibodies or Protein A/G needed for an assay are not available labelled with PE, commercial labelling kits that require only limited hands-on time can be used, such as the Lightning-Link antibody labelling kits. Since these kits are normally optimised for conjugation to antibodies, differences in molecular weights of the target molecules need to be taken into account.
10. The optimal dilution of PE-labelled secondary antibody or Protein A/G has to be determined for each assay. It will depend on various factors like the source of the antibody or the labelling efficiency. In our experience, the optimum is somewhere in the range of 1000 to 10.000× dilution.
11. Depending on the goal of the desired serological test, a peptide panel can consist of peptides to capture antibodies against multiple pathogenic species, or a panel can consist of multiple peptides derived from one pathogenic species (multiple targets, or same target from multiple strains/serotypes).
12. A peptide panel can contain known seroreactive peptides, or peptides that need to be screened for seroreactivity, or a mix of both. As bead-based assays are flexible and can be assembled prior to each assay, over the course of time beads with non-functional peptides can be omitted and beads with other peptides can be added.
13. Suppliers may provide a service where multiple peptides are synthesized on a small scale, without purification after synthesis. These peptides are much cheaper and do function in antibody capture, but less efficiently [18]. The amount acquired may be sufficient to investigate their seroreactivity in a

suspension array, but as signals with purified peptides are higher, for a final assay purified peptides (produced at a larger scale) may be preferred.

14. Luminex beads can photobleach when they are prolongedly exposed to light. This should be taken into account at each step in all protocols given here. It is advised to keep beads protected from light as much as possible, e.g. during storage and incubation. When the beads are in microcentrifuge tubes, wrapping the tubes in aluminium foil will protect them sufficiently. During the incubation steps of the actual assay, beads in plates can be protected by using black microplates and covers. Plates can also be wrapped in aluminium foil. Beads can be stored in the fridge in amber coloured microcentrifuge tubes. During handling of beads, exposure to light is not a problem.
15. This protocol is written for the production of enough beads for a little more than 200 assays, assuming a bead-recovery of approx. 80% ($\frac{200000*80\%}{750} \approx 213$). If a bigger batch of beads is needed, the described protocol can be upscaled by multiplying relevant volumes, keeping the concentrations of reagents as stated.
16. For end-over-end incubation it is important to rotate at a speed that is high enough to allow the beads to disperse through the liquid. Too fast rotation needs to be avoided as it will cause pelleting of beads due to the centrifugal force.
17. A biotin block of potentially free avidins is performed as a precaution to prevent migration of peptides from one bead set to another in a multiplex assay, a phenomenon described in literature [27]. However, in pilot experiments we never found evidence for immobilized peptides moving to another bead set. In singleplex assays however we did see that a biotin block can improve signals slightly [17]; the reason for this is unclear.
18. In the research/development phase, we often use a limited number of bead sets and design parallel multiplex assays which use the same bead sets, but loaded with different peptides. For example, if 27 peptides are to be tested, three parallel 10-plexes can be set up for which the same 10 bead sets are used, i.e. 9 bead sets with peptides and 1 bead set with the internal negative control. By doing so, only 10 different bead sets have to be purchased instead of 28 (27 peptides +1 internal control) thereby reducing costs. When parallel multiplex assays are performed, each assay with a bead mix is performed in a different well. When the internal negative control is incorporated in each bead mix, the average of the parallel assays can be used for determining the serum-specific background of each serum [5].

19. Both pre-treatment of sera with neutravidin and presence of Luminex xMAP blocking buffer present in Dilution Buffer help in reducing assay background. In Fig. 1 we present two examples that show the effect of neutravidin and the Luminex xMAP blocking buffer in a Luminex assay. For both sera non-seroreactive peptides were investigated, using the protocols described here.
20. In this setup serum is diluted 1:400 during incubation with beads. Use of more concentrated serum can be necessary when signals are low, but too concentrated serum can lead to a prozone effect, causing low signals despite high concentration of serum antibodies. In addition, background at low serum dilutions can also be high and obscure interpretation of low signals. Therefore it is advisable to find a dilution that is useable for all sera, by looking at the ratio between positive and negative signals (P/N ratio). In ELISA this is done with a positive and a negative sample, in a bead-based multiplex assay with peptides as antigens, this can be done for each bead set within a well by dividing signals on peptide beads by the signal on the INC bead ('relative signal', *see* **Note 23**). In Fig. 2, an example is shown in which the effect of serum dilution on the raw signal seems limited, but the effect on the relative signal is significant. Ideally, a large set of characterized samples is tested to investigate the validity of choices made.
21. In these protocols, we aim to start with approximately 750 beads per bead set per well. After investigating a minimum of 50 to 100 beads per set for the amount of fluorescent label per bead, the system calculates the median fluorescent intensity (MFI) for each bead set. MFI values are in fact the assay results (cf. OD in ELISA) for each bead set in the bead mix.
22. In this chapter, we describe the use of an internal negative control for establishing cutoffs, but other ways of setting cutoffs are also possible, depending on the antigens used, the purpose of the assay and the availability of known samples (tested for example with a golden standard assay). Traditionally, for ELISA cutoffs are established based on for example (i) observations with known negative sera, where the average optical density (OD) plus 2 or 3 times the standard deviation (SD) is taken as cutoff, or (ii) the presence of two distinct populations, with a low OD (seronegative) and high OD (seropositive), allowing a cutoff between these two groups, or (iii) by testing known seropositives and seronegatives and analyse resulting data by receiver operating characteristic (ROC) analysis [3, 19]. Whatever procedure is used, there are likely to be false-negatives and false-positives. Consequences of incorrect results may affect the choice for a cutoff. For example if false negatives are an issue (e.g. for Trichinella in pigs [24]),

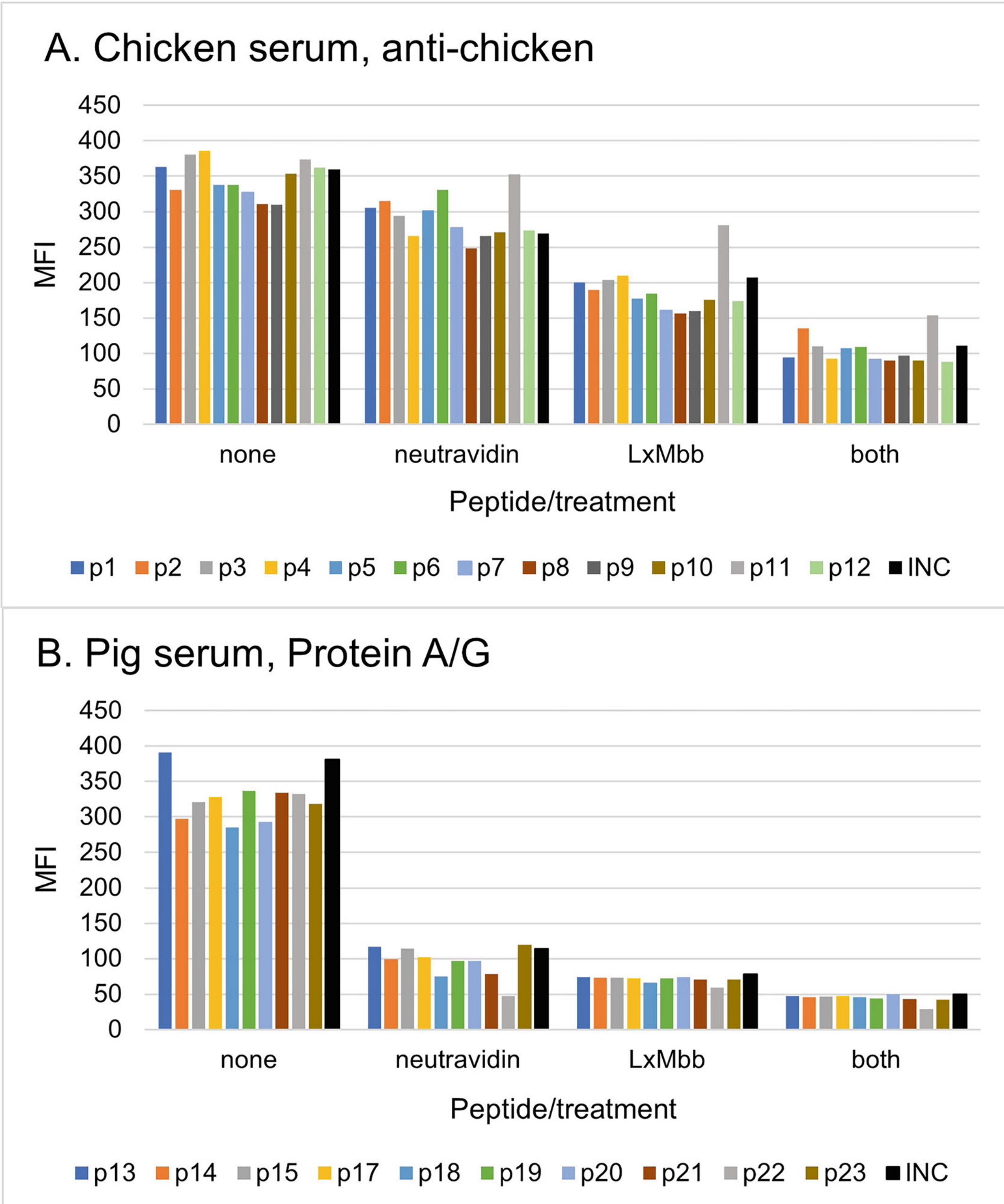

Fig. 1 Effect of blocking strategies on the assay background. Pre-treating sera and the presence of a blocking agent during capture of serum antibodies reduces the background that is detected on (non-seroreactive) peptides immobilized on beads. Experiments were done with chicken serum (seronegative for influenza A virus) and pig serum (seropositive for influenza virus H1N1) with resp. anti-bird or Protein A/G as detector. No blocking was performed (none) or blocking was performed with neutravidin, Luminex xMAP blocking buffer (LxMbb), or a combination of both. Two series of peptides that are not seroreactive were investigated (p1-p12 and p13-p23)*. The internal negative control, i.e. the 'empty peptide' (INC) was tested with both sera, and is in all cases presented as a black bar

lower cutoffs can be chosen based on an ROC analysis, with as a result no false negatives, but more false positives. In case of Trichinella in pigs, (false) seropositives are verified with an appropriate confirmatory test. For bead-based serology, we have also used another approach where, within a multiplex assay, a particular number of the bead sets with the lowest signals where regarded as negative (no serum antibodies captured) and used to calculate sample specific cutoffs (average MFI + 5 times SD); here we assumed that all antigens would have a similar background as they were variants of the same proteins, produced and purified with identical procedures [6]. In other work using five different carbohydrates as antigens, background hugely differed for each antigen. This required an alternative approach: cutoffs were determined using *estimated* numbers of seronegatives based on historic data, in combination with ROC analyses [28].

23. Using an internal negative control allows the detection of the background that is specific for a sample, within the same reaction mixture and in the same well as all other signals on the other bead sets of the bead mix. This makes it a true internal control that can be used to calculate the background that is specific for that sample. The internal negative control (INC) is a bead set that is conjugated with a non-reactive peptide, or an 'empty' peptide. This is a compound consisting of the same elements as the peptides (N-terminal biotin separated from the peptide by a spacer, with a C-terminal amide group) but without a peptide sequence. The signal on the INC bead represents the serum-specific background. This value can be used to calculate signal to noise ratios (S/N ratio or relative signal: $\frac{\text{MFI of peptide bead}}{\text{MFI of INC bead}}$), serum-specific cutoffs (a multiple of the serum-specific background), or to normalize data (subtraction of serum-specific cutoff (or background) from signals on peptide beads). When parallel multiplex assays are performed with a serum, the background of that serum can be established by averaging the MFI acquired on the INC beads present in the

Fig. 1 (continued) For chicken serum (panel A), pre-blocking sera with neutravidin modestly reduced background. The effect of LxMbb halved background, whereas the combination further reduced background to around 100 MFI. For pig serum (panel B), both pre-blocking of sera with neutravidin and the presence of LxMbb reduced background to, or slightly under, 100 MFI. The combination of both neutravidin and LxMbb brought the assay background further down to ca. 50 MFI. The strong background reduction by neutravidin treatment of sera, when Protein A/G is used for detection of captured antibodies, is consistent with earlier observations [17]

* Peptides were derived from the nucleoprotein of Influenza A virus and are part of a larger set of peptides (unpublished) predicted to contain residues that are part of linear epitopes using BepiPred [29] (of which updated versions have become available)

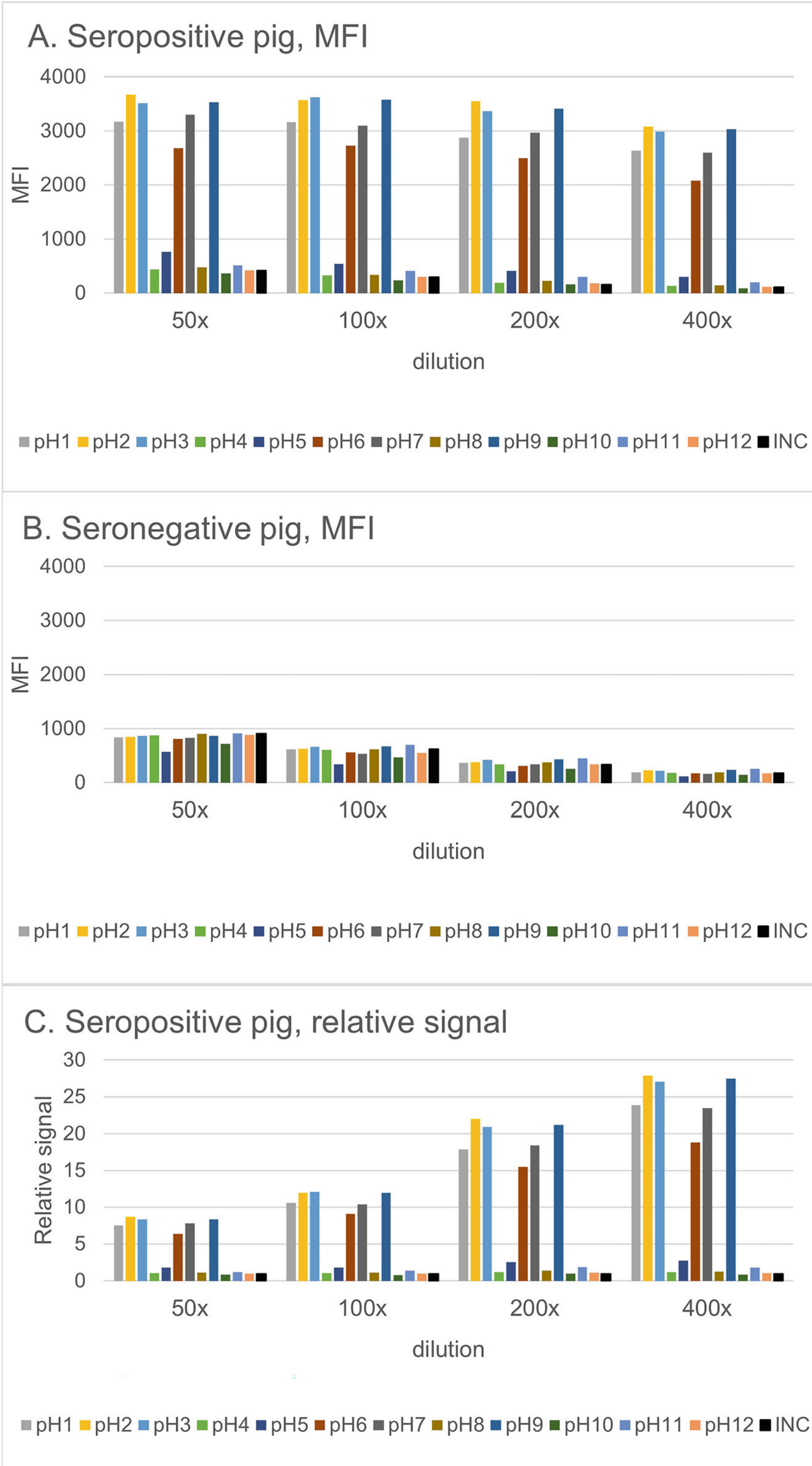

Fig. 2 Effect of serum dilution on raw signal and relative signal. Sera of pigs seropositive or seronegative for Hepatitis E virus (HEV) were tested in different dilutions, using PE-Protein A/G (1:1000) as detector for serum antibodies captured on immobilized peptides (pH 1–12)* and the internal negative control (INC). Of the

different bead mixes that were tested in parallel with that particular serum [5].

24. In earlier work on bead-based serology using peptides as antigens, we observed that three times the background worked well as cutoff for all control sera that were available at the time, even from different animal species [5]. For each new assay, this needs to be evaluated.
25. Serum-specific cutoff values can also be used to normalize data acquired for one serum by simply subtracting the cutoff value from the signals acquired on all peptide bead sets (tested in the same assay with the same serum). This will result in a virtual cutoff of 0 for each serum and enables presentation of results from different sera in one graph.

Acknowledgments

Thanks to Aart Davidse, Alex Bossers, Balazs Brankovics, Bernie Moonen, Dre Kampfraath, Renate Hakze-van der Honing, Ruth Bossers-de Vries, and Wim van der Poel for sera and identification of peptides.

Funding

This work was supported by the Dutch Ministry of Agriculture, Nature and Food Quality (Grant WOT-01-002-039), and in part by ERRAZE@WUR.

Fig. 2 (continued) 12 peptides tested, MFIs on 6 peptides are high at all dilutions for a seropositive pig, and decrease slightly at higher dilutions (panel A). Signals on the non-seroreactive peptides and the INC also decrease with increasing dilutions. The seronegative pig serum does not result in high MFIs on the 6 seroreactive peptides (panel B), and signals decrease with increasing dilutions. In contrast to raw signals (MFI), relative signals (MFI on peptide bead sets divided by MFI on the INC bead set) increase at increasing serum dilution (panel C)

* Peptides were derived from ORF2 of Hepatitis E virus (HEV) (unpublished); 20-mers were selected based on sequence differences of recognized genotypes. A subset of these peptides is seroreactive

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Chapter 4

On-Array Citrullination of Protein Microarrays

Jacob Skallerup, Thomas B. G. Poulsen, and Allan Stensballe

Abstract

Protein microarrays are a powerful tool for investigating autoantigens that drive autoimmune diseases. However, in many cases, autoantibody reactivity is directed against slightly modified proteins, generating neo-autoantigens, rather than targeting native antigens. Citrullination, a post-translational modification, is a key example, playing a central role in the pathogenesis of rheumatoid arthritis (RA). Excessive activity of peptidylarginine deiminases (PAD) leads to the formation of neo-autoantigens that trigger an autoimmune response. This chapter describes a detailed protocol for on-array citrullination of protein microarrays. By incubating the arrays with a PAD-containing buffer, the protocol attempts to replicate the formation of citrullinated autoantigens that occurs in vivo and contributes to RA. This approach enables high-throughput identification of citrullinated autoantigens, providing a platform to identify novel autoantigens and explore their role in RA pathology.

Key words Anti-citrullinated protein antibody (ACPA), Autoantibody profiling, Citrullination, Citrulline, Microarray, Neo-antigen, Neo-autoantigen, Peptidylarginine deiminase (PAD), Protein microarray, Rheumatoid arthritis

1 Introduction

In this chapter, we present a protocol for on-array citrullination of protein microarrays. This protocol is optimized using i-Ome Discovery microarrays (Sengenics, Singapore), but we expect it to be compatible with other protein microarray platforms as well. The specific microarray used in this protocol, contains 1631 proteins spotted in quadruplicate across the slide. Each protein on the microarray is expressed in its native state, and remains functional. This is accomplished using KREX technology, in which the proteins are tagged with a biotin-carboxyl carrier protein, ensuring that only proteins that are correctly folded are immobilized on the microarrays, *see* Fig. 1 [1–3].

Protein microarrays provide a powerful platform to study antigen-antibody interactions relevant to various diseases with an autoimmune aspect [4]. These arrays can feature thousands of

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_4,

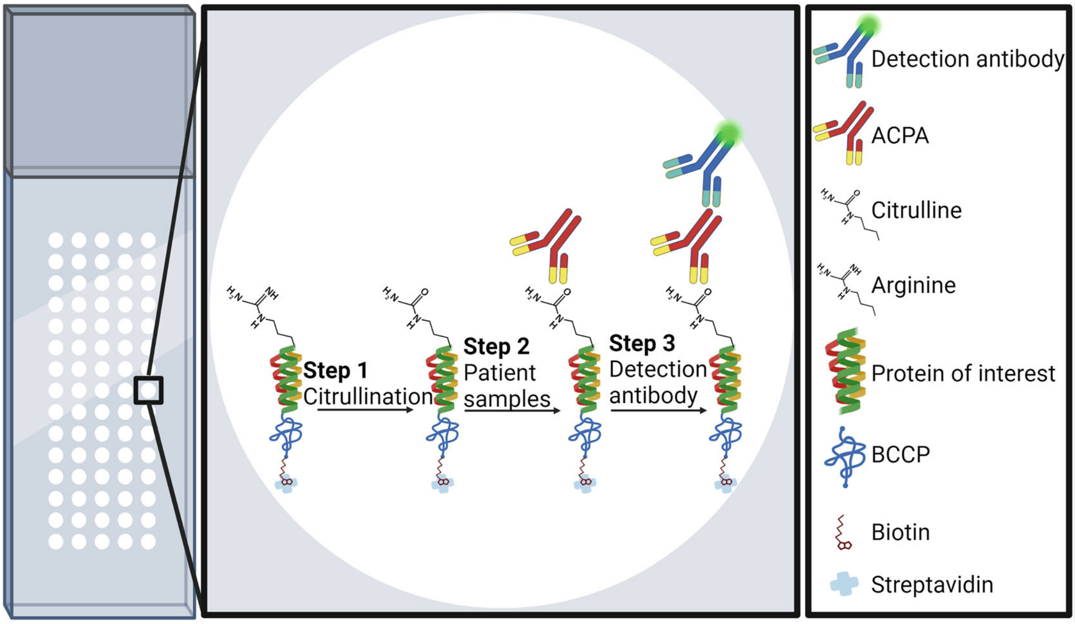

Fig. 1 Overview of the protocol for on-array citrullination of protein microarrays. The specific microarrays used for this protocol contain proteins that are only immobilized on the microarray if they are correctly folded. This is achieved using KREX technology. Each protein of interest is linked to a biotin carboxyl-carrier protein (BCCP, as shown in the figure). The BCCP can only bind biotin if it is properly folded, and only then can the biotin bind to the streptavidin-coated microarray surface, ensuring that only correctly folded proteins are immobilized. In **step 1**, proteins are citrullinated by incubation with a citrullination buffer containing PAD enzymes. In **step 2**, the citrullinated protein arrays are incubated with plasma or serum from rheumatoid arthritis (RA) patients containing anti-citrullinated protein antibodies (ACPAs, as shown in the figure). Finally in **step 3**, the antibodies in the patient sample that have bound to the immobilized proteins on the microarray are detected using a fluorescently labelled anti-human antibody, after which the microarrays are ready to be scanned

different proteins, making them ideal for broad interaction discovery studies. Alternatively, they can focus on a smaller set of proteins for validation purposes. This approach allows more samples to be tested per microarray, thus maintaining high throughput of the method despite a narrower focus [4]. Figure 3b shows two examples of hybridization cassettes that subdivide microarrays into smaller sections, enabling the screening of 16 samples (upper part of the image) or 8 samples (lower part of the image) on a single microarray.

While the nature of microarrays as a platform makes it ideal for the study of autoantigens in autoimmune diseases, in many cases, autoreactivity is not driven by an immune reaction towards native antigens but rather against neo-autoantigens [5, 6]. These neo-autoantigens can arise from post-translational modifications, such as citrullination, causing the immune system to mistakenly recognize them as foreign and facilitate an autoimmune response [6]. A prime example of this is rheumatoid arthritis (RA), an autoimmune disease that primarily affects the synovial membrane in joints but with the potential of multi-organ involvement [7]. In RA, one of the key drivers of disease development is believed to be excessive citrullination, a post-translational modification that causes modified proteins to behave as neo-autoantigens [6, 7]. Citrullination related to RA onset is thought to result from a complex interplay between genetic factors and environmental triggers, such as smoking and infection with *Porphyromonas gingivalis* or *Aggregatibacter actinomycetemcomitans* [6, 7].

In vivo, citrullination is performed by peptidylarginine deiminases (PADs). PADs are a family of five enzymes that convert arginine residues on proteins into citrulline in a calcium-dependent reaction. Citrullination is part of normal physiological processes such as gene regulation by modifying histones and as a result altering accessibility of genes for transcription. Especially relevant in this context, PADs have been shown to play an important role in altering the transcriptional activity of genes involved in neutrophil extracellular trap (NET) formation by decondensing chromatin [6, 8]. However, as mentioned, excessive PAD activity may cause formation of neo-autoantigens, an essential part of the pathogenesis of RA [6, 7].

In this protocol, we aim to replicate the process of citrullination in vitro by citrullinating already-printed protein microarrays. An overview of the entire protocol can be seen in Fig. 1. We achieve this by subjecting the microarrays to a PAD containing buffer. We use PAD2 or PAD4 due to their implication in autoimmune diseases, specifically RA [6]. PAD2 is widely expressed in various tissues, while PAD4 is mostly found in neutrophils, playing a role in NET formation, as mentioned earlier [6, 8]. To illustrate the effectiveness of this protocol and to highlight the relevance of citrullination in the study of RA pathology, Fig. 2 showcases the increase in reactive targets when screening RA patient plasma against citrullinated microarrays as compared to untreated microarrays. As one would expect, this increase is most prominent for patients that are anti-cyclic citrullinated peptide (anti-CCP) positive.

2 Materials

1. Protein microarray.
2. MilliQ water.
3. 10× Phosphate Buffered Saline (PBS).
4. Microarray reaction tray.
5. Pipette controller and 5 mL serological pipettes.
6. Laboratory aspirator.

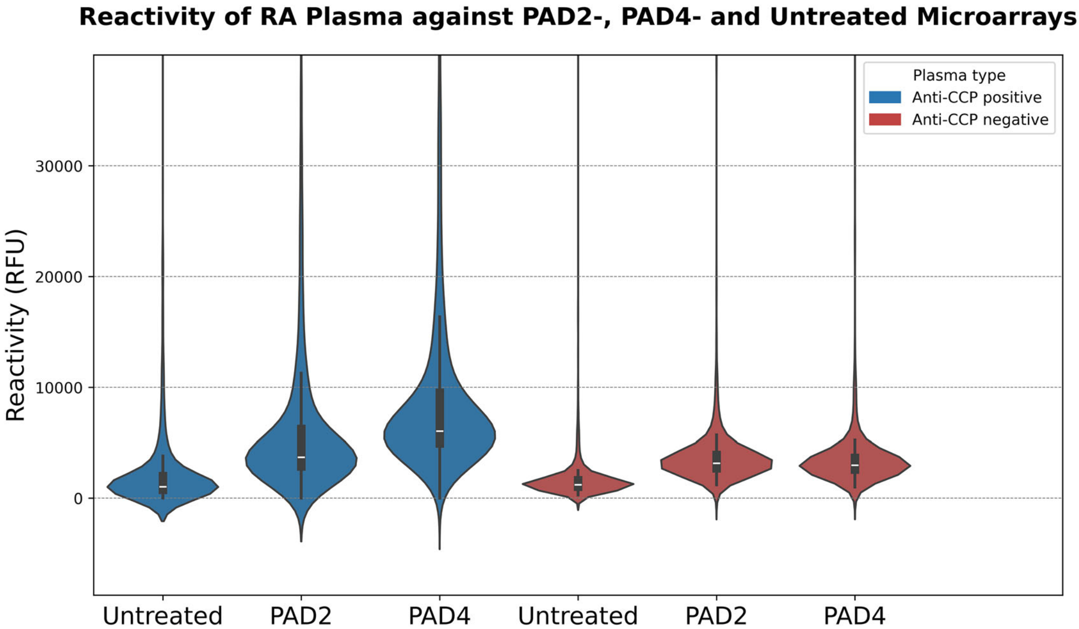

Fig. 2 Violin plot displaying the distribution of autoantibody reactivity, measured in relative fluorescence units, across six microarrays tested with plasma pools from two groups of rheumatoid arthritis (RA) patients: anti-cyclic citrullinated peptide (anti-CCP) positive (blue) and anti-CCP negative (red). The y-axis is capped at 40,000 RFU to improve readability, though a few data points per array reach the saturation limit of 65,000 RFU. Each of the two plasma pools were tested against microarrays treated in three different conditions: PAD2-treated, PAD4-treated, and untreated. The untreated microarrays were incubated in citrullination buffer for 3 h, similar to the treated arrays, but without enzyme in the buffer. This figure highlights the increased autoantibody reactivity in RA samples, particularly in the anti-CCP positive group, against the citrullinated microarrays, suggesting the potential for enhanced biological relevance when incorporating citrullination in RA investigations. All intensity data were normalized using the absorbance of standard control spots present on each microarray

7. Pap jars.
8. 15 mL conical tubes.
9. 50 mL conical tubes.
10. Orbital shaker.
11. Centrifuge (able to handle 50 mL conical tubes at 1200× *g*).
12. Centrifuge (able to handle plasma or serum sample tubes at 13000× *g*).
13. Serum Albumin Buffer (SAB): 0.1% Triton X-100, 0.1% Bovine Serum Albumin (BSA) in 1× PBS.
14. Citrullination buffer: 1.2 μg/mL PAD2 or PAD4 enzyme (Cayman Chemicals, Ann Arbor, MI, USA), 1 mM dithiothreitol (DTT), 10 mM calcium chloride ($CaCl_2$), 100 mM Tris–hydrochloride (Tris–HCl).
15. Polyclonal rabbit anti-human IgG (Dako, Santa Clara, CA, USA), conjugated to Cy3 (GE Healthcare, Chicago, Ill, USA).

3 Methods

In this section, the method for citrullination of *one* protein microarray is described, however, every step can be scaled up and done for multiple microarrays simultaneously (*see* **Note 1**).

Important
The microarrays are light-sensitive throughout the entire protocol. Even before the addition of detection antibodies, there are light-sensitive control spots. Hence, the microarray should be kept dark as much as possible by always keeping the microarray reaction tray wrapped in foil.

3.1 Preparation and Citrullination of the Microarray

1. Prepare the microarray reaction tray for the first cleaning cycle by filling a chamber in the reaction tray with cold 1× PBS, see reaction tray in Fig. 3d (for performing multi-array experiments, *see* **Note 1** describing the use of a slide-staining dish with a rack, Fig. 3c).
2. Remove the microarray from storage pap jar, *see* Fig. 3a, let storage buffer drip off the array as much as possible and onto a collection tissue or cloth. Gently scrape the excess buffer collected towards the bottom edge of the array by running the edge along the rim of the pap jar.
3. Place the microarray in the slot of the reaction tray containing cold PBS, make sure the proteins are facing up (*see* **Note 2**). Agitate the reaction tray a little by hand and make sure the buffer is floating across the upper surface microarray (*see* **Note 3**). Incubate for 5 min at room temperature on an orbital shaker, shaking at 50 RPM.
4. Repeat the above washing step once, carefully aspirating and discarding the PBS already in the reaction tray using a pipette controller (Fig. 3e) or a laboratory aspirator (Fig. 3f), *see* **Note 4**. Carefully dispense 4 mL of fresh PBS into the reaction tray using a pipette controller. Again, shake for 5 min on the orbital shaker at room temperature and 50 RPM.
5. Wash the microarray by adding 4 mL of SAB buffer to the reaction tray, shake for 5 min at 50 RPM on an orbital shaker at room temperature. Repeat this step twice in total.
6. Dry the microarray by transferring it to a 50 mL conical tube (the microarray should be upright in the tube), and centrifuge at 1200× *g* for 2 min at room temperature. *See* **Notes 5** and **6** for tips on easy removal of the microarray from the reaction tray and the conical tube after centrifugation.

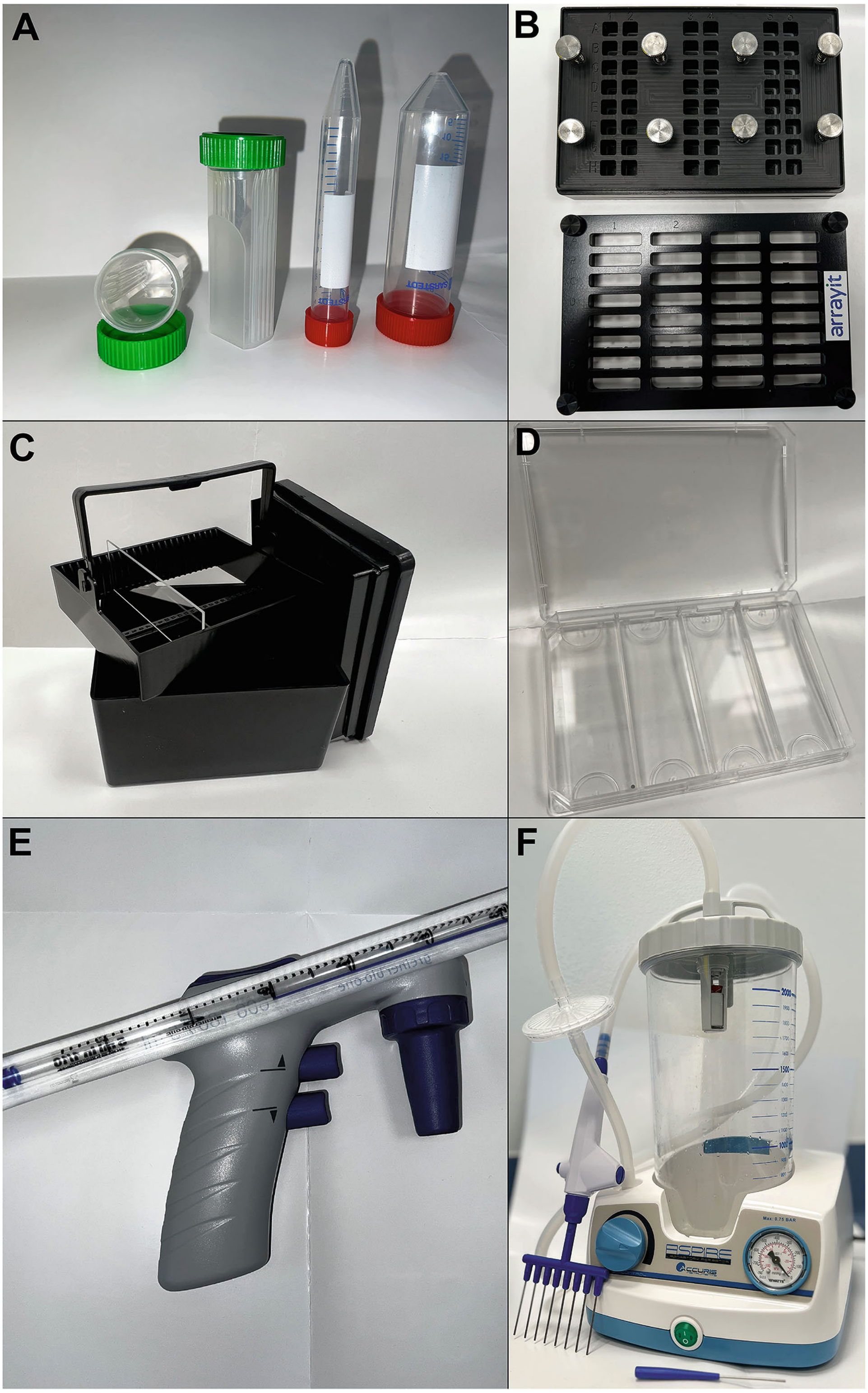

Fig. 3 An overview of equipment referenced throughout the protocol. Photograph (**a**) Cylindrical tubes used in the protocol, with pap jars (left) for storing microarrays before and after the protocol, and conical tubes (right) for buffer mixing and microarray drying via centrifugation. Photograph (**b**) Hybridization cassettes for smaller,

7. While the microarray is being dried, fill a chamber in the reaction tray with 4 mL citrullination buffer.
8. When centrifugation is complete, transfer the microarray from the conical tube to the reaction tray slot containing the citrullination buffer.
9. Place the reaction tray on the orbital shaker and shake at 50 RPM for 3 h at room temperature.
10. With approximately 30 min left of the incubation in **step 9**, remove the serum or plasma sample from the −80 °C freezer and allow it to thaw at room temperature.
11. When citrullination buffer incubation is complete, aspirate and discard the buffer using a pipette controller as described in previous steps.
12. Wash the microarray by adding 4 mL of SAB buffer to the reaction tray. Shake for 5 min at 50 RPM on an orbital shaker at room temperature. Repeat this step once more. After the final wash, leave the buffer in the reaction tray until beginning the first step of Subheading 3.3.

3.2 Preparation of Serum or Plasma Sample

During the final washing step of the previous section, prepare the sample to be analyzed. This protocol is optimized for plasma or serum samples, which needs to be diluted 1:200 in SAB buffer:

1. Vortex the thawed plasma or serum sample thoroughly, then centrifuge the sample at 13,000× *g* for 3 min to pellet any debris at the bottom of the sample tube.
2. Transfer 4.5 mL SAB buffer into a 15 mL conical tube.
3. Dilute 22.5 μL plasma or serum into the buffer from **step 2** (1: 200 dilution), aspirating from the top of the sample tube to avoid the pellet formed in **step 1**.
4. Vortex the diluted plasma or serum to ensure homogeneous mixing with the SAB buffer.
5. Keep the diluted sample at room temperature until incubation with the citrullinated protein microarray.

Fig. 3 (continued) multi-replicate microarrays, enabling autoantibody screening of multiple patient samples on a single microarray. Photograph (**c**) A slide-staining dish with a rack, useful for washing and antibody incubation when running larger experiments with multiple microarrays. Photograph (**d**) Microarray reaction trays for smaller experiments and patient sample incubation. Photograph (**e**) Pipette controller with a 5 mL serological pipette for adding and removing buffers from the reaction tray. Photograph (**f**) Laboratory aspirator, which can be used as an alternative to the pipette controller for buffer removal. It is especially useful when working with hybridization cassettes as it enables removal of buffers from multiple chambers simultaneously with a controlled vacuum pressure

3.3 Performing the Assay

1. Empty the SAB buffer remaining in the reaction tray from **step 12** in Subheading 3.1.
2. Carefully transfer 4 mL of the 1:200 diluted plasma or serum sample onto the microarray in the reaction tray.
3. Incubate the reaction tray for 2 h on an orbital shaker, shaking at 50 RPM at room temperature.
4. After incubation, remove the diluted sample from the reaction tray. Wash the microarray by adding 4 mL of SAB buffer to the reaction tray and shake for 5 min at 50 RPM at room temperature.
5. Repeat the above step two more times but incubate for 20 min during each wash.
6. While the microarray is shaking during the wash cycles in **step 4**, prepare the detection antibodies by adding 4.5 mL SAB buffer to a 15 mL conical tube. Add 1.25 ug/mL of detection antibody and keep wrapped in foil at room temperature until the wash steps are complete. For considerations if using multiple detection antibodies, *see* **Note 7**.
7. Once the wash cycles are complete, fill the reaction tray with 4 mL of the prepared antibody solution and incubate for 2 h at 50 RPM at room temperature.
8. After incubation, discard the antibody buffer and wash three times by adding 4 mL SAB buffer to the reaction tray. For each wash, incubate for 5 min, shaking at 50 RPM at room temperature.
9. Discard the remaining SAB buffer and perform three quick rinses with 4 mL MilliQ water, gently agitating the reaction tray by hand.
10. Perform one more wash with MilliQ water but this time shaking at 50 RPM at room temperature for 15 min.
11. Dry the microarray by transferring it from the reaction tray into a 50 mL conical tube in an upright position and centrifuge for 2 min at 1200× *g*.
12. Using tweezers, carefully remove the microarray from the conical tube, and place it in a pap jar, ready for scanning (*see* **Note 8**).

4 Notes

1. For larger multi-array experiments consider performing the washing and antibody staining steps in a slide-staining dish with a rack, *see* Fig. 3c. These racks hold up to 25 microarrays simultaneously, reducing the workload of emptying and

refilling the reaction tray for each microarray. However, do not use the slide-staining dish for the steps immediately after sample incubation to prevent cross-contamination between samples.

2. For the specific microarrays used in this protocol, proteins are printed on the barcode-labeled side of the microarrays, which can also be identified by a black dot located at the bottom right of the slide. Be sure the printed side faces up during all incubation and washing steps to avoid damaging the proteins.
3. Before each incubation step with a buffer in the reaction tray, avoid the microarrays floating on top of wash and incubation buffers by swirling the reaction tray gently by hand and ensure you see liquid floating across the top of the microarray.
4. When using the pipette controller or laboratory aspirator to aspirate and dispense liquid into the reaction tray with the microarrays present, direct the tip away from the microarray itself and onto the edges of the reaction tray. Aspirate and dispense very slowly to avoid damaging the proteins on the array surface.
5. For the steps that involve lifting the array out of the reaction tray, always do it while there is still liquid present in the tray to avoid the arrays sticking to the bottom of the tray.
6. When extracting the microarray from the conical tube, press gently on the side of the tube opposite the edges of the array. This will slightly widen the tube and make it easier to remove the slide without damaging it.
7. Multiple detection antibodies: It is possible to use multiple detection antibodies, such as anti-IgG and anti-IgA by adding multiple antibodies to the anti-body incubation buffer. Each antibody should be tested to assess the optimal concentration for signal intensity. A good starting point would be 1.25 μg/mL antibody of each antibody in the buffer. Ensure that the fluorophores conjugated to the antibodies do not overlap and verify that the antibodies do not cross-bind to each other to avoid signal interference.
8. If artifacts appear when you scan the microarrays, attempt to remove any remaining contaminants on the microarray surface by washing in a 50 mL conical tube filled with MilliQ water shaking at 50 RPM on an orbital shaker for 30 min.

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Chapter 5

High-Throughput Post-Translational Modification Analyses by Reverse Phase Protein Array

Zhongcheng Shi, Michael Hoang Nguyen, Yuan Yao, and Shixia Huang

Abstract

Post-translational modifications play a crucial role in regulating protein functions by chemically modifying amino acids without altering underlying protein sequences. Protein modifications are highly involved in cellular signal transduction pathways that require swift changes between active and inactive states. Various methods, including reverse phase protein array (RPPA), have been developed to comprehensively assess the proteomic profile. RPPA technology, a robust antibody-based platform, can detect not only protein expression but also modifications, such as phosphorylation, methylation, and acetylation. This chapter presents the detailed RPPA protocol for profiling post-translational modifications, including protein phosphorylation using whole cell lysates and histone modifications using purified histones.

Key words Antibody-based proteomics, Reverse phase protein array, Post-translational modification, Phosphorylation, Histone, High throughput

1 Introduction

Proteins serve as the ultimate executors of various cellular functions biologically and pathologically. The complexity of the cell requires innumerable proteins that are controlled by a limited number of genes. Post-translational modifications (PTMs) affecting the structure and dynamics of proteins significantly expand protein diversity by proteolytic cleavage or addition of a modifying group, such as acetyl, phosphoryl, glycosyl, and methyl groups, to one or more amino acids in proteins [1–3]. Therefore, alongside profiling the expression level of proteins, investigating protein modifications has emerged as a new focal point for researchers examining cell functions, such as proliferation and differentiation.

More than 300 different types of PTMs were discovered to take place after protein synthesis [4]. The dbPTM [5], a comprehensive PTM database, reveals 24 major PTMs with more than 80 experimentally verified modification sites, in which the top-5 most

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_5,

common modifications are phosphorylation, acetylation, ubiquitylation, succinylation, and methylation. The most common amino acids where the PTMs occur are serine, lysine, threonine, tyrosine, and cysteine [6]. For these PTMs, protein phosphorylation is among the most important protein modification that mediates signal transduction from the cell surface to the cytosol; and exerting effects to the nucleus as well [7, 8]. In the nucleus, histone post-translation modifications, one of the most important mechanisms of epigenetic changes, play important roles in chromatin accessibility, switching between transcriptionally active and inactive states, regulating the activity of the engaged transcriptional machinery, and even modulating the splicing of pre-mRNA to create various protein isoforms [9–11].

The modifications of the core histones (H2A, H2B, H3, and H4) include methylation, acetylation, phosphorylation, ubiquitylation, and sumoylation. Mass spectrometry enables discovery of novel proteins and unbiased quantification of multiple post-translational modifications (PTMs) [12–15]. However, it faces some challenges in PTM analysis, such as sensitivity issues in detecting the low abundance of phosphorylated proteins, the stability of the modifications, and the mass shift in the peptide molecular weight and throughput limitations [15–17]. In recent years, RPPA, an antibody-based technique, has emerged as a complementary approach for PTM analysis by using different antibodies to detect specific PTMs with high sensitivity in a high throughput manner [17–19].

The high throughput capacity of RPPA is demonstrated by its ability to simultaneously analyze up to thousands of samples on a single slide and to detect hundreds of protein targets using validated antibodies through instrument automation. RPPA has been widely utilized for pathway investigation in cancer research, serving as a discovery tool to identify novel protein functions in cancer development and to pinpoint potential therapeutic targets [17, 20–28]. In addition, RPPA is valuable for validation of protein expression results by other methods, including mass spectrometry [17, 29].

In this chapter, we elucidate the detailed methodology of RPPA for detection of phosphorylation in cell signaling proteins using whole cell lysates and detection of changes of histone PMTs using purified histones.

2 Materials

2.1 Whole Cell Lysate Preparation

1. 1× phosphate-buffered saline (PBS, Thermo Fisher Scientific, Waltham, MA)

2. RPPA lysis buffer: Tissue Protein Extraction Reagent (TPER) (Life Technologies Corporation, Carlsbad, CA) buffer supplemented with 300 mM NaCl and a cocktail of protease and phosphatase inhibitors (Roche, Pleasanton, CA).
3. Bicinchoninic assay (BCA, Thermo Fisher Scientific).
4. Novex Tris-Glycine SDS Sample Buffer (2×) (Thermo Fisher Scientific).
5. β-mercaptoethanol (Sigma-Aldrich, St. Louis, MO).
6. Spectramax 340PC Plate Reader (Molecular Devices, San Jose, CA).

2.2 Histone Preparation for Histone PTM Detection

1. 1× phosphate-buffered saline (PBS)
2. Nuclear isolation buffer (NIB): 15 mM Tris-HCl pH 7.5, 60 mM KCl, 15 mM NaCl, 5 mM $MgCl_2$, 1 mM $CaCl_2$, 0.25 M sucrose, 0.3% NP-40 and a cocktail of protease and phosphatase inhibitors.
3. 10% (W/V) NP-40 (Sigma-Aldrich)
4. 0.2 M hydrochloric acid (HCl, Thermo Fisher Scientific)
5. 1 g/mL aqueous Trichloroacetic Acid (TCA, Thermo Fisher Scientific)
6. Pure acetone (Thermo Fisher Scientific), 4 °C.
7. Acetone with 0.1% (v/v) HCl, −20 °C.
8. Novex Tris-Glycine SDS Sample Buffer (2×) (Thermo Fisher Scientific).
9. β-mercaptoethanol (Sigma-Aldrich).

2.3 RPPA Slide Printing

1. 384-well microarray plates (Molecular Devices)
2. Aushon 2470 arrayer (Quanterix, Billerica, MA) for printing.
3. ONCYTE ®AVID Nitrocellulose-coated glass slides (Grace Bio-labs, Bend, OR).

2.4 Total Protein Staining

1. Microscope slide staining dish and slide holder.
2. Sypro Ruby staining reagent (Invitrogen, Eugene, OR).

2.5 Antibody Labeling

1. Autostainer Link 48 (Dako/Agilent, Santa Clara, CA) for antibody labeling.
2. Reagent bottles: 5, 25, or 50 mL (Dako/Agilent).
3. Reagent bottle holders and slide racks (Dako/Agilent).
4. Nitrocellulose-coated glass slides printed with whole cell lysates or purified histones.
5. Biotin-blocking system (Dako/Agilent): includes avidin and biotin to block endogenous biotin.

6. Vectastain Elite ABC-HRP Kit, Peroxidase (Vector labs, Newark, CA): Prepare a 0.1% BSA in PBS solution. Add Reagent A and Reagent B from the kit in a 1:50 dilution with the PBS.
7. Antibody diluent (Dako/Agilent).
8. Biotinylated secondary antibody H + L (Vector Labs): dilute 1: 7500 in antibody diluent (*see* **Note 1**).
9. Tyramide Signal Amplification plus biotin kit (Akoya Biosciences, Marlborough, MA): Dissolve TSA-biotin with 300 μL of DMSO and dilute the TSA-biotin reagent at 1:250 dilution in the amplification diluent provided.
10. IRDye 680 Streptavidin (Li-Cor Biosciences, Lincoln, NE): dilute the dye 1:50 in prepared 1% BSA in PBS solution.
11. 1× I-Block (Applied Biosystems, Bedford, MA): prepare a 0.2% I-Block solution in 1× PBS with 0.2% Tween-20. Add the I-Block to the PBS solution with a stir-bar and heat the solution to 95 °C until the I-Block dissolves. Once the solution cools to room temperature, add the Tween-20 and mix.
12. 3% Hydrogen peroxide (H_2O_2, Thermo Fisher Scientific).
13. 10× TBS-T (Dako/Agilent): Dilute to 1× with ultrapure water. The resulting solution has a composition of 50 mM Tris–HCl, 300 mM NaCl, 0.1% Tween-20, 0.001% preservative, and a pH of 7.6.

2.6 Slide Scanning

1. GenePix 4400AL Microarray Scanner (Molecular Devices) 4400A attached with SL50 Slide Loader (Molecular Devices).
2. GenePix Pro 7.2 Microarray Acquisition & Analysis Software (Molecular Devices) for image acquisition.

2.7 Image Analysis

1. TIF files from slide scanning.
2. GenePix Pro 7.2 Microarray Acquisition & Analysis Software (Molecular Devices) for image analysis.

3 Methods

The RPPA workflow comprises several key steps: protein sample preparation, slide printing with protein lysates or purified histones, antibody labeling, slide scanning, image analysis, and data normalization and analyses (Fig. 1). Prior to the RPPA experiment, antibodies are validated [30, 31]. Total protein staining is performed to assess printing quality and for total protein normalization.

3.1 Protein Sample Preparation

Protein lysate sample preparation is critical for RPPA analysis. To assess protein phosphorylation and its total protein counterpart, whole cell lysates are used from either tissue samples or cultured

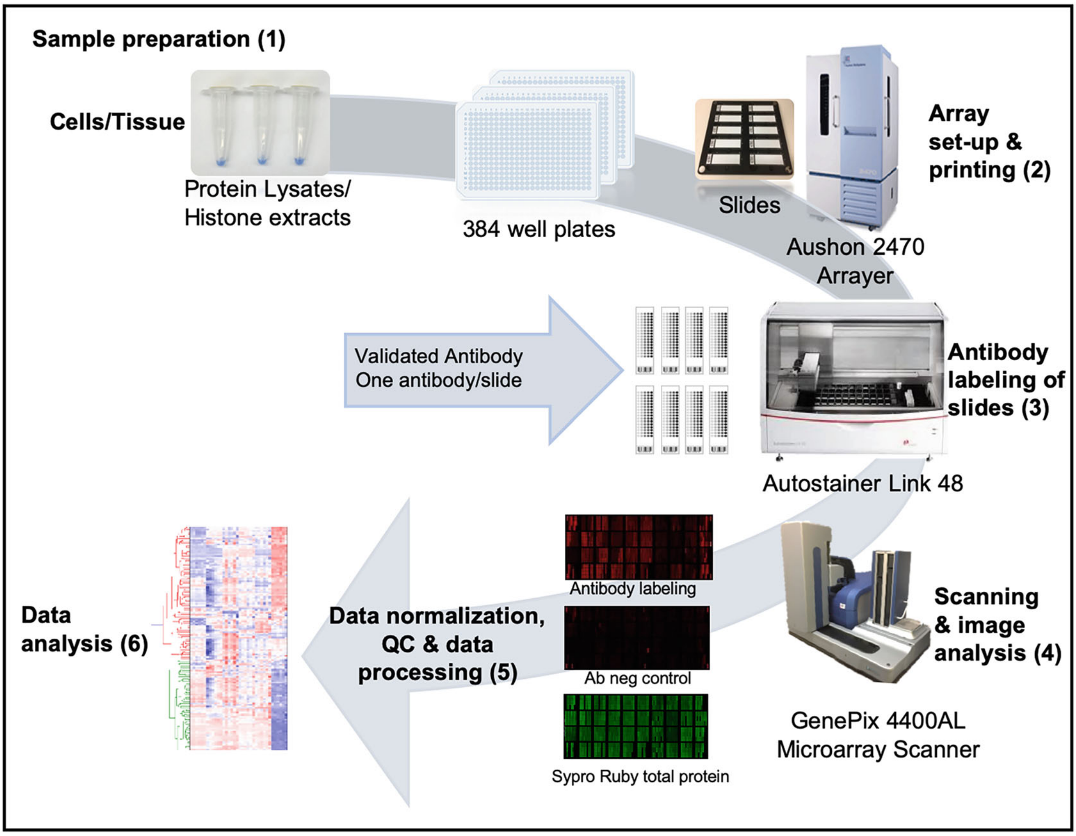

Fig. 1 The RPPA workflow: Following whole cell lysates extraction or histone purification from cells or tissue, denatured samples are transferred to 384-well microarray plates. Individual slides are printed with the protein samples and subsequently labeled with validated antibodies. These slides are then scanned at various PMT settings. Normalization is performed by subtracting negative control signal at the same setting and normalized by group-based total protein signal. The normalized data are subsequently analyzed based on individual experimental designs. "Ab neg" refers to negative control slide (without primary antibody). (Reproduced with permission from *Ref* [30])

cells. In this chapter, we used an example in which whole cell lysates were prepared from human breast ductal carcinoma in situ (DCIS) cells treated with E2 together with a progestin agonist. Preparation of whole cell lysates from tissues was previously described [30]. For histone modification analysis, purified histone samples are preferred and as described [17]. The protein concentrations of the lysates or histones vary based on cell number or tissue size and cell/tissue types.

3.1.1 Whole Cell Lysate Preparation

1. Wash the cultured cells three times with ice-cold PBS and remove PBS completely.
2. Add RPPA lysis buffer over cells (200 μL/~5 × 10^6 cells) (*see* **Note 2**), scrape the cells, and then transfer cell suspension to a

1.5 mL tube. Vortex the tube for 15 s and incubate on ice for 30 min. During the incubation, vortex the tube briefly every 10 min.

3. Centrifugate the lysates at 14,000 *g* for 15 min at 4 °C, and then transfer the supernatants to fresh tubes.
4. Repeat **step 3** once or multiple times, as needed (*see* **Note 3**).
5. Determine protein concentration of cell lysates by BCA or Bradford assay.
6. Adjust the concentrations of the protein lysates to 1 μg/μL with RPPA lysis buffer and add equal volume of SDS Sample Buffer with 2.5% β-mercaptoethanol to make the final concentration of 0.5 μg/μL.
7. Denature samples for 5 min at 95 °C and store at −80 °C.

3.1.2 Histone Purification

1. Wash the cultured cells three times with ice-cold PBS. Afterward, collect cells to a 1.5 mL tube by scraping using a cell scraper. Subsequently, centrifuge the tubes at 1000 *g* for 5 min at 4 °C. Remove PBS completely.
2. For every 100 μL of cell pellets, add 1 mL of nuclear isolation buffer and mix by gently pipetting.
3. Incubate on ice for 5 min.
4. Centrifuge at 600 *g* for 5 min at 4 °C to pellet nuclei and discard the supernatant with a pipet.
5. Wash the nuclear pellet twice to remove NP-40 by gentle resuspension in 1 mL of nuclear isolation buffer without NP-40, and centrifuge at 600 *g* for 5 min at 4 °C. Pellet nuclei at max speed (15,000 to 21,000 *g*) for 5 min after the second wash.
6. Gently resuspend the nuclear pellet with 500 μL of 0.2 M HCl and rotate on an end-over-end rotator for 2 h at 4 °C.
7. Centrifuge for 5 min at max speed (15,000 to 21,000 *g*) at 4 °C and transfer the supernatants containing histones to a new tube.
8. Add 125 μL of cold (4 °C) 1 g/mL trichloroacetic acid (TCA) slowly to the supernatant (final TCA concentration 20%) with continuous and vigorous mixing and incubate on ice for 5 min to precipitate histones.
9. Centrifuge for 5 min at max speed (15,000 to 21,000 *g*) at 4 °C and discard the supernatant.
10. Wash the histone pellets with 200 μL cold (−20 °C) acetone containing 0.1% (v/v) HCl.
11. Centrifuge for 2 min at max speed (15,000 to 21,000 *g*) at 4 °C and carefully remove acetone.

12. Wash the histone pellets with 200 μL of cold (4 °C) pure acetone without HCl and repeat the centrifugation to remove the acetone.
13. Air-dry final histone pellets at room temperature overnight, and store as dry pellets at −80 °C.
14. Before use, dissolve the histone pellet with 20–50 μL of ddH_2O.
15. Use 1 μL of the dissolved histone to verify the quality of the isolated histones by SDS-PAGE analysis with a 4–20% gradient Tris-glycine gel and Coomassie Brilliant Blue staining. The concentration of purified histones can be determined at this step by comparison with the band density of a protein marker on an SDS-PAGE gel stained with Coomassie blue (*see* **Note 4**).
16. Prepare RPPA samples by diluting the histones with RPPA lysis buffer, 2× SDS Sample Buffer and 2.5% β-mercaptoethanol to obtain a final concentration of 0.25–0.5 μg/μL, and denature samples for 5 min at 95 °C.

3.2 Validation of PTM Detection Antibody

The efficacy of PTM detection using RPPA depends on the sensitivity and specificity of corresponding antibodies. As previously described, for the validation of antibodies against phosphorylated proteins, immunoblotting analysis is performed for each antibody using known positive and negative control samples (e.g., lysates from cells w/wo treatments) prepared in house or purchased commercially [31]. Immunoblotting using a validated antibody should show the bands at the correct molecular size (multiple bands may appear for certain protein with multiple isoforms).

The antibodies against histone PTMs are validated by RPPA with synthetic peptides containing chemically modified PTMs at the specific amino acid residues corresponding to endogenous histone PTMs, including acetylation, methylation (mono-, di-, and trimethylation), and phosphorylation (*see* **Note 5**). For RPPA, each peptide with 5-point twofold serial dilutions (0.05 to 0.003125 μg/μL) is prepared in SDS sample buffer with 2-mercaptoethanol and denatured by heating at 95 °C for 5 min [31]. The peptides are included as positive controls for quality evaluation of histone PTM antibodies during printing, together with experimental samples.

3.3 Printing of RPPA Slides

The RPPA platform uses the Aushon 2470 Arrayer with a 40-pin (up to 48 pins) configuration to spot samples onto nitrocellulose-coated slides (*see* **Note 6**). The control samples are routinely spotted with the experimental samples on the same RPPA slides for quality checking purposes [17, 30, 31].

1. A sample map is prepared in a Microsoft Excel spreadsheet (*see* **Note 7**). With the help of the internal RPPA Setup Tool plate and layout utility, the 384-well location files of all samples in excel are generated using the map file [30]. The instrument software can also generate a sample map with assistance from the vendor's software application specialist. The Aushon 2470 Arrayer generates a GenePix Array List (GAL) file containing information of each sample spot based on the location excel files. The GAL file contains sample identity with detailed location information for each spot. Check the consistency of sample spot information between the GAL file using GenePix software and in the excel sample map file.
2. The frozen protein samples and peptides, prepared in the SDS sample buffer with β-mercaptoethanol, are thawed at room temperature, then heated at 95 °C for 5 min. After a brief centrifugation, they are transferred to 384-well microarray plates according to the location design.
3. Clean pins and the pin holder before printing (*see* **Note 8**).
4. Program the arrayer with three technical replicate spots for each sample and two depositions per spot. To minimize carry-over between samples, pins are washed in between two samples by immersing each of them in flowing water for 2 s. After the wash cycle, five sequential dips are performed in the sample well plate to ensure consistent spotting between washes. During the printing process, the internal humidity of the instrument is maintained at 70%.
5. Load plates with samples and platens with slides into the arrayer (*see* **Note 9**).
6. Print one slide and perform total protein staining on this slide for quality checking (*see* **Note 10**).
7. Proceed to finish printing all slides. The total number of replicate slides varies based on the number of antibodies, the number of negative control slides, and the number of slides used for total protein staining (*see* **Note 11**). We routinely print 350 slides for an RPPA experiment with ~300 antibodies.

3.4 Total Protein Staining

Currently, one slide out of every 20 printed slides is selected for total protein staining based on the printing order, such as #10, #30, #50, #70, #90, #110, and so on. Total protein staining is used for normalization of the fluorescent intensity from the adjacent 19 slides with antibody labeling. For example, #10 is used for #1-#20 slides; #30 is used for #21–#40. For the 350 slides we print for the RPPA experiment, we routinely stain total protein for about 20 slides (*see* **Note 12**), including the first and last slides in the printing order.

1. Hydrate the slides on a slide holder in a container filled with ultrapure water covering the entire slide for 5 min.
2. Remove the slide holder with slides and drain excess water.
3. Place the slide holder with slides into the container with Sypro Ruby staining reagent and cover the container with aluminum foil. Incubate the slides with Sypro Ruby on the shaker for 15 min.
4. Remove the slide holder with slides from Sypro Ruby and wash the slides in a container filled with ultrapure pure, cover with aluminum foil, and place on shaker for 5 min.
5. Repeat the destaining step with ultrapure water 3 times.
6. Air dry slides in a dark room (*see* **Note 13**).

3.5 Antibody Labeling

Antibody labeling is performed on an autostainer, Autostainer Link 48, and it has a capacity of up to 34 slides in our process, which is determined by the number of reagents needed and the number of reagent bottle slots available in the instrument. Typically, it includes 32–33 primary antibody slides and 1–2 negative control slides. Primary antibody dilution for RPPA is optimized based on binding sensitivity prior to the experiment. Antibody labeling is carried out with a catalyzed signal amplification system at room temperature (*see* **Note 14**).

1. Before the labeling run, three steps of preparation are incorporated: (1) creating a protocol and entering it in the autostainer (refer to #7 immediately below); (2) calculating the volume of reagents required for each reagent including primary and secondary antibodies; (3) creating an Excel file containing information to align the antibody with the RPPA slide barcode.
2. Label the reagent bottles and scan the barcodes using the autostainer-linked barcode scanner to have the system identify each bottle and associate it with a specific antibody or reagent.
3. Prepare reagents and transfer them to the corresponding reagent bottles (*see* **Notes 15** and **16**).
4. Place the reagent bottles in the reagent racks and place them into the machine.
5. Carefully place each RPPA slide to the holder position on the slide rack (*see* **Note 17**) and place the racks with slides into the machine.
6. Prime the autostainer with water and TBS-T buffer. Begin staining.
7. Run the pre-determined antibody staining protocol below for the Autostainer:

(a) TBS-T (5 min).
(b) I-Block (15 min × 2).
(c) TBS-T rinse.
(d) H_2O_2 (5 min).
(e) Avidin block (10 min).
(f) TBS-T rinse.
(g) Biotin block (10 min).
(h) TBS-T rinse (2×).
(i) Primary antibody (30 min).
(j) TBS-T (5 min).
(k) Biotinylated secondary antibody (15 min).
(l) TBS-T (5 min).
(m) SABC (Streptavidin-Biotin Complex).
(n) TBS-T (5 min).
(o) TSA reagent (15 min).
(p) TBS-T (5 min).
(q) Streptavidin-IRDye-680 (LiCor) (15 min).
(r) ddH_2O (1 min).

8. Once the antibody labeling steps above are completed, remove slides from rack individually and place each one carefully in a glass slide holder inside a glass container filled with water. Wash the slides with agitation (10–20 times) and transfer the slides to a new container for a second wash. This process effectively removes any residual fluorescent dye and salts on the top or bottom of the slides.
9. Air dry slides in a dark environment.

3.6 Slide Scanning

1. Set up the GenePix 4400 AL scanner at an appropriate photomultiplier tube (PMT) setting with 100% laser power.
2. Place slides with the nitrocellulose membrane facing downwards in the slide loader.
3. Scan Sypro Ruby stained slides at 532 nm wavelength with 380 PMT setting.
4. Scan negative control slides and antibody-labeled slides at 635 nm wavelength with 400, 460, 500, 550, or 600 PMT setting (*see* **Note 18**).
5. Check the quality of the TIFF files from the scanned slides using the GenePix 7 software. Rescan slides at different PMT settings if needed.

3.7 Image Analysis

The TIFF images generated from Subheading 3.6 are analyzed by GenePix 7 software. Within the software, each individual spot on the array is called a feature, and the circular outlines called feature indicators given by the software feature can be adjusted based on size and position. Feature indicators are the primary unit of analysis in GenePix. A collection of feature indicators is called a block. In our configuration, one pin prints one block of spots. Therefore, a 40-pin configuration generates a total of 40 blocks. The goal of image analysis is to closely align the features with the feature indicators. At the conclusion of the image analysis, a .GPR result file is generated for each image, containing all the numeric data for each antibody and each sample or control spot on the slide. The image analysis software settings are pre-optimized for spot size and distances designed during the printing process.

1. Open GenePix 7 software.
2. Open a TIFF image file. Import the GAL file prepared in Subheading 3.3.
3. Align the feature indicators in the correct block locations first and then adjust each feature indicator to each feature (corresponding spot) as closely as possible.
4. Click the Analyze icon.
5. Save the settings of grid aligned file as a GPS file.
6. Perform batch image analysis on TIF files using the GPS file generated from **step 5**.
7. Check the grid alignment in GPR files generated from batch analysis using GenePix software. If necessary, adjust the grid, re-analyze the files, and save the GPR files again.

3.8 Data Normalization

The numerical data in the GPR data file displays the fluorescence signal intensity (SI) of each spot of protein samples on the slides, which are used for data normalization.

The antibody signal intensity (SI) of each spot is subtracted by the corresponding SI of negative control and then normalized to the corresponding SI of total protein. The normalized antibody SI (N) is calculated using the following equation $N = [(A-C)/T]*M$, where A is the antibody SI, C is the negative control SI, T is the SI of total protein, and M is the median SI of total protein of the spots within the same group. Thorough quality control data is gathered and assessed for each antibody across various PMT settings, primarily aimed at identifying and flagging potential problematic antibodies. The detailed calculation and the extensive QC process were previously described [30].

3.9 Data Analyses

The normalized data is used for subsequent statistical analysis. Each sample is printed in triplicate, and the triplicate data is used for quality assessment and the data from each sample is summarized into a single data point for subsequent analyses. Generally, antibodies displaying low signal intensity across all samples (e.g., lower than 200, or determined by researcher's assessment based on the relative antibody signal vs negative control signal for the given group of samples) and/or high variation among technical replicates (e.g., CV% $>$ 25%) for most samples are excluded from further analyses. The remaining antibody SI values are subjected to statistical analyses, which include analysis of variance (ANOVA) for multiple group comparison and t-test for two-group comparison with significance P value <0.05 and adjusted p value (FDR) ≤ 0.1 [30].

In our previous study with Coarfa et al. [30], we utilized RPPA to identify protein pathways (phosphorylation and total proteins) in analyzing the effects of steroid hormones in an estrogen receptor (ER) and progesterone receptor (PR) positive human ductal carcinoma in situ (DCIS) cell line. The cells were subjected to treatment with vehicle (ethanol) or E2 (1 nM E2) together with a progestin agonist (10 nM R5020 (17 alpha, 21-dimethyl-19-nor-pregna-4, 9-diene-3,20-dione)) for either 6 or 24 h in triplicate experimental cultures. Subsequently, whole cell lysates were prepared for RPPA analyses. Figure 2a illustrated changes upon treatment with E2 and R5020 for 6 or 24 h compared to those of the ethanol control, including increased phospho-p38 Thr180/Tyr182 and the decreased phospho- AKT Ser473 (Fig. 2a & b) modifications [30].

In another study by Wang et al. [17], histone PTMs were analyzed by RPPA in primary human dermal fibroblasts and induced pluripotent stem cells (iPSCs) using purified histones from both cell lines. RPPA analyses revealed significant changes for multiple histone PTMs (Fig. 2c), such as increased euchromatin (active) histone PTMs (e.g., H3K18ac) and decreased heterochromatin (repressive) histone PTMs (e.g., H3K27me3) for histone H3 in the iPSCs than those in the differentiated fibroblast cells (Fig. 2d) [17].

4 Notes

1. Secondary antibody dilutions should be optimized based on the observed antibody signal and background signal for most primary antibodies.
2. For other specific cell lines or primary cells, pre-testing for optimal buffer-to-cell ratio is recommended to achieve the desired concentration.
3. More centrifugation cycles (at 14,000 *g* for 15 min at 4 °C) might be needed to obtain a completely clear supernatant,

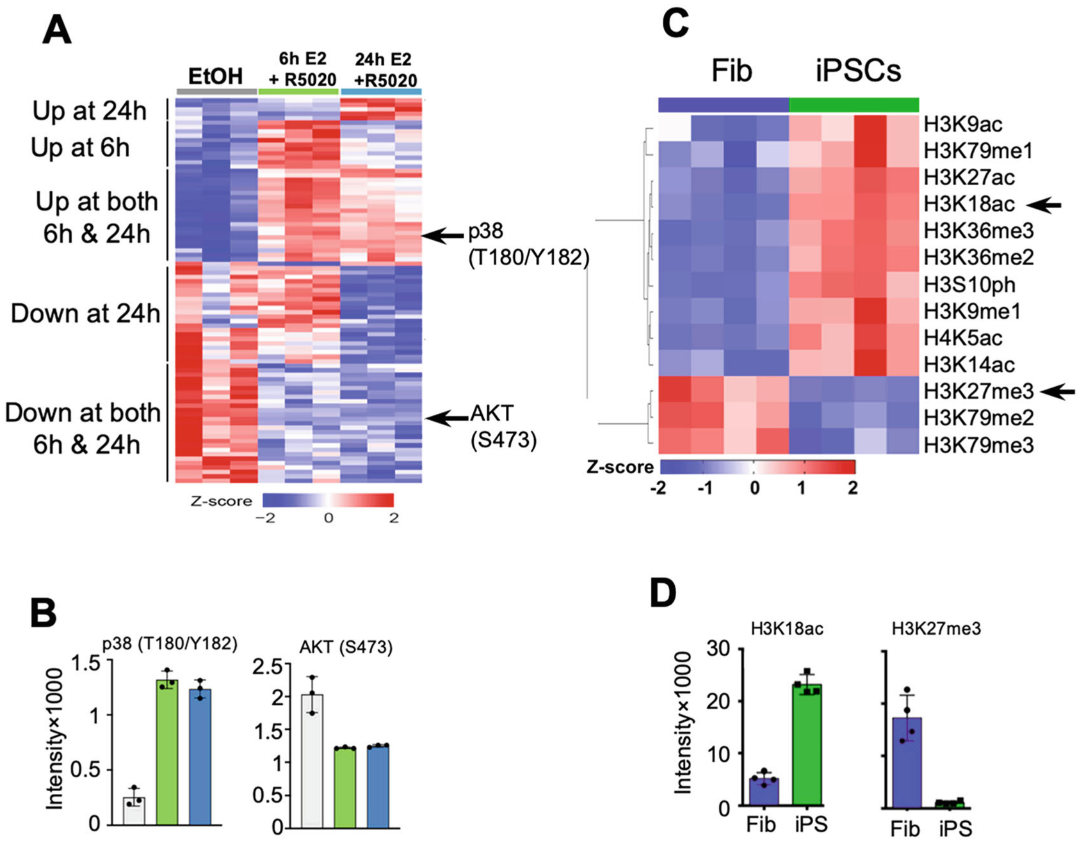

Fig. 2 Profiling of protein phosphorylation and histone PTMs. (**a**) Heatmap representation of protein changes in DCIS cells treated with E2 and R5020 or with ethanol control for 6 or 24 h. Expression values were scaled by converting intensities to z-score values. (**b**) Bar charts of normalized RPPA signal of an example of a hormone induced phospho-protein, p38 Thr180/Tyr182 and an example of a hormone repressed phospho-protein, AKT Ser473. EtOH, ethanol (Reproduced with permission from *Ref* [30]). (**c**) Heat map of changes in histone PTMs in iPSCs compared to fibroblasts. (**d**) Bar charts of normalized RPPA signal intensities for histone PTM antibodies of an active mark H3K18ac, a repressive mark H3K27me3, and increased phosphorylation of H3S10 in iPSCs compared to fibroblasts. Data are expressed as means $\pm$ SD from four replicates in each experimental group. Unpaired t-tests, adjusted p value (FDR) ≤ 0.1, and a cut-off of a 1.5-fold difference were performed to evaluate differences between iPSCs and Fibroblast cells. Reproduced with permission from the Copyright Clearance Center (CCC) for *Ref* [17]

depending on different cell/tissue types. For example, in our laboratory, 2–3 centrifugation cycles are routinely used for cell lines, 5 centrifugation cycles for tissue samples such as tumor samples, and 7–8 centrifugation cycles for fat tissues.

4. The yield of purified histones varies across different types of cells or tissues. In general, one million cells or 1 mg tissues yield approximately 5–10 μg histone proteins.
5. Generally, biotinylated (tagged) peptides are bound to streptavidin plates or slides [32], but we previously found that

non-tagged peptides can also bind to nitrocellulose membrane-coated slides [17], making it feasible to use free peptides on RPPA slides. In this chapter, the non-tagged peptides are used.

6. The different array configurations in a 40-pin format are utilized based on the total number of samples. Each sample is printed in triplicate, allowing for a maximum of 2880 spots to be printed on each slide in our setup. Adjustments to the configuration could accommodate more spots if needed.
7. If samples from multiple projects are printed on the same slide, the samples of the same project are usually grouped together on the map for enhanced visualization.
8. Minor surface contaminants such as salts and organic films may accumulate on pins throughout the printing process, potentially diminishing printing performance. Thus, regular pin cleaning is performed. In addition, the accumulation of debris or contaminants in the pin holder can cause certain pins to become stuck. Often, they may print inconsistent spots or puncture through the nitrocellulose membrane on the slide while printing. To clean the pins, sonicate them in water to remove water-dissolved dirt, and then clean them with pure acetone to remove oily residue. Pure acetone is used for cleaning the pin holder.
9. While mounting the slides on the platen, glass debris may inadvertently land onto the slides. This can be easily removed with a compressed gas can before loading the platen into the arrayer.
10. As part of our standard procedure, we print the first slide and stain it with Sypro Ruby for total protein assessment as a quality check. If the total protein staining reveals small or shifted spots from one pin, it is essential to thoroughly clean both the pin and the corresponding pin holder slot. Subsequently, print one slide for quality testing or clean pins and re-print a test slide until all the spots exhibit consistent staining.
11. We print extra slides to ensure an adequate supply for additional staining, particularly if certain antibodies require repeated staining due to high background or other issues.
12. Antibody labeling for hundreds of antibodies is carried out in multiple batches due to the instrument capacity, while total protein staining is conducted throughout the whole printing run. The protein concentration from slide #1 to slide #350 gradually increases during 3–5 days of printing due to evaporation, even in a humidity-controlled environment (70%). Although the relative protein concentrations among samples within a given group remain constant regardless of the printing order, we normalize the antibody signals with total protein for every 20 slides from the first to the last slide as a precaution.

However, based on our preliminary analyses, we have found that normalization using a smaller number of slides for total protein staining, such as 1 for every 50 slides or more, still provides reliable data. Consequently, we may adjust this number in the future.

13. Since Sypro Ruby is light sensitive, all slides that have been stained with Sypro Ruby should be covered with aluminum foil or other light protection material.
14. Drawing from past experience, we typically start with the antibody dilution for RPPA that is approximately four times the concentration used for Western Blot. For example, if 1:1000 dilution is used for western blot, 1:250 is initially used for RPPA. Subsequently, we optimize the concentration/dilution based on RPPA and negative control results for that antibody. Although the amplification system is not used in some RPPA platforms [33], it is beneficial for detection of proteins with low abundance. To maximize the sensitivity of the RPPA platform, adopting the amplification system is recommended. However, this approach may lead to saturated signals for the proteins with high abundance or antibodies with high binding signal, which can be addressed by decreasing the PMT setting.
15. Take into account the varying dead volumes for each reagent bottle size. Add an additional 0.7–2 mL reagent to the bottles based on the dead volume indicated by the manufacturer for each bottle size. This precaution prevents the staining from stopping midway due to insufficient reagent detected. To enhance efficiency, all common buffers (i.e., TBST, antibody diluent, etc.) can be preloaded into bottles for all batches at one time without adding reagents that are either sensitive to light or needed to be prepared freshly, such as Li-Cor, primary antibody, secondary antibody, ABC, and TSA. Store all loaded bottles at 4 °C and add the additional required reagents on the day of staining.
16. Break any bubbles present in the reagent bottles before loading them in the autostainer using a pipet, paying special attention to BSA-containing buffers, which tend to generate bubbles easily.
17. To achieve even staining for the slides, ensure that every slide slot is leveled with used slides. If a slide is uneven in a particular position, relinquish that slot position on the rack or change the rack to ensure all slide slots are even.
18. Based on previous slide scanning tests for low and high expression proteins, we have found that the 460, 500, and 550 PMT settings provide a reliable and observable signal intensity. However, in some instances, multiple spots in a sample group may have saturated signals (white pixels in spot) at the

460 PMT setting for certain antibody-stained slides, which requires rescanning at the lower 400 PMT setting along with the respective negative control slide. Conversely, if the fluorescent spot signal intensities are too low at 550 PMT, the slides can be rescanned at the higher 600 PMT setting.

Acknowledgments

This work was supported in part by a CPRIT (Cancer Prevention and Research Institute of Texas) Core Facility Award RP210227, NIH NCI-Cancer Center Support Grant P30 CCSG (P30CA125123) and NIH/NIEHS P42 ES027725 to the Antibody-based Proteomics Core/Shared Resource (SH) and NIH S10 award (S10OD028648, SH).

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Chapter 6

Optimization of Serum and Microarray Concentrations for a Semiquantitative Multiplex Indirect Immunoassay

Patricia Quaranta, Rocío Paz-González, Selva Riva-Mendoza, Pablo Domínguez-Guerrero, Francisco J. Blanco García, Cristina Ruiz-Romero, Lucía Lourido, and Valentina Calamia

Abstract

The versatility of protein microarrays provides a variety of possibilities to explore proteomic profiles, allowing researchers to investigate multiple proteins simultaneously with high efficiency. In this study, we employed xMAP® technology to carry out a multiplex indirect semiquantitative immunoassay using the MILLIPLEX® HCYTAAB-17K kit. This kit is designed with an integrated microarray containing 3 positive controls, 1 negative control, and 14 analytes.

In this chapter, we provide a detailed protocol for optimizing the serum conditions required to perform the immunoassay. In addition, we have optimized the commercial microarray itself, focusing on reducing non-specific binding and minimizing false positives. Through these improvements, we aim to increase both the sensitivity and specificity of the assay, ensuring more robust and reproducible results in proteomic profiling. The optimization also helps to extend the usability of the kit, making it more versatile for future experiments.

Key words Bead microarray, Luminex, Serum, Rheumatoid arthritis, Biomarker discovery, Anti-cytokine antibodies

1 Introduction

Rheumatoid arthritis (RA) is a chronic systemic autoimmune inflammatory disease. Cytokines play a crucial role in the RA pathology as they are considered the main responsible for the synovial membrane inflammation characteristic of RA [1]. RA is distinguished by a deregulation of the immune system, leading to the formation of antibodies against self-antigens (autoantibodies), which promote the autoimmune processes associated with the disease [2].

The RA pathology can be influenced positively or negatively by the formation of anti-cytokine antibodies. The anti-cytokine

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_6,

antibody's role not only depends on their activity but also on the intrinsic function of the target cytokine. Therefore, it is essential to define the different functions and significance that anti-cytokine antibodies can exhibit in the context of RA [3].

In this work, xMAP® technology was employed to perform a multiplex indirect immunoassay. Bead-based protein array technology is an alternative method to traditional planar arrays. This technology has been described as an effective method for discovering disease-associated protein profiles. This type of immunoassay enables the detection of antibodies in serum samples through the use of antigens immobilized on a color-coded bead array. Subsequently, the antigen-antibody interaction is detected using an anti-human immunoglobulin G (IgG) antibody labeled with the fluorophore phycoerythrin [4].

We have used the MILLIPLEX® HCYTAAB-17K kit to perform the multiplex indirect immunoassay. This kit is integrated with a microarray with 3 positive controls, 1 negative control, and 14 analytes. For this experiment, we optimized the dilution of serum samples for the autoantibody profiling and the bead microarray concentration to minimize false positives and extend the kit's usability. We conducted several tests with different sample dilutions. The MILLIPLEX® HCYTAAB-17K kit protocol recommends a 1:100 dilution, but we also tested a more concentrated condition (1:20) and a more diluted condition (1:1000). Additionally, we evaluated a 1:250 dilution, which had been optimized in previous studies using antigen bead arrays [5]. When optimizing the sample conditions, we observed an elevated number of magnetic beads per region. Furthermore, we optimized the concentration of the commercial microarray. We assessed the microarray concentration according to the protocol and tested a twofold dilution (1:2). We checked whether changing the microarray concentration would affect the semiquantitative results. After achieving the microarray optimization by dilution, we were able to extend the usability of the kit and reduce the non-specific binding resulting into an increase of the signal used for the semiquantitative results.

2 Material

2.1 Reagents Supplied in the Kit

1. MagPlex® magnetic microspheres: Human Cytokine Autoantibody Magnetic Bead Panel (Cat. n° HCYTAAB-17K).
2. Assay Buffer.
3. Wash Buffer (1X): 60 mL of 10X Wash Buffer (0,05% Proclin) with 540 mL deionized water and stored at 2–8 °C for up to 1 month.
4. Human Cytokine Autoantibody PE-IgG Conjugate.

2.2 Materials Required

1. Serum samples for dilutions.
2. MAGPIX Drive fluid® PLUS (Ref.40-50019).

2.3 Expendable

1. Mixing Bottle.
2. PCR Plates.
3. 96 well half-area flat-bottom polystyrene plates.
4. Plate sealer.
5. Absorbent towels.
6. Aluminum Foil.

2.4 Equipment

1. Sonicator.
2. Vortex mixer.
3. Plate shaker.
4. Plate magnet.
5. MAGPIX Luminex®.

3 Methods

For this study, the MILLIPLEX® HCYTAAB-17K kit in combination with Luminex xMAP® technology, was used to perform a semiquantitative multiplex immunoassay targeting anti-cytokine antibodies (Fig. 1). The assay included 3 positive controls, 1 negative control, and 14 analytes (Tables 1 and 2).

Positive controls are characterized by a decreasing concentration of immobilized human IgG. Positive control 1 contains the highest concentration of immobilized human IgG, followed by progressively lower concentration in positive control 2 and 3. The negative control is a set of beads without immobilized human IgG, serving as a baseline reference.

3.1 Beads Array Preparation

1. Sonicate each of the 19 individual vials for 30 s, followed by vortexing for 1 min. Each Luminex Magnetic Bead region is stored in a separate vial (Tables 1 and 2).
2. Add 60 μL of each bead region (50X concentration) into the mixing bottle (*see* **Note 1**) and bring the final volume to 6 mL with Assay Buffer (Array 1/2) (*see* **Note 2**). The microarray may be stored at 2–8 °C for up to 1 month.

3.2 Sample Preparation

1. Thaw the serum samples (*see* **Note 3**).
2. Mix the samples by vortex and centrifuge for 10 min at 10.000 g prior to use to remove pellet insoluble components.

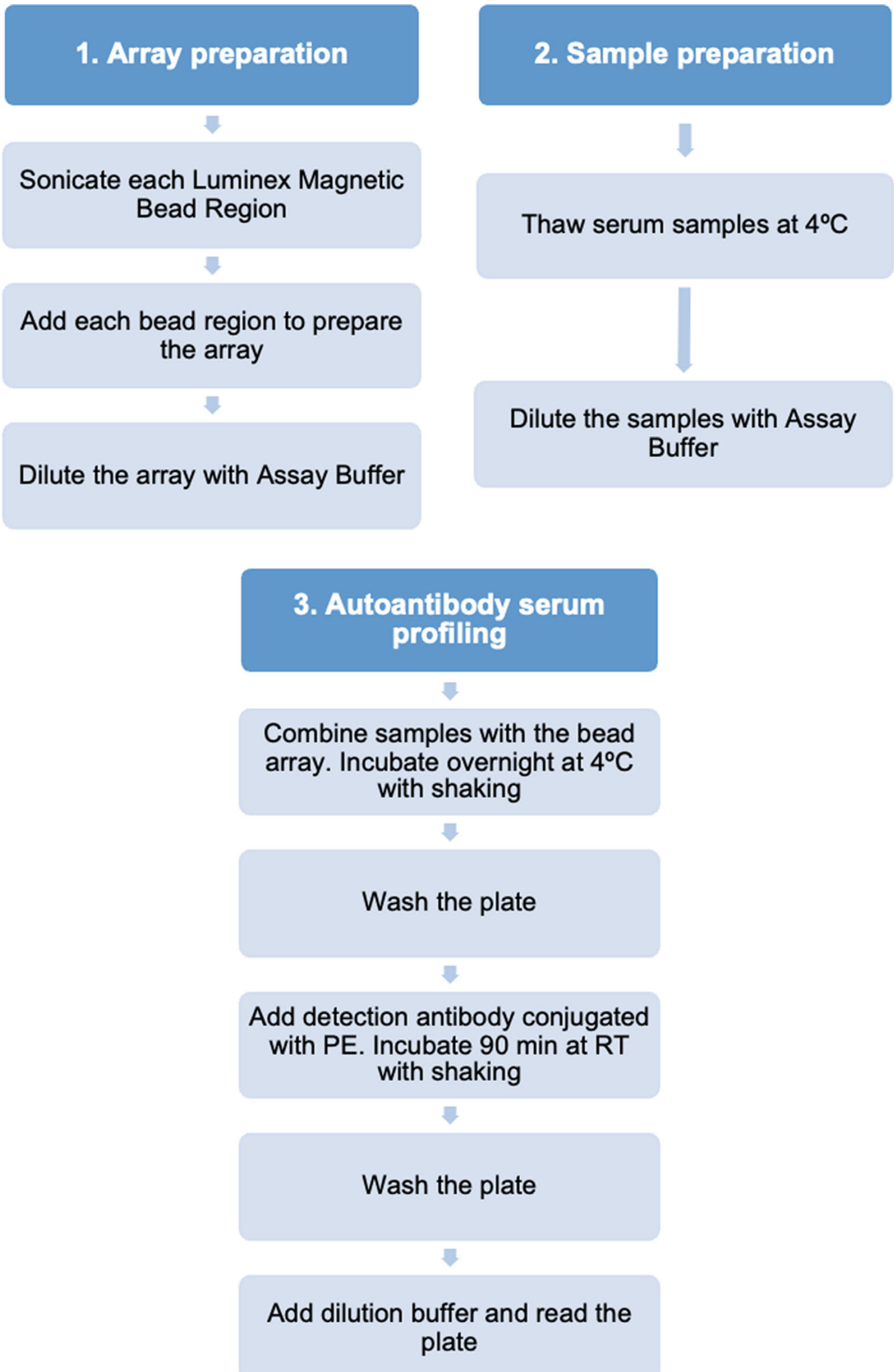

Fig. 1 Workflow overview

Table 1
Control antigen-immobilized magnetic beads

Bead name	Luminex magnetic bead region	Customizable 14 antigen Cat.
Control 1 beads magnetic	12	CB1-MAG
Control 2 beads magnetic	13	CB2-MAG
Control 3 beads magnetic	14	CB3-MAG
Negative control beads magnetic	15	NCB-MAG

Table 2
Human cytokine autoantibody antigen-immobilized magnetic beads

Bead/antigen name	Luminex magnetic bead region	Customizable 14 antigen Cat.
Human IFNß beads magnetic	25	HAAIFNB-MAG
Human IL-22 beads magnetic	30	HAAIL22-MAG
Human IL-12p40 beads magnetic	36	HAAIL12P40-MAG
Human IL-6 beads magnetic	54	HAAIL6-MAG
Human IL-15 beads magnetic	56	HAAIL15-MAG
Human IL-17A beads magnetic	62	HAAIL17A-MAG
Human IL-17F beads magnetic	64	HAAIL17F-MAG
Human G-CSF beads magnetic	65	HAAGCSF-MAG
Human IL-10 beads magnetic	72	HAAIL10-MAG
Human BAFF beads magnetic	73	HAABAFF-MAG
Human IFNγ beads magnetic	75	HAAIFNG-MAG
Human IL-1α beads magnetic	76	HAAIL1A-MAG
Human IL-8 beads magnetic	77	HAAIL8-MAG
Human IL-18 beads magnetic	78	HAAIL18-MAG

3. Transfer 50 μL of each sample into the respective wells in a PCR plate (*see* **Note 4**).
4. Optimize the dilution in serum samples, several dilution factors were tested under the following conditions: 1:20, 1:100, 1:250, 1:1000 (*see* **Notes 5** and **6**).

3.3 Assay Procedure

1. Add 200 μL of wash buffer into each well of the 96-well plate half-area flat-bottom polystyrene (*see* **Note 7**). Seal with a plate sealer and mix on a plate shaker at room temperature for 10 min.

2. Decant the wash buffer and remove the residual Buffer by inverting the plate and tapping into absorbent towels.
3. Add 25 μL of the microarray (*see* **Note 8**).
4. Transfer 25 μL of sample from the PCR plate diluted into the 96-well plate half-area flat-bottom polystyrene (*see* **Note 6**).
5. Add 25 μL of Assay Buffer in each well (*see* **Note 9**). Final volume per well is 75 μL (*see* **Note 10**).
6. Seal the plate with an opaque plate sealer and wrap the plate with foil (*see* **Note 1**).
7. Incubate on a plate shaker overnight at 2–8 °C (*see* **Note 11**).
8. Eliminate the contents from the wells by decantation (*see* **Note 12**).
9. Wash the beads with 3 × 200 μL Wash Buffer (*see* **Note 13**).
10. Add 50 μL of Human Autoantibody PE-IgG Conjugate into each well (*see* **Note 14**).
11. Seal the plate with a plate sealer and wrap the plate with foil (*see* **Note 1**).
12. Incubate 90 min at room temperature (20–25 °C) on a plate shaker.
13. Eliminate the contents from the wells by decantation (*see* **Note 12**).
14. Wash the beads with 3 × 200 μL Wash Buffer (*see* **Note 13**).
15. Add 150 μL of Drive Fluid if using MAGPIX® to all wells (*see* **Note 15**).
16. Run plate on Luminex® 200TM, HTS, FLEXMAP 3D® OR MAGPIX® with xPONENT® software (*see* **Note 16**).

4 Notes

1. Try to minimize the light exposure, as the internal fluorescence of the beads could be bleached. Use an opaque mixed bottle and during incubation, protect the plates with an opaque sealer.
2. We recommend preparing the commercial microarray diluted 1:2 with Assay Buffer (Fig. 2). After analyzing a serum sample tested with the microarray at a 1X as final concentration and diluted 1:2 with Assay Buffer, no decrease in median fluorescence intensity (MFI) was observed. As shown in Fig. 3, the MFI remained consistent across the microarray, and the microarray dilution did not result in a significant reduction in MFI. On the contrary, in some cases such as, IL-6, IL-17A, IL-17F, and IL-1α, the MFI was higher in the diluted microarray. This

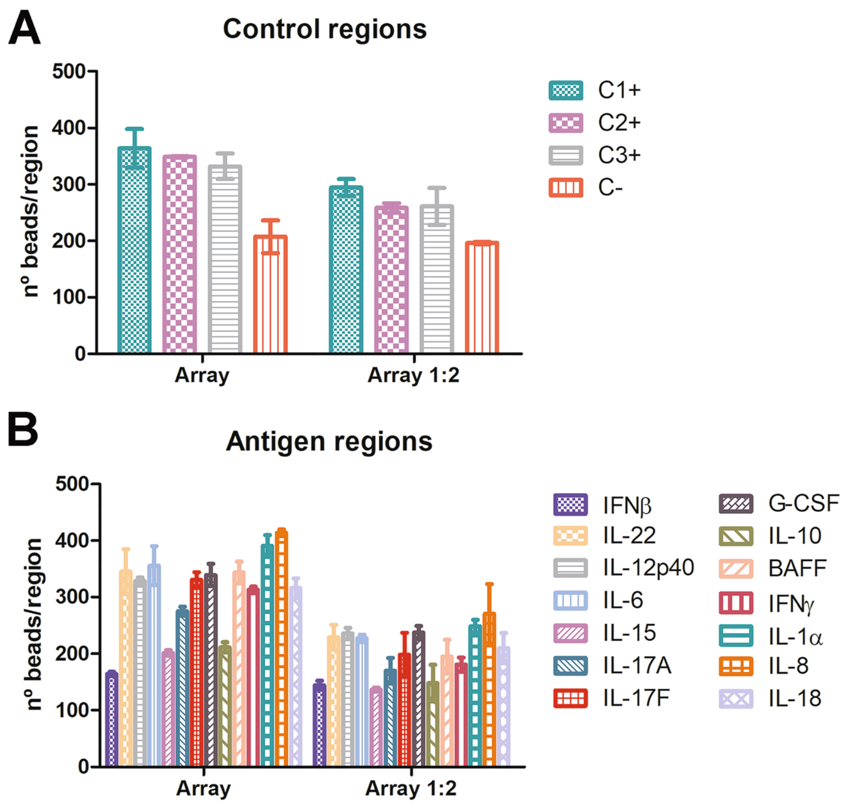

Fig. 2 Number of beads (mean ± SD) per region in a serum sample analyzed using different microarray concentration: according to the manufacturer instructions and 1:2 diluted array. After dilution, the number of magnetic beads per region decreased as expected. (**a**): Control regions. (**b**): Panel of 14 analytes

is likely due to the lower number of beads per region, which reduces non-specific binding and amplifies the MFI signal.

3. We have found that thawing samples overnight at 4 °C is a more practical approach.
4. We suggest designing a plate layout in which samples are randomly positioned to avoid any patterns that could influence the result.
5. The optimization of the dilution in serum samples was conducted under different conditions (1:20, 1:100, 1:250, and 1: 1000). Dilute the serum sample by transferring 5 μL of serum samples into a PCR plate with 95 μL of assay buffer to achieve a 1/20 dilution. Following, add 20 μL from the 1:20 diluted sample to a new PCR plate with 80 μL of assay buffer to create a 1:100 dilution. From the 1100 dilution, transfer 40 μL to another PCR plate with 80 μL of assay buffer to obtain a 1: 250 dilution. Finally, add 25 μL of the 1:250 diluted sample into a PCR plate with 9 μL of assay buffer to reach a 1:1000 dilution.

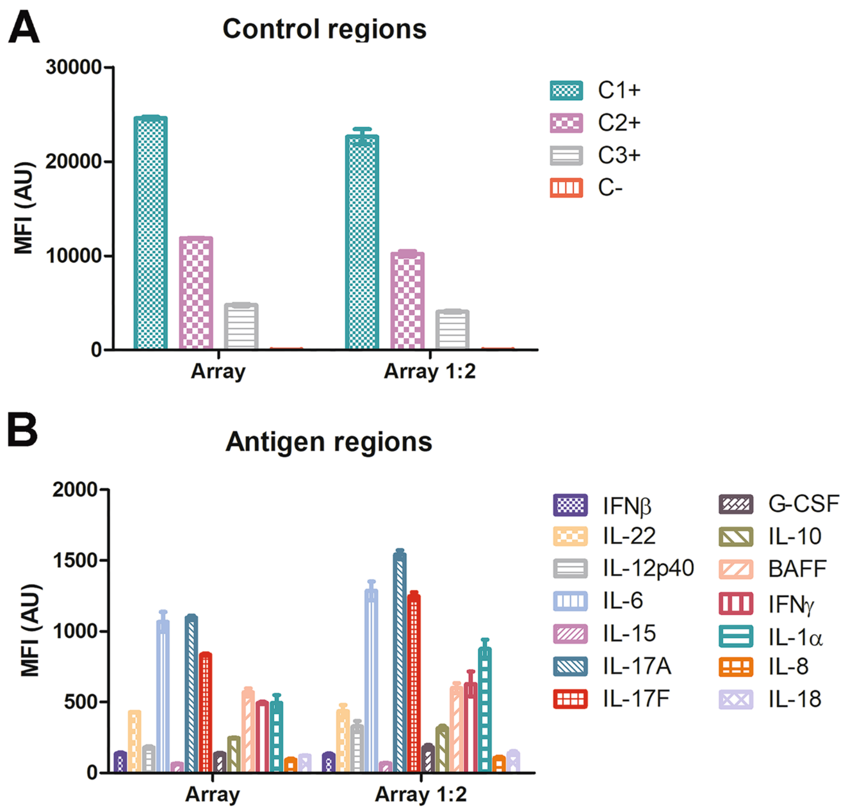

Fig. 3 Median fluorescence intensity (MFI) in arbitrary units (U.A.) for a serum sample analyzed using different microarray concentration: according to the manufacturer instructions and 1:2 diluted array. The MFI remained consistent across the different array conditions. Interestingly, the dilution of the microarray did not significantly decrease the MFI. For some analytes, such as IL-17A, the MFI increased when using the diluted microarray compared to the undiluted version: Diluting the microarray enhanced its sensitivity leading to an overall increase in MFI. (**a**): Control regions. (**b**): Panel of 14 analytes. Mean values ± SD are represented

In the 1:20 dilution, the MFI in the negative control was approximately 200 arbitrary units (AU). This elevated value suggests the presence of non-specific binding, leading to the rejection of the 1:20 dilution.

The average MFI values in 1:250 condition were below 200 AU for most analytes in the microarray, with the highest MFI observed at approximately 500 AU. For analytes such as IL-6, IL-17A, and IL-17F. On the other hand, the 1/1000 dilution exhibited a very low average MFI across the microarray, with the highest values reaching around 200 AU. Due to the low MFI values associated with the 1:1000 and 1:250 conditions, both were rejected.

In the 1:100 dilution, most analytes in the microarray presented MFIs ranging from 400 to 600 AU, with some values exceeding 1000 AU. For IL-6 and IL-17A. The lowest values were around 200 AU. For INFβ, IL-13p40, IL-15,

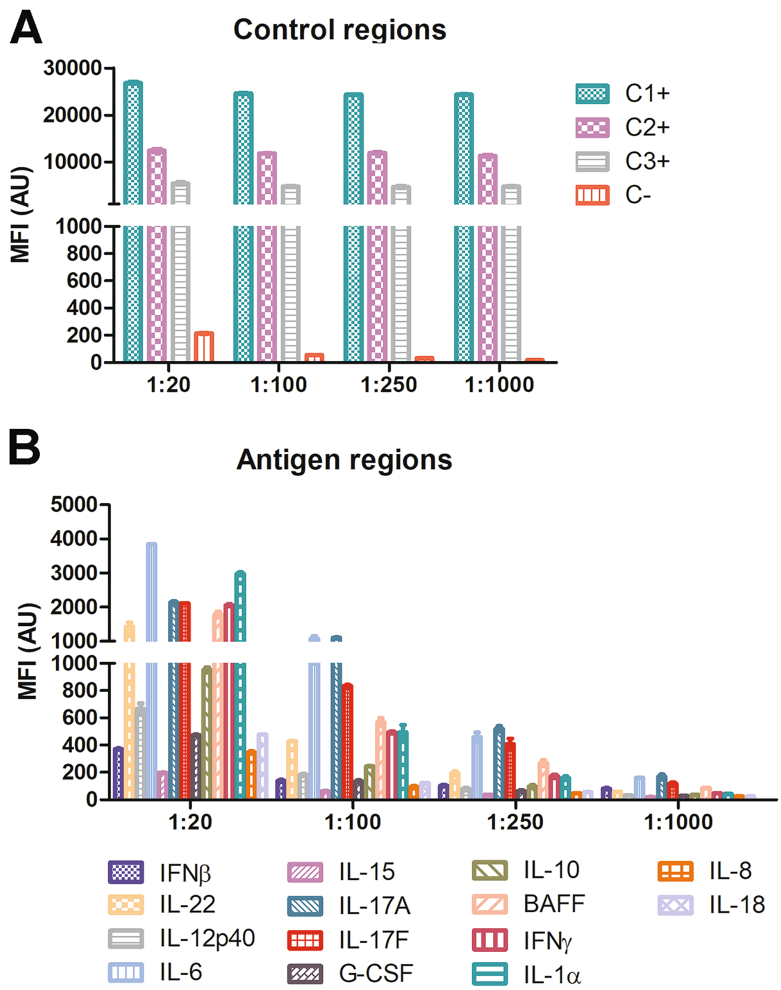

Fig. 4 Median fluorescence intensity (MIF) in arbitrary units (AU) for a pooled serum tested at different dilutions: 1:20, 1:100, 1:250, and 1:1000. (**a**): The elevated MFI show for the 1:20 negative control suggests the presence of non-specific binding, leading to the rejection of this dilution. (**b**): Due to the low MFI values associated with the 1:250 and 1:1000 sample dilutions, both were rejected. Therefore, among the different conditions tested, the 1:100 dilution produced the highest MFI values for the antigens in the microarray maximizing the signal/noise ratio. (**a**): Control regions. (**b**): Panel of 14 analytes. Mean values ± SD are represented

IL-8, and IL-18. Therefore, we recommend using the 1:100 dilution factor, as it provided a decreasing MFI value in the positive controls and a low MFI in the negative control. Among the different conditions tested, the 1:100 dilution produced the highest MFI values for the antigens in the microarray (Fig. 4).

6. We propose using the epmotion 96 by Eppendorf® for transferring volumes between plates since automating pipetting system ensures reproducibility.
7. We recommend adding the wash buffer with repetitive multichannel pipette capable of delivering 50 μL to 1200 μL.
8. We advise to mix the bottle before transferring the array into a 96-well half-area flat-bottom polystyrene plate. Use a repetitive pipette capable of delivering 5 μL to 100 μL to reduce the likelihood of variability.
9. We propose to use a repetitive multichannel pipette capable of delivering 5 μL to 100 μL to add the assay buffer.
10. We recommend adding the bead array, serum samples, and Assay Buffer in that specific order.
11. We propose incubating the assay overnight (16–18 h) at 2–8 ° C to increase affinity of the sample with the array. Alternatively, incubate for 2 h at room temperature (20–25 °C).
12. Place the 96-well half-area flat-bottom polystyrene plate into the plate magnet and incubate for 1 min to facilitate reattachment of the magnetic beads to the magnet. Decant the place attach with the magnet to eliminate the contents from the wells.
13. All washing steps described are performed using a 96-well plate magnet. Add the wash buffer off the magnet using a repetitive multichannel pipette capable of delivering 50 μL to 1200 μL. Afterwards, place the plate on the magnet and allow the array to soak for 1 min to facilitate reattachment of the magnetic microbeads to the magnet.
14. Allow the PE-IgG Conjugate to warm up to room temperature (20–25 °C) prior to the addition with a multichannel pipette capable of delivering 5 μL to 100 μL.
15. Resuspend the beads on a plate shaker for 5 min.
16. We recommend adjusting the settings based on the bead IDs listed in Tables 1 and 2 and modifying the following parameters: count a minimum of 50 beads per region and use 90 μL per well. For subsequent analysis, we suggest using the median fluorescence intensity (MFI).

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Part II

Protein Microarray Data Analysis

Check for updates

Chapter 7

Protein Microarray Analysis with GenePix

Rodrigo Barderas, Ana Montero-Calle, and Pablo San Segundo-Acosta

Abstract

Protein microarrays are powerful proteomics tools for high-throughput analysis. They allow to screen simultaneously large numbers of proteins for their binding specificity, enzymatic activity, and abundance in biological samples, among other uses. One widely used platform for protein microarray analysis is GenePix, which offers instrumentation and software solutions for data acquisition and analysis. Here, we provide an overview of the use of GenePix for protein microarray analysis, exploring its applications and advantages.

Key words Protein microarrays, Antibody microarrays, Phage microarrays, Screening, Proteomics, GenePix

1 Introduction

Protein microarrays consist of miniature arrays of immobilized proteins on a solid support, such as glass or membrane slides. These arrays enable the parallel analysis of thousands of proteins in a single experiment, making them valuable tools for various applications in biomedical research [1, 2].

Key applications of protein microarrays are related to the study of protein-protein interactions [3], the analysis of humoral immune responses in health and disease [1, 4–11], or the analysis of cytokines, immunomodulators, and growth factors as biomarkers of disease [12–15]. By immobilizing hundred to thousands of proteins on the array surface, the arrays can be probed for interactions with other proteins, nucleic acids, small molecules, or even entire cells [1, 16]. This allows for the systematic mapping of protein-interaction networks, identification of novel protein-binding partners, and characterization of signaling pathways.

Another important application of protein microarrays is the analysis of protein expression profiles [1, 16]. By probing the array with complex biological samples, such as cell lysates, tissue

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_7,

extracts, or body fluids, the abundance of specific proteins can be quantified, and biomarkers associated with physiological conditions or disease states can then be discovered [2, 7–11, 15, 17].

GenePix is a leading platform for protein microarray analysis, offering software solutions for state-of-the-art instrumentation and data acquisition and analysis [18]. The GenePix system typically consists of a microarray scanner equipped with laser excitation sources and high-resolution imaging detectors, along with software for image capturing of high-quality fluorescence images of protein microarrays, analysis, and data processing. The scanner uses precision optics and advanced detection mechanisms to capture fluorescence signals emitted by fluorescently labeled proteins bound to the array surface. This allows for the quantification of protein binding events with high sensitivity and a wide dynamic range. Additionally, GenePix software provides powerful tools for data analysis and visualization besides image acquisition. Background subtraction, normalization, and statistical analysis might also be performed to extract meaningful information.

GenePix technology finds a wide range of applications in protein-microarray-based proteomics analysis for interactomic studies, protein expression profiling, and biomarker discovery [5–8, 18–20]. In drug discovery and development, the use of protein microarrays interrogated with small molecule libraries or drug candidates followed by their analysis with GenePix has allowed to identify potential drug targets, survey for compound binding affinities, and analyze drug-induced changes in protein expression or activity in parallel [21–23]. In clinical diagnostics, the development of multiplexed protein microarray assays for disease detection and monitoring can be used to detect specific antibodies, antigens, or cytokines in patient samples, allowing for the diagnosis and prognosis of diseases such as cancer, autoimmune disorders, and infectious diseases [1, 12, 16].

In summary, GenePix technology is the most widely used software for image capturing, image analysis, and data processing prior to further analysis with different bioinformatic approaches, offering advanced instrumentation and software solutions to accelerate discovery efforts in proteomics, drug discovery, and clinical diagnostics. In this context, we report here basic protocols for the analysis of protein microarrays with GenePix for the characterization of disease-associated markers.

2 Materials

1. Hybridized nitrocellulose, plastic, or glass protein microarrays with fluorophores for protein-protein interaction, humoral-immune responses, or small-molecule targeting analyses.

2. Axon Molecular Devices GenePix 4000B microarray scanner (Molecular Devices), or any compatible microarray scanner.
3. GenePix Pro 7.1 (Molecular Devices).
4. Bioinformatic tools (Pomelo II, MultiExperiment Viewer, or Babelomics Suite, among others).

3 Methods

GenePix offers a wide range of features that allow users to extract, analyze, and manage data efficiently (*see* **Note 1**). This includes automated grid alignment, flexible data-extraction options, and powerful data-analysis tools (*see* **Note 2**). The first step in protein microarray analysis involves the scanning of the microarray slide to generate a *.tiff image. This image contains all the spots of the array representing the intensity of each protein printed on the microarray. GenePix Pro 7.1 software is then used to align a grid (*.gal file) over the image to match each spot with the corresponding printed protein. Once the whole grid is aligned, GenePix extracts the data from each spot converting the intensity to a numerical value (*.gpr file) (Fig. 1). The software also calculates a range of other metrics,

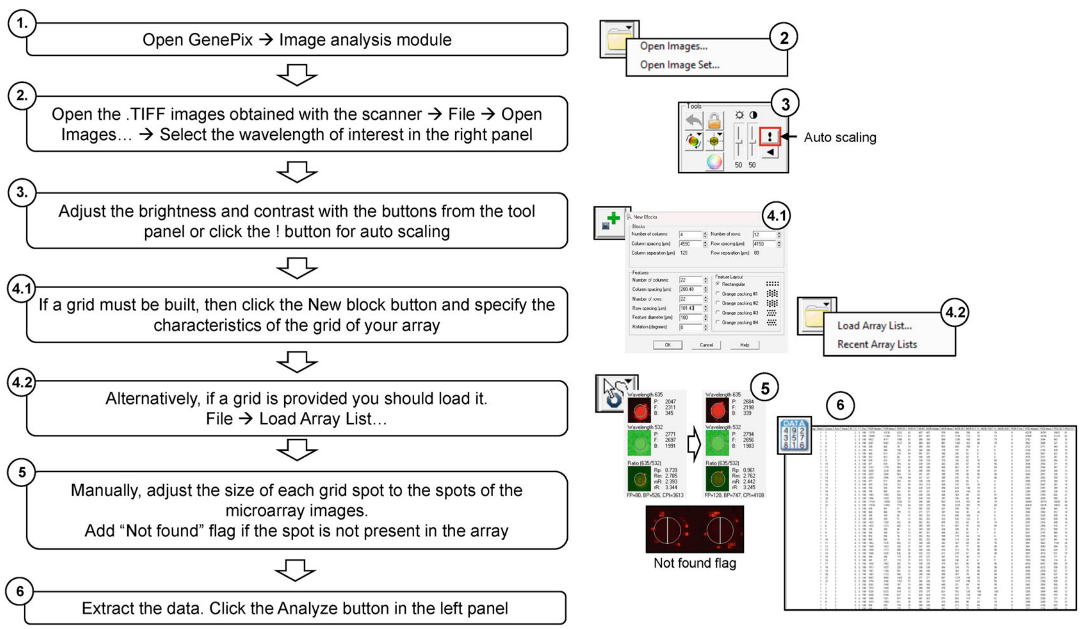

Fig. 1 Workflow for the analysis of protein microarrays with GenePix
After loading the *.TIFF images obtained with the microarray scanner using the GenePix program, a grid must be added to adjust the position and the size of the spots according to the fluorescence signal intensity obtained. Once all grid spots are correctly aligned with the spots of the microarrays, the signal intensity of each spot is obtained, which would be used for subsequent analyses

such as the background intensity and the signal-to-noise ratio, which can provide additional insights into the quality of the data. After extraction, data can be analyzed either with GenePix using a variety of statistical tools, or by means of web-based bioinformatics tools such as Pomelo II, Babelomics suite, or MultiExperiment Viewer [24–26] (*see* **Note 3**).

3.1 Protein Microarray Image Acquisition

1. Launch GenePix Pro 7.1 software. Open the software on your computer.
2. Place the protein microarray slide into the GenePix scanner (i.e., GenePix 4000B microarray scanner), according to the manufacturer's instructions. The scanner must be turned on 30 min before use to allow lasers to warm-up and reach optimal temperature and stability. Laser preheating allow for proper laser operation and a stable and consistent light emission.
3. In the GenePix Pro 7.1 software navigate to the image acquisition module.
4. Set the scanning parameters such as laser intensity, photomultiplier tube (PMT) gain, and scan resolution in *Preview* mode (*see* **Note 4**).
5. Capture images of the microarray slide by clicking on the *Scan* button to get the images at 532 nm for the green laser and 635 nm for the red laser (*see* **Note 5**).

3.2 Protein Microarray Image Analysis

1. After acquiring the *.tiff images, open the image analysis module of GenePix.
2. Load the acquired images.
3. Define the microarray grid by specifying the location of each spot on the array. Alternatively, if already defined or provided by the manufacturer, load the *.gal file with the grid (*see* **Note 6**).
4. Align the grid over the microarray image (*see* **Note** 7). Use the software tools to move the grid. Adjust the left top and the bottom right spots at the beginning is the easiest way to perfectly align the grid to every spot in the microarray.
5. As some variations along the different protein microarrays used in the experiment can occur, it is mandatory to align the grid per each microarray. Grid spots can be moved individually to ensure the complete alignment with the spots on the microarray. Also adjust the size of each spot to minimize the background signal in each spot. This alignment might be saved as an *.gps file, where you can come repeatedly until users are completely satisfied with the alignment of the grid (Fig. 2).

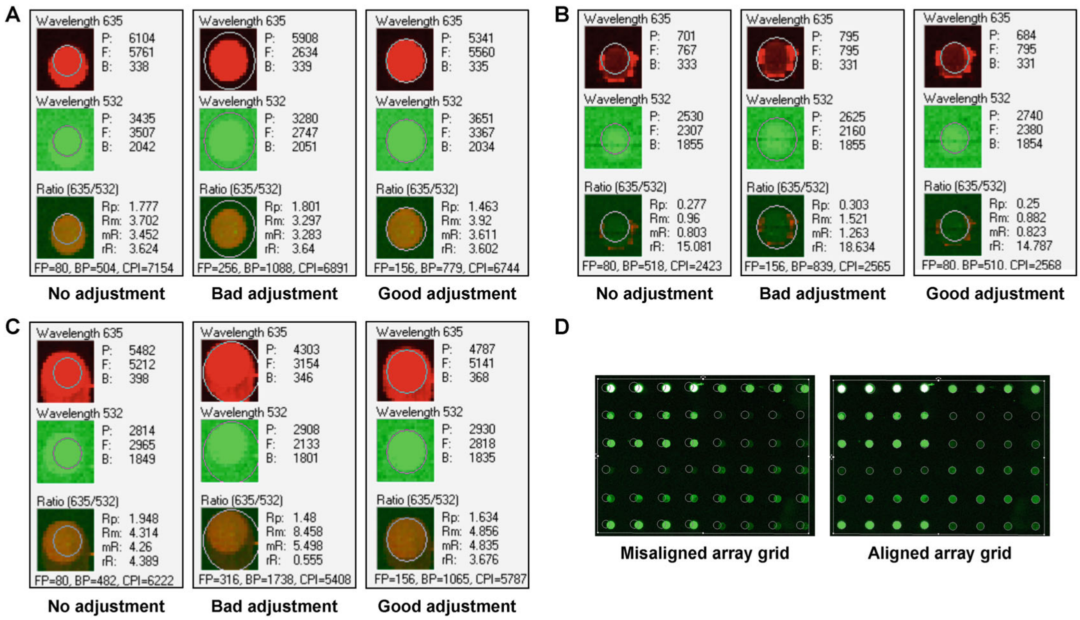

Fig. 2 Adjustment of grid spots to the spots on the microarrays
(**a**, **b**, and **c**), represent different type of spots that can be found on protein microarrays after scanning. The size and position of the grid spots might match the position of the spots printed on the microarrays, and signals around the spot should not be selected for quantification, which in most cases are associated to signal saturation. (**d**) An image of a subarray of a high-density protein microarray as seen in the GenePix software with the array grid misaligned to the spots (left) and the same subarray after manually or automatically adjusted using GenePix (right)

6. Extract the data once the grid is completely aligned. Click in the *Analysis* menu and select *Extract data*. This step will convert the intensity of each spot to a numerical value (*see* **Note 8**).
7. Save the results. Clicking on the *File* menu and selecting *Save as* will allow users to save the results as a *.gpr file in the specified folder.

3.3 Data Processing

1. Once the spot intensities have been quantified and normalized, and results have been saved as a *.gpr file, data from GenePix software might be exported and used directly into a spreadsheet (i.e., Microsoft Excel) or another data analysis software for further analysis (*see* **Note 9**).
2. Perform statistical analysis to identify differentially expressed proteins or to compare protein expression levels across different conditions or experimental groups. Several web-based approaches might be used (i.e., cloud-based applications supporting analysis, visualization, and stratification of microarray data such as Pomelo II, Babelomics suite, or MultiExperiment Viewer -WebMeV-) [24–28].

3. Generate different graphical representations of the results such as heatmaps (i.e., supervised or unsupervised clustering), scatter plots, or volcano plots to better visualize the data and identify patterns or trends (*see* **Note 10**).
4. Interpret the results in the context of the biological relevance to select those specific proteins or hits to be further validated by orthogonal approaches or functionally in vitro, ex vivo, or in vivo.

3.4 Quality Controls and Validation

Two important aspects to consider regarding protein microarray analyses are quality control and validation. These critical stages in protein microarray experiments must be performed to ensure the reliability and reproducibility of the results. Regarding quality control (*see* **Note 11**), there are several technical aspects to be surveyed during experimentation:

1. Array quality. This includes the quality of the printed spots, the spot morphology, and the uniformity of protein immobilization.
2. Sample quality. This includes the purity, concentration, and integrity of the protein samples.
3. Labeling efficiency. If the experiment involves labeled proteins or labeling of protein extracts with fluorescent dyes, it should be mandatory to check the efficiency of the labeling process by measuring the signal-to-noise ratio in the microarray images.
4. Reproducibility. This point involves running technical and biological replicates to check for variability in the results.

Regarding validation, one important aspect to consider relies on the lack of validation on omics studies focused on the dataset side. In this sense, due to the potential for the presence of false-positive identifications in protein microarray-based proteomics approaches (as occurs in all high-throughput techniques), results must be validated using alternative orthogonal approaches to ensure accuracy of the generated data.

In this context, three key steps are needed to complete a protein microarray experiment from the analysis to the validation:

1. Perform quality controls during the whole protein microarray experiment to ensure the reliability and reproducibility of the data.
2. Validate the results using alternative methods or orthogonal approaches such as western blotting (WB), enzyme-linked immunosorbent assay (ELISA), immunohistochemistry, Luminex technology, biosensors, or Halotag-beads based immunoassays [8, 9, 29–34] (*see* **Note 12**).

3. Critically compare the results obtained from the protein microarray analysis with those obtained from the validation assays to confirm the accuracy and reliability of the findings.

In summary, in this chapter we have intended to give to the readers an effective protocol to get protein microarrays images, perform the alignment of the grid, and to extract data with GenePix software. This would lead to get valuable insights into protein interactions, expression profiles, and biomarker discovery upon application of the protocol to the differently designed protein microarrays experiments by the potential readers. However, the exact steps may vary depending on the GenePix software version used and the specific type of microarray surface. Finally, always ensure to validate your results and, if possible, contact with a bioinformatics prior to the experiment and during data analysis.

4 Notes

1. In addition to its analysis capabilities, GenePix might also provide features for managing microarray data. The users might store their data in a structured format, making data easy to be retrieved and shared. The software also supports the MIAME (Minimum Information About a Microarray Experiment) standard [35], which would allow to share raw microarray data with the scientific community.
2. One of the advantages of GenePix is its flexibility, which makes it suitable for a wide range of protein microarray applications. The software allows users to customize data extraction process (i.e., users might choose to extract data from a specific region of interest within each spot, or adjust the threshold to distinguish between signal and background).
3. It would be recommended to analyze the data with more than two different programs. These programs use algorithms which perform different statistical analyses, offering potentially different results. Restricting the validation process to the proteins that are found to be statistically significant by several programs might facilitate the process.
4. Scanning the arrays at a minimum of two different laser intensities, such as 100% and 10%, might enhance the subsequent analysis of low and high intensity spots. To obtain a preview of the microarray scans, set up laser intensities at a resolution higher than 20 μm. For final image acquisitions, the scan should be ideally performed at 5 μm. Alternatively, a resolution as high as 10 μm can also be used.
5. The microarray slide should be scanned at a different laser intensity depending on the nature of the microarray (i.e.,

nitrocellulose, plastic, or glass). Nitrocellulose microarrays usually present a larger background than glass and plastic microarrays because of unspecific binding of the dyes. Thus, nitrocellulose slides are usually scanned at lower intensities than glass and plastic ones to get better signal-to-noise ratios.

6. If using a commercial protein microarray, the manufacturer usually provides a *.gal file for each lot. This file contains the information of the grid to get the correspondence between each spot and the protein to be analyzed.
7. The software will automatically try to align the grid over the microarray (use F5 in the keyword). However, if the automatic alignment (F5) is not adequate, you should carefully adjust it manually.
8. The software performs background subtraction to remove non-specific signal from the surrounding area of each spot, so in case the background is much higher in some specific regions of the microarray the software takes into account these differences in the background to each spot.
9. Users might analyze the data with GenePix Pro 7.1 software. To this end, users might click on the *Analysis* menu and select *Analyze Data* to open a new window and perform different statistical analyses on the data. However, it is most usual to extract the data from the protein microarray with GenePix and perform the statistical analysis with other software. In this sense, it would be recommended to try two or three different software tools to focus the validation on those spots more consistent across the different software tools.
10. For obtaining plots, MeV is highly recommended for the display of high resolution supervised and unsupervised clusters, among other analyses.
11. For quality control of protein microarrays, microarray scanners and image analysis software might be used to assess the protein microarray quality. To assess for protein or protein extracts quality to be used in the protein microarrays, several methods such as SDS-PAGE, WB, and/or mass spectrometry might be used. Finally, to assess reproducibility, statistical measures such as the coefficient of variation might be used to confirm the reproducibility of the results among spots and replicates.
12. While most of the validation is performed on dysregulated proteins, it would be also recommended to validate the absence of dysregulation in several proteins for consistency of the results regarding upregulated proteins, downregulated proteins, and non-dysregulated proteins.

Acknowledgments

This work was supported by the financial support of PI20CIII/00019 and PI23CIII/00027 grants from the AES-ISCIII program cofounded by FEDER funds to R.B, and the financial support of grant PID2022-140307OB-I00 funded by MCIN/AEI/10.13039/501100011033 and by “ERDF A way of making Europe”. P.SS-A. was supported by a Sara Borrell Postdoctoral contract of the AESIRRHH 2021–2022 call.

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Chapter 8

Evaluating Cross-Reactivity in Custom-Made Multiplex Bead-Based Antibody Microarrays: A Methodological Approach

Rocío Paz-González, Patricia Quaranta, Selva Riva-Mendoza, Pablo Domínguez-Guerrero, Francisco J. Blanco García, Cristina Ruiz-Romero, Valentina Calamia, and Lucía Lourido

Abstract

Multiplex antibody microarray assays based on suspension microsphere technology have emerged as a powerful tool in both basic and translational biomedical research, enabling the simultaneous quantification of multiple proteins within a single assay. Developing custom-designed multiplex assays offers the advantage of flexibility beyond predefined biomarker panels; nonetheless, addressing cross-reactivity remains a crucial step to ensure assay reliability and accuracy results. This chapter provides a comprehensive, step-by-step methodology for evaluating cross-reactivity in multiplex antibody microarrays, detailing techniques to validate and optimize assay performance. Additionally, it explores strategies for mitigating cross-reactivity to enhance the precision and interpretability of assay results.

Key words Cross-reactivity, Multiplex microarray, Antibody microarray, Suspension bead array

1 Introduction

Multiplexed analysis enables the simultaneous detection of several analytes in a single assay, thereby reducing time, labor, and costs by minimizing the amount of sample and reagents required compared to single-reaction methods. It allows for higher throughput and better cross-comparability within and across experiments, as all assay components are processed at the same time under identical conditions. In addition to target approaches, the ability to theoretically measure over 500 analytes simultaneously makes multiplex technology a powerful tool for screening, providing a comprehensive perspective of diseases and a holistic understanding of physiological pathways and mechanisms [1]. Therefore, multiplexing supports the progressive shift towards personalized medicine,

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_8,

enabling molecular fingerprinting of diseases by identifying biomarker profiles [2].

Bead-based array immunoassays also known as suspension microarrays, such as those sold by Luminex, BD Biosciences, Thermo, Abcam, and other companies, are well-established platforms for multiplexed quantification in both clinical and research studies [3]. Luminex has become the leader in bead-based multiplex solutions, with its xMAP® Technology—based on labelled magnetic beads or microspheres—being the most widely implemented platform across a wide range of applications, including transplant medicine, biomarker discovery and validation, pathogen detection, drug discovery and vaccine development [4].

While many commercially manufactured kits are designed for the simultaneous analysis of predefined biomarker panels, researchers frequently require the flexibility to develop custom microarrays for specific biomarker sets tailored to their own research objectives. Luminex multiplex sandwich immunoassays are conducted by incubating in suspension capture antibodies immobilized on distinct microsphere regions, with a biological sample. This is followed by adding a cocktail of detection antibodies, designed to bind exclusively to their target analytes (Fig. 1). However, a critical step in multiplex assays is the evaluation of cross-reactivity once both capture and detection antibodies are mixed in solution [5]. The susceptibility of multiplex sandwich suspension assays to cross-reactivity increases quadratically with the number of targets, restricting the potential scalability of multiplexing [6].

Cross-reactivity occurs when antibodies bind to non-target molecules, leading to inaccurate and irreproducible results [7]. It could increase background noise, thereby reducing the assay's detection limit; whereas in the worst case, it may result in false positives, raising the risk of misleading outcomes. Cross-reactions can occur in different ways, such as through antibody—antibody or antibody-antigen non-specific interactions. The scenarios of possible cross-reactivity between capture antibody (cAb), detection antibody (dAb), and analytes are summarized in Fig. 2 [6, 8]. Addressing this issue, it is crucial to ensure the reliability and accuracy of quantification in multiplex assays. This chapter details methods for evaluating cross-reactivity in the development of custom-made multiplex suspension protein microarrays, along with strategies to minimize it.

2 Materials

This section details the consumables and equipment required to perform a multiplex sandwich assay using Luminex xMAP technology. However, the protocol described in this chapter can also be applied to other suspension protein array platforms.

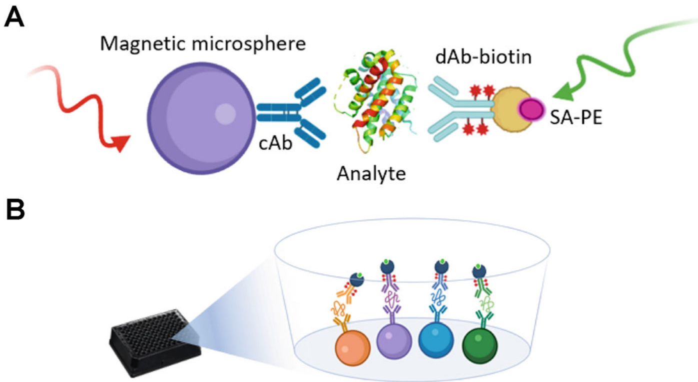


Fig. 1 Luminex xMAP Technology. (**a**) The xMAP Luminex technique is a suspension sandwich immunoassay based on magnetic microspheres (beads) as the solid phase for the microarray. These microspheres are available in 500 distinct fluorescent colors (bead regions), each carrying a detection reagent on its surface for discrimination purposes. This capability enables the simultaneous detection of up to 500 analytes in a single assay. Each bead region is coated with a capture antibody specific to a particular analyte of interest. In a subsequent step, the bead regions, each coated with different capture antibodies, are incubated together with the biological sample in a single experiment. Thereafter, a biotinylated detection antibody (dAb-biotin) cocktail is introduced, along with streptavidin (SA), which binds to the bead complex. Finally, a phycoerythrin (PE) fluorescent reporter is coupled to the complex to generate the signal. (**b**) An ideal multiplex assay performance, in the absence of cross-reactivity, displays each analyte sandwiched between the corresponding capture and detection antibodies

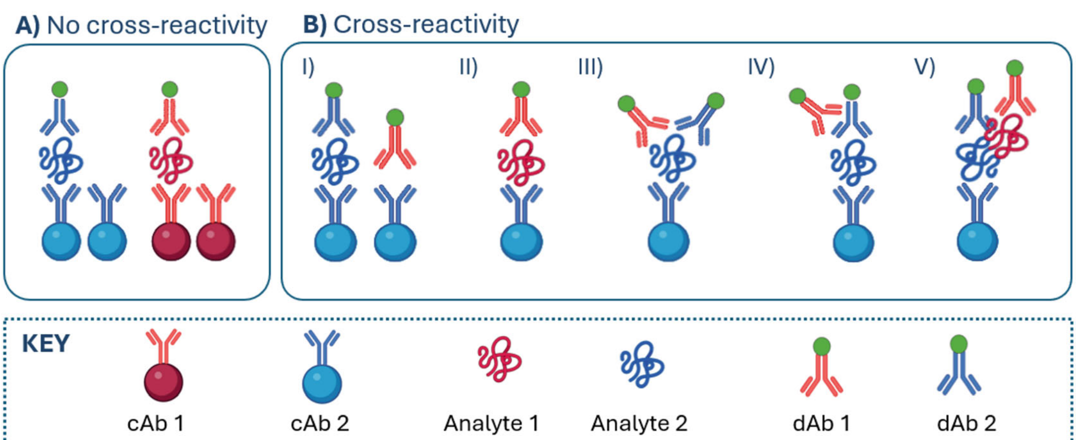


Fig. 2 Potential scenarios of cross-reactivity in a suspension sandwich microarray. (**a**) Ideal scenario in the absence of cross-reactivity. (**b**) Potential scenarios of cross-reactivity leading to false positive signal when: the non-matched capture antibody (cAb) (blue) reacts with detection antibody (dAb) (red) in the absence of the target analyte (I), cAb (blue) binds with no target protein (red) (II), dAb (red) binds with no target protein (blue) (III), dAb (red) binds dAb (blue) (IV) and, protein-protein interactions of two proteins targeted on the multiplex (V)

2.1 Consumables

1. Beads: MagPlex magnetic microspheres. Protect from light exposure.
2. Plates: 96-well dark flat-bottom polystyrene plates.
3. Matched antibody pairs and recombinant protein. Detection Antibody must be biotinylated.
4. Sample dilution buffer: 1% Serum Bovine Albumin in PBST, stored at +4 °C up to 1 week.
5. Wash buffer: 0.05% Tween 20 in 1× PBS, pH 7.4 (PBST), stored at +4 °C.
6. Assay buffer: 0.05% Tween 20 in 1× PBS, pH 7.4 (PBST), stored at +4 °C.
7. Detection solution: R-Phycoertythrin modified streptavidin diluted to 0.5 μg/mL in PBST. Protect from light exposure.
8. Running buffer: PBST.

2.2 Equipment

1. Plate shaker.
2. Plate magnet.
3. Luminex xMAP® instruments: MAGPIX.
4. Software: MAGPIX xPONENT.

3 Methods

Since this chapter focuses on evaluating cross-reactivity, the preceding steps for building a multiplex assay based on Luminex xMAP technology are not covered in detail. These initial steps include coating the capture antibodies onto independent microsphere regions and verifying the efficiency of the coupling. The methodology for running the assay is not described here. The proper execution of these stages is crucial for the overall success of the assay.

3.1 Developing a Standard Curve for Each Single Analyte

The first step requires the optimization of the standard curve for each analyte individually within the multiplex format (*see* **Note 1**). The accuracy of a multiplex sandwich immunoassay heavily relies on the development of a well-optimized standard curve. To achieve this, several key steps should be followed:

3.1.1 Prepare Serial Dilutions

Start by preparing a serial dilution of known concentrations of the recombinant protein in an appropriate dilution buffer (*see* **Note 2**). Typically, this involves a 1:2 or 1:3 dilution series to cover a wide dynamic range of protein concentrations. The goal is to observe upper signal saturation and background noise to establish an appropriate concentration threshold for standard curve development (*see* **Note 3**).

3.1.2 Choose the Right Range

The concentration range for the standard curve should encompass the expected concentrations of the analyte in the samples. Adjust the highest and lowest points of the curve to ensure sufficient coverage of the analyte for low and high-abundance samples (*see* **Note 4**). Ensure a strong upper signal and low background, representing the highest and lowest concentrations, then perform the necessary serial dilutions to fit an 8-point standard curve. Perform all standards in technical replicates to account for variability, ensuring that the coefficient of variation (CV) between replicates remains below 15% (*see* **Note 5**).

3.1.3 Curve-Fitting Model

Fit the standard data to a suitable mathematical model, often a 4- or 5-parameter logistic regression. Thereafter, validate the linearity by ensuring that the coefficient of determination (R^2) is close to 1. The highest concentration point from the optimized standard curve of each analyte will be used for cross-reactivity testing (detailed in Subheading 3.2).

3.2 Cross-Reactivity Test

To evaluate all potential cross-reactivity scenarios involving the capture antibody (cAb), detection antibody (dAb), and analytes in sandwich suspension-format assays, three independent experiments are performed (Fig. 3) (*see* **Note 6**). Cross-reactivity tests are conducted using the highest concentration of the recombinant protein, as determined after optimizing the single standard curve for each analyte in Subheading 3.1.2 (*see* **Note 7**). Therefore, the reference signal for no cross-reactivity is the one obtained using the highest concentration of the recombinant protein of the single standard curve. For cases where more than two analytes are multiplexed, the same protocol applies, incorporating the reagents for all the analytes intended for quantified simultaneously (*see* **Note 8**).

3.2.1 Test 1: Cross-Reactivity Test of Capture Antibodies

- Test 1 assesses whether cross-reactivity arises from non-specific binding between the analytes and the cAb.
- Each analyte is incubated individually in an independent assay with its corresponding dAb and a mixture of all the capture antibodies potentially included in the multiplex assay (Fig. 3a).

3.2.2 Test 2: Cross-Reactivity Test of Detection Antibody

- This test determines whether cross-reactivity occurs due to non-specific bindings when all the dAb included in the multiplex assay are mixed.
- Each single analyte is incubated in a separate assay with a mix of both cAb and dAb (Fig. 3b).

3.2.3 Test 3: Cross-Reactivity Test of Target Analyte

- This test evaluates the overall cross-reactivity within the multiplex setup.

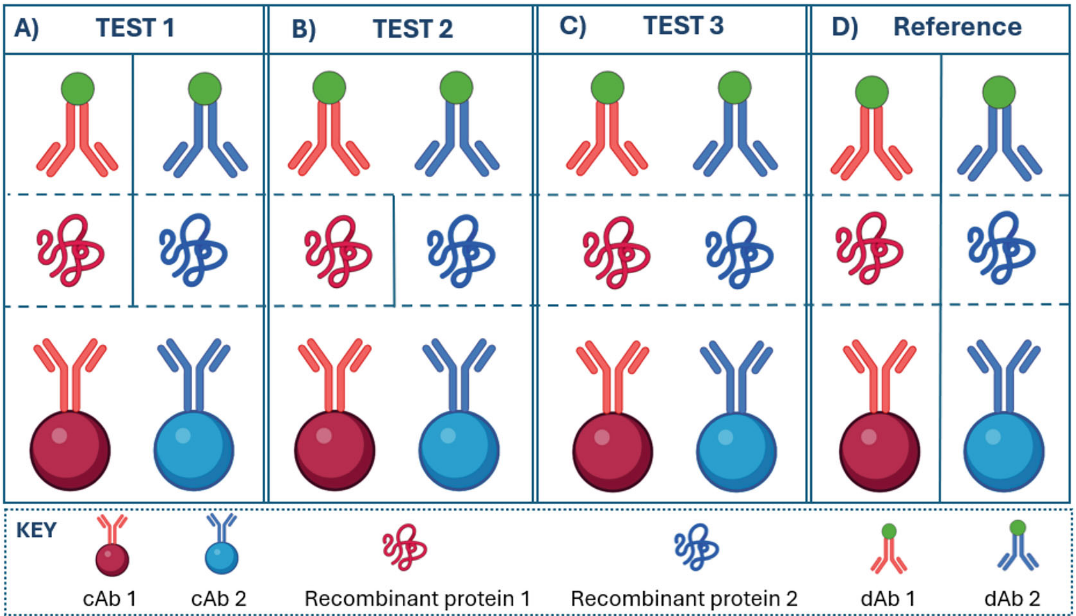


Fig. 3 Overview of cross-reactivity tests. (**a**) Test 1: Assessment of non-specific interactions of capture antibodies (cAb). A cocktail of cAbs is incubated with its corresponding recombinant protein and detection antibody (dAb) independently for each analyte. (**b**) Test 2: Assessment of non-specific interactions of detection antibodies (dAb). A cocktail of cAbs is incubated with its corresponding recombinant protein and a cocktail of dAbs, independently for each analyte. (**c**) Test 3: Multiplex assay to assess analyte interferences. A cocktail of cAbs, recombinant proteins, and dAbs is incubated together in a single assay. (**d**) Single-plex assay for signal reference. The horizontal dashed line marks the reagents used in each assay step (cAb, recombinant protein, dAb), while the vertical solid line indicates the reagents that must be mixed per analyte for each test to assess cross-reactivity in a 2-plex

- The custom-designed multiplex assay is performed by simultaneously incubating the cocktail of cAbs, dAbs, and analytes (Fig. 3c).

3.2.4 Running the Cross-Reactivity Assay

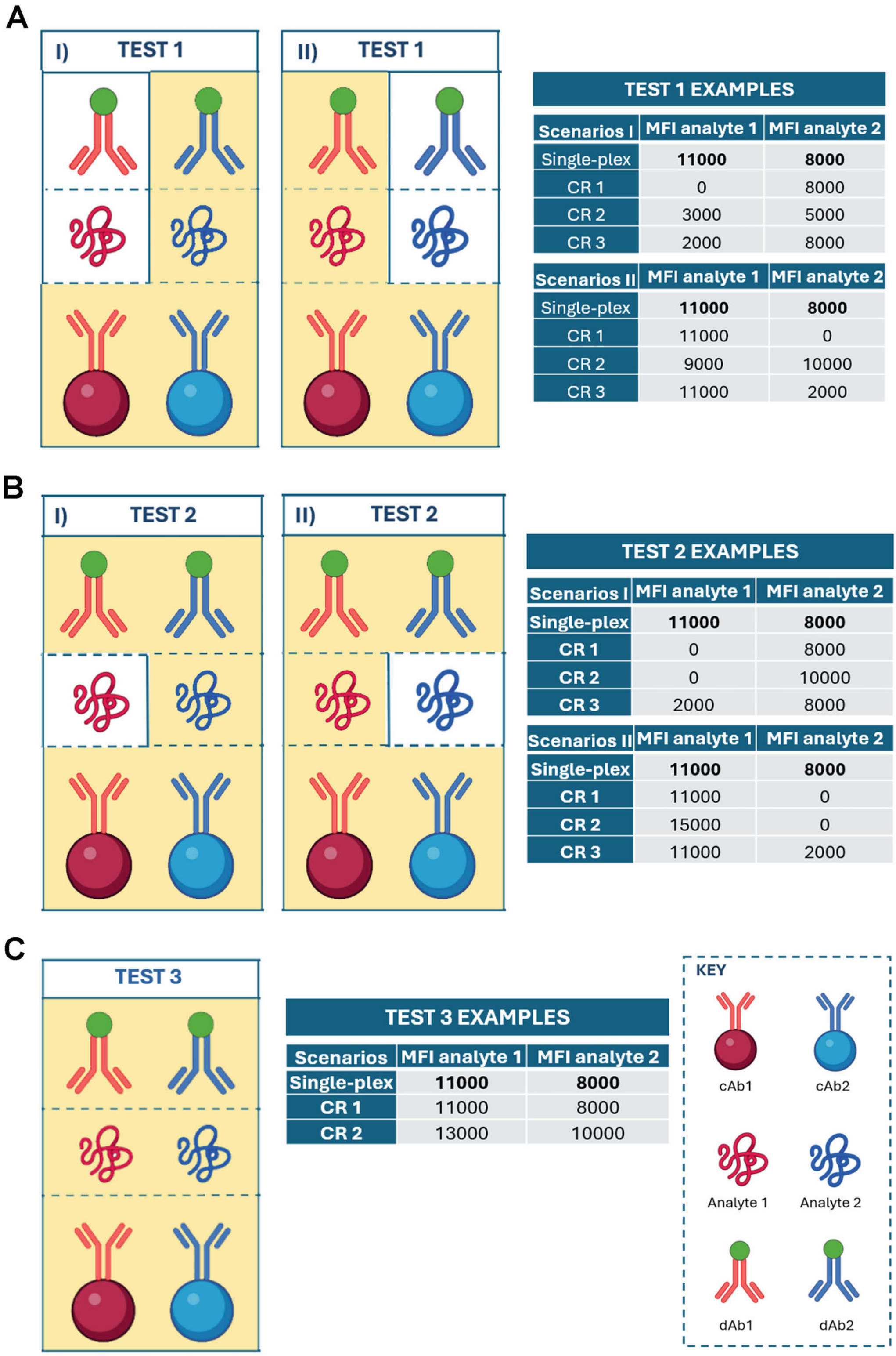
Finally, the cross-reactivity analysis must include Test 1, Test 2, and Test 3 run simultaneously alongside single-plex assay for each analyte as a reference to ensure comparison of results across conditions.

3.3 Data Analysis and Interpretation

To interpret cross-reactivity results each test must be independently assessed using single-plex signal as a reference for each analyte. To conclude that no cross-reactivity is present, the multiplex signal obtained for each analyte must not differ by more than 10% from the single-plex assay results.

3.3.1 Test 1 Interpretation

If cross-reactivity is detected in test 1, it can be attributed to non-specific binding by the cAbs (Fig. 4a).



TEST 1 EXAMPLES

Scenarios I	MFI analyte 1	MFI analyte 2
Single-plex	**11000**	**8000**
CR 1	0	8000
CR 2	3000	5000
CR 3	2000	8000

Scenarios II	MFI analyte 1	MFI analyte 2
Single-plex	**11000**	**8000**
CR 1	11000	0
CR 2	9000	10000
CR 3	11000	2000

TEST 2 EXAMPLES

Scenarios I	MFI analyte 1	MFI analyte 2
Single-plex	**11000**	**8000**
CR 1	0	8000
CR 2	0	10000
CR 3	2000	8000

Scenarios II	MFI analyte 1	MFI analyte 2
Single-plex	**11000**	**8000**
CR 1	11000	0
CR 2	15000	0
CR 3	11000	2000

TEST 3 EXAMPLES

Scenarios	MFI analyte 1	MFI analyte 2
Single-plex	**11000**	**8000**
CR 1	11000	8000
CR 2	13000	10000

Fig. 4 Examples of Cross-Reactivity Scenarios. The mean fluorescence intensity (MFI) for each analyte in the single-plex serves as the reference signal. In the example provided, the highest concentration of recombinant protein for Analyte 1 (red) yields an MFI of 11,000, while Analyte 2 (blue) reaches an MFI of 8000. Cross-

3.3.2 Test 2 Interpretation

If no cross-reactivity is detected in test 1 but it is observed in test 2, the cross-reactivity is likely related to non-specific binding by the dAbs (Fig. 4b). Furthermore, if cross-reactivity is detected in both Test 1 and Test 2 with differing signals, both cAbs and dAbs are involved in non-specific interactions.

3.3.3 Test 3 Interpretation

If cross-reactivity was not detected in test 1 and test 2 but it was observed in test 3, it indicates the presence of protein-protein interactions interfering with the sandwich format (Fig. 4c). The signal obtained with Test 3 is the one you would expect when performing the multiplex assay.

3.4 Strategies to Minimize Cross-Reactivity

- To reduce the likelihood of cross-reactivity from cAb, it is recommended to use monoclonal antibodies. Currently, monoclonal antibodies generated by classical hybridoma techniques constitute the most widely utilized capture ligands [6].
- Reducing reagent concentrations can decrease cross-reactivity by up to five-fold. To minimize dAb-related cross-reactivity, lower the concentration of the dAb. Typically, 1 μg/mL is used in Luminex xMAP experiments. Verify that the signal remains consistent in the single-plex assay when reducing the dAb concentration and repeat the cross-reactivity test with the adjusted lower dAb concentration.
- Efforts to mitigate cross-reactivity include longer washing steps to remove weak interactions.

3.5 Summary

Multiplex microarrays have become widely used in both translational and basic research due to their ability to simultaneously quantify multiple biomarkers in a single assay. The key advantage of custom-designed multiplex assays is their flexibility, allowing researchers to tailor biomarker panels to their specific study needs. However, one of the critical challenges in developing multiplex assays from scratch is the evaluation of cross-reactivity, especially

Fig. 4 (continued) reactivity scenario 1 (CR 1) illustrates the ideal outcome, with no cross-reactivity observed. The reagents used in each test are highlighted in yellow. A) Interpretation of Cross-Reactivity in Test 1 Due to Non-Specific Binding of Capture Antibodies: Analyte 1 produces a false positive signal, while a reduction in the signal for Analyte 2 is observed, suggesting that cAb1 is binding non-specifically to Analyte 2 and competing with cAb2 for its target (CR 2 in Scenario I); analyte 1 shows a false positive signal, while the signal for Analyte 2 remains stable, indicating that cAb1 is directly binding to dAb2 (CR 3 in Scenario I); cAb2 may bind non-specifically to Analyte 1 (CR 2 in Scenario II) or directly bind to dAb1 (CR 3 in Scenario II).B) Interpretation of Cross-Reactivity in Test 2 Due to Non-Specific Binding of Detection Antibodies: dAb1 binds to Analyte 2 or to dAb2, resulting in no signal for Analyte 1 and an increased signal for Analyte 2 (CR 2 in Scenario I); analyte 1 shows a false positive signal while the signal for Analyte 2 keep the reference, suggesting that dAb1 is binding to cAb1 in the absence of recombinant protein 1 (CR 3 in Scenario I); dAb2 may bind to Analyte 1 or dAb1 (CR 2 in Scenario II) or bind to cAb2 in the absence of recombinant protein 2 (CR 3 in Scenario II).C) Interpretation of Cross-Reactivity in Test 3 Due to Protein-Protein Interactions: If no cross-reactivity is detected in Tests 1 or 2, but the signal varies in Test 3, this is likely due to protein-protein interactions when using suspension microsphere technology such as the offered by Luminex. To obtain accurate and reliable results, it is essential to identify and address the extent of any non-specific bindings among assay reagents—capture antibodies, analytes, and detection antibodies—which require testing the entire multiplex assay against each target alone.

In this chapter, we have outlined a step-by-step methodological approach to evaluate cross-reactivity in custom-made multiplex assays, providing a robust framework for identifying which assay component is responsible for the non-specific bindings. Additionally, we provide key strategies to mitigate cross-reactivity and enhance multiplexing without comprise assay performance. These steps are crucial for achieving accurate and reproducible results, which are essential for successful biomarker discovery and validation in multiplex assay platforms.

4 Notes

1. The final volume of each reagent should be established by the user based on the platform needs and samples availability. Likewise, the concentration of the dAb should be established beforehand.
2. To enhance accuracy, the standard curve should be generated in a matrix similar to that of the actual samples. When using serum or plasma, it is advisable to spike the standards into an assay buffer containing 1% Bovine Serum Albumin (BSA) to minimize background interference. For developing custom-made Luminex xMAP assays with serum or plasma samples, we recommend using a buffer composed of Phosphate Buffered Saline (PBS), 1% BSA, and 0.05% Tween 20.
3. In initial trials, especially if there is no prior experience with the analyte, it is most efficient to create a standard curve with multiple points (generally no more than 12) using 1:2 or 1:3 dilutions depends of the stock concentration. This approach allows for a comprehensive evaluation of the recombinant protein behavior across a broad concentration range, enabling the observation of both signal saturation and background noise.
4. If the dynamic ranges of the analytes you want to quantify differ significantly, it may not be practical to use a multiplex assay, as you will need to perform multiple sample dilutions.
5. It is recommended to conduct each assay of the tests in triplicate.
6. In addition to cross-reactivity among the immunoassay components, heterophilic antibody interference can also lead to false results [9]. False positives arise where the heterophilic antibodies bridge capture and detection antibodies in the absence of analyte. False negatives occur where the capture or

detection antibody is directly bound by the heterophilic antibodies and blocked from binding to the analyte. The procedure to reduce the heterophilic antibody interference is described in Lourido L. et al [10].

7. The absence of cross-reactivity at the maximum recombinant protein concentration implies that at lower concentrations will also remain unaffected.
8. For cases where more than two analytes are multiplexed, the same protocol applies but incorporating the reagents for all the analytes intended for quantified simultaneously in each test. In the event that cross-reactivity is observed in any of the tests, it is essential to identify the specific analyte responsible. This is achieved by repeating the test that yielded a positive result, sequentially adding one analyte at a time until the source of the cross-reactivity is determined.

Acknowledgments

Redes de Investigación Cooperativa Orientadas a Resultados en Salud (RICORS), Red de Enfermedades Inflamatorias (REI) (RD24/0007/0026). Centro de Investigación Biomédica en Red-Bioingeniería, Biomateriales y Nanomedicina (CB06/01/0040). Instituto de Salud Carlos III (ISCIII). RPG is supported by Xunta de Galicia fellowship funded by Axencia Galega de Innovacion (RD24/0007/0026).

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Chapter 9

SomaScan Bioinformatics: Normalization, Quality Control, and Assessment of Pre-Analytical Variation

Julián Candia

Abstract

SomaScan is an aptamer-based proteomics assay designed for the simultaneous measurement of thousands of human proteins with a broad range of endogenous concentrations. In its most current version released on November 1, 2023, the 11K SomaScan assay v5.0 is capable of measuring 10,776 human proteins covering major biological processes and disease areas, including cardiology, inflammation, neurology, and oncology. Here, I review bioinformatic approaches to perform normalization, quality control, and variability assessments.

Key words SomaScan, Bioinformatics, Normalization, Quality control, QC, Pre-analytical variation, PAV, SomaSignal Test

1 The SomaScan Assay

SomaScan [1, 2] is a highly multiplexed, aptamer-based assay capable of simultaneously measuring thousands of human proteins broadly ranging from femto- to micromolar concentrations. This platform relies upon a new generation of protein-capture SOMAmer (*Slow Offrate Modified Aptamer*) reagents [3]. SOMAmers are based on single-stranded, chemically modified nucleic acids, selected via the so-called SELEX (*Systematic Evolution of Ligands by EXponential enrichment*) process, which is designed to optimize high affinity, slow off-rate, and high specificity to target proteins. These targets extensively cover major molecular functions including receptors, kinases, growth factors, and hormones and span a diverse collection of secreted, intracellular, and extracellular proteins or domains. In recent years, SomaScan has increasingly been adopted as a powerful tool for biomarker discovery across a wide range of diseases and conditions, as well as to elucidate their biological underpinnings in proteomics and multi-omics studies [4–15].

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_9,

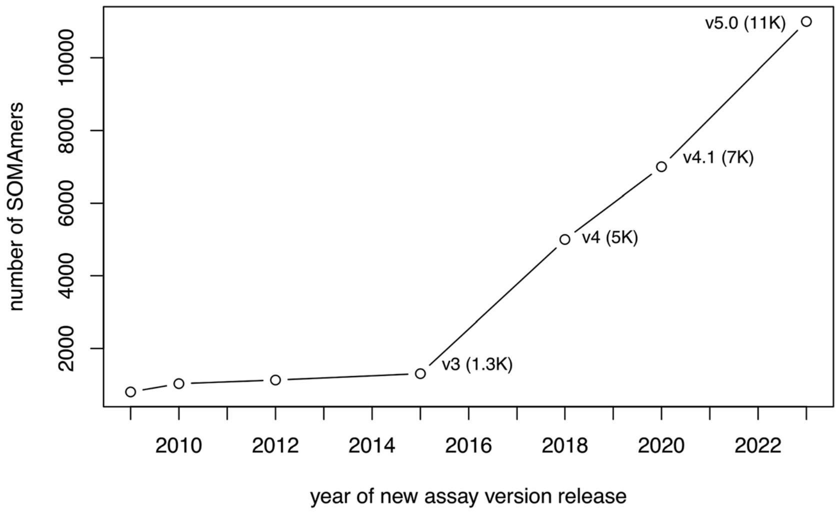

Fig. 1 Timeline of SomaScan assay versions. Approximate timeline of assay version release dates showing the growth in the number of SOMAmers in each assay. Since 2015, the assay has been growing at a steady, approximately linear rate

Concurrently with its wider adoption, SomaScan has expanded its proteome coverage by increasing the number of SOMAmers included in different versions of the assay, from roughly 800 SOMAmers in 2009, to 1100 in 2012, 1300 in 2015, 5000 in 2018, 7000 in 2020, and the most recent 11,000 protein assay available since late 2023 (Fig. 1). The first independent analysis of SomaScan normalization procedures and their variability was published by our teams at the U.S. National Institutes of Health (NIH) on the 1.1K and 1.3K assays [16], later followed by technical reports from other laboratories [17–21] and a recently updated assessment based on the 7K assay [22].

Table 1 shows the distribution of SOMAmer types in each SomaScan assay version. While most SOMAmers target human proteins, a small fraction of them targets mouse proteins. The remaining SOMAmers are different types of control, including twelve HCE (*Hybridization Control Elution*) SOMAmers used in the hybridization control normalization step (described below), as well as a few non-cleavable SOMAmers, non-biotin SOMAmers, spuriomers (designed as random, nonspecific sequence motifs), and legacy SOMAmers targeting proteins from other species. In order to cover a broad range of endogenous concentrations, SOMAmers are binned into different dilution groups, namely 20% (1:5) dilution for proteins typically observed in the femto- to picomolar range (which comprise about 80% of all human protein SOMAmers in the assay), 0.5% (1:200) dilution for proteins typically present in nanomolar concentrations (slightly below 20% of human protein SOMAmers in the assay), and 0.005% (1:20,000) dilution for proteins in micromolar concentrations (about 2–3% of human

Table 1
Distribution of SOMAmer types

SOMAmer type vs. assay	11K	7K	5K	1.3K
Human protein	10,776	7289	4979	1302
Mouse protein	233	233	228	–
Hybridization control elution	12	12	12	12
Non-biotin	10	10	10	–
Non-cleavable	4	4	4	–
Spuriomer	20	20	20	–
Other	28	28	31	8

protein SOMAmers in the assay). Although the proportions of SOMAmers across dilution groups have remained quite stable from version 5K onward, it should be noticed that version 1.3K and earlier implemented a different scheme based on 40%, 1%, and 0.005% dilution groups. The human plasma or serum volume required is 55 μL per sample.

SOMAmers are uniquely identified by their "SeqId," but the relation between SOMAmers and annotated proteins is not one to one. In the 11K assay, for instance, the 10,776 SOMAmers that target human proteins are mapped to 9609 unique UniProt IDs, 9550 unique Entrez gene IDs, and 9604 unique Entrez gene symbols. In the 7K assay, the 7289 SOMAmers that target human proteins are mapped to 6383 unique UniProt IDs, 6378 unique Entrez gene IDs, and 6383 unique Entrez gene symbols. Among cases where two or more SOMAmers share the same annotated target, the observed RFU correlation distribution is bimodal, with peaks around $r \approx 0$ and $r \approx 1$ [22]. While the $r \approx 1$ peak simply indicates the expected redundancy of SOMAmers that bind to the same target, the $r \approx 0$ peak may be due to SOMAmers that bind to different proteoforms annotated under the same protein target name, although they may also be due to artifacts such as cross-reactivity and nonspecific binding. All of the SOMAmers in the 7K and 5K SomaScan assays, as well as most of those in the 1.3K version, are included in the newest 11K assay (Fig. 2). Data processing and delivery by SomaLogic is in the `adat` format, which is described in SomaLogic's public GitHub repository (https://github.com/SomaLogic) and has remained consistent across assay versions. The data normalization procedures described below are equally applicable to all assay versions.

The experimental workflow, summarized in Fig. 3, consists of a sequence of steps, namely: (1) SOMAmers are synthesized with a fluorophore, photocleavable linker, and biotin; (2) diluted samples are incubated with dilution-specific SOMAmers bound to

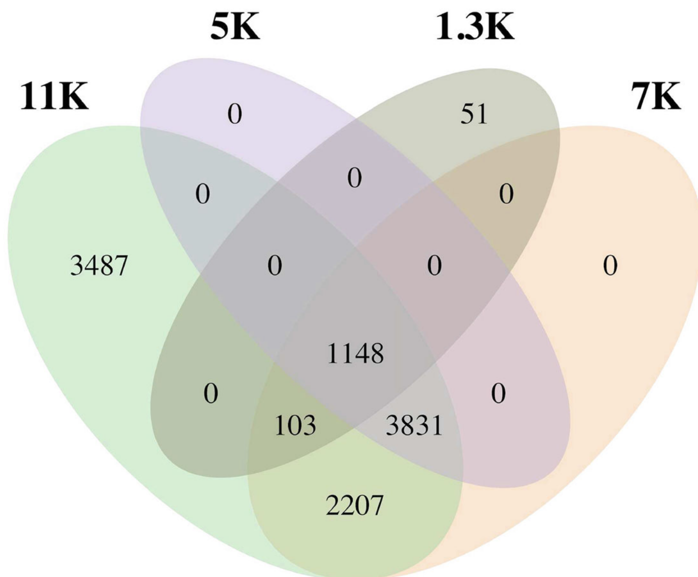

Fig. 2 Human protein SOMAmers across SomaScan versions. Venn diagram showing the SOMAmer overlap (based on "SeqId" identifiers) between the 1.3K, 5K, 7K, and 11K SomaScan assays

streptavidin beads; (3) unbound proteins are washed away, and bound proteins are tagged with biotin; (4) UV light breaks the photocleavable linker, releasing complexes back into solution; (5) nonspecific complexes dissociate, while specific complexes remain bound; (6) a polyanionic competitor is added to prevent rebinding of nonspecific complexes; (7) biotinylated proteins (and bound SOMAmers) are captured on new streptavidin beads; and (8) after SOMAmers are released from the complexes by denaturing the proteins, fluorophores are measured following hybridization to complementary sequences on a microarray chip. The fluorescence intensity detected on the microarray, measured in RFUs (*Relative Fluorescence Units*), is assumed to reflect the amount of available epitope in the original sample.

Samples are organized in 96-well plates, each plate consisting of buffer wells (reagent blanks with no sample material added), calibrator, and QC samples (which are provided by SomaLogic from pooled healthy donor controls), and the experimental samples of interest. SomaScan users may be interested in adding their own set of control samples to the plate design, which would allow them to bridge across studies independently from SomaLogic-provided controls. On the one hand, SomaLogic's controls may change over time, making it difficult to compare between studies run at different times. On the other hand, owning a set of control samples may allow users to use other omics technologies, including other proteomic assays such as mass spectrometry and Olink [23], for further validation. As a standard practice, our labs at the NIH have

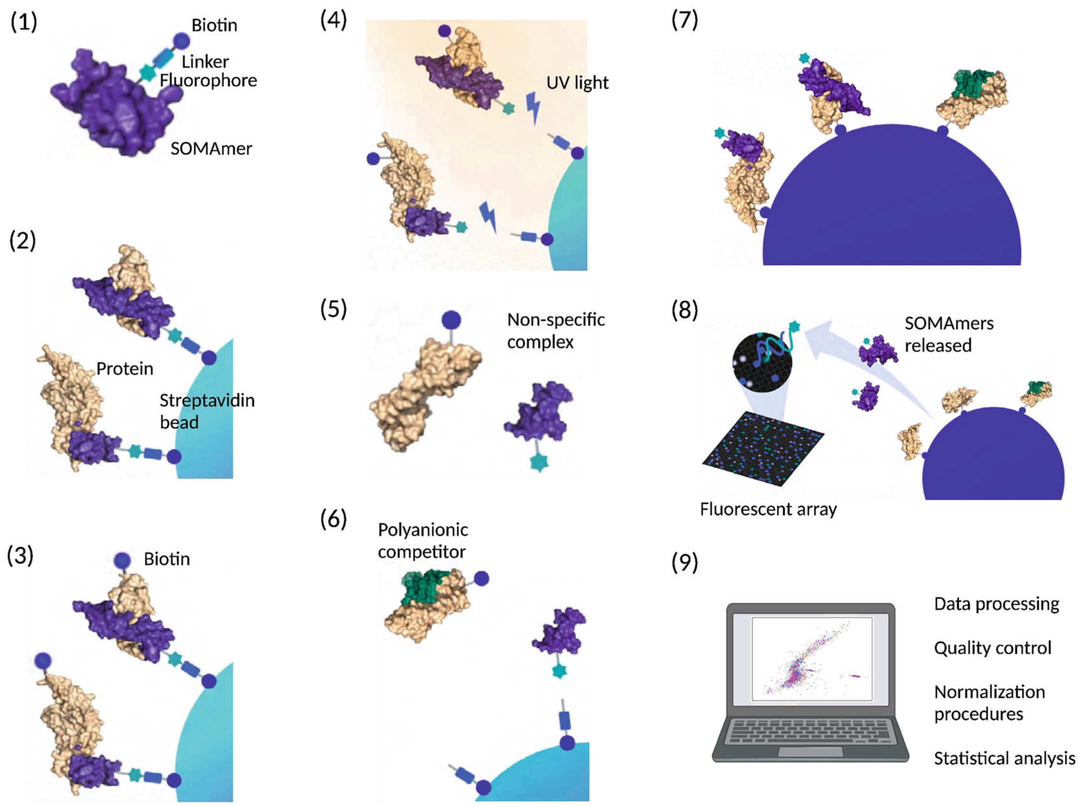

Fig. 3 Workflow of the SomaScan assay. (1) SOMAmers are synthesized with a fluorophore, photocleavable linker, and biotin; (2) diluted samples are incubated with dilution-specific SOMAmers bound to streptavidin beads; (3) unbound proteins are washed away, and bound proteins are tagged with biotin; (4) UV light breaks the photocleavable linker, releasing complexes back into solution; (5) nonspecific complexes dissociate, while specific complexes remain bound; (6) a polyanionic competitor is added to prevent rebinding of nonspecific complexes; (7) biotinylated proteins (and bound SOMAmers) are captured on new streptavidin beads; and (8) after SOMAmers are released from the complexes by denaturing the proteins, fluorophores are measured following hybridization to complementary sequences on a microarray chip. Upon completion of all experimental steps, (9) the bioinformatic analysis proceeds. Adapted from SomaLogic's Technical Note SL00000572 and created with BioRender

implemented the use of a control sample derived from pooled healthy donors, which is run on every plate with three to four technical replicates per plate.

2 Data Normalization Procedures

Raw data, as obtained after aggregation from slide-based hybridization microarrays, exhibit intraplate nuisance variance due to differences in loading volume, leaks, washing conditions, etc., which is then compounded with batch effects across plates. In order to account for intra- and interplate variability of buffer, calibrator, QC, and experimental samples, here we consider a sequence of

Table 2
Summary of normalization steps

Step	Name	Abbreviation	Scale factors
0	Raw	Raw	None
1	Hybridization control normalization	hyb	Well-specific
2	Median signal normalization on calibrators	hyb.msnCal	Calibrator- and dilution-specific
3	Plate-scale normalization	hyb.msnCal.ps	Plate-specific
4	Interplate calibration	hyb.msnCal.ps.cal	Plate- and SOMAmer-specific
5	Median signal normalization on all sample types	hyb.msnCal.ps.cal.msnAll	Well-specific (grouped by type)

steps, whose main elements are hybridization normalization, median signal normalization, plate-scale normalization, and interplate calibration. Each of these steps generates scale factors at different levels: plate-specific, by SOMAmer dilution group, SOMAmer-specific, by sample type, and combinations thereof. Besides removing technical variability, these scale factors can be used as quality control flags at the plate, sample, and SOMAmer levels [8]. A summary of these data normalization procedures, explained below in detail, is shown in Table 2. To keep a consistent mathematical notation in the expressions below, wells are indicated by Latin subindices and SOMAmers by Greek subindices. Plate index, sample type, and SOMAmer groupings are denoted by super-indices. We use *well* and *sample* interchangeably, although it should be noticed that no sample material is added to buffer wells; also, depending on context, *sample* may refer specifically to the experimental samples of interest, thus excluding control sample types such as buffer, calibrator, and QC.

As we describe each normalization step, we also include R code chunks that implement each of those steps. We assume that the raw `adat` file is available to the user and that it has been split into three different plain-text files, namely: (i) "samples.txt," with headers and one sample per row, providing metadata information on each well in the study; (ii) "somamers.txt," with headers and one SOMAmer per row, providing metadata information on each SOMAmer in the assay; and (iii) "RFU.raw.txt," without headers, samples as rows and SOMAmers as columns, providing the measured raw RFU values for each well and SOMAmer in the study. The ordering of samples and SOMAmers in "RFU.raw.txt" must be consistent with that of "samples.txt" and "somamers.txt," respectively. Files in the `adat` format can be opened and manipulated programmatically via custom scripts, using the `SomaDataIO` R package available from SomaLogic's public GitHub repository a

PlateId	PlatePosition	SampleId	SampleType	SampleMatrix	Barcode	ExtIdentifier
P0031168	A1	5302-15	Sample	EDTA plasma	S1545380	EXID40000006690911
P0031168	A10	4875-1	Sample	EDTA plasma	S1545524	EXID40000006690909
P0031168	A11	824-29	Sample	EDTA plasma	S1545540	EXID40000006690925
P0031168	A12	4922-2	Sample	EDTA plasma	S1545442	EXID40000006690906
P0031168	A2	4913-2	Sample	EDTA plasma	S1545396	EXID40000006690763
P0031168	A3	1550-11	Sample	EDTA plasma	S1545412	EXID40000006690800
P0031168	A4	1768-4	Sample	EDTA plasma	S1545428	EXID40000006690756
P0031168	A5	7828-1	Sample	EDTA plasma	S1545444	EXID40000006690765
P0031168	A6	5619-11	Sample	EDTA plasma	S1545460	EXID40000006690936
P0031168	A7	7862-1	Sample	EDTA plasma	S1545476	EXID40000006690753

b

SeqId	SomaId	Target	UniProt	EntrezGeneID	EntrezGeneSymbol	Organism	Type	Dilution
10000-28	SL019233	CRBB2	P43320	1415	CRYBB2	Human	Protein	20
10001-7	SL002564	c-Raf	P04049	5894	RAF1	Human	Protein	20
10003-15	SL019245	ZNF41	P51814	7592	ZNF41	Human	Protein	0.5
10006-25	SL019228	ELK1	P19419	2002	ELK1	Human	Protein	20
10008-43	SL019234	GUC1A	P43080	2978	GUCA1A	Human	Protein	20
10010-10	SL014943	BECN1	Q14457	8678	BECN1	Human	Protein	20
10011-65	SL019246	OCRL	Q01968	4952	OCRL	Human	Protein	20
10012-5	SL014669	SPDEF	O95238	25803	SPDEF	Human	Protein	20
10013-34	SL025418	Fc_MOUSE	Q99LC4			Mouse	Protein	20
10014-31	SL007803	SLUG	O43623	6591	SNAI2	Human	Protein	20

c

748.8	325.4	217.1	572	720.7	357.6	2323.9	1382.1	421.7	859.7
577.5	385.4	325.7	634.8	731	413.9	2847.4	2004	484.7	936.9
526.8	318.6	203	581.3	618	317.6	2510.5	1370.7	412.9	796.9
508.4	422.7	213.1	641.2	688.2	340.4	2366	1317.6	393.7	850.5
450.5	290.5	202.2	580.2	2294.6	277.9	2108	1210.6	356.6	694.6
621.3	456	204.3	663.7	599	814.1	2486	1362.2	445.9	861.5
658.4	508.4	226.7	821.2	681	440.9	3236.4	1772.6	505.3	1013.5
571.8	385.6	221.3	612.3	633.5	373.1	2849.8	1890	446.4	852.3
482	312.1	191	575.3	565.7	334	2792	1286.4	390.7	776.7
5986.7	300.4	206.7	538.1	651.4	307.3	2293.8	1271.2	407.5	762.7

Fig. 4 Input data files for the normalization pipeline. (**a**) "samples.txt," with headers and one sample per row, provides metadata information on each well in the study (including buffer, QC, calibrator, and experimental samples). (**b**) "somamers.txt," with headers and one SOMAmer per row, provides metadata information on each SOMAmer in the assay (including controls such as Hybridization Control Elution (HCE) SOMAmers). (**c**) "RFU.raw.txt," without headers, samples as rows and SOMAmers as columns, provides the measured raw RFU values for each well and SOMAmer in the study (in the same order as the sample and SOMAmer metadata files)

(https://github.com/SomaLogic), or using previously developed open-source tools [24, 25]; however, it is often simple enough to open the `adat` file in Excel or similar software, then copy-paste the three separate regions corresponding to sample metadata, SOMAmer metadata (that needs to be transposed in order to display SOMAmers row-wise) and RFU matrix, and save them separately as tab-delimited plain-text files. Figure 4 shows an example of the input data files required for this normalization pipeline.

The following code chunk reads the input datasets and sets up the variables needed to implement each of the normalization steps described below. Notice that, for the sake of clarity, we avoid using variable names that overlap with names of core R functions (e.g., `sample`). The code presented here was written with an emphasis on readability but not optimized for computing performance. Since `adat` files are much smaller than typical datasets generated by other omics and bioinformatic tasks, further optimization was not deemed necessary.

```
norm = c("raw","hyb","hyb.msnCal","hyb.msnCal.ps","hyb.msnCal.ps.cal","hyb.msnCal.ps.cal.msnAll")
n_norm = length(norm)
RFU = vector("list",n_norm)

sampl = as.matrix(read.table("samples.txt",header=T,quote="",comment.char="",sep="\t"))
somamers= as.matrix(read.table("somamers.txt",header=T,quote="",comment.char="",sep="\t"))
RFU[[1]] = as.matrix(read.table("RFU.raw.txt",header=F,sep="\t"))

n_sampl = nrow(sampl)
n_somamer = nrow(somamer)
plate = unique(sampl[,"PlateId"])
n_plate = length(plate)

dil = c(0.005,0.5,20)
dil_lab = c("0_005","0_5","20")
n_dil = length(dil)
```

2.1 Hybridization Control Normalization (hyb)

Hybridization control normalization is designed to adjust for nuisance variance on the basis of individual wells. Each well contains $n_{HCE} = 12$ HCE (*Hybridization Control Elution*) SOMAmers at different concentrations spanning more than three orders of magnitude. By comparing each observed HCE probe to its corresponding reference value and then calculating the median over all HCE probes, we obtain the scale factor for the i-th well, i.e.,

$$SF_i = \text{median}\left\{\frac{RFU_{\alpha}^{HCE,ref}}{RFU_{i\alpha}^{HCE,obs}}\right\}_{\alpha=1,\ldots,n_{HCE}}. \qquad (1)$$

Notice that this normalization step is performed independently for each well; once the scale factor is determined, all SOMAmer RFUs in the well are multiplied by the same scale factor. Instead of using an external reference, a plate-specific internal reference can be determined by the median across the $n_s = 96$ wells in the plate, i.e.,

$$RFU_{\alpha}^{HCE,ref} \equiv \text{median}\left\{RFU_{i\alpha}^{HCE,obs}\right\}_{i=1,\ldots,n_s}. \qquad (2)$$

The following code chunk implements the hybridization control normalization step:

```
sel_HCE = somamer[,"Type"]=="Hybridization_Control_Elution"
n_HCE  = sum(sel_HCE)
RFU_HCE = RFU[[1]][,sel_HCE]

RFU[[2]] = RFU[[1]]
SF_hyb = rep(NA,n_sampl)

HCE_ratio = matrix(rep(NA,n_sampl*n_HCE),ncol=n_HCE)
for (i_plate in 1:n_plate) {
    sampl_sel = sampl[,"PlateId"]==plate[i_plate]
    hyb_intra_ref = apply(RFU_HCE[sampl_sel,],2,median)
    HCE_ratio_plate = HCE_ratio[sampl_sel,]
    SF_hyb_plate = SF_hyb[sampl_sel]
    RFU_plate = RFU[[2]][sampl_sel,]
    for (i_sampl in 1:sum(sampl_sel)) {
        HCE_ratio_plate[i_sampl,] = hyb_intra_ref/RFU_plate[i_sampl,sel_HCE]
        SF_hyb_plate[i_sampl] = median(HCE_ratio_plate[i_sampl,])
        RFU_plate[i_sampl,] = SF_hyb_plate[i_sampl]*RFU_plate[i_sampl,]
    }
    HCE_ratio[sampl_sel,] = HCE_ratio_plate
    SF_hyb[sampl_sel] = SF_hyb_plate
    RFU[[2]][sampl_sel,] = RFU_plate
}
```

2.2 Median Signal Normalization on Calibrators (hyb.msnCal)

Median signal normalization is an intraplate normalization procedure performed within wells of the same sample class (i.e., separately for buffer, QC, calibrator, and experimental samples) and within SOMAmers of the same dilution group. It is intended to remove sample-to-sample differences in total RFU brightness that may be due to differences in overall protein concentration, pipetting variation, variation in reagent concentrations, assay timing, and other sources of variability within a group of otherwise comparable samples. Since RFU brightness differs significantly across SOMAmers, median signal normalization proceeds in two steps. First, the median RFU of each SOMAmer is determined (across all samples of the same sample type) and sample RFUs are divided by it. The ratio corresponding to the i-th sample and α-th SOMAmer is thus given by

$$r^{gd}_{i\alpha} = RFU_{i\alpha}/\text{median}\left\{RFU_{j\alpha}\right\}_{j=1,\ldots,n^g_s}, \tag{3}$$

where indices g and d denote sample type and SOMAmer dilution groupings, respectively. Then, the scale factor associated with the i-

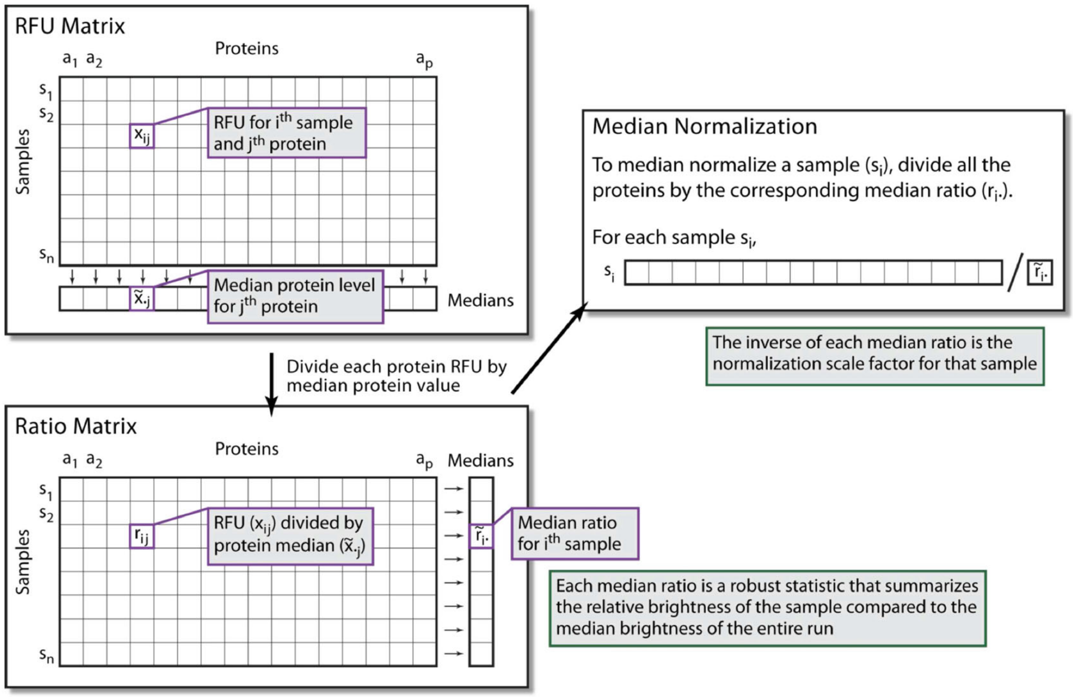

Fig. 5 Median signal normalization. An RFU sub-matrix is defined by all samples of the same class (i.e., separately for buffer, QC, calibrator, and experimental samples) and all SOMAmers within the same dilution group. For each SOMAmer, the median RFU level is calculated across all samples (top left). By dividing each RFU measurement by the corresponding SOMAmer median, we obtain the ratio matrix; then, for each sample, a median ratio is calculated (bottom left). The median normalization scale factor associated to each sample is the inverse of the median ratio. To median-normalize each sample, all SOMAmers within the target dilution group are multiplied by its corresponding scale factor. Adapted from image courtesy of Darryl Perry (SomaLogic)

th sample is determined as the inverse of the median ratio for that sample across all SOMAmers in the dilution group:

$$SF_i^{gd} = 1/\text{median}\left\{ r_{i\alpha}^{gd} \right\}_{\alpha=1,\ldots,n_{SOMAmer}^d}. \quad (4)$$

To median-normalize the i-th sample, then all its SOMAmer RFUs in the same dilution group are multiplied by this scale factor. This procedure is illustrated by Fig. 5. As discussed in our previous work [16], performing median signal normalization on experimental samples *before* interplate calibration presents the risk of enhancing plate-to-plate differences. Thus, in this step, we restrict median signal normalization to calibrators only.

The following code chunk implements the median signal normalization step on calibrators:

```
RFU[[3]] = RFU[[2]]
SF_medNormInt = matrix(rep(1,n_sampl*n_dil),ncol=n_dil)

sampl_type = c("Calibrator")
n_sampl_type = length(sampl_type)

for (i_plate in 1:n_plate) {
    sel1 = sampl[,"PlateId"]==plate[i_plate]
    for (i_sampl_type in 1:n_sampl_type) {
        sel2 = sampl[,"SampleType"]==sampl_type[i_sampl_type]
        indx_sampl = which(sel1&sel2)
        for (i_dil in 1:n_dil) {
            sel_somamer = as.numeric(somamer[,"Dilution"])==dil[i_dil]
            dat = RFU[[3]][indx_sampl,sel_somamer,drop=F]
            dat2 = dat
            for (i in 1:ncol(dat2)) {
                dat2[,i] = dat2[,i]/median(dat2[,i])
            }
            for (i in 1:nrow(dat)) {
                SF_medNormInt[indx_sampl[i],i_dil] = 1/median(dat2[i,])
                dat[i,] = dat[i,]*SF_medNormInt[indx_sampl[i],i_dil]
            }
            RFU[[3]][indx_sampl,sel_somamer] = dat
        }
    }
}
```

2.3 Plate-Scale Normalization (hyb.msnCal.ps)

Plate-scale normalization aims to control for variance in total signal intensity from plate to plate. No protein spikes are added to the calibrator; the procedure solely relies on the endogenous levels of each protein within the set of calibrator replicates.

For the α-th SOMAmer on the p-th plate,

$$SF_{\alpha}^{p} = RFU_{\alpha}^{Cal,ref} / \text{median}\left\{ RFU_{i\alpha}^{Cal,obs} \right\}_{i=1,\ldots,n_{Cal}}^{p}. \tag{5}$$

Calibration scale factors may be pinned to an external reference, but here we utilize an internal reference determined by the median across all calibrators on all n_p plates, i.e.,

$$RFU_{\alpha}^{Cal,ref} \equiv \text{median}\left\{ RFU_{i\alpha}^{Cal,obs} \right\}_{i=1,\ldots,n_{Cal}}^{p=1,\ldots,n_p}. \tag{6}$$

In order to correct the overall brightness level of the p-th plate, we calculate the plate-scale factor as the median of SF_{α}^{p} across all SOMAmers, i.e.,

$$SF^{p} = \text{median}\left\{ SF_{\alpha}^{p} \right\}_{\alpha=1,\ldots,n_{SOMAmer}}. \tag{7}$$

For all wells on the p-th plate and all SOMAmers, RFUs are multiplied by the plate-scale factor SF^p.

The following code chunk implements the plate-scale normalization step:

```
RFU[[4]] = RFU[[3]]

SF = matrix(rep(NA,n_plate*n_somamer),ncol=n_somamer)
sel_cal = sampl[,"SampleType"]=="Calibrator"
cal_interplate_ref = rep(NA,n_somamer)
for (i_somamer in 1:n_somamer) {
    cal_interplate_ref[i_somamer] = median(RFU[[4]][sel_cal,i_somamer])
    for (i_plate in 1:n_plate) {
        sel_plate = sampl[,"PlateId"]==plate[i_plate]
        cal_intraplate_ref = median(RFU[[4]][sel_cal&sel_plate,i_somamer])
        SF[i_plate,i_somamer] = cal_interplate_ref[i_somamer]/cal_intraplate_ref
    }
}
SF_plateScale = apply(SF,1,median)

for (i_plate in 1:n_plate) {
    sel_plate = sampl[,"PlateId"]==plate[i_plate]
    RFU[[4]][sel_plate,] = RFU[[4]][sel_plate,]*SF_plateScale[i_plate]
}
```

2.4 Interplate Calibration (hyb.msnCal.ps.cal)

Following plate-scale normalization, we recalculate SOMAmer- and plate-specific scale factors via Eqs. (5)–(6). Separately for each SOMAmer and plate, all wells on the plate are corrected by the recalculated SF^p_α.

The following code chunk implements the interplate calibration step:

```
RFU[[5]] = RFU[[4]]

SF_cal = matrix(rep(NA,n_plate*n_somamer),ncol=n_somamer)
sel_cal = sampl[,"SampleType"]=="Calibrator"
cal_interplate_ref_N = rep(NA,n_somamer) # new interplate reference
for (i_somamer in 1:n_somamer) {
    cal_interplate_ref_N[i_somamer] = median(RFU[[5]][sel_cal,i_somamer])
    for (i_plate in 1:n_plate) {
        sel_plate = sampl[,"PlateId"]==plate[i_plate]
        cal_intraplate_ref = median(RFU[[5]][sel_cal&sel_plate,i_somamer])
        SF_cal[i_plate,i_somamer] = cal_interplate_ref_N[i_somamer]/cal_intraplate_ref
        RFU[[5]][sel_plate,i_somamer] = RFU[[5]][sel_plate,i_somamer]*SF_cal[i_plate,i_somamer]
    }
}
```

2.5 Median Signal Normalization on All Sample Types (hyb.msnCal.ps.cal.msnAll)

At this stage, after correcting plate-to-plate variability to the fullest extent possible, median signal normalization (described in step 2 and Fig. 5) can be performed separately on each sample type. This step yields the final, fully normalized dataset. The following code chunk implements the median signal normalization step on all sample types:

```
RFU[[6]] = RFU[[5]]
SF_medNormFull = matrix(rep(1,n_sampl*n_dil),ncol=n_dil)

sampl_type = c("QC","Sample","Buffer","Calibrator")
n_sampl_type = length(sampl_type)

for (i_sampl_type in 1:n_sampl_type) {
    indx_sampl = which(sampl[,"SampleType"]==sampl_type[i_sampl_type])
    for (i_dil in 1:n_dil) {
        sel_somamer = as.numeric(somamer[,"Dilution"])==dil[i_dil]
        dat = RFU[[6]][indx_sampl,sel_somamer,drop=F]
        dat2 = dat
        for (i in 1:ncol(dat2)) {
            dat2[,i] = dat2[,i]/median(dat2[,i])
        }
        for (i in 1:nrow(dat)) {
            SF_medNormFull[indx_sampl[i],i_dil] = 1/median(dat2[i,])
            dat[i,] = dat[i,]*SF_medNormFull[indx_sampl[i],i_dil]
        }
        RFU[[6]][indx_sampl,sel_somamer] = dat
    }
}
```

Finally, the following code chunk saves all sample-, SOMAmer-, and plate-based metadata (including the scale factors used in the normalization procedure) along with the RFU matrices at each normalization step:

```
header = c(colnames(sampl),"SF_hyb",paste0("SF_msnCal_d",dil_lab),paste0("SF_msnAll_d",dil_lab))
output = rbind(header,cbind(sampl,SF_hyb,SF_medNormInt,SF_medNormFull))
write(t(output),ncol=ncol(output),file="samples_SF.txt",sep="\t")

header = c(colnames(somamer),"Cal_Interplate_Ref_Pass1","Cal_Interplate_Ref_Pass2")
output = rbind(header,cbind(somamer,cal_interplate_ref,cal_interplate_ref_N))
write(t(output),ncol=ncol(output),file="somamers_SF.txt",sep="\t")

header = c("Plate","SF_plateScale",paste0("SF_cal_",somamer[,"SeqId"]))
output = rbind(header,cbind(plate,SF_plateScale,SF_cal))
write(t(output),ncol=ncol(output),file="plates_SF.txt",sep="\t")

for (i_norm in 2:n_norm) {
    outfile = paste0("RFU.",norm[i_norm],".txt")
    write(t(RFU[[i_norm]]),ncol=n_somamer,file=outfile,sep="\t")
}
```

It should be pointed out that SomaLogic currently delivers normalized output files (as plain-text files of extension `adat`) that follow similar steps as those described here, but using external references [8]. These references may be outdated (because they are specific to the control samples used), not necessarily representative of the target samples of interest (because they utilize a fixed pool of healthy human control samples for the last normalization step), and are not delivered with the `adat` files provided to the customer (therefore precluding any attempt of independent analysis). In contrast, the process described here relies solely on internal references derived from the measured data. In our study of nearly 1800 experimental samples from the Baltimore Longitudinal Study on Aging (BLSA), we found that fully normalized datasets using internal versus external references were highly concordant. For each human protein SOMAmer in the plasma 7K assay, we calculated the Spearman's correlation between the fully normalized RFU values from the `adat` file provided by SomaLogic using external references (a file designated with the "hybNorm.medNormInt.plateScale.calibrate.anmlQC.qcCheck.anmlSMP" suffix) and the full normalization described here using internal references ("hyb.msnCal.ps.cal.msnAll"). The distribution of correlation estimates over all 7289 human protein SOMAmers is shown in Fig. 6; the distribution median is $r = 0.996$. It should be added that, due to Spearman's correlation invariance, these results remain valid under any type of monotonic (e.g., logarithmic) RFU transformations.

SomaLogic's standard data quality reports utilize scale factors from the normalization procedure to flag samples, SOMAmers, and plates that do not pass preestablished acceptance criteria.

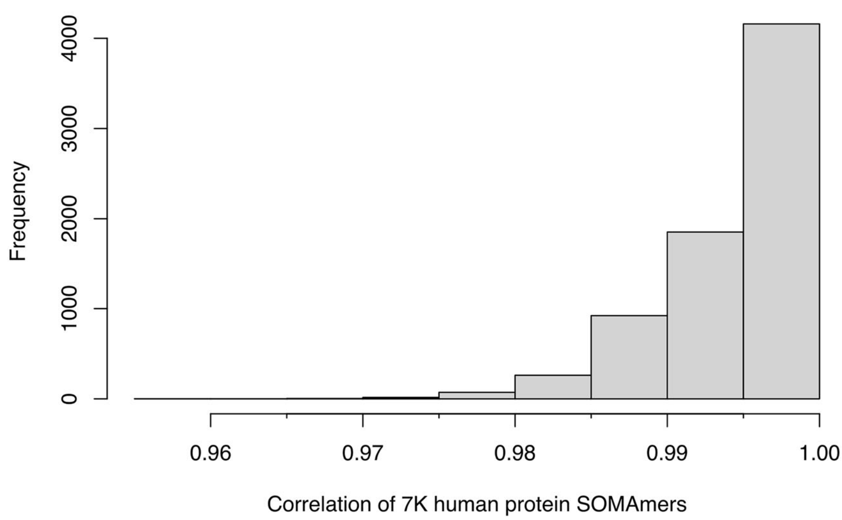

Fig. 6 Concordance between normalization approaches. Distribution of Spearman's correlation estimates for the 7289 human protein SOMAmers in the 7K plasma SomaScan assay, calculated over nearly 1800 human donor samples from the BLSA [22]. For each SOMAmer, the correlation was calculated between the fully normalized RFU values from SomaLogic's pipeline using external references and the full normalization described here using internal references

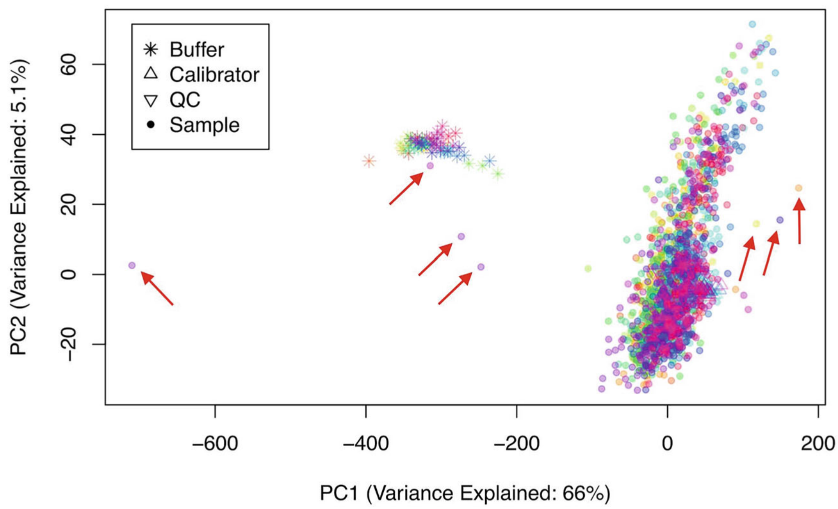

Fig. 7 PCA as quality control and data exploration tool. PCA performed on 2050 raw (not normalized) samples, including 68 buffers, 110 calibrators, 66 QC, and 2806 BLSA samples distributed across 22 plates. Each plate is shown by a different color. The red arrows show seven samples flagged due to their out-of-range normalization scale factors

Additional standard approaches that SomaScan users may implement are dimensional reduction techniques such as Principal Component Analysis (PCA), which serves as both a quality control and exploration tool to gain insight on the data. Figure 7 shows PCA performed on 2050 raw (not normalized) samples, including 68 buffers, 110 calibrators, 66 QC, and 2806 human donor samples from the BLSA [22]. Seven samples, flagged in SomaLogic's data quality report due to their out-of-range normalization scale factors, are shown by red arrows. In this case, PCA helps to confirm that those samples indeed appear as outliers. Four of them, in fact, appear closer to the reagent-only buffer wells than the large cluster formed by calibrator, QC, and the remaining biological samples. Since this study spans 22 plates, shown by different colors, PCA may provide visual hints of plate effects; inspecting and comparing PCA plots with different normalizations provides more clues to explore whether those potential plate effects were effectively removed by the normalization process [22]. It should be noticed that other bioinformatic tools, developed to assess and remove batch effects for other omics, may be useful in the context of SomaScan data processing, for instance: ComBat [26], implemented in the sva R package [27], a very well-established method for batch correction in microarray and RNA-seq data; guided PCA [28]; and multi-MA normalization [29], among others. This important topic certainly deserves further investigation.

3 Pre-analytical Variation (PAV) SomaSignal Tests (SSTs)

Pre-analytical variation (PAV) due to sample collection, handling, and storage is known to affect many analyses in molecular biology. By implementing data modeling techniques similar to those previously developed to find SomaScan signatures associated with clinical phenotypes [8], SomaLogic has developed a novel set of the so-called SomaSignal Tests (SSTs) to assess pre-analytical variation due to different sample processing factors, including fed-fasted time, the number of freeze-thaw cycles, time-to-decant, time-to-spin, and time-to-freeze. While it is not always possible to control sample collection, processing, storage, and handling to ensure consistency, PAV-SSTs were designed to quantitatively estimate whether and to what extent samples may have been impacted by these nuisance factors.

In previous work [22], we implemented a framework to assess technical variability based on measurements performed on duplicate biological interplate pairs from samples obtained from $n_{dupl} = 102$ human participants in the BLSA. Panels (a)–(e) of Fig. 8 show, for each PAV metric, the magnitude of the difference

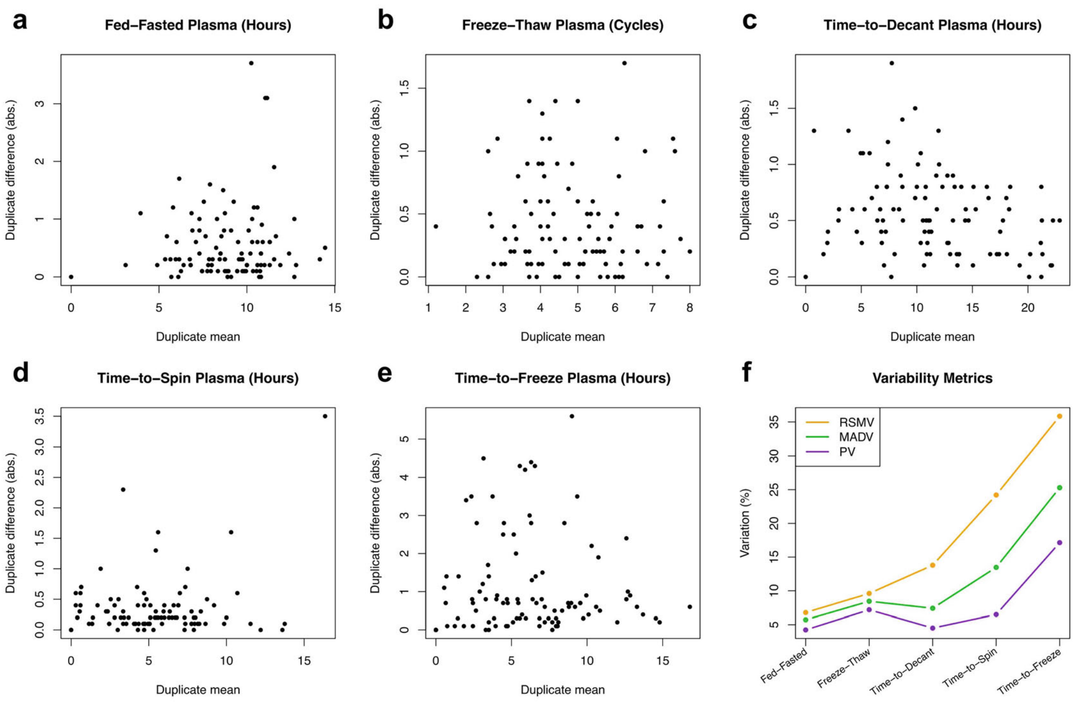

Fig. 8 Assessment of the robustness of PAV-SST estimates using BLSA technical replicates. (**a**–**e**) Duplicate difference vs. duplicate mean for different PAV-SST estimates, as indicated. Each circle represents one pair of technical duplicates from the same human donor sample. (**f**) Variation of PAV-SST estimates determined by different metrics, namely: Root-Mean-Squared Variation (RMSV), Mean Absolute Difference Variation (MADV), and Percentile Variation (PV)

between estimates from each duplicate pair, $\Delta x_i = |x_{i,1} - x_{i,2}|$, as a function of their average, $\overline{x_i} = \left(x_{i,1} + x_{i,2}\right)/2$. Next, we will leverage this framework to generate an independent assessment of the robustness of SomaLogic's PAV estimates. Let us first recall that, based on the scaled relative differences, defined as $D_i = (\Delta x_i / \overline{x_i})/\sqrt{2}$, different variability estimates may be built, namely: (i) the Root-Mean-Squared Variation (RMSV) metric, defined as

$$RMSV = \sqrt{\frac{1}{n_{dupl}} \sum_{i=1}^{n_{dupl}} D_i^2} \times 100\% \;, \tag{8}$$

(ii) the Mean Absolute Difference Variation (MADV) metric, defined as

$$MADV = \sqrt{\frac{\pi}{2}} \frac{1}{n_{dupl}} \sum_{i=1}^{n_{dupl}} |D_i| \times 100\% \;, \tag{9}$$

and (iii) the Percentile Variation (PV) metric, defined as

$$PV = \frac{1}{2}\left(P_{84}\{D_i\}_{i=1,\ldots,n_{dupl}} - P_{16}\{D_i\}_{i=1,\ldots,n_{dupl}}\right) \times 100\% \;, \tag{10}$$

where P_{84} and P_{16} are the 84$^{\text{th}}$ and 16$^{\text{th}}$ percentiles in the distribution of scaled relative differences. If the scaled relative differences are normally distributed, these variability estimates are equivalent; the magnitude of the differences between these estimates is indicative of the magnitude and severity of deviations from normality. Armed with these various definitions, Panel Fig. 8f shows the percent variation across PAV estimates. As expected by theoretical considerations, the PV metric is the least sensitive to distribution tails and therefore yields the lowest variation. We observe that, with the exception of the time-to-freeze estimate, which shows a PV variation of about 15%, the PV variation for the remaining four PAV metrics lies approximately within the 5–7% range.

4 Summary and Conclusions

With the addition of the latest v5.0 (11K) version released on November 1, 2023, SomaScan galvanized its role as one of the leading technologies in high-throughput human proteomics. Going back to the v3 (1.3K) version about eight years prior, the assay has shown a steady, approximately linear growth in protein coverage. Several independent teams, including ours at the NIH, have explored the assay's sensitivity and variability [16–

22]. Furthermore, multiple studies have investigated SomaScan's specificity, cross-reactivity, and orthogonal assay reproducibility [4, 5, 30–32]. Less attention, however, has been dedicated to explore normalization procedures. Typically, SomaLogic provides one or several data files in `adat` format, which are normalized using their internal analysis pipeline. This pipeline, however, has historically remained under SomaLogic's control and operated to a large extent as a black box, providing SomaScan users only with descriptive explanations of the steps involved in the process. SomaLogic's normalization procedures, moreover, have evolved overtime, switching from internal references to external references, altering the order in which median normalization was implemented, and splitting the interplate calibration step into plate-level rescaling (referred to as plate-scale normalization) followed by SOMAmer- and plate-specific normalization. To the best of our knowledge, no software version control of these changes have been shared with SomaScan users, making it difficult for the task to compare or integrate studies run years apart, perhaps even using different versions of the assay.

As an early contribution to the nascent field of SomaScan bioinformatics, our NIH labs have undertaken the task to reconstruct the steps involved in a normalization procedure that, without the use of SomaLogic's proprietary external references, is capable of reproducing normalized RFU values concordant with those from SomaLogic's pipeline [16, 22]. The latest version of our normalization pipeline is described here in detail, including its R code implementation, and available from a public repository. The ability to run independent normalization analyses is important for various reasons. On the one hand, the set of external references (based on a fixed pool of healthy human control samples) may be inappropriate to study individuals and populations with large deviations from a healthy plasma proteome. As noted by Lopez-Silva et al. [33] in their study of chronic kidney disease, SomaLogic's normalization could potentially attenuate the strength of associations by reducing more extreme values if they are associated with a clinical outcome, perhaps contributing to the attenuated prognostic associations they observed with SomaScan. Pietzner et al. [20] report that using SomaScan data without a normalization step applied to correct for unwanted technical variation and to make data comparable across cohorts, a higher median correlation with results from Olink was observed, along with substantial differences in the association with various phenotypic characteristics. On the other hand, users may be interested in using their own set of control samples to bridge across studies, thereby needing to calibrate different studies using those controls. More generally, users may be interested in adapting the normalization process according to their studies' characteristics and objectives (for instance, splitting the median normalization procedure into different sample subclasses identified by clinical phenotype). Therefore, we believe that presenting this independent normalization pipeline empowers SomaScan end users to tailor the normalization procedures to their own individual needs.

Following our discussion of normalization approaches, we briefly described quality control procedures based on combining flags from outlier normalization scale factors (such as those provided in SomaLogic's standard data quality reports) with Principal Component Analysis (PCA), a linear dimensional reduction technique. Alternatively, nonlinear dimensional reduction methods such as Uniform Manifold Approximation and Projection (UMAP) and t-distributed Stochastic Neighbor Embedding (t-SNE) could be used for the purpose of quality control and data exploration. In addition, we suggested that bioinformatic tools developed to assess and remove batch effects in data from other omics, such as ComBat [26], Surrogate Variable Analysis (SVA) [27], guided PCA [28], and multi-MA normalization [29], may be useful techniques that deserve further investigation.

Finally, we investigated SomaLogic's SomaSignal Test (SST) models, which were recently developed to estimate a variety of Pre-analytical Variation (PAV) metrics, including fed-fasted time, the number of freeze-thaw cycles, time-to-decant, time-to-spin, and time-to-freeze. Leveraging technical duplicates from our study of human donors from the Baltimore Longitudinal Study on Aging (BLSA), we performed an independent evaluation of the statistical variation of the PAV estimates. Within the limited scope of our assessment, we conclude that PAV-SST estimates appear to be robust (percentile variation in the 5–7% range, with the exception of time-to-freeze, which is about 15%) and may be used during biomarker evaluations to exclude samples due to nuisance effects related to sample collection and processing, and/or to incorporate these estimates as model covariates to control the downstream impact of those unwanted effects.

Acknowledgements

This work was supported entirely by the Intramural Research Program of the National Institute on Aging (NIA).

Competing Interests The author declares no competing interests.

Data and Software Availability Anonymized datasets and R source code for the normalization pipeline described here are available on the Open Science Framework repository, osf.io/srgef. DOI 10.17605/OSF.IO/SRGEF.

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Chapter 10

Proteomics by qPCR Using the Proximity Extension Assay (PEA)

Julio Manuel Martínez-Moreno, Adrián Llamas-Urbano, Nuria Barbarroja, and Carlos Pérez-Sánchez

Abstract

The *Proximity Extension Assay (PEA)* is an innovative technology developed by *Olink Proteomics* that has revolutionized proteomics research. This method enables the simultaneous detection of hundreds of proteins using a minimal sample volume (1 microliter), combining the specificity of antibody-based immunoassays with the sensitivity of quantitative PCR (qPCR). The PEA process relies on the binding of two antibodies, each conjugated with unique DNA sequences, to a target protein. Upon proximity, these DNA sequences hybridize, forming a molecular barcode that is subsequently amplified by PCR, ensuring high specificity and minimizing cross-reactivity—an issue commonly encountered in multiplexed immunoassays. This chapter provides a detailed overview of the technical and methodological aspects of PEA, including the incubation, extension, amplification, and detection steps, as well as the quality control measures implemented in the *Olink® NPX Signature* software for data analysis. This technology has been applied in more than 2000 peer-reviewed studies, demonstrating its potential for biomarker discovery in diagnostics, prognostics, and personalized medicine across various diseases. The integration of PEA into biomedical research continues to enhance the precision and reproducibility of proteomics studies.

Key words Proximity extension assay, PEA, Proteomics, qPCR, Olink, Protein array

1 Introduction

The Proximity Extension Assay (PEA) is an innovative technology developed by the Swedish company, Olink Proteomics, which has revolutionized proteomics research. This cutting-edge method allows for the simultaneous analysis of thousands of proteins using just a tiny amount of blood (1 microliter), taking proteomics to new heights.

PEA combines the precision of an antibody-based immunoassay with the efficiency of qPCR, ensuring exceptional levels of specificity, sensitivity, and dynamic range. The process involves a dual-recognition immunoassay, where two antibodies labeled with unique DNA oligonucleotides bind to a target protein, bringing

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_10,

them into close proximity. This proximity enables the DNA oligonucleotides to hybridize, forming a unique double-stranded DNA "barcode" that correlates with the initial concentration of the target protein. This barcode is then amplified through PCR.

The dual antibody recognition and DNA-coupled measurement of PEA ensure unparalleled specificity, eliminating issues commonly seen in multiplexed immunoassays such as antibody cross-reactivity. This specificity is a key feature of PEA, setting it apart from other technologies.

For data analysis, Olink provides the Olink® NPX Signature software, which converts qPCR Ct values into Normalized Protein eXpression (NPX), a logarithmic unit that reflects protein concentration. This software minimizes technical variations and enhances result interpretation. Olink's technology includes a quality control system that monitors assay and sample performance, with internal controls added to each sample to ensure protocol integrity.

Olink offers a range of 15 panels known as Target 96, each containing 92 proteins linked to different biological pathways (https://olink.com/products/olink-target-96). Custom panels can also be created to validate specific biomarkers. This groundbreaking technology has been featured in over 2000 peer-reviewed studies (https://olink.com/knowledge/publications).

Our team at Cobiomic Bioscience SL has processed more than 10,000 samples over the past 2 years. Additionally, utilizing this technique, we have investigated the molecular pathways involved in the pathogenesis of two autoimmune and inflammatory diseases, such as rheumatoid arthritis and psoriatic arthritis. Our research has focused on identifying patient stratification signatures, diagnostic biomarkers, and biomarkers for response to therapies [1–6]. Within this chapter, we offer precise technical insights into the protocol and analysis of this cutting-edge technique.

2 Materials

2.1 Incubation Step

Ensure the laboratory (pre-PCR room) is set up for the incubation step by organizing pipettes and necessary consumables. Allow the reagents and controls to acclimate to room temperature before starting.

The required materials are as follows:

1. Pipettes for necessary volumes.
2. Pipette tips: Sterile and suitable for your pipettes.
   - Three pipette tips of 10 μL.
   - Four pipette tips of 100/200 μL.
   - One pipette tip of 1000 μL.
   - 104 multichannel pipette tips of 10 μL.

3. Prepared and thawed 96-well plate with samples.
4. Reagents (vortex and spin all reagents before use):
   - Incubation Solution (Target 96 panel).
   - Incubation Stabilizer (Target 96 panel).
   - A and B probes (Target 96 panel).
5. Controls (vortex and spin before use):
   - Negative and Inter-plate Controls (Target 96 panel).
   - Pooled sample controls.
6. Two 8-well strips.
7. One new 96-well plate, preferably with a full skirt.
8. One microcentrifuge tube (1–1.5 mL).
9. Two adhesive films.
10. One vortex mixer.
11. One plate mixer.
12. One microcentrifuge.
13. One plate spinner.
14. Refrigerator for overnight incubation at 4 °C.

2.2 Extension and Amplification Step

Ensure the laboratory (pre-PCR room) is set up for the extension and amplification step by organizing pipettes and necessary consumables. Allow the PEA Solution to acclimate to room temperature.

The required materials are as follows:

1. Pipettes for necessary volumes.
2. Pipette tips: Sterile and suitable for your pipettes.
   - Two pipette tips of 1000 μL.
   - Two pipette tip of 200 μL.
   - Eight multichannel pipette tips of 200 μL.
3. Reagents:
   - PEA Solution (Target 96 panel).
   - PEA Enzyme (Target 96 panel).
   - PCR Polymerase (Target 96 panel).
4. DNase- and RNase-free water of high purity.
5. One centrifugal tube (15 mL).
6. One multichannel pipette reservoir.
7. One adhesive film.
8. One vortex mixer.
9. One plate mixer.

10. One microcentrifuge.
11. One plate spinner.
12. PCR machine (Thermocycler).

2.3 Detection Step

Ensure the laboratory (post-PCR room, different from the pre-PCR room) is set up for detection step by organizing pipettes and necessary consumables.

The required materials are as follows:

1. 96.96 Dynamic Array IFC chip and Two syringes with control line fluid for priming the chip.
2. One 8-well strip.
3. One 96-well plate.
4. One microcentrifuge tube (1.5 mL).
5. Two pipette tips (10 μL).
6. One pipette tip (100/200 μL).
7. Two pipette tips (1000 μL).
8. Three boxes + eight multichannel pipette tips (10 μL).
9. Reagents:
   - Detection Solution (Target 96 panel).
   - Detection Enzyme (Target 96 panel).
   - PCR Polymerase (Target 96 panel).
   - Primer Plate (Target 96 panel).
10. *DNase-* and *RNase-free* water of high purity.
11. Two adhesive films.
12. One plate mixer.
13. Extension products.
14. Olink Signature Q100 instrument.

3 Methods

3.1 Incubation Step (See Note 1)

1. Thaw the samples.
2. After defrosting the Negative Control, Inter-plate Control, and Sample Control, vortex and briefly spin them. Then add 5 μL of each control to an 8-well strip according to the following order: Sample control (A, B), Negative control (C, D, E), and Inter-plate Control (F, G, H).
3. Prepare the Incubation Mix in a 1.5 mL tube by adding the following reagents, avoiding foaming when pipetting. Vortex and spin each reagent before use:

- 280 μL of Incubation Solution.
- 40 μL of Incubation Stabilizer.
- 40 μL of A probe.
- 40 μL of B probe.

4. Vortex and spin down the Incubation Mix.
5. Then transfer 47 μL of the Incubation Mix to each well of a new 8 well strip.
6. Transfer 3 μL of Incubation Mix to the bottom of the wells of a new 96-well plate by reversing pipetting (*see* **Notes 2** and **3**) using a multichannel pipette without changing the tips. This plate is referred to as the Incubation Plate.
7. Vortex the plate with samples using a plate mixer at 2000 rpm for 30 s or a vortex with flat bottom platform for 20–30 s at full speed.
8. Spin down the liquid at 400–1000× *g*, for 1 min at room temperature.
9. Using a multichannel pipette, transfer 1 μL of each sample by forward pipetting (*see* **Note 4**) to the bottom of the wells of the Incubation plate. Transfer 1 μL of the prepared 8-well strip with controls to column 12 of the plate by forward pipetting too (*see* **Note 4**). Change pipette tips between every column.
10. Incubate the Incubation Plate overnight for 16–22 h at +4 °C after sealing the Incubation Plate with an adhesive plastic film and spinning at 400–1000× *g* for 1 min at room temperature (*see* **Notes 5** and **6**). During the overnight incubation, proteins will attach to their targets.

3.2 Extension and Amplification Step (See Note 1)

After 16–22 h of incubation, specific DNA reporter sequences for each target protein are obtained using regular PCR in the extension and amplification step.

1. Prepare the PCR machine with the PEA program to 50 °C and pause the program.
2. Prepare the Extension Mix in a 25 mL tube adding the following reagents and avoiding foaming when pipetting. Vortex and spin each reagent before use (*see* **Note** 7):
   - 9385 μL High Purity Water.
   - 1100 μL PEA Solution (bring to room temperature before use).
   - 55 μL PEA Enzyme.
   - 22 μL PCR Polymerase.
3. Vortex the Extension Mix (*see* **Note 8**) and spin down the Incubation Plate at 400× *g* for 1 min at room temperature (*see* **Note 5**).

4. Pour the Extension Mix into a reservoir and carefully remove the adhesive film from the Incubation Plate.
5. Transfer 96 μL of Extension Mix to the upper parts of each of the well walls of the Incubation Plate by reversing pipetting (*see* **Notes 2** and **3**) using a multichannel pipette without changing tips, ensuring no come into contact with the contents of the wells.
6. Seal the plate with an adhesive plastic film and vortex thoroughly using a plate mixer at 2000 rpm for 30 s before spinning it (*see* **Notes 6** and **9**).
7. Take the plate to the thermocycler in the PCR room and resume the PEA program (*see* **Note 10**).
8. Once the PCR PEA program is complete (approximately 1 h and 55 min), proceed to the detection step.

3.3 Detection Step (See Note 1)

After the extension and amplification step, each biomarker is measured by high throughput real-time qPCR on the Olink Signature Q100 system in the final Detection step.

1. Priming the IFC chip:
   - Remove the protective film from the bottom of the chip.
   - After ensuring that the valve is fully open with the seal pressed down and to the side (*see* **Note 11**), insert the syringe tip into the valve with the chip tilted, then slowly inject a syringe of control line fluid into each IFC accumulator.
   - Press "Target 96" on the home screen of the Olink Signature Q100 instrument, place the chip on the interface with the barcode facing forward, close the lid, and press start (*see* **Note 12**).
2. Preparing the Detection Mix:
   - During the priming of the IFC, prepare the Detection Mix in a 1,5 mL tube adding the following reagents (vortex and spin down each reagent before use):
     - 550 μL of Detection Solution.
     - 230 μL of High purity water.
     - 7.8 μL of Detection Enzyme.
     - 3.1 μL of PCR Polymerase.
3. Vortex the Detection Mix and spin briefly. Then transfer 95 μL of the mix to each well of an 8-well strip.
4. Using a multichannel pipette without changing the tips, add 7.2 μL of the Detection Mix to each well of a new 96-well plate named the Sample Plate by reverse pipetting (*see* **Notes 2** and **3**).

5. Remove the Extension products from the thermocycler (*see* **Note 6**), vortex for 30 s and spin down the liquid for 3 min.
6. Using a multichannel pipette, transfer 2.8 μL from the extension products to the Sample Plate by forward pipetting (*see* **Note 4**), and changing the tips between different columns.
7. Seal, vortex, and spin the Sample Plate at 400 g for 1 min at room temperature.
8. Vortex the Primer Plate and spin down the Sample Plate and Primer Plate together.
9. Withdraw the primed chip from the Olink Signature Q100.
10. Remove the adhesive film from the Primer Plate, avoiding contamination between wells.
11. Position the chip with the barcode oriented towards the left side. Load the primers on the left side and the samples on the right side of the 96.96 Dynamic Array IFC according to the following instructions:
    - Using a multichannel pipette, load both primer and sample plates by reverse pipetting (*see* **Note 2**), transferring 5 μL of each well of columns 1–6 of the plate to columns 1–6 of the chip, starting with the first well of the first row. Then transfer from columns 7–12 of the plate to columns 1–6 of the chip, starting with the first well of the second row.
12. After loading, search and remove any bubbles with a syringe needle, avoiding contamination.
13. Place the chip in the Olink Signature Q100 instrument and select "Start Run."
14. The technical process overview is depicted in Fig. 1.

3.4 Data Analysis Using Olink® NPX Signature

The input data for Olink NPX Signature consists of run files from the Olink Signature Q100. Following quality control and normalization using the Extension Control, IPC, and a bridging factor, the output data are produced in NPX values.

3.4.1 Create a Study

To create a study, follow these steps:

1. Click on "Create Study."
2. Select the Product.
3. Enter the required information: Study Name, along with Sample Type and Notes.
4. Click "Create."

3.4.2 Add Run Data to a New Study

To add run data to a new study, follow these steps:

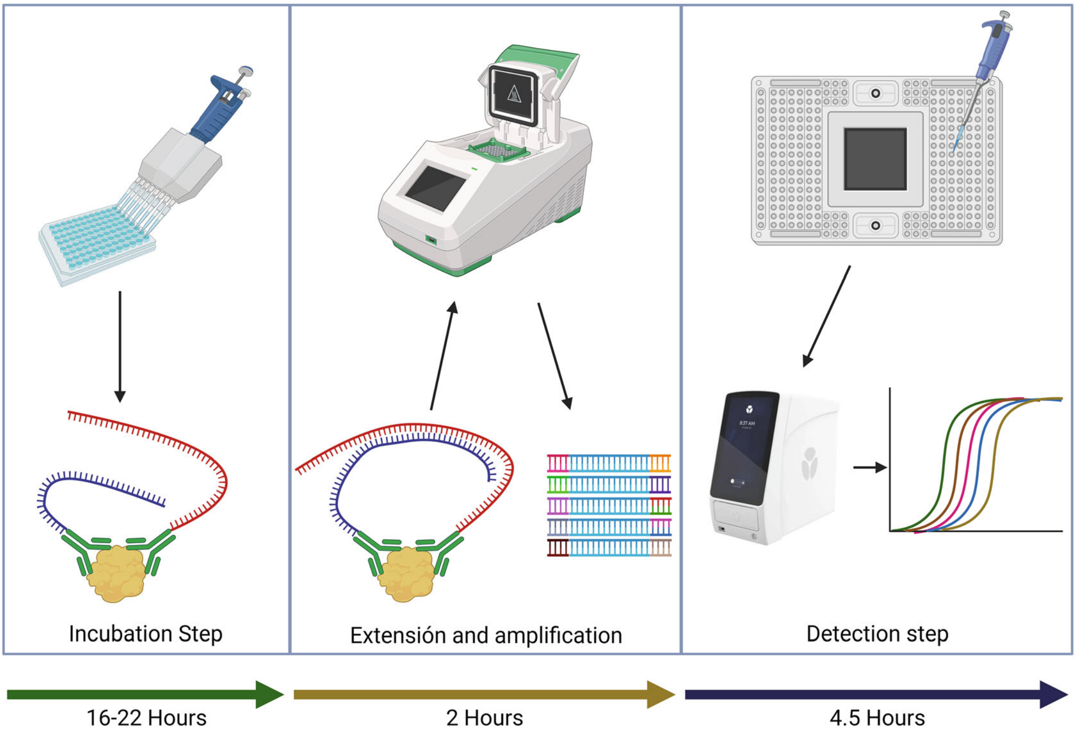

Fig. 1 Summary of the technical workflow using Proximity Extension Assay

- To import run data directly from an Olink Signature Q100 instrument to Olink NPX Signature, select "Olink Q100 Connect." Choose the desired run(s) and press "Next."
  - Press "Next" again, then verify panel, data file version, and plate layout information. Click "Add." The Study Overview will appear.
- To import run data files from your computer or a local server, select "Run Data Files."
  - Choose the desired run data file(s). Valid options include Olink Signature Q100 export files (.q100 or .zip), and exported heatmap or table results files in .csv format from Fluidigm Real-Time PCR analysis software.
- Add plate names for Plates 1 and 2.
- Select Sample Plate Dispensing: Left and Right or Even and Odd.

If duplicate sample names are found, Olink NPX Signature will prompt an alert message box. To ensure proper data handling during import, unique sample and assay names are required (*see* **Note 13**).

After the selected runs are validated, click "Next." Choose the correct Panel and Panel Data File for each imported run. Then, decide whether to use the default plate layout or to map the layout based on a plate manifest. Finally, click "Add."

3.4.3 Enter Study Information

Study information can be entered in Study → Details. Adding study details is optional but can be useful for including additional information in the Analysis Report.

You can enter information in any of the following fields to be displayed in the Analysis Report:

- Sample Type.
- Customer Info.
- Business Development Manager.
- Lab Info.
- Report Notes.
- QC Notes for Analysis Report.

3.4.4 Verify Plate Layout and Sample Annotation

1. Go to Plate QC → Plate Layout to view the plate layout for each Olink panel and plate.
2. In the Plate Details section, ensure that the correct panel data file version is selected from the drop-down menu. This data file version will be used consistently for all runs of the same Olink panel within the study.
3. Confirm that the plate layout is accurate. If you need to change the layout, select one or multiple wells and choose the appropriate Sample Type from the list above the plate layout.

   The available types are (*see* **Note 14**):

   - Sample (S).
   - Sample Control (SC).
   - Negative Control (NEG).
   - Inter-plate Control (IPC).
   - N/A.
4. Repeat **steps 2** and **3** for all imported plates before proceeding.

3.4.5 Perform Quality Controls

1. Perform the QC for each plate separately using the Sample QC tab.
2. Conduct a quality assessment for all plates using the other views, and then make an overall assessment.
3. Navigate to the Assay QC tab to review the values of internal controls and assays for each sample. Use the drop-down menu above the plate layout to change assays. Look for specific patterns that could indicate technical errors.

4. On the Table tab, examine the patterns of data below the LOD across sample plates in the study. You can also remove specific assays or individual data points from a study or sample plate.
5. Go to the Heatmap tab to identify outlier samples and assess the homogeneity of Negative Controls and IPCs. Use the drop-down menu to change the sample type.

In the Detectability tab, evaluate the detection limits for each assay and sample plate. The optimal LOD range is ±2.5 NPX. Determine if a deviating LOD on one sample plate affects detectability on other sample plates.

On the Z-scores tab, analyze differences between sample plates or distributions within a plate.

3.4.6 The Acceptance Criteria for Passing QC are as Follows

- *Run Quality Control (QC):*
  - The standard deviation for Incubation Control 1 should be less than 0.2 NPX.
  - The standard deviation for Incubation Control 2 should be less than 0.2 NPX.
  - The standard deviation for Detection Control should be less than 0.2 NPX.
  - The number of flagged samples should not exceed 1/6 of the total samples on the plate, which equates to a maximum of 15 flagged samples on a full plate.
- *Sample Quality Control (QC):*
  - The deviation for Incubation Control 2 should be within ±0.3 NPX from the plate median.
  - The deviation for Detection Control should be within ±0.3 NPX from the plate median.
- *Assay LOD:*
  - Recommended Value: LOD <2.5 NPX.
    - Comment: If LOD exceeds 2.5, please contact Olink Support for guidance and perform a Plate ANOVA.
  - Recommended Value: ≤10 assays.
    - Comment: No warning.
  - Recommended Value: ≥10 assays.
    - Comment: This may be due to insufficient randomization or other issues.
- *Row/Column:*

- Recommended Value: ANOVA <30 assays.
  - Comment: No warning.
- Recommended Value: 30–60 assays.
  - Comment: Potential randomization issue. If the samples are completely randomized, ensure that intensity normalization has been applied.
- Recommended Value: 60–90 assays.
  - Comment: Re-run the plate or ensure the issue is not due to a poorly executed run.

- *%CV Intra:*
  - Recommended Value: <15%.
    - Comment: Verify the annotation of the Sample Controls and ensure they pass QC. Examine the row pattern for rows A and B to ensure that neither row is an outlier. If the remaining data on the plate pass all QC analyses and assessments, you may fail one or both control samples.
- *%CV Inter:*
  - Recommended Value: <25%.
    - Comment: Verify the annotation of the control samples and ensure they pass QC. If intensity normalization has been used, check that the randomization assumption is valid by comparing it with IPC normalization.
- *Detectability:*
  - Recommended Value: >75% of the samples.
    - Comment: If detectability is below 75% of the samples in a proteomic study, the data may be less reliable and exhibit greater technical variability, making interpretation and validation of results difficult.

3.4.7 Export Data

1. In the main window, select "NPX Value" from the status bar located in the upper right corner.
2. Go to the Study tab and, under Study Options, choose the output option for NPX values below LOD. The default setting is "Actual Value," which displays the actual measured value for each data point.
3. In the Export Data view, adjust the settings to change the format, layout, or to add additional data to the export.
4. On the Data tab, select the desired runs to be included in the report and click "Generate."
5. In the Save dialog, enter a filename and choose a location, select an export format (XLXS or CSV), and click the "Save" button.

3.4.8 Considerations

- Samples flagged by the analysis will appear in red text, and NPX values below the maximum plate LOD will be highlighted with a red background.
- Samples, assays, or data points marked as failed in the software will be reported as "No Data."

When exporting NPX results to an Excel file, the data presentation will be as follows:

- Displayed as NPX values.
- Cells with values below the maximum plate LOD will have a red background.
- Data for samples with a QC warning will be indicated in red text. Exercise caution with data from samples that do not pass QC.
- The maximum plate LOD value for each assay will be listed on a separate row beneath the sample data, labeled as "LOD."
- The missing data frequency for each assay will be provided, showing the percentage of samples with values below the maximum plate LOD.

3.4.9 Create an Analysis Report

After thoroughly reviewing all the data, you can generate an Analysis Report for the study using Olink NPX Signature. The Analysis Report will include detailed information and QC parameters for the study.

To create the Analysis Report:

1. Navigate to Export → Report.
2. Select all the panels you want to include in the Analysis Report.
3. Click "Create" to save the Analysis Report as a PDF document. A preview of the document will be displayed.

The data analysis process overview is depicted in Fig. 2.

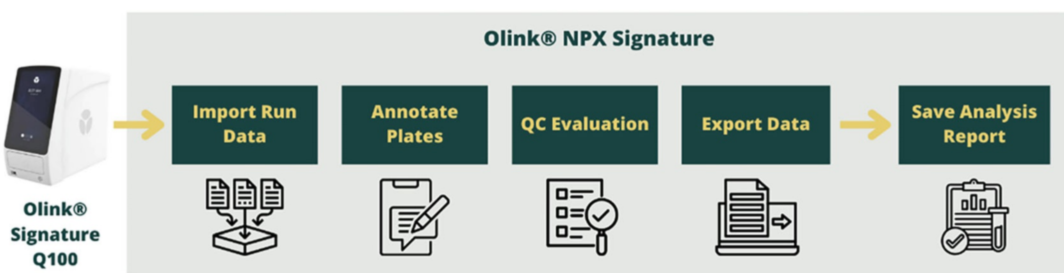

Fig. 2 Summary of the data analysis workflow using Olink® NPX Signature

4 Notes

1. Always verify at each step that the correct volume has been drawn into all pipette tips.
2. For reverse pipetting, press the operating button past the first stop, immerse the tip into the solution at an appropriate depth based on the set volume, and slowly release the operating button. After removing the tip from the solution, dispense the liquid into the receiving vessel by gently pressing the operating button to the first stop.
3. Before distributing the mix, pipette up and down a few times to pre-condition the pipette tips and ensure consistent distribution.
4. For forward pipetting, press the operating button to the first stop, immerse the tip into the solution at a suitable depth for the set volume, and slowly release the operating button. After removing the tip from the liquid, dispense it into the receiving vessel by gently pressing the operating button to the first stop, then to the second stop to expel any remaining liquid.
5. Do not vortex the Incubation Plate.
6. Double-check that all wells in the plate contain equal volume. Record any deviations, as this information will be crucial for accurate interpretation during the statistical analysis phase.
7. Keep the PEA Enzyme and the PCR Polymerase in a freezing block or on ice.
8. The Extension Mix must be used within 30 min of preparation.
9. Ensure that all wells are securely sealed to prevent samples evaporation.
10. The time between adding the Extension Mix to the incubation plate and starting the PCR (from **steps 5** to **8**) should not exceed 5 min. This is critical to maintaining assay sensitivity.
11. Gently operate both retention valve stems with the syringe caps to ensure smooth, unrestricted movement up and down.
12. While the chip is priming, begin preparing the Detection Mix.
13. If needed, sample names can be modified directly in Olink NPX Signature by importing a manifest in the Plate Manifests view.
14. Calibrator, Negative Control, and Sample Control must be properly annotated to perform calibration normalization and QC.

Acknowledgments

CPS and NB were supported by "Ramón y Cajal" Program of MCIN (RyC-2021-033828-I and RyC-2017-23437) and ALU by the Industrial PhD Program of AEI (DIN2022-012766).

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Part III

Novel Applications of Protein Microarrays

Chapter 11

Multi-array Technology for Secreted Cytokine Profiling in a Nano-aerosol Exposed 3D Human Alveolar Cell Model

An Jacobs and Inge Nelissen

Abstract

Multi-array technology allows for multiplex profiling of protein biomarker panels through sandwich immunoassays with high sensitivity, excellent reproducibility, and wide dynamic range. We describe here the detection of secreted cytokine levels in media of an in vitro stimulated human alveolar cell model that vary in a dose- and time-dependent manner. Depending on the analyte concentrations, several sample dilutions can be tested on a single multi-spot plate. Thus, the technology provides an efficient method for broad biomarker profiling.

Key words Multi-array technology, Electrochemiluminescence, Immunoassay, Meso Scale Discovery, Multiplex

1 Introduction

Multiplex immunoassays have been adopted in biochemistry laboratories worldwide to profile panels of biomarkers such as cytokines, intracellular signaling proteins, and antibodies in a biological sample. Compared to Enzyme-linked immunosorbent assay (ELISA), they offer an efficient method for parallelization, requiring less sample volume, reagents and time, with major impact in biomedical and health research, drug development, and (companion) diagnostics [1–5]. Multi-array technology combining electrochemiluminescence and patterned arrays, developed and manufactured by Meso Scale Discovery (MSD) [6], offers multiplexing capability with high sensitivity, excellent reproducibility, and wide dynamic range (4 logs). Samples are added to multi-spot plates, which are pre-coated with capture antibodies immobilized on independent and well-defined spots of the working electrode surface (Fig. 1a). Target analytes are bound, and detected by adding antibodies conjugated with electrochemiluminescent SULFO-TAG™ labels in an appropriate buffer, and reading out

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_11,

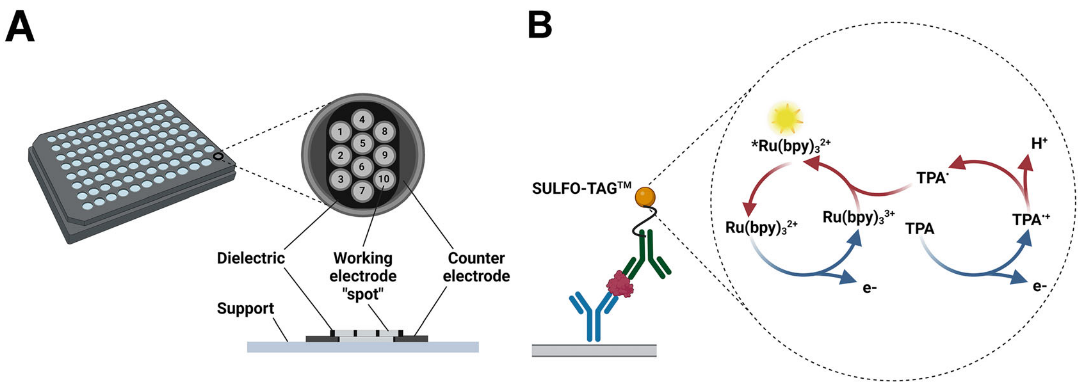

Fig. 1 MSD multi-array technology. (**a**) Diagram of a 96-well 10-spot plate showing the numbering of the different spots or working electrodes. (**b**) Electrochemiluminescence detection principle. Immobilized antibodies on the working electrode capture the target analytes, which bind SULFO-TAG™ labeled detection antibodies. SULFO-TAG™ labels are electrochemically excited to emit intense light when the multi-spot plate electrode surfaces are energized by an MSD instrument. Light intensity is then measured to quantify analytes in the sample. (Created with BioRender.com)

the electrochemiluminescence signal after applying a voltage to the plate electrodes which causes the captured labels to emit light (Fig. 1b). The read-out instrument measures the intensity of emitted light which is proportional to the amount of analyte present in the sample. These sandwich immunoassays can measure individual or multiple targets in a single sample, and allow for a broad matrix tolerance, including serum, plasma, urine, cerebrospinal fluid, saliva, and cell culture media. They are available as easy-to-build, personalized multiplex assays (U-PLEX kits) or high-performance, validated assays (V-PLEX kits). The latter kits are validated following fit-for-purpose principles [7] and MSD design control procedures.

Here, we demonstrate the use of a V-PLEX kit for simultaneous profiling of 10 pro-inflammatory cytokines in culture media of a 3D tetra-culture human alveolar cell model exposed at the air-liquid interface to three doses of silver nanowires aerosols for 6 and 24 h [8]. Although the levels of five cytokines were below the lower limits of quantification (LOQ) of the respective assays in minimally (twofold) diluted medium samples from untreated cells or at the lowest treatment dose, a dose-dependent, transient acute pro-inflammatory profile could be recorded (Fig. 2). While most cytokine levels remained well below the respective upper LOQ, interleukin (IL)-8 levels were consistently and significantly higher. In such case, testing of different sample dilutions to fall within the quantitative range of the respective assays, i.e., between the lower and upper LoQ, is warranted. Thus, we show here that the multi-array platform enables to efficiently quantify high- and

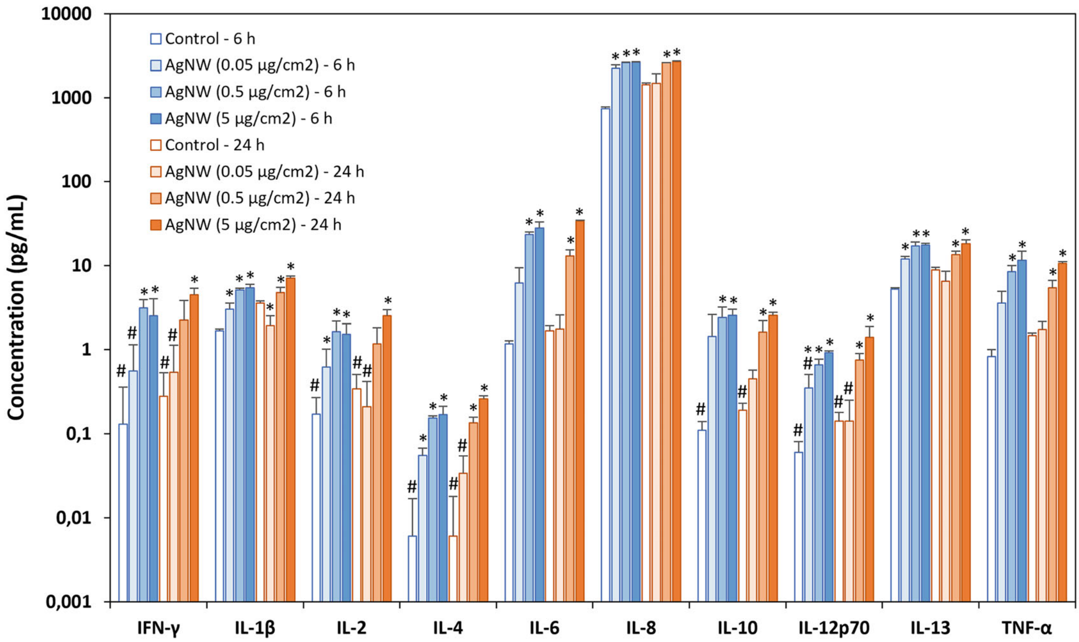

Fig. 2 Proinflammatory cytokine secretion levels in basolateral cell culture media of a 3D tetra-culture alveolar cell model after exposure to 0.05, 0.5, and 5 μg/cm^2 silver nanowires (AgNW) aerosols, obtained using a MSD Proinflammatory Panel 1 (human) V-PLEX kit. Data represent the mean ± standard deviation of three biological replicates, all samples were measured in technical duplicates on the same 10-spot plate. Asterisks (*) indicate values that are statistically different from their negative control when $p < 0.05$, based on one-way ANOVA followed by Dunnett's post-hoc test. Hashtags (#) indicate levels below the lower limit of quantification for the specific cytokine. (Data are reproduced and modified from Ref. [8], permitted by the Creative Commons license (https:/creativecommons.org/licenses/by/4.0/))

low-abundance targets in a single sample, and reliably profile dynamic cell responses.

2 Materials

The MSD V-PLEX kits include most of the specific reagents listed below (*see* **Note 1**), indicated with prefix 'MSD.'

2.1 Calibrators, Controls, and Samples

1. MSD Multi-analyte calibrator blend (lyophilized): Store at 2–8 °C (*see* **Note 2**).
2. Multi-analyte controls (lyophilized), supplied as MSD Panel Control Pack: Store at 2–8 °C (*see* **Note 3**).
3. MSD Diluent 2: Store below −10 °C (*see* **Note 4**).
4. Samples: fresh or stored at ≤ −10 °C until needed (*see* **Note 5**).
5. Polypropylene microcentrifuge tubes.

2.2 Multiplex Sandwich Immunoassay

1. Deionized water: resistivity of 18 MΩ-cm at 25 °C.
2. MSD Multi-spot plate: Store at 2–8 °C.
3. MSD Diluent 3: Store below −10 °C (*see* **Note 4**).
4. MSD SULFO-TAG™ detection antibodies (50× solution): Store at 2–8 °C. For one plate of a 10-plex assay kit, combine 60 μL of each detection antibody with 2400 μL of Diluent 3 (*see* **Note 6**).
5. Wash buffer: 1× phosphate-buffered saline (PBS), 0.05% Tween-20. For one plate, combine 300 mL of PBS with 150 μL of Tween-20. Store at room temperature.
6. MSD Read buffer T (4× solution): Store at room temperature. For one plate, combine 10 mL of Read Buffer T with 10 mL of deionized water to obtain a 2× working solution (*see* **Note 7**).
7. Plate washing equipment: automated plate washer or multi-channel pipette.
8. Adhesive plate seals (3 per plate).
9. Microtiter plate shaker (rotary) capable of shaking at 500–1000 rpm.
10. MSD MESO QuickPlex SQ 120 assay read-out instrument, or equivalent.

3 Methods

Carry out all procedures at room temperature.

3.1 Calibrator, Control, and Sample Preparation (See Note 8)

1. Prepare 7 calibrator solutions plus a zero calibrator for up to 4 replicates, by adding 1000 μL of Diluent 2 to the lyophilized calibrator vial to yield the highest calibrator (Calibrator 1, *see* **Notes 2** and **9**). After reconstituting, invert at least 3 times (do not vortex), let the solution equilibrate at room temperature for 15–30 min, and then vortex briefly using short pulses. Prepare the next calibrator by transferring 100 μL of the highest calibrator to 300 μL of Diluent 2. Mix well by vortexing. Repeat fourfold serial dilutions 5 additional times to generate 7 calibrators. Use Diluent 2 as the zero calibrator.
2. Add 250 μL of Diluent 2 to the lyophilized controls vial (*see* **Note 3**). Wait for a minimum of 15–30 min without agitation before diluting the controls twofold in Diluent 2. Vortex briefly using short pulses.
3. Dilute samples with Diluent 2: For human serum, plasma, and urine samples, MSD recommends a minimum twofold dilution, by adding 60 μL of sample to 60 μL of Diluent 2. You may conserve sample volume by in-plate dilution (*see* **Note 10**) or

using a higher dilution. Tissue culture supernatants may require additional dilution based on stimulation and analyte concentrations in the sample (*see* **Note 11**).

3.2 Multiplex Sandwich Immunoassay

1. MSD V-PLEX plates are pre-coated with capture antibodies and exposed to a proprietary stabilizing treatment to ensure the integrity and stability of the immobilized antibodies. Plates may be used as delivered, no additional preparation is required (*see* **Note 12**).
2. Wash the plate 3 times with at least 150 μL/well of Wash buffer (*see* **Notes 13** and **14**). Add 50 μL of prepared samples, calibrators, or controls per well, each in duplicate (*see* **Note 15**). Seal the plate with an adhesive plate seal and incubate at room temperature with shaking for 2 h (*see* **Notes 16** and **17**).
3. Wash the plate 3 times with at least 150 μL/well of Wash buffer (*see* **Note 18**). Add 25 μL of detection antibody solution to each well. Seal the plate with an adhesive plate seal and incubate at room temperature with shaking for 2 h.
4. Wash the plate 3 times with at least 150 μL/well of Wash buffer. Add 150 μL of 2× Read buffer T to each well (*see* **Note 19**), and proceed to read the plate on an MSD instrument (*see* **Note 20**), such as MESO QuickPlex SQ 120.
5. Data from plate runs are automatically exported to a text file in a format that is compatible with the current version of the DISCOVERY WORKBENCH® data analysis software. The software allows you to calculate the percentage recovery of the calibrators and controls, and mean calculated concentration (± standard deviation) of cytokines in the samples based on the respective fitted calibrator curves (*see* **Note 21**).

4 Notes

1. Do not mix or substitute reagents from different sources or different kit lots. Lot information is provided in the lot-specific Certificate of Analysis (CoA).
2. Reconstituted calibrator (Calibrator 1) is not stable when stored at 2–8 °C, but may be stored frozen at ≤ −70 °C and is stable through three freeze-thaw cycles. For the lot-specific concentration of each calibrator, refer to the CoA supplied with the kit.
3. Controls can be included to evaluate the precision of the assay, and are made by spiking calibrator into a non-species specific matrix at three well-defined (low, mid, high) concentrations within the quantitative range of the assay. Reconstituted controls are stable for 10 days at 2–8 °C. For long-term storage,

reconstituted controls must be stored at ≤ −70 °C and are stable through three freeze-thaw cycles. The concentration of the controls is provided in the lot-specific CoA.

4. Bring frozen diluent to room temperature in a 24 °C water bath.
5. Use published guidelines for human sample collection, storage, and handling. Remove visible particulates by centrifugation at 2000× *g* for 3–20 min (depending the sample type) prior to using or freezing. Repeated freeze-thaw of samples is not recommended.
6. Prepare the detection antibody solution immediately prior to use. Avoid prolonged exposure of detection antibody (stock or diluted) to light. During the antibody incubation step, plates do not need to be shielded from light except for direct sunlight.
7. To reduce foam forming, prepare the read buffer T during the detection antibody incubation step and let it rest for at least 1 h. You may keep excess diluted Read Buffer T (2×) in a tightly sealed container at room temperature for up to 1 month.
8. Prepare calibrators, samples, and controls in low protein bind polypropylene microcentrifuge tubes; use a fresh pipette tip for each dilution; vortex after each dilution before proceeding.
9. For individual assays that do not saturate at the highest calibrator concentration, the calibration curve can be extended by creating a more concentrated highest calibrator. In such a case, follow the steps below using 250 μL instead of 1000 μL of Diluent 2 when reconstituting the lyophilized calibrator.
10. Dilute-in-Plate: To limit sample handling, you may dilute samples and controls in the plate. For twofold dilution, add 25 μL of assay diluent to each sample/control well, and then add 25 μL of neat control or sample. Calibrators should not be diluted in the plate; add 50 μL of each calibrator directly into empty wells.
11. If using non-ionic detergents (e.g., Triton X-100) in your samples, keep the total concentration below 1% since the detergent can affect the binding of the antibodies.
12. Preparation of a custom designed plate, such as U-PLEX, involves coating the provided plate with linker-coupled capture antibodies on activated spots.
13. When using an automated plate washer, rotate the plate 180 degrees between wash steps to improve assay precision. Gently tap the plate on a paper towel to remove residual fluid after washing.
14. Washing the plate prior to sample addition is an optional step that may provide greater uniformity of results for certain assays.

Analytical parameters including limits of quantification, recovery of calibrators and controls, and sample quantification, are not affected by washing the plate prior to sample addition.

15. Do not touch the pipette tip on the bottom of the wells when pipetting into the MSD plate. For empty wells, pipette to the bottom corner. Avoid bubbles in wells at all pipetting steps by using reverse pipetting, as bubbles may lead to variable results.
16. Assay incubation steps should be performed between 20 and 26 °C to achieve the most consistent signals between runs. Plate shaking should be vigorous, with a rotary motion between 500 and 1000 rpm. Binding reactions may reach equilibrium sooner if you use shaking in the middle of this range (~700 rpm) or above.
17. Extending incubation of the samples overnight at 2–8 °C may improve sensitivity for some assays. If an incubation step needs to be extended, leave the sample or detection antibody solution in the plate to keep the plate from drying out.
18. For tissue culture samples, you may simplify the protocol by eliminating one of the washing steps. After incubating diluted sample, calibrator, or control, add detection antibody solution to the plate without decanting or washing the plate.
19. Incubation in Read Buffer T is not required before reading the plate. Make sure that Read Buffer T is at room temperature when added to a plate, and avoid introducing bubbles as they may interfere with signal detection. Do not shake the plate after adding Read Buffer T. To improve inter-plate precision, read the plate as soon as possible after adding Read Buffer T and keep time intervals consistent between adding Read Buffer T and reading the plate.
20. Remove all plate seals prior to reading the plate. When running a partial plate, seal the unused sectors to avoid contaminating unused wells. Sealed, partially used plates may be stored up to 30 days at 2–8 °C in the original foil pouch with desiccant.
21. Best quantification of unknown samples can be achieved by generating a calibration curve for each plate using a minimum of two replicates at each calibrator concentration. If assay results are above the top of the calibration curve, dilute the samples and repeat the assay.

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Chapter 12

Use of Humanized NFAT-Dsred RBL Reporter for Detection of Allergen-Specific IgE Sensitization in Human Serum in 96-Well Microarray Format

Daniel Wan, Marcos J. C. Alcocer, and Franco H. Falcone

Abstract

We have established a workflow allowing printing of allergens and control proteins into the wells of a MaxiSorp 96-well plate. Allergens are printed e.g., in a 4 × 4 pattern into each well of the plate. Prior to adding to the printed plates, a humanized rat basophil leukemia (RBL)-derived fluorescent reporter cell line (e.g., RBL NFAT DsRed) is sensitized overnight with the IgE-containing serum to be tested. Next, the humanized reporter cell line is added to the wells with the printed allergens and incubated for 24 h in a cell standard culture incubator. If the serum sample contains enough IgE specific for the printed allergens, NFAT-dependent signal transduction will occur, leading to synthesis of intracytosolic DsRed protein during an overnight incubation. Fluorescence can then be measured using a microplate reader using the well scan function.

Key words Reporter system, DsRed, RBL, IgE, Allergy, NFAT-DsRed, Array

1 Introduction

Type I (or immediate type) hypersensitivity is an allergic immune response mediated by Immunoglobulin E (IgE) antibodies, which bind to the high affinity IgE receptor FcεRI found mainly on mast cells and basophils, but also on other cell types. Engagement of receptor-bound IgE by cognate, multivalent allergens results in fast degranulation and release of preformed as well as de novo synthesized mediators such as histamine [1], resulting in the well-known symptoms of allergy. Because IgE is at the centre of (type I) hypersensitivity reactions, the measurement of allergen-specific IgE is an important parameter and one of the three pillars of allergy diagnosis [2]. Traditionally, allergen-specific IgE is measured by immobilizing the allergens on a solid phase, incubating with the IgE-containing fluid (e.g., serum) and detecting IgE binding with a labelled secondary antibody. Alternatively, sensitization of mast

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_12,

cells in the skin can be assessed in vivo by means of skin prick tests or variations thereof. Our group proposed an alternative format in which allergens are printed onto protein arrays and probed with human basophils, after the latter have been sensitized with the blood to be tested [3]. In this system, activation of basophils is measured by monitoring surface expression of CD63. However, using basophils is logistically challenging, despite the existence of fast protocols for isolation from peripheral blood [4]. High affinity IgE-receptor expressing cell lines would be very advantageous, provided that these cell lines are able to bind human IgE and IgE-dependent activation can be measured.

We and others have previously developed a series of humanized IgE reporter systems derived from the rat basophil leukemia cell line RBL-2H3 [5–7]. This cell line has been used for decades in studies of IgE-dependent signaling and its inhibition, as it contains a fully functional tetrameric high affinity IgE receptor FcεRI [8]. However, most of these reporter cell lines are not suitable for use in allergen arrays, as their activation results in degranulation and release of a soluble reporter protein (e.g., RFP) or cytosolic induction of luciferase, the latter requiring cell lysis for enzymatic measurement. We have therefore created a humanized reporter cell line in which IgE-dependent activation results in the synthesis of cytosolic fluorescent protein (GFP or DsRed), which then allows measurement of cell activation 'on the spot' in the array format [9]. The NFAT-DsRed cell line was further improved by co-transfecting the human FcεRI gamma chain, as this results in higher surface expression of the IgE binding human alpha chain of FcεRI [10]. This improved cell line was used in array formats on printed slides [11]. However, combining cells requiring an overnight incubation with printed arrays, i.e., needing continuous incubation with cell culture medium, poses technical challenges. We therefore adapted the technique to the use of standard MaxiSorp 96-well plates, allowing easy incubation of cells and measurement using fluorescent microplate readers with well scanning function (Fig. 1).

2 Materials

All reagents and supplies should be sterile. Follow local waste disposal regulations.

2.1 Reagents

1. RBL medium: minimum essential medium (MEM), 10% v/v heat-inactivated fetal bovine serum, 100 U/mL penicillin, 100 μg/mL streptomycin, 2 mM L-glutamine (if not already present in MEM). For maintenance, the medium can contain Phenol Red. Use Phenol red-free formulation when measuring the plates. Store at 4 °C.

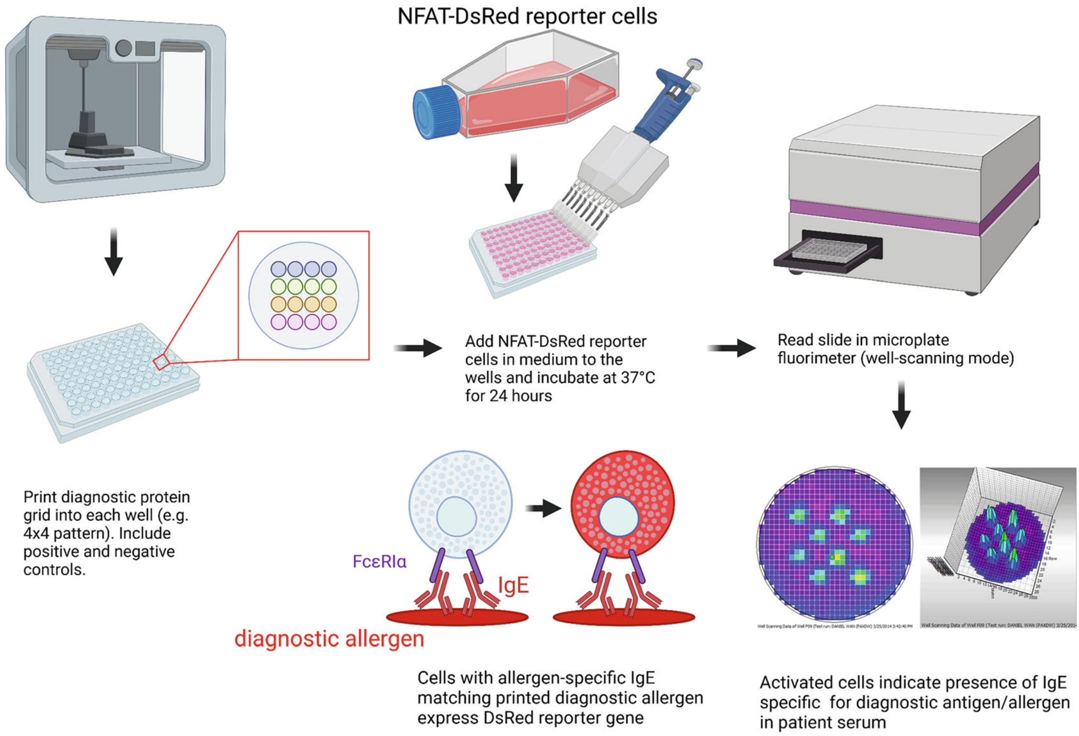

Fig. 1 General workflow of the technique. An array printer is used to print allergens, as well as positive/negative controls into the wells of a 96-well MaxiSorp plate. NFAT-DsRed (FCER1G) cells are sensitized overnight with the serum to be tested. The next day, 100 μL of reporter cell suspension are added to the wells and incubated for 24 h. Cells sensitized with IgE matching the printed allergen will bind to the allergen spots and produce cytosolic DsRed protein. The red fluorescence can then be measured in a microplate reader using the well scan function. Activation can be expressed as volume under the curve (created with BioRender)

2. For cell line maintenance: RBL medium supplemented with 1 mg/mL G418 sulfate and 20 μg/mL blasticidin S (NFAT-DsRed) or RBL medium supplemented with 1 mg/mL G418 sulfate and 20 μg/mL blasticidin S and 600 μg/mL Hygromycin B (NFAT-DsRed FCER1G).
3. NFAT-DsRed and NFAT DsRed FCER1G reporter cell lines are available from the authors through a Material Transfer Agreement (MTA).
4. G418 (for NFAT-DsRed and NFAT DsRed FCER1G cells).
5. Blasticidin S (for NFAT-DsRed and NFAT DsRed FCER1G cells).
6. Hygromycin B (for NFAT DsRed FCER1G cells).
7. Dulbecco's Phosphate Buffered Saline (DPBS) without Ca^{2+} or Mg^{2+}.
8. 0,05% v/v Tween-20/DPBS.

9. Trypsin-EDTA: 0.25% Trypsin and 1 mM EDTA in Hank's Balanced Salt Solution (HBSS), without Ca^{2+}/Mg^{2+}. Phenol red is optional.
10. Cell culture-grade DMSO for freezing cells (*see* **Note 1**).
11. Patient sera.
12. Allergens.
13. Human serum albumin (HSA) or bovine serum albumin (BSA) as negative control or for blocking.
14. Anti-human IgE (e.g., polyclonal goat-anti-human IgE).
15. 500 mM calcium chloride solution ($CaCl_2$).

2.2 Supplies

1. Graduated pipettes and pipette tips.
2. 25 cm^2/75 cm^2 cell culture flasks.
3. 50 mL centrifuge tubes.
4. MaxiSorp 96-well ELISA plates.
5. Cell scrapers (optional).

2.3 Equipment

1. Pipetters for graduated pipettes/tips.
2. Water bath.
3. Class II microbiological safety cabinet.
4. Cell culture incubator: humidified incubator set to 37 °C and 5% CO_2.
5. Heat block.
6. Haemocytometer or automated cell counter.
7. Centrifuge capable of holding 50 mL tubes.
8. Array printer. We used sciFLEXARRAYER S11 (Scienion, Germany).
9. Microplate reader for fluorescence with well scanning function (e.g., CLARIOStar Plus from BMG Labtech, Germany, or Tecan SPARK from Tecan Trading AG, Switzerland).

3 Method

3.1 Array Printing into 96-Well Plates

Allergens to be tested are printed into the wells of a 96-well plate in the desired pattern (e.g., 4 × 4 dots) using an array printer. Always include suitable positive and negative controls (*see* **Note 1**).

1. If sterility is an issue, you can sterilize the 96-well plate by exposing it to a short wavelength UV-source (e.g., a transilluminator) e.g., for 20 min.
2. Position your plate in the Array printer.

3. Carefully choose the parameters for printing according to your equipment, leaving some safety margin from the walls of the wells (*see* **Note 2**).
4. As positive control, use polyclonal anti-human IgE at 100 μg/mL and all other proteins (allergens, negative control) at 1 mg/mL, all diluted in PBS. Use 3 nL to print each spot, in triplicate. If a larger amount is desired, you can deposit a higher number of droplets.
5. The next day, wash the printed MaxiSorp plates three times with 0,05% v/v Tween-20/DPBS.
6. Block with 100 μL 1 mg/mL HSA (or BSA, *see* **Note 1**) for 1 h at room temperature.
7. Wash the wells again three times with 0,05% v/v Tween-20/DPBS, followed by one final wash with DBPS to remove traces of Tween-20.
8. The plate is now ready for use.

3.2 General Cell Culture

Always work in a class II laminar flow cabinet using sterile techniques.

1. If starting from frozen stocks, partially defrost a vial of cells by warming in hand, then add 1 mL pre-warmed RBL medium to fully defrost.
2. Transfer cells to a 25 cm^2 flask containing 8 mL RBL medium, and place in a cell culture incubator overnight.
3. When the cells are confluent in the 25 cm^2 flask (this may take several days, depending on viability of frozen cells), transfer cells to a 75 cm^2 flask by detaching adherent cells using trypsin-EDTA (*see* **steps 2**–**11** of Subheading 3.3 Sensitization below).
4. Maintain cells in 10 mL RBL medium supplemented with 1 mg/mL G418 sulfate and 20 μg/mL blasticidin S (NFAT-DsRed) in a 75 cm^2 flask, in the cell culture incubator. If working with the NFAT-DsRed FCER1G line, also add 600 μg/mL Hygromycin B, (*see* **Note 3**).
5. Passage cells when confluent using trypsin-EDTA (*see* **steps 2**–**10** of below). Replacing the flask with 20% cells will result in confluence after 3 days. Replace with 10% cells for confluence (*see* **Note 4**) after 4 days (refresh the medium after 2 or 3 days). Top up the level of liquid in the flask to 10 mL.
6. Freeze stocks of cells in larger batches i.e., from several flasks. Centrifuge cells at 300× *g* for 5 min, resuspend in RBL medium with 10% v/v DMSO to obtain a cell concentration of 5×10^6 cells/mL. Work quickly since DMSO is toxic to cells at room temperature. Aliquot 1 mL cells/cryovial, transferring cryovials to a Mr. Frosty™ (Nalgene) container filled with

isopropanol, and place in a −80 °C for several hours or overnight. After freezing, transfer the cryovials to a liquid nitrogen Dewar flask for long-term storage. Defrost a vial to check the viability of the batch (*see* **Note 5**).

3.3 (Optional) Treatment of Serum before Sensitization

1. If necessary, heat human serum in an Eppendorf tube at 56 °C for 5 min (*see* **Note 6**) on a heat block and then place it on ice.

3.4 Sensitization

Cells are sensitized with the sera to be studied for 16–20 h by incubating them in RBL medium with the desired dilution of previously heat-inactivated serum (heat treatment optional).

1. Pre-warm RBL medium to 37 °C in a water bath.
2. Remove medium from 75 cm^2 flask of confluent cells.
3. To the flask, add 10 mL DPBS without Ca^{2+} or Mg^{2+}.
4. Tilt the flask to wash the cells with DPBS.
5. Discard the DPBS in the flask.
6. Add 2 mL trypsin-EDTA to the flask (*see* **Note 7**).
7. Tilt the flask to ensure that the cells are exposed to the trypsin-EDTA solution.
8. Incubate the flask for 5–15 min at 37 °C in a cell culture incubator, until clumps of cells are visibly detached (*see* **Note 8**).
9. Add 8 mL pre-warmed RBL medium to the flask and resuspend the cell suspension.
10. Transfer the cells to a 50 mL tube. Some cells can be left behind in the flask for passaging (*see* Subheading 3.1 General cell culture).
11. Count a sample of cells in a haemocytometer or automated cell counter and calculate the total cell number.
12. Pellet the cells by centrifuging at 300× *g* for 5 min.
13. Resuspend the cells to a concentration of 3.5×10^5 cells/350 μL using RBL medium containing a 1:50 (or 1:100) dilution of the serum to be tested.
14. Pipette 350 μL cells/well of a 24-well plate.
15. Incubate the plate overnight in the cell culture incubator.
16. The next day, remove the medium from the wells and wash once with 350 μL DPBS w/o Ca^{2+}/Mg^{2+} before adding 70 μL phenol red-free Trypsin/EDTA (*see* **Note 9**).
17. Put back the plate into the incubator for approximately 5 min to ensure detachment of cells from the plate.

18. Once cells have detached, add 280 μL phenol red-free RBL medium supplemented with 500 mM $CaCl_2$ (yielding a final concentration of ~20 mM Ca^{2+}).
19. With a pipette, transfer the cell suspension from each well to a sterile 1.5 mL microcentrifuge tube.
20. Pipette 100 μL of this cell suspension into appropriate wells of the prepared MaxiSorp plate, and place in a cell culture incubator.
21. 24 h later, intracellular fluorescence in each well can be measured (*see* **Note 10**).

3.5 Measurement of Activation

3.5.1 Fluorescence Assay with NFAT-DsRed

Plates can be measured with a plate reader capable of reading the wells with a well scanning function (see Fig. 2). If using e.g., a ClarioStar Plus or PHERAstar FS reader, use the matrix well-scanning function, with 540 nm excitation and 590 nm emission filters.

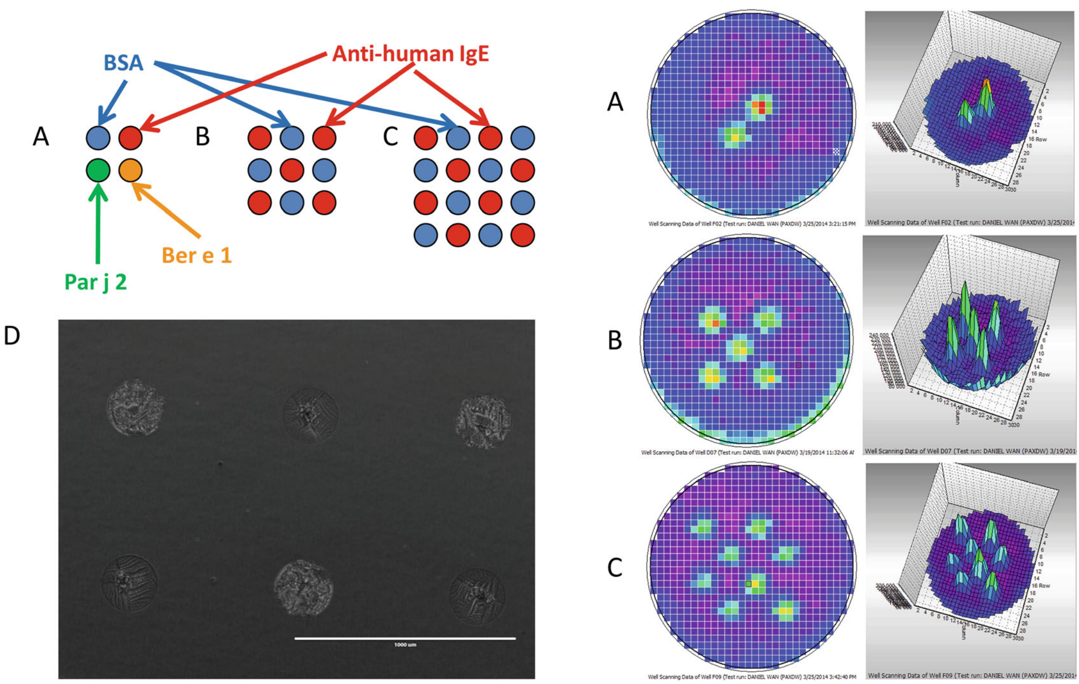

Fig. 2 Examples of results obtained with NFAT DsRed cells sensitized with serum of a *Parietaria judaica* (spreading pellitory)-allergic patient. (**a**) 2 × 2 printing pattern including a positive control (red, polyclonal anti-human IgE), a negative control (blue, BSA), Par j 2 recombinant allergen (green) and recombinant allergen Ber e 1 (orange, negative control). Activation is only seen with the positive control and the Par j 2 sample. (**b**) 3 × 3 alternating pattern with polyclonal anti-human IgE (red) and the negative control BSA (blue), and (**c**) 4 × 4 alternating pattern with polyclonal anti-human IgE (red) and the negative control BSA (blue). The corresponding well scan measurements are shown on the right. (**d**) Successful deposition of allergen and controls can be verified (but not quantified) by conventional light microscopy of the dry spots. Spots have a diameter of approximately 270 μm

4 Notes

1. As positive control, we use polyclonal goat-anti-human IgE, which will cross-link allergen-specific IgE independently of its specificity. The optimal amount of anti-IgE will depend on the type (species, clonality, affinity, etc.) used and needs to be determined empirically. Optimum concentrations need to be determined by testing the anti-IgE antibody over a wide range of concentrations (e.g., at least 3 orders or magnitude). Particular care should be taken regarding the choice of a negative control. For example, if working with sera from a cohort of patients allergic to beef, using BSA (a.k.a. Bos d 6) would not be a recommended choice. In this case, use the more expensive human serum albumin instead.
2. Hitting the wall of the plate while attempting to print at the bottom of the well will result in broken pins, which will be very expensive. We were able to print 5 × 5 patterns per well, leaving a good safety margin away from the well walls. This will allow you to test a minimum of 25 × 96 = 2400 samples per printed plate.
3. These antibiotics are needed to preserve the transfected cDNAs encoding the reporter system (NFAT-DsRed; Blasticidin S), the human FcεRI alpha chain (G418) and the human FcεRI gamma chain (Hygromycin). Antibiotics can be omitted for short periods of time to reduce costs, but this needs to be weighed against the risk of losing the transgenes in the longer term.
4. Splits larger than 1/10 split (10% cells) are not recommended for routine passaging as this can cause "colonies" to form and unhealthy cells in the middle of the cell clusters and can delay growth.
5. For convenience, we routinely use commercial cryopreservation media rather than self-made conditioned medium. For example, we have very good experience with AMSBIO's Cellbanker I (for own storage needs) or serum-free Cellbanker 2 for cryovials sent to other labs abroad.
6. Some human sera can be cytotoxic for RBL cells. This becomes visible under the microscope after the overnight sensitization. Cytotoxicity can be avoided by diluting the serum samples 1: 100. Alternatively, a 5 min incubation step at 56 °C is enough to remove most of the complement activity responsible for cytotoxicity, without harming the ability of IgE to bind to the human IgE receptor. This treatment allows use of more serum during sensitization, e.g., 20% serum v/v concentration (1:5 dilution). Alternatively, the addition of 2 M glucose during heat treatment stabilizes the structure of IgE during heating

allowing treatment at −56 °C for 30 min without noticeable losses of IgE binding. In our hands most sera, however, do not require heat inactivation, hence the heat treatment is described as optional.

7. There is no need to worry about degradation of the IgE antibodies by trypsin, as IgE becomes quite resistant to tryptic cleavage upon binding to the high affinity IgE receptor [12].
8. Make sure any visible clumps are broken up, by pipetting up and down. We routinely use a 5 mL serological pipette for this purpose, as these have a narrower tapered end. Make sure you do not create bubbles when pipetting up and down by always leaving some medium in the pipette when expelling the liquid and not taking up any air when aspiring it, as this will damage the cells. You can speed up the detachment process by repeatedly knocking the side of the flask with your knuckles, holding it tight with the other hand. Be careful, however, as the use of too much force can split the flask if knocking its bottom. Also keep the neck area of the flask free from cells and medium, particularly if reusing the same flask. Cell scrapers can be used to help detach the cells instead of trypsin-EDTA. Simply refresh RBL medium and scrape the cells, ensuring these are always covered by medium. There is no need for a DPBS wash. However, excessive or badly performed scraping will affect the viability of the cells.
9. Phenol red in cell culture medium interferes with DsRed fluorescence measurement. Use Phenol red–free medium instead.
10. Fluorescence will be stable for 48–72 h; however, cells will start deteriorating once they have become overconfluent.

Acknowledgments

We would like to thank Dr. Adriano Mari, Centri Associati di Allergia Molecolare (CAAM, Italy) for providing us with serum of Par j 2 allergic patients.

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Chapter 13

A Competitive Ligand-Binding Assay for Simultaneous Detection of Neutralizing Antibodies to Both Arms of a Bispecific Drug

Vitaly Ablamunits, Soma Basak, Rosemary Lawrence-Henderson, Teresa M. Caiazzo, and John Kamerud

Abstract

Bispecific monoclonal antibodies gain their attractiveness as a novel class of drugs that are capable of neutralizing two or more cytokines simultaneously or activate hetero-dimeric cell receptors. Monitoring appearance of neutralizing antibodies (NAbs) to these drugs is a challenging task. Here, we report a competitive ligand- binding assay (CLBA) for detection of NAbs to a bispecific candidate drug using a multiplex Meso Scale Discovery (MSD) platform, which allows for detection of NAbs to both drug arms in the same sample. The assay has sensitivity better than 250 ng/mL, is tolerant to the presence of drug at concentration > 600 μg/mL, and to the level of soluble target(s) >400 ng/mL. Thus, multiplex approach provided by MSD platform can be successfully used for development of NAb assays in competitive ligand-binding assay format.

Key words Bispecific drug, Competitive ligand-binding assay, Neutralizing antibodies, U-plex MSD

1 Introduction

Cancers and chronic inflammatory diseases involve a number of molecules that might be important pharmaceutical targets. Novel large molecule drugs such as bi-specific antibodies have been considered advantageous compared to a combination of monospecific drugs [1, 2]. Preclinical experiments and clinical trials require monitoring of immune responses to such molecules since antidrug antibodies (ADA) may impair pharmacokinetics and cause adverse events [3, 4]. NAbs are ADA that can directly block the pharmacological effects of such drugs and may result in situations when patients stop responding to the treatment. In most cases, NAbs are detected using cell-based assays that are recommended by US FDA and EMA [5, 6]. In some instances, however, NAbs can be detected using CLBA, in which a labeled drug is prevented from

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_13,

binding to an immobilized target in the presence of NAbs [7]. CLBAs are considered an acceptable approach for detecting Nabs when the mechanism of action (MOA) of the drug is purely dependent on binding of the target by the drug, such as when the drug is an antagonist of a cytokine and binding of the cytokine blocks the activity of this ligand to its receptor. Here, we report development of an assay to detect NAbs to a bispecific drug candidate designed for treatment of an autoimmune inflammatory disease. One arm of this drug targets inflammatory mediator 1, and the other arm—mediator 2 [8].

In addition to detecting the presence of Nab, it was desired to determine the specificity of Nabs towards domain 1, which targets mediator-1, and domain 2, which targets mediator-2. Rather than develop two separate assays, we endeavored to use a multiplexed approach, which would allow for simultaneous detection of NAbs with different specificity in the same sample.

To develop the NAb assay, we used the multiplex Meso Scale Discovery platform (U-plex MSD) that allows simultaneous detection of several different analytes in the same well of a 96-well plate due to the ability to bind analyte-specific reagents to a specific spot area on the bottom of the well using proprietary spot-specific linkers [9]. The details on the method are presented below.

2 Materials

2.1 Reagents

1. The bispecific drug, proprietary monoclonal Ab biotinylated or conjugated with ruthenium (Ru); or unlabeled for drug tolerance assessment. For conjugation procedures, refer to Paragraph 3.
2. Normal human serum (pooled, mixed gender), and individual sera from healthy donors of both genders.
3. Target 1 and Target 2 biotinylated to capture NAbs; unlabeled targets may be used to investigate target tolerance.
4. Monoclonal NAbs to each of the drug arms (custom made) for positive controls (PCs) (*see* Paragraph 3).
5. Mouse antihuman antibodies specific to Target 1 or 2 to deplete excess target.
6. Streptavidin (SA)-coated blocked magnetic beads; Sera-Mag SpeedBeads (GE Healthcare, PA, USA).
7. Pierce™ Streptavidin Coated High Binding Capacity 96-Well Plate with SuperBlock (Thermo Fisher Scientific) or similar.
8. MSD Gold SULFO-TAG NHS-Ester™ (Meso Scale Discovery ®, Meso Scale Diagnostics, LLC, MD, USA).

9. EZ-Link™Sulfo-NHS-LC biotin, No-Weigh™ Format (Thermo Fisher Scientific, IL, USA).
10. Phosphate-buffered Saline, Calcium- and Magnesium-free (PBS-CMF) pH 7.2 with and without 1% or 4% Bovine Serum Albumin.
11. 1 M Tris buffer, pH 9.0.
12. 300 mM Glycine buffer, pH 2.5.
13. U-plex Development kit (Meso Scale Discovery).
14. Slide-A-Lyzer Cassettes varying size and Molecular Weight Cut Off (MWCO) (Thermo Fisher Scientific, IL, USA).
15. Bio-Gel P6 medium (Bio Rad, CA, USA).
16. Stabilizing Buffer for conjugated Ru reagent storage.

2.2 Instruments and Disposable Plasticware

1. ThermoMixer C (Eppendof AG, Hamburg, Germany).
2. Orbital plate shaker, vortex, Hulu mixer.
3. Refrigerators, deep freezers.
4. Plate and tube magnetic bead separators.
5. Single- and multi-channel pipettes with adjustable volumes.
6. Five, 14, and 50 mL polypropylene tubes, 96 well plates, pipette tips, disposable serological pipettes.
7. MSD plate reader with Methodical Mind software.

3 Methods

3.1 Conjugation of Drug or Targets with Biotin

Drug and targets, formulated in phosphate-buffered saline, calcium and magnesium free (PBS-CMF),

pH 7.2, are conjugated with EZ-Link™Sulfo-NHS-LC biotin (Thermo Fisher Scientific, IL, USA) at excess molar challenge ratios of 20:1 or 5:1 (biotin:protein), respectively. The reactions are incubated at ambient temperature for 2 h, subsequently quenched with 1 M Tris, pH 9.0 and dialyzed into PBS-CMF, pH 7.2 to remove unbound biotin by passive diffusion using a Slide-A-Lyzer™ (Thermo Fisher Scientific, 10,000 or 3500 MWCO size of starting material determined) for 18–24 h at 4 °C with multiple buffer exchanges per manufacturer's recommendation.

3.2 Conjugation of Drug with Ru

Modification of drug with Ru at molar challenge ratio of 12:1 with MSD Gold SULFO-TAG NHS-Ester™ Ru (Meso Scale Discovery ®, Meso Scale Diagnostics, LLC, MD, USA) is performed at ambient temperature for 2 h, subsequently quenched with 1MTris, pH 9.0 and utilizing Bio-Gel P6 medium (Bio Rad, CA, USA) processed into a stabilizing buffer to remove unbound Ru [10]. All reagent conjugates are stored at −70 °C or below.

3.3 Production of Custom Antidrug Monoclonal NAbs*

1. Hyper immunize mice with the drug.
2. Select animals that responded by testing serum samples in ELISA (optional).
3. Isolate spleen cells and fuse them with myeloma cell line to create hybridomas.
4. Clone hybridomas by limiting dilutions and select the best responder clones.
5. Select monoclonal antibodies with neutralizing activity for anti-target −1 arm (PC-1) and anti-Target-2 arm (PC-2). Test to ensure there is no cross-reactivity between PC-1 and Target −2, and vice versa (*see* **Note 2**).
6. Determine working concentrations of PC-1 and PC-2 (*see* **Notes 3** and **6**).

* A detailed methodology for generation of monoclonal antibodies is outside of the scope of this Methods paper, because of its complexity (*see* **Note 7**).

3.4 U-Plex Plate Coating with Targets

U-plex development kit consists of U-plex plates with 10 spots per well, two spot-specific streptavidin-coupled linkers targeting spots 1 and 10, respectively, Stop Solution, and Read buffer.

1. Prepare biotinylated targets 1 and 2 at concentration of 10 μg/mL in PBS/1%BSA.
2. Mix 800 μL of each target with 1200 μL of linker 1 and 10, respectively, incubate for 30 min at room temperature (RT).
3. Add 800 μL Stop Solution and incubate for additional 30 min.
4. Combine 2600 μL of each reagent and add 20.8 mL of Stop Solution added to result in 26 mL coating mixture. This will be enough for 5 plates.
5. Store the coating mixture for up to 7 days at 4 °C for subsequent use on the day of experiment, or use for plate coating immediately.
6. On the day of experiment, add 50 μL/well of biotinylated target coating reagent to U-plex MSD plate and incubate on an orbital shaker at 450 rpm, at RT for >60 min.

3.5 Enrichment of NAbs by Solid-Phase Extraction and Acid Dissociation (Optional) [11, 12]

This step is needed when the sample contains drug at concentrations that inhibit detection of NAbs (*see* **Note 4**). If not required, go to the Subheading 3.7.

1. Calculate the volume of SA-coated magnetic beads (10 mg/mL) based on 25 μL/sample.

2. Wash the beads twice with 2 mL assay buffer on a tube magnet, resuspend in 1800 mL of assay buffer and add 200 μL of biotinylated drug at 1 mg/mL.
3. Incubate the suspension for 30 min at RT on a rotating Hulu mixer.
4. Pellet the beads on a magnet, rinse with 2 mL of 300 mM Glycine, pH 2.5 for 2 min, wash and resuspended in four times the initial bead volume of assay buffer.
5. Pipette the beads into a 96 well polypropylene plate, 100 μL/well, pellet on a plate magnet and remove assay buffer.
6. Add 50 μL/well Tris, pH 9.0 to the plate with beads.
7. Dilute PCs, negative control (here, Drug controls, or DC) and samples 10-fold with 300 mM Glycine, pH 2.5, incubate at RT for 15 min on an orbital shaker.
8. Transfer 200 μL/well of acidified controls/samples to the plated beads. Seal the plate and incubate for 30 min at RT on a ThermoMixer C (Eppendof AG, Hamburg, Germany) shaking at 800 r.p.m.
9. Pellet the beads on a plate magnet, wash twice with the assay buffer, and add 100 μL Glycine pH 2.5 to elute NAbs for 15 min.
10. Transfer 75 μL of the eluate to another plate with 25 μL/well Tris pH 9.0 to neutralize.

3.6 Removal of Excess Target (Optional)

This step is needed when the sample contains target at concentrations that affect detection of NAbs. If not required, go to Subheading 3.7. Soluble target(s) in the sample will bind to the Drug-Ru conjugate and will reduce its binding to immobilized target. This will create false-positives in samples negative for NAbs (*see* **Note 5**).

1. Prepare mouse anti-target 1 and(or) 2 biotinylated antibody at concentration 10 μg/mL in PBS.
2. Add anti-target antibody to a SA-coated polystyrene plate, 100 μL/well.
3. Incubate the plate on an orbital shaker for 60 min and wash the plate.
4. Add 85 μL neutralized eluate from Subheading 3.5, **step 10** to the plate and incubate for 30 min at RT.
5. Collect 70 μL of the samples and continue as described in Subheading 3.7.

3.7 Detection of Drug-Neutralizing Activity

1. Dilute Drug-Ru to 20 ng/mL and add 70 μL/well into a new polypropylene plate.

2. Add 70 μL of samples/controls from Subheading 3.5, **step** 10 or Subheading 3.6, **step** 5 to the plate with Drug-Ru and incubate for 30 min on a shaker at 450 rpm protected from light to allow the NAbs to interact with the drug.
3. Wash target-coated U-Plex MSD plate from Subheading 3.4, **step 6**. three times with PBS-CMF.
4. Add 50 μL of samples/controls to the MSD -plate in duplicates, incubate protected from light for 40 min on a shaker.
5. Wash the MSD plate 3 times with PBS-CMF.
6. Add 150 μL of MSD Read buffer and read the plate on a Sector Imager 600 MSD reader.
7. Calculate mean relative light units (RLU) from replicates for data from Spot 1 and Spot 10.
8. Calculate Signal to Drug control or Drug control to signal ratio for each NAb specificity:

   S/DC = mean RLU of PC or samples/RLU of assay DC, or

   DC/S = RLU of assay DC/mean RLU of PC or samples.
9. Calculate % Inhibition for each NAb specificity, if desired:

   % INHIB = (1 − S/DC) ∗ 100
10. Report the data (*see* **Note 1**).

4 Notes

1. In the described method, RLU signal decreases with increased concentration of NAbs, since NAbs prevent Drug-Ru from binding to the target (Fig. 1).
2. As with any other method, an investigator is encouraged to test that each individual reagent fits to the purpose, i.e., that the

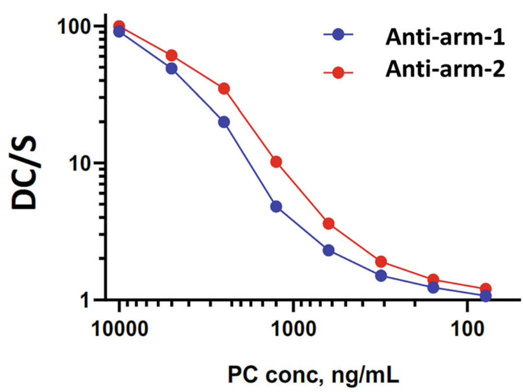

Fig. 1 NAb1 and NAb2 dose-response curves expressed as DC/S demonstrating inverse correlation between the signal (S) and the NAb concentration

drug specifically recognizes the targets, and PCs are specific to the arms of the drug with no cross-reactivity.

3. Final concentrations of the PCs will be different in each particular method, as sensitivity of the assay will depend on affinities of PC NAbs.
4. The need for sample acidification to dissociate NAb-Drug complexes is common when notable drug tolerance is required, and this step may reveal PCs that are not resistant to this treatment and need to be replaced. So, it is highly recommended to have a set of different monoclonals as candidate PCs before assay development is initiated. Depending on the required level of drug tolerance, which in turn depends on the drug levels in circulation, enrichment for Drug-free NAbs may be performed using Drug-biotin on SA-coated magnetic beads, or SA plates, or drug passively absorbed on plates; in some cases, this step may be completely omitted. The presence of unlabeled drug in the sample will compete with NAbs, increase RLU, and create false negatives [11, 12].
5. Circulating levels of targets may interfere with the assay by preventing Drug-Ru from binding to the plate-bound targets, decrease RLU, and create false positives. In our assay, target tolerance for one of the arms was too low and required target removal using anti-target antibody adsorbed on a polystyrene plate. This antibody cannot be the drug itself to avoid depletion of NAbs from the sample.
6. In our hands, the assay demonstrated sensitivity better than 250 ng/mL at minimal required dilution 1:10, drug tolerance exceeding 600 ug/mL, and target tolerance of at least 400 ng/mL with no prozone and no matrix interference [8]. Figure 1 shows an example of expected dose response curves expressed as DC/S ratios.
7. Generation of custom monoclonal antibodies and selection of NAbs is a complicated task. For those that are not familiar with the methodology, we advise to outsource this part of work to a commercial provider of this service.

Acknowledgments

The authors appreciate the contribution of our colleagues Glenn Miller, Renee Ramsey, and Lee Walus for their technical expertise.

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Chapter 14

Alternative Matrices for Protein Biomarker Analysis by qPCR Using Proximity Extension Assay (PEA)

Adrián Llamas-Urbano, Julio Manuel Martínez-Moreno, Carlos Pérez-Sánchez, and Nuria Barbarroja

Abstract

Advancements in proteomics and biomarker discovery require technologies capable of high-throughput, multiplexed, and minimally invasive protein detection. The Proximity Extension Assay (PEA), developed by Olink Proteomics, meets these demands by integrating dual-recognition immunoassays with quantitative PCR (qPCR), enabling the precise quantification of multiple proteins with exceptional sensitivity and specificity. While PEA is commonly applied to serum and plasma, its adaptability allows for the analysis of alternative biological matrices, including cerebrospinal fluid, urine, synovial fluid, bone marrow, amniotic fluid, blister fluid, conditioned media, interstitial fluid, and extracellular vesicles, among others. These matrices offer valuable opportunities for biomarker discovery, precision medicine, and translational research, particularly in diseases where conventional sampling is challenging. This chapter details the methodological considerations required for processing these alternative biofluids, ensuring their compatibility with PEA technology. It provides best practices for sample collection, processing, and quality control to maximize the accuracy and reproducibility of proteomic analyses. Additionally, it highlights PEA's ability to detect low-abundance proteins, which is critical for identifying novel biomarkers in complex biological samples.

Key words Proximity extension assay, Proteomics, qPCR, Olink, Protein array, Biomarkers

1 Introduction

In today's biomedical landscape, a profound understanding of the molecular mechanisms underlying various diseases necessitates advanced technological tools that enable precise and efficient analysis of key biomolecules. Protein detection and quantification play a crucial role in biomedical research and the development of new therapies. In this context, the Proximity Extension Assay (PEA) technology, developed by the Swedish company, Olink Proteomics OLINK, has emerged as an innovative solution that combines high sensitivity and specificity in protein detection.

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_14,

PEA combines the precision of antibody-based immunoassays with the efficiency of quantitative PCR (qPCR), delivering exceptional levels of specificity, sensitivity, and dynamic range in protein detection. The technique involves a dual-recognition immunoassay where two antibodies, each labeled with unique DNA oligonucleotides, bind to specific epitopes on the same target protein, bringing them into close proximity. This proximity enables the DNA oligonucleotides to hybridize and, through an extension reaction, form a unique double-stranded DNA "barcode" that directly correlates with the initial concentration of the target protein. This barcode is then amplified and quantified using real-time PCR, allowing for highly sensitive and multiplexed protein analysis.

One of the most remarkable advantages of PEA technology is its minimal sample requirement: Only one microliter of biological material is needed for analysis. This is especially beneficial in studies where the sample quantity is limited or obtaining it is invasive. Additionally, the versatility of the technique allows for the use of various types of biological samples, including serum, plasma, cerebrospinal fluid, urine, and tissues, expanding its applicability across different areas of medicine and research.

PEA is commonly utilized in samples like serum or plasma. In this chapter, we will delve into the primary alternative matrices that can be examined using this technology with minimal volumes (Fig. 1), thereby revolutionizing the conventional approach to proteomics.

2 Materials

2.1 Buffer Compatibility with Olink

Buffers are essential in the preparation of cell and tissue lysates, protein elution from solid supports, and in lavage solutions. When formulating buffers, it is crucial to consider the following: (i) proteins must remain in their native state to ensure effective antibody recognition, and (ii) avoid reagents that could interfere with PCR processes.

Recommendations for Buffer formulation in PEA analysis are detailed below.

2.2 Detergents

- Using excessive detergent concentrations can cause protein denaturation.
- For ionic detergents like sodium dodecyl sulfate (SDS) and deoxycholic acid (DCA or deoxycholate), the concentration should be kept at or below 0.1%.
- Non-ionic detergents, such as Tween® 20, Triton™ X-100, and NP-40, should not exceed 1% concentration.

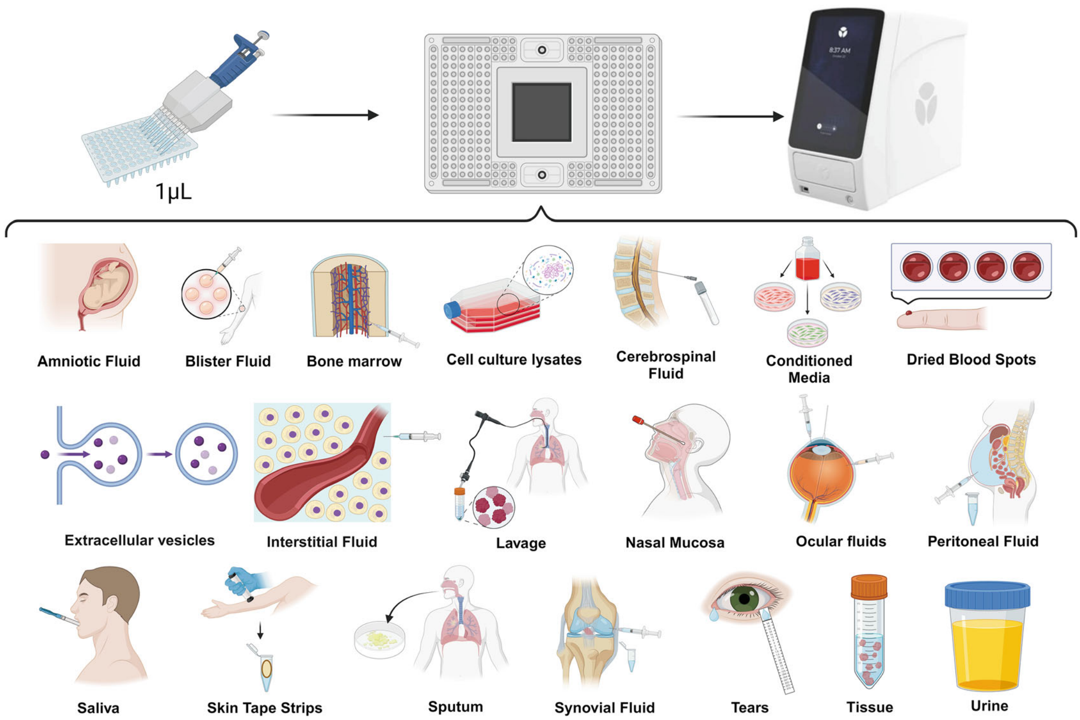

Fig. 1 Alternative Matrices. Overview of the different sample matrices that can be analyzed with the "Proximity Extension Assay" from Olink, using just one microliter

2.3 Other Denaturants to Avoid

- Dithiothreitol (DTT) concentrations higher than 1 mM.
- Alcohols.
- Excessive heat.
- Urea.
- Heavy metal salts.

2.4 Salts

- High salt concentrations can lead to protein aggregation and precipitation.
- Concentrations should be maintained at or below 250 mM for NaCl, 25 mM for KCl, and 10 mM for $MgCl_2$.

2.5 pH

- Buffer pH should be maintained close to physiological levels, typically within the 7–8 range.

2.6 Commercial Buffers

- It is recommended to use T-PER™ Tissue Protein Extraction Reagent (#78510) and M-PER™ Mammalian Protein Extraction Reagent (#78501) from Thermo Fisher for effective protein extraction.

- For Bio-Plex® Cell Lysis Kit (Bio-Rad #17130411), add the following components just before use: Factors 1 and 2 (included in the kit), along with a protease inhibitor cocktail.
- RIPA buffer can be prepared as follows: 50 mM Tris-HCl (pH 7.4), 150 mM NaCl, 1 mM EDTA, 1% Triton X-100, and 0.1% deoxycholic acid (DCA). Based on our tests with commercial brands, Millipore RIPA Lysis Buffer (#20–188) is compatible with Olink technology, whereas Sigma RIPA Buffer (#R0278) does not perform as well.
- To prepare NP-40 lysis buffer, use 1× Tris-EDTA buffer (pH 8.0), 1% NP-40, 0.1% Triton X-100, 0.1% sulfobetaine, 150 mM NaCl, and 1× protease inhibitor cocktail.

2.7 Diluents

- When normalizing samples to a specific protein concentration, it is important to use the original lysis/elution buffer to maintain consistency and accuracy.
- For pre-dilution of samples in Olink panels or alternative matrices, the Olink Diluent should be used to ensure proper dilution and compatibility with the assay.

2.8 PCR Inhibitors

- EDTA should be ≤25 mM to avoid sequestration of Mg^{2+}.
- Urea.
- Heme.

2.9 Protease and Phosphatase Inhibitors

- The Roche cOmplete™ Mini Protease Inhibitor Cocktail (#11836153001) is highly recommended. One tablet can be dissolved in 10 mL of lysis buffer. Alternatively, you can prepare a 10× stock solution by dissolving one tablet in 1 mL of distilled water or PBS, or a 7× stock by dissolving it in 1.5 mL. The stock solution can be stored at 4 °C for up to 2 weeks or at −20 °C for up to 12 weeks. Use a 1× final concentration of the inhibitor cocktail and avoid exceeding this concentration.
- General concentrations of compatible protease inhibitors include: 1 mM PMSF, 10 mM AEBSF, 8 mM aprotinin, 0.2 mM leupeptin, 0.4 mM bestatin, 0.15 mM pepstatin, and 0.15 mM E-64.
- Although Olink assays do not detect phosphorylation sites, phosphatase inhibitors such as NaF (5–10 mM) and Na_3VO_4 (1–2 mM) are acceptable. Additionally, phosphatase inhibitor cocktails like PhosSTOP™ can be used.

3 Methods

In this section, specific procedures for the analysis by PEA of 20 different biological matrices are detailed. *See* **Note 1.**

3.1 Amniotic Fluid

As a biofluid rich in proteins and fetal cells, amniotic fluid is ideal for analysis using techniques like PEA. Initially, amniotic fluid consists mostly of water and electrolytes, but by week 13, it also contains proteins, carbohydrates, lipids, and urea, all of which contribute to fetal growth. PEA's high sensitivity and specificity are particularly valuable for identifying biomarkers related to fetal development, intrauterine infections, and pregnancy complications. This makes it a powerful tool for both biomedical research and prenatal diagnostics.

3.1.1 Recommendations for Sample Preparation

- Amniotic fluid should be collected following established clinical best practices to ensure sample integrity.
- Freshly collected samples remain stable for a limited time at room temperature but should ideally be stored on ice or at 4 °C to preserve quality.
- To remove cells and insoluble material, samples should be centrifuged for 10 min at a minimum of 500× *g*.
- Avoid using samples contaminated with blood, as this can compromise the analysis.
- Aliquots of the processed samples should be stored at −80 °C for long-term preservation.

See Pre-Dilution Strategies in **Notes 2** and **22.**

3.2 Blister Fluid

Wound exudate, also known as blister fluid, is produced as a result of vasodilation during the early stages of healing, and it is primarily composed of water, along with electrolytes, nutrients, waste products, proteins such as inflammatory mediators, proteases (MMPs), and growth factors, as well as cells like neutrophils, macrophages, and platelets. This fluid forms a protective barrier that aids tissue repair and fights infection. It is typically clear or light yellow, but a red or pink color indicates blood contamination, while cloudiness or a green hue can signal infection. Several studies have used PEA technology to analyze the proteomic profile of blister fluid, identifying key biomarkers associated with inflammation and healing. It is important to record the appearance and viscosity of the fluid, as these characteristics provide valuable insights into the wound's condition. Blisters should not be punctured unless necessary, and should be protected with clean dressings, as proper management of blister fluid accelerates recovery and prevents complications [1, 2].

3.2.1 Recommendations for Sample Preparation

Wounds can result from accidents, medical procedures, insect and animal bites, infection, and disease conditions. In addition, they come in different sizes and shapes. Therefore, it is not possible to recommend a single sample preparation protocol. Please follow these general guidelines for use with Olink analysis:

- Samples should be collected as liquid biofluids. Normalization can be done either by volume or by adjusting the protein concentration to 0.5 mg/mL.
- There are various methods to aid in sample collection, such as syringes, vacuum/suction, drainage, or use of absorbent materials.
- Immediately after collection, samples should be placed on ice for short-term storage or at −80 °C for long-term storage.
- It's recommended to use a protease inhibitor cocktail, such as Roche cOmplete™ Mini (#11836153001). Dissolve one tablet in 10 mL of lysis buffer for immediate use, or prepare a 10× stock by dissolving in 1 mL of distilled water or PBS. Use a 1× final concentration and avoid exceeding this.
- Samples should be collected in Eppendorf LoBind® microcentrifuge tubes and spun at high speed for 5 min to pellet cells and particulates and stored at −80 °C until analysis.
- For dilutions, it is recommended to use Olink Diluent.

  See Pre-Dilution Strategies in **Notes 3** and **22.**

3.3 Bone Marrow

Bone marrow is a spongy tissue found inside long bones, such as the hips and thighs, responsible for producing red blood cells, white blood cells, and platelets. It contains a fluid portion, which can be extracted by aspiration, and a solid portion, which is obtained through biopsy. It plays a key role in diagnosing conditions like anemia, metastatic cancer, and blood disorders, and is also used in transplants due to its hematopoietic stem cells. Recently, the PEA technology has been employed in various scientific studies, making a significant impact within the scientific community due to its ability to detect multiple biomarkers with high precision, greatly enhancing the research and diagnosis of bone marrow-related diseases [3, 4].

3.3.1 Recommendations for Sample Preparation

- Bone marrow should be collected following established clinical best practices to ensure sample quality.
- A specific protocol for bone marrow aspiration from the posterior superior iliac spine has been outlined, with key considerations including:
  - The presence of spicules (tiny bone fragments) confirms that the marrow cavity has been successfully accessed and that the aspiration has been performed correctly.
  - Most spicules are typically found in the first 1–2 mL of the initial withdrawal.

  - It is recommended to aspirate only 1–2 mL to avoid hemodilution, which could interfere with accurate proteomic analysis.
  - If the bone marrow collection process is slow or challenging, pre-flushing syringes with an anticoagulant may help facilitate the procedure.
- Freshly collected samples are stable at room temperature for a limited time but should ideally be placed on ice or stored at 4 °C to preserve integrity.
- Samples should be centrifuged for 10 min at 2000× *g* to remove cells and insoluble debris.
- Aliquots should then be stored at −80 °C for long-term preservation.

See Pre-Dilution Strategies in **Notes 4** and **22.**

3.4 Cell Culture Lysates

Cell lysates, containing intracellular components such as proteins, lipids, and nucleic acids, are vital for studying cellular functions, molecular interactions, and signaling pathways. Numerous studies have successfully employed PEA technology in these lysates, leading to the identification of key biomarkers and a detailed understanding of molecular pathways. This has significantly advanced research in areas such as cancer, autoimmune disorders, and neurodegenerative diseases. Additionally, cell lysates are crucial for evaluating cellular responses to external stimuli, making them essential for both therapeutic development and advanced biomedical research [5, 6].

3.4.1 Recommendations for Sample Preparation

- Follow the manufacturer's guidelines for using lysis buffers or kits. Keep the lysis buffer cold, perform the lysis on ice, and centrifuge at 4 °C.
- Wash adherent or suspended cells twice with cold PBS or HBSS to remove any remaining culture medium containing FBS or growth factors.
- Typically, add about 100 μL of lysis buffer per 10^6 cells, ensuring that the final protein concentration is above 0.5 mg/mL for proper normalization.
- For adherent cells, after removing the wash solution, add the lysis buffer directly on top of the cells, swirl gently to distribute evenly, and incubate on ice for 10 min. Use a cell scraper to collect the lysates and transfer them into a LoBind® Eppendorf tube using a pipette.
- For cells in suspension, wash the cells by low-speed centrifugation (500× *g* for 5–10 min at 4 °C), then remove excess wash solution by carefully aspirating it from the surface of the pellet. Add the lysis buffer to the pellet, mix gently by vortexing or

pipetting, and incubate on ice for 5–10 min to complete the lysis.

- After cell lysis, centrifuge the samples at high speed and 4 °C to remove any debris.
- Use standard methods such as BCA, Bradford, Lowry, or Nanodrop assays to estimate the protein concentration in the clarified lysates.
- Store aliquots of the lysate at −80 °C or colder.

See Pre-Dilution Strategies in **Notes 5** and **22.**

3.5 Cerebrospinal Fluid (CSF)

As a clear, colorless fluid that surrounds the brain and spinal cord, CSF provides valuable insights into nervous system pathology and neurological or infectious diseases. However, any samples showing signs of blood contamination should be documented, and it's generally recommended not to use them. Blood-derived proteases can degrade proteins, heme can interfere with PCR processes, and blood-derived proteins may obscure accurate measurements of the CSF proteome, affecting the reliability of the analysis [7, 8].

3.5.1 Recommendations for Sample Preparation

- Freshly collected samples are stable for a brief period at room temperature but should be stored on ice or at 4 °C to maintain their quality. Adding protease inhibitors is not required.
- Protein concentration in these samples typically ranges from 0.15 to 0.45 μg/μL. To ensure optimal results, it's important to minimize exposure to room temperature before proper refrigeration.
- To remove cells and insoluble materials, samples should be centrifuged for 10 min at speeds exceeding 500× *g*.
- The aliquots should be stored at a temperature of −80 °C.

See Pre-Dilution Strategies in **Notes 6** and **22.**

3.6 Conditioned Media

The medium used to nourish in vitro cultured cells, known as conditioned media or cell culture supernatant, is of great interest to researchers because it contains secreted proteins. This medium can originate from homogeneously cultured cells, such as clonal primary cells or immortalized cell lines, or from heterogeneous cultures, like organoids or ex vivo biopsies. Conditioned media offers valuable insights into cellular activity and interactions with the environment, making it essential for studies on protein secretion, metabolism, and biomarker discovery. Additionally, it has been pivotal in research using PEA technology, which enables the detection of low-abundance proteins. To ensure purity and reliability in analysis, it is crucial to remove cells and cellular debris through high-speed centrifugation before storage or further study [9].

3.6.1 Recommendations for Sample Preparation

- Some media formulations include growth serum additives, such as fetal bovine serum (FBS) or fetal calf serum (FCS). Olink recommends using heat-inactivated serum to minimize cross-reactivity with proteins closely related to human proteins.
- After carefully collecting conditioned media from cells, high speed centrifugation at 4 °C should be used to remove cells, cell debris, and other particulates.
- Samples should aliquoted and stored at −80 °C.
- Additionally, phenol red may interfere with internal controls in the Target 96 assay, so its presence should be considered during QC analysis. However, phenol red does not affect the Target 48, Flex, Focus, or Explore assays.

See Pre-Dilution Strategies in **Notes** 7 and **22.**

3.7 Dried Blood Samples (DBS)

The use of DBS is a convenient and less invasive method for collecting and storing blood, particularly in infants and in environments where refrigeration is not readily available. These samples preserve biomolecules such as DNA and metabolites at room temperature, making them valuable for diagnostics, disease monitoring, and genetic research. Moreover, recent studies have utilized this biofluid in investigations employing PEA technology, providing new insights into the detection of low-abundance proteins [10, 11].

3.7.1 Recommendations for Standard DBS Sample Preparation

- Prepare 10 mL of elution buffer consisting of 1× PBS with 0.05% TWEEN 20 and 1× protease inhibitors.
- Using a 3 mm punch tool, punch through the section of the collection card where the blood has fully penetrated the filter paper. Place the punch into a LoBind microcentrifuge tube.
  - Important: To prevent contamination, clean the punch tool and tweezers between samples to avoid cross-contamination.
- Add 20 μL of elution buffer to the microcentrifuge tube for each 3 mm punch (approximately 3 μL dried blood).
- Ensure the punch is fully submerged in the diluent and can move freely within the tube.
- Seal the tubes and place them on a shaker for 1 h at room temperature, set to around 600 rpm (or enough to create visible movement of the buffer).
- After shaking, transfer the eluates to new tubes or a 96-well plate.
- Immediately freeze and store the samples at −80 °C.

See Pre-Dilution Strategies in **Notes 8** and **22.**

3.8 Extracellular Vesicles

Membranous structures of varying sizes (ranging from nanometers to micrometers) contain membrane and luminal proteins, which play crucial roles in both health and disease. There are different classes, such as exosomes, microvesicles, macrovesicles, and apoptotic bodies, which can be distinguished by their size or cell surface biomarkers. These vesicles are essential for cellular communication and the transport of molecules, making them a highly valuable biofluid for use as biomarkers in the diagnosis and prognosis of various diseases, including cancer and neurodegenerative disorders [12].

Normalization of EV Samples

1. For biofluids, normalization is recommended based on the starting volume.
2. In in vitro studies, samples can be normalized by the number of cells producing EVs [13].
3. When using total protein concentration for normalization, 0.5 mg/mL is advised.
4. EV particle count can be quantified using various methods, such as Nanoparticle Tracking Analysis and flow cytometry [14].

3.8.1 Recommendations for Sample Preparation

Purification of EVs

- There are various methods for isolating EVs including ultracentrifugation, density gradient centrifugation, size exclusion chromatography, polymer-based precipitation, filtration and immunological separation [15, 16].

General Guidelines for Lysing Purified EVs

- Keep lysis buffer cold, perform lysis on ice, and centrifugations at 4 °C.
- Since different isolation techniques yield varying particle sizes and numbers, the volume of lysis buffer should be optimized experimentally, starting with a smaller amount to achieve a protein concentration of approximately 0.5 mg/mL for optimal detection.
- If samples are turbid, centrifuge at high speed at 4 °C to remove debris.
- If necessary, protein concentration of clarified lysates can be estimated using standard techniques.

See Pre-Dilution Strategies in **Notes 9** and **22.**

3.9 Interstitial Fluid

Filling the spaces between cells and distinct from blood plasma, interstitial fluid facilitates nutrient exchange, cellular communication, and the removal of metabolic waste. It is essential for homeostasis and reflects changes in the cellular environment, making it valuable for biomedical research and biomarker monitoring in diseases like cancer and diabetes. Unlike blood, it does not clot, making it a promising option for continuous clinical monitoring and minimally invasive techniques. Internal tissues can be sampled through microdialysis, where a catheter with a dialysis membrane is used to collect proteins [17, 18].

3.9.1 Recommendations for Sample Preparation

- Interstitial fluid should be collected following established clinical best practices. Normalize samples by volume for consistency.
- Freshly collected samples are stable for a short duration at room temperature but should be stored on ice or at 4 °C if possible. (It's recommended to use a protease inhibitor cocktail, such as Roche cOmplete™ Mini (#11836153001). Dissolve one tablet in 10 mL of lysis buffer for immediate use or prepare a 10× stock by dissolving in 1 mL of distilled water or PBS. Use a 1× final concentration and avoid exceeding this.
- Samples should be centrifuged for 10 min at ≥500× *g* to remove cells and insoluble material.
- Aliquots should be stored at −80 °C.

See Pre-Dilution Strategies in **Notes 10** and **22.**

3.10 Lavage Samples

Collected by introducing and removing a buffered saline solution into an internal body cavity, lavage samples allow the collection of cells, proteins, and other molecules for analysis. Bronchoalveolar lavage is the most commonly used technique, particularly for studying lung diseases, but nasal, uterine, and other cavity lavage samples can also be analyzed. These samples are valuable for clinical research and diagnostics, providing direct access to biological material from the affected site, enabling the analysis of biomarkers, cytological studies, and microbiological cultures. However, variability in collection and processing can impact results, making it essential to follow standardized protocols [19].

3.10.1 Recommendations for Sample Preparation

- Ensure the lavage solution follows the recommended salt concentrations (≤250 mM NaCl, ≤25 mM KCl, and ≤10 mM $MgCl_2$) and maintains a physiological pH of 7–8.
- After collection, keep the samples on ice or at 4 °C.
- Centrifuge the samples at high speed to remove cells and particulates.
- Finally, aliquot the supernatant and store it at −80 °C.

See Pre-Dilution Strategies in **Notes 11** and **22.**

3.11 Nasal Mucosa

Collected from the inner lining of the nasal cavity, nasal mucosa samples are a valuable resource for clinical and diagnostic studies. They contain epithelial cells, proteins, mucins, and immune components, making them a key biofluid for investigating respiratory infections, allergies, and inflammatory diseases. These minimally invasive sampling techniques are also useful for studying lung cancer, as well as pulmonary, inflammatory, and infectious diseases such as asthma, allergic rhinitis, and tuberculosis. Additionally, they allow for the detection of viruses, bacteria, and biomarkers associated with various pathologies, including COVID-19. Collection is performed using nasal swabs, offering an accessible method for disease monitoring and research into local immune mechanisms [20].

3.11.1 Recommendations for Sample Preparation (Nasal Swabs)

- Prepare 10 mL of elution buffer (PBS with 0.05% Tween-20 and 1% BSA) with 1× protease inhibitors and store on ice.
- Insert one IVALON® Post-Op Sinus Packin (#Q770530) nasal swab into each nostril in the region of the middle meatus for 5 min.
- Remove swabs from nostrils and transfer to two 15 mL tubes with 3 mL of elution buffer. Push the swabs to the bottom of the tubes with tweezers so that they are completely immersed.
- Cap the tubes and incubate for 2 h on ice or 4 °C.
- Using tweezers, transfer the 2 swabs from the elution buffer into one 5 cc syringe barrel. Transfer both the remaining eluates and the 5 cc syringe with swabs into a new 15 mL conical.
- Centrifuge the 15 mL tube with syringe barrel at 1500× *g* for 15 min at 4 °C.
- Discard the 5 cc syringe barrel with swabs into a medical waste container.
- Aliquot eluates into 0.5 mL or 1.5 mL LoBind tubes and store at −80 °C.

See Pre-Dilution Strategies in **Notes 12** and **22.**

3.12 Ocular Fluids

Composed of aqueous humor (AH) and vitreous humor (VH), these fluids are essential for visual health and play a crucial role in the study of eye diseases. AH, found in the front part of the eye, provides nutrients and removes waste, while VH, a gel-like substance between the retina and lens, helps maintain the eye's shape and supports the retina. Both fluids are valuable for investigating infectious, inflammatory, and degenerative eye conditions such as glaucoma and diabetic retinopathy. Due to its high viscosity, VH requires clarification techniques like high-speed centrifugation or centrifugal filters for accurate analysis [21, 22].

3.12.1 Recommendations for Sample Preparation

- Ocular fluids should be collected using best practice clinical guidelines.
- Freshly collected samples are stable for a short duration at room temperature but should be stored on ice or at 4 °C if possible.
- Aliquots should be stored at −80 °C.

See Pre-Dilution Strategies in **Notes 13** and **22.**

3.13 Peritoneal Fluid

Is a serous liquid produced by the peritoneum that lines and protects the abdominal organs. Under normal conditions, it is clear or pale yellow but can become milky, cloudy, or bloody depending on the underlying disease. Its analysis is crucial for diagnosing conditions such as peritonitis, ascites, liver cirrhosis, infections, and cancers, particularly ovarian cancer. This biofluid is collected via paracentesis and is studied through cytological, biochemical, and microbiological analyses to detect infections, inflammation, malignant cells, and evaluate specific abdominal biomarkers [23].

3.13.1 Recommendations for Sample Preparation

- Peritoneal fluid should be collected following established clinical best practices. Freshly collected samples are stable at room temperature for a limited time but should ideally be stored on ice or at 4 °C to maintain their integrity. (It's recommended to use a protease inhibitor cocktail, such as Roche cOmplete™ Mini (#11836153001). Dissolve one tablet in 10 mL of lysis buffer for immediate use or prepare a 10× stock by dissolving in 1 mL of distilled water or PBS. Use a 1× final concentration and avoid exceeding this).
- Samples are normalized by volume or standardized to 1 mg/mL.
- After collection, centrifuge the samples for 10 min at ≥500× *g* to remove cells and insoluble material.
- Store aliquots at −80 °C for long-term preservation.

See Pre-Dilution Strategies in **Notes 14** and **22.**

3.14 Saliva

As a valuable biofluid for clinical studies, it offers a non-invasive, low-cost collection process. It contains a wide range of biomolecules, including hormones, proteins, DNA, and metabolites, making it useful for investigating infectious, endocrine, immunological diseases, and cancer. Consistency in collection, processing, and storage is crucial, ensuring that samples are taken from a standardized location in the mouth. Absorptive techniques should be avoided unless their impact on the salivary proteome has been thoroughly assessed. This biofluid is particularly well-suited for analyzing disease biomarkers and conducting large-scale population health studies, offering an accessible and repeatable method for diagnostic and research purposes [24].

3.14.1 Recommendations for Sample Preparation

- Ensure the individual avoids food, drinks, gum, nicotine, and dental products for 1 h before sample collection.
- Collect samples at the same time each day to minimize diurnal variations, and normalize them based on either volume (standard method) or total protein concentration (0.5 mg/mL), depending on the study's requirements.
- Collect samples at the same time daily to reduce diurnal variations.
- Instruct the individual to rinse their mouth with water before sampling.
- Five minutes after rinsing, ask the person to spit into a 50 mL plastic tube placed on ice. Collect a total of 5 mL of saliva within a 30-min window.
- Add a protease inhibitor cocktail to reach a final concentration of 1× and mix thoroughly.
- Centrifuge the samples at 2500–3000× *g* for 15 min at 4 °C. If separation is incomplete, repeat centrifugation for another 20 min at the same temperature.
- Transfer the supernatant to a 15 mL tube on ice and vortex it.
- Aliquot the sample into LoBind microcentrifuge tubes and freeze at −80 °C for storage.
- Before running the Olink assay, thaw the samples to room temperature, vortex them, and centrifuge for 10 min at ≤500× *g*. Retain the supernatant for the incubation step.

  See Pre-Dilution Strategies in **Notes 15** and **22.**

3.15 Skin Tape Strips

As a minimally invasive method, tape stripping is used to remove the outermost layer of the skin, the stratum corneum. It involves applying adhesive tape to the skin and abruptly removing it, allowing for the collection of superficial cells for analysis. This technique is widely used in dermatological and cosmetic research to assess the efficacy of topical formulations, dermal toxicology, and the study of skin conditions such as dermatitis and psoriasis. It also helps in analyzing the skin barrier, lipid and protein composition, and drug absorption. Due to its non-invasive nature, simplicity, and low cost, tape stripping is ideal for longitudinal and large-scale studies without causing significant discomfort to patients [25].

3.15.1 Recommendations for Sample Preparation

- Collect a D-Squame tape strip by applying it to the skin with a standardized pressure of 225 g/cm^2 for 10 s, then quickly remove it.

- Samples are normalized by tape strip area or by protein concentration (ideally 0.5 mg/mL, though lower levels are common).
- Prepare RIPA buffer, with or without protease inhibitors, in a 15 mL tube and keep it on ice.
- Set up four 1.5 mL tubes with 300 μL of RIPA buffer each and place them on ice.
- Cut the tape strip into quarters and transfer each piece into a separate 1.5 mL tube using tweezers, ensuring the strips are pushed to the bottom of the tubes.
- Sonicate the samples for 15 min in an ice bath (0–4 °C) using an ultrasonic bath.
- Combine the eluates from all tubes into a new 1.5 mL tube.
- Centrifuge the combined sample at high speed for 5 min at 4 °C to remove any particulates.
- Transfer the supernatant to a fresh 1.5 mL tube.
- Aliquot the samples and store them at −80 °C for long-term preservation.

See Pre-Dilution Strategies in **Notes 16** and **22.**

3.16 Sputum

Is a thick type of mucus that is a thick type of mucus that is produced in the lungs. Sputum can be collected by either non-invasive or invasive methods. It can be used to investigate infectious and inflammatory airway disease such as bronchitis, pneumonia, tuberculosis, COPD and cystic fibrosis [26].

3.16.1 Recommendations for Sample Preparation

- Advise the person to drink plenty of water the night before collection to help loosen secretions and make it easier to cough up sputum. Instruct them to avoid drinking, eating, chewing gum, using nicotine, or dental hygiene products for 1 h before sample collection.
- To ensure consistency, collect samples at a standard time, preferably early in the morning when sample quality is highest, to avoid diurnal shifts in mucus production. Normalize the samples based on total protein concentration, ideally aiming for 0.5 mg/mL.
- The mouth should be free of foreign matter. If dentures are worn then they should be removed.
- Instruct the person to rinse their mouth with tap or sterile water and spit it out. Then, have them open a pre-weighed 50 mL conical tube, hold it close to their mouth, and deposit the mucus into the tube. Keep the sample on ice and calculate the sputum weight afterward.
- Add protease inhibitor cocktail to a 1× final concentration.
- Incubate for 15 min while vortexing to break up the mucus.

- Transfer 1 mL to a microcentrifuge tube and centrifuge at high speed for 5 min at 4 °C.
- Aliquot the supernatant and store at −80 °C.

See Pre-Dilution Strategies in **Notes 17** and **22.**

3.17 Synovial Fluid

Synovial fluid is a thick liquid located between the joints that acts as a shock absorber and helps to alleviate friction. Proteomic analysis of synovial fluid can aid in the investigation of certain diseases such as arthritis, particularly septic or crystal-inducing arthritis [27].

3.17.1 Recommendations for Sample Preparation

- Synovial fluid should be collected using best practice clinical guidelines.
- Freshly collected samples are stable for a short duration at room temperature but should be stored on ice or at 4 °C if possible.
- Samples should be centrifuged for 10 min at ≥2000× *g* to remove cells and insoluble material.
- If necessary, samples can be diluted 1:4 with Olink diluent to reduce viscosity. All samples should be treated equally regardless of viscosity.
- It is not recommended to include samples which appear to have blood contamination.
- Aliquots should be stored at −80 °C.

See Pre-Dilution Strategies in **Notes 18** and **22.**

3.18 Tears

Tears primarily consist of water but also contain salt, fatty oils, and over 1500 unique proteins. Tear samples can be collected by Schirmer strips, like a standard technique used in ophthalmology clinics and is a type of filter paper which is placed partially under the eyelid within the lower cul-de-sac [28].

3.18.1 Recommendations for Sample Preparation

- Perform the tear sampling with Schirmer strips according to the manufacturer's instructions.
- Using sterile nitrile gloves, roll the Schirmer strip up and transfer to a Spin-X centrifuge tube with sterile tweezers. Push to the bottom. Proceed immediately to processing.
- Prepare stock of elution buffer in 15 mL conical.
- Add 300 μL of elution buffer to the Spin-X centrifuge tube with Schirmer strip.
- Incubate samples at room temperature for 10 min.
- Centrifuge tubes at 16000× *g* for 10 min at room temperature.
- Aliquot eluates into 0.5 mL or 1.5 mL LoBind tubes and store at −80 °C.

See Pre-Dilution Strategies in **Notes 19** and **22.**

3.19 Tissue Samples

Are fragments of biological material obtained from living or deceased organisms, widely used in scientific, medical, and diagnostic research. These samples can originate from various tissue types, such as skin, muscle, liver, or tumors, and play a key role in studies spanning molecular biology to personalized medicine. They enable the analysis of DNA, RNA, proteins, and other biomolecules. Proper collection, handling, and storage are essential to maintain data integrity, with common preservation methods including freezing or formalin fixation for future analysis [29].

3.19.1 Recommendations for Sample Preparation

- Tissue samples are generally more challenging to lyse than in vitro cell cultures, requiring additional mechanical methods such as homogenizers, bead mills, or sonicators.
- Always follow the manufacturer's instructions for using specific instruments and commercial lysis buffers.
- Keep the lysis buffer cold, conduct the lysis process on ice, and perform centrifugation at 4 °C.
- The tissue-to-lysis buffer ratio should typically range from 1: 10 to 1:50 g/mL.
- Start with a higher tissue-to-buffer ratio to maintain sufficient protein concentration and avoid excessive dilution.
- Monitor the samples during mechanical lysis to ensure they are breaking down properly. You may need to increase the time or intensity of homogenization, blending, or sonication, or add more lysis buffer if necessary.
- Once lysis is complete, centrifuge the samples at high speed and at 4 °C to remove debris.
- Determine the protein concentration of the clarified lysates using standard techniques like BCA, Bradford, Lowry, or Nanodrop assays.
- Normalize the samples to a consistent protein concentration, such as 0.5 mg/mL, using the same lysis buffer.
- Store aliquots of the lysed samples at −80 °C or lower for long-term preservation.

See Pre-Dilution Strategies in **Notes 20** and **22.**

3.20 Urine

Urine is a widely used biofluid in biomedical research due to its accessibility, non-invasive collection, and ability to reflect the physiological and pathological state of the body. It contains a broad range of compounds, such as metabolites, electrolytes, hormones, proteins, and nucleic acids, making it a key resource for studies in metabolomics, proteomics, and molecular diagnostics. However, due to its variable contents and high levels of salts and other metabolites, urine can potentially interfere with Olink's PEA technology. Therefore, the standard procedure for using urine in Olink assays requires diluting samples at least 1:4 with Olink Diluent in all panels, except for the Target 96 Cardiometabolic panel, which requires a minimum dilution of 1:40. Despite these considerations, urine's ease of storage and processing makes it ideal for longitudinal and population studies, proving useful in the investigation of kidney diseases, infections, metabolic disorders, and biomarker monitoring across various pathologies [30].

3.20.1 Recommendations for Sample Preparation

- Collect first-morning urine (more concentrated and consistent). An alternative approach would be to take aliquots from a combined 24 h collection.
- Fresh urine samples should be stored at 4 °C until processing.
- Roche cOmplete™ Mini Protease Inhibitor Cocktail (#11836153001) is highly recommended.
- Aliquot urine samples and store them at −80 °C for long-term preservation.
- Just prior to running Olink, samples should be thawed to room temperature, vortexed and centrifuged for 10 min at ≥500× *g* to remove insoluble material. The supernatant is retained for the Incubation step.

See Pre-Dilution Strategies in **Notes 21** and **22.**

4 Notes

1. To evaluate protein assays at risk for hook, it is recommended to run a few samples from each study group at two additional dilutions. It is not necessary to include biological replicates or to add protease inhibitors. Technical replicates can be included for better estimation of CVs when using an alternative matrix.
2. Pre-dilutions for amniotic fluid (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

CAM cardiometabolic panel, *CRE* cell regulation, *CVDII* and *CVDIII* cardiovascular, *DEV* development, *IMO* immune-oncology, *INF* inflammation, *IRE* immune response, *MET* metabolism, *NEU* neurology, *NEX* neuro exploratory, *ODA* organ damage, *ONCII* and *ONCIII* oncology

3. Pre-dilutions for blister fluid (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII

(continued)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:400	1:4	1:4	1:40	1:40	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4

4. Pre-dilutions for bone marrow (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

5. Pre-dilutions for cell culture lysates (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

6. Pre-dilutions for cerebrospinal fluid (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

7. Pre-dilutions for conditioned media (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

8. Pre-dilutions for dried blood spots (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

9. Pre-dilutions for extracellular vesicles (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

10. Pre-dilutions for interstitial fluid (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

11. Pre-dilutions for lavage samples (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

12. Pre-dilutions for nasal mucosa (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

13. Pre-dilutions for ocular fluids (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

14. Pre-dilutions for peritoneal fluid (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

15. Pre-dilutions for saliva (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

16. Pre-dilutions for skin tape strips (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

17. Pre-dilutions for sputum (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

18. Pre-dilutions for synovial fluid (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4

19. Pre-dilutions for tears (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

20. Pre-dilutions for tissue samples (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:100	1:1	1:1	1:10	1:10	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1

21. Pre-dilutions for urine (target 96 panels)

CAM	CRE	CVDII	CVDIII	DEV	IMO	INF	IRE	MET	NEU	NEX	ODA	ONCII	ONCIII
1:40	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4	1:4

22. Pre-dilution strategies for target 48 panels

Amniotic fluid	**1:1**
Blister fluid	**1:4**
Bone marrow	**1:1**
Cell culture lysates	**1:1**
Cerebrospinal fluid	**1:1**
Conditioned media sample	**1:1**
Dried blood spots	**1:1**
Extracellular vesicles	**1:1**
Interstitial fluid	**1:1**
Lavage sample	**1:1**
Nasal mucosa	**1:1**
Ocular fluid	**1:1**
Saliva	**1:1**
Skin tape strip	**1:1**
Sputum	**1:1**
Synovial fluid	**1:4**
Tears	**1:1**
Tissue samples	**1:1**
Urine	**1:1**

Acknowledgments

CPS and NB were supported by Ramón y Cajal Program of MCIN (RyC-2021-033828-I and RyC-2017-23437) and ALU by the Industrial PhD Program of AEI (DIN2022-012766).

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Chapter 15

Cytokine Antibody Microarray-Based Proteomic Strategies for Characterizing the Dysregulation of Cytokines Disease-Specific

Rodrigo Barderas, Pablo San Segundo-Acosta, and Ana Montero-Calle

Abstract

The understanding of disease mechanisms and the search for potential therapeutic targets have driven the focus of proteomics studies towards the identification of specific protein profiles and molecular pathways disease associated. Protein microarrays offer high-throughput alternatives to mass spectrometry, with peptide, phage, and recombinant protein microarrays providing different options for studying disease-specific alterations. Commercially available antibody microarrays, particularly cytokine arrays, enable the simultaneous analysis of multiple cytokine expression levels, offering valuable insights into disease processes, drug efficacy, and biomarker discovery. Indeed, cytokine arrays outperform mass spectrometry methods due to their ability to analyze cytokines at an extremely low concentration in parallel in a high-throughput manner. Here, we outline protocols for cytokine analysis using cytokine antibody microarrays, detailing sandwich approaches and direct or indirect incubation methods for the characterization of disease-associated cytokine profiles.

Key words Cytokines, Screening, Cytokine array, Antibody microarray, Proteomics immunomodulators, Growth factors

1 Introduction

For the identification of protein profiles, specific targets, and molecular pathways to understand the clinical behavior of chronic diseases like cancer or neurodegenerative disorders [1–19], and other less prevalent conditions [20–23], efforts have predominantly focused on employing high-throughput molecular biology methods. The notion that the specific dysregulation of proteins, protein complexes, cytokines, immunomodulators, growth factors, and alterations in signature networks and pathways are associated with unique diseases is driving proteomics forward. Indeed, defining specific protein expression patterns that reflect the disease

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
https://doi.org/10.1007/978-1-0716-4595-6_15,

would enable early diagnosis and better prognosis, offering personalized medicine modalities and paving the way for the identification of new intervention targets and improved patient management.

Although mass-spectrometry-based proteomics approaches have been widely used in the last years for the identification of disease associated biomarkers, another potential solution for identifying such protein profiles lies in the use of protein microarrays [24–29]. Protein microarrays are high-throughput methods used to track protein interactions, protein activities, and to determine their function on a large scale. Their main advantage lie in the fact that large numbers of proteins can be tracked in parallel. At the time they were developed and implemented, they replaced other traditional proteomics techniques such as 2D-gel electrophoresis, which is mostly obsolete nowadays, or chromatography, which were time-consuming, labor-intensive, required specific equipment, and ill-suited for the analysis of low abundant proteins. Nowadays, protein microarray-based proteomics represents a valuable tool for enhancing our understanding of physiological and pathological mechanisms, for identifying both dysregulated protein profiles and abnormalities, and for identifying and characterizing humoral immune responses associated with diseases. Since protein microarrays derived information is targeted, where the proteins to be surveyed are known a priori, in contrast to most methodologies implemented with mass spectrometry based proteomics, protein microarray-derived data complement the data obtained by mass-spectrometry and both approaches can be performed at a time to survey specific disease conditions [16].

Among the different protein microarray approaches, peptide microarrays and phage microarrays offer alternatives to study disease-specific alterations at the protein level [11, 16, 20, 29–31]. The latter are printed with specific peptides and proteins—produced in the pathological tissue under study due to specific mRNA alterations— displayed on the surface of M13 or T7 phages. This allows for identifying protein alterations related to the disease of study, such as aberrant peptides or proteins, alternative splicings, frameshifts, or point mutations, which would otherwise be missed by screening protein microarrays printed with wild type natural proteins [11, 20, 30, 32]. The multiplexing capacities of phage microarrays allow for simultaneously surveying several thousands of probes using small amounts of sample.

However, despite these interesting approaches, the most widely used protein microarrays are those commercially available, which contain recombinant proteins or antibodies at a high density [29]. In this sense, there are now available in the market protein microarrays containing recombinant proteins -HuProt- [33, 34], or unique fragments of human proteins up to 150 amino acids in length (Protein-epitope signature tag -PrESTs-) [16, 35, 36], covering almost the entire human proteome (considering one protein per gene), or antibody microarrays containing from several hundreds to thousands of antibodies, which are of particular interest to survey for changes in cytokines, immunomodulators, and growth factors [7, 29, 31, 37].

Cytokines are proteins secreted in response to specific biological stimuli, which possess an important role in angiogenesis, inflammation, cell growth, and cell differentiation. In addition, cytokines encompass a wide range of sizes and structures, and are typically present at an extremely low concentrations in biological samples, making their detection and quantification challenging. Because these proteins play a key role in several diseases, their identification is an interesting tool for the elucidation of the signaling pathways associated to a disease and for biomedical discovery, which in most cases requires the simultaneous detection of numerous cytokines. Although mass spectrometry methods are powerful for the in-depth identification and quantification of complex proteomes, cytokine antibody microarrays are particularly useful over mass spectrometry methods for the simultaneous determination of multiple cytokine-expression levels in parallel [7, 38, 39]. This is particularly advantageous because multiple cytokines often share the same upstream signal pathway, and one cytokine can regulate other cytokine-expression levels [38, 39]. Therefore, profiling cytokine expression in certain types of diseases may help identifying potential molecular targets involved in disease processes. Moreover, by profiling cytokine expression in response to drugs, the efficacy of medicines can be monitored [38, 39]. Additionally, new biomarkers may be discovered as a result of profiling cytokine-expression levels in patient specimens [4, 7].

In this context, we here report the protocols needed for the analysis of cytokines by means of antibody microarrays either in sandwich approaches using glass or membrane antibody microarrays without the need of labelling, or by the direct incubation of protein samples labeled with biotin or a fluorophore. The screening, data analysis, and validation methods are depicted in the chapter for the characterization of disease-associated cytokine profiles.

2 Materials

This section summarizes all common materials and reagents needed along the protocol for the screening and analysis of cytokine antibody microarrays followed by the specific materials and solutions used for the specific procedures in subsequent indicated subheadings.

2.1 Reagents

1. Phosphate-Buffered Saline pH 7.4: PBS. 137 mM NaCl, 2.7 mM KCl, 8 mM Na_2PO_4, and 1.5 mM KH_2PO_4.
2. Washing Solution: 0.1% PBST. 999 mL PBS containing 1 mL Tween-20.
3. 1% Bovine Serum Albumin (BSA): 1 g BSA and 100 mL PBS.
4. 10% BSA: 10 g BSA and 100 mL PBS.
5. 1,4-Dithiothreitol: DTT (*see* **Note 1**).
6. MilliQ water.
7. Double-distilled water: ddH_2O.
8. Commercial high-density cytokine microarrays.
9. Tissue, plasma, serum samples, or any biological fluid (*see* **Note 2**).
10. Cell pellets or conditioned media (secretomes) (*see* **Note 2**).
11. Probe buffer: 1% BSA, 0.05% Triton X-100, 5% Glycerol, 5 mM $MgCl_2$, 0.5 mM DTT in PBS; 10 g. BSA, 500 μL Triton X-100, 50 mL glycerol 100%, 1.02 g $MgCl_2$, 0.078 g DTT, and 950 mL PBS (*see* **Note 1**).
12. Blocking solution: 1% BSA, 0.1% Tween-20 in PBS; 10 g BSA, 1 mL Tween-20, and 999 mL PBS (*see* **Note 1**).
13. For sandwich microarrays, biotinylated or fluorescence labeled (Bodipy, Cyanine, or Alexa Fluor dyes) antibodies cocktail is needed (*see* **Note 3**).

2.2 Equipment

1. Heating and drying oven.
2. 1.7 mL Eppendorf tubes.
3. 15 mL Falcon tubes.
4. 50 mL Falcon tubes.
5. QuadriPERM cell Culture Vessel.
6. Orbital shaker/oscillating rocker/roller.
7. 0.22 μm pore filters.
8. Slide hybridization polycarbonate chamber.
9. Falcon centrifuge 5810 R.
10. Microtube centrifuge.
11. Nanodrop 2000C.
12. Microarray scanner: GenePix 4000B or ImageQuant 800.
13. Microarray analysis software: GenePix Pro 7.1, or ImageJ.
14. Bioinformatic tools (Pomelo II, MultiExperiment Viewer, or Babelomics Suite, among others).

2.3 Cytokine Antibody Microarrays

High-density commercial Cytokine Antibody Microarrays (i.e., PEPperCHIP Human Cytokine Microarray or the Human L8000 Array (RayBiotech)) allowed for the unbiased detection through direct biotin or fluorescence labeling of samples or through the analysis by means of sandwich assays using a cocktail of biotin- or fluorescence-labeled antibodies of cytokines, immunomodulators, or growth factors.

Specific antibodies are printed either on nitrocellulose membranes or glass slides along with positive and negative control proteins as IgG or BSA. Each array is composed of several subarrays with about 100 μm diameter spots, being each probe printed in duplicate onto the slide. Customized cytokine antibody microarrays with specific panels of antibodies against custom-dependent cytokines, growth factors, or immunomodulators are also available attending to researchers' needs and provided from different biotech's (i.e., RayBiotech).

2.4 Protein Samples

Protein samples to be used for the identification of differentially expressed cytokines could be: (i) serum or plasma samples (i.e., detection of differentially expressed cytokines in response to a specific disease), (ii) any biological fluid different from serum or plasma (i.e., aqueous humors, nipple aspirate, urine, …), (iii) cell or tissue protein extracts (i.e., pathological and healthy tissues to identify disease-specific cytokines), or (iv) conditioned medium from cell cultures (i.e., cancer cells with different metastatic abilities to identify cytokines related with metastasis).

If neither primary nor secondary antibodies are used, protein samples must be labeled, either with biotin (indirect assay) or with fluorophores (direct assay).

2.4.1 Protein Extraction

1. 4 mM EDTA PBS.
2. Lysis buffer: RIPA. 150 mM NaCl, 1.0% IGEPAL CA-630, 0.5% sodium deoxycholate, 0.1% SDS, 50 mM Tris–HCl, pH 8.0 (Sigma-Aldrich).
3. Protease inhibitor cocktail (MedChemExpress).
4. Phosphatase inhibitor cocktail (MedChemExpress).
5. Cell or tissue lysis buffer: RIPA buffer supplemented with protease inhibitor and phosphatase inhibitor cocktails (MedChemExpress).
6. 16-gauge and 18-gauge needle syringes or TissueLyser.
7. Precast SDS-PAGE gels (Invitrogen, BioRad, Merck …).
8. Coomassie brilliant blue R-250 (*see* **Note 2**).

2.4.2 Biotin Labeling

1. Sterile-filtered PBS.
2. MilliQ water.

3. Labeling buffer: PBS, DMSO, EZ-Link NHS-Biotin (Thermo Fisher Scientific).
4. Dimethyl sulfoxide: DMSO (Merck).
5. Stop solution: 1 M Tris pH 8.0 (Thermo Fisher Scientific).
6. Floating dialysis rack.
7. Fluorescence (Bodipy, Cyanine, or Alexa Fluor dyes) labeled streptavidin (RayBiotech) or HRP-conjugated streptavidin (Thermo Fisher Scientific) (*see* **Note 3**).

2.4.3 Fluorophore Labeling

1. Fluorophores NHS ester (GE Healthcare).
2. Extraction/labeling solution (E0655-30ML, Sigma-Aldrich).
3. Stop solution: Tris 1 M pH 8.0 (Thermo Fisher Scientific).
4. Labeling Buffer: PBS pH 8, DMSO, EZ-Link NHS-Biotin (Thermo Fisher Scientific).
5. SigmaSpin column (Ref. S0185, Sigma-Aldrich).
6. NanoDrop System (ThermoFisher Scientific).

2.5 Detection

The detection of antibody-antigen (cytokine) interactions might be performed by using fluorophores or HRP-labeled reagents (Fig. 1). If the cytokines of interest have been labeled with a fluorophore, such as Bodipy, Cyanine, or Alexa Fluor dyes, additional reagents or any steps prior to be probed in the arrays are not necessary, since the fluorescence signal is directly detected after the scanning of the cytokine antibody microarrays. However, when the cytokines of interest are unlabeled, tagged, or untagged, a secondary incubation with fluorescence- or biotinylated-primary antibodies is necessary to detect their protein levels. In addition, when biotinylated samples or biotinylated antibodies are used, a subsequent incubation with fluorescence labeled (Bodipy, Cyanine, or Alexa Fluor dyes) or HRP-conjugated streptavidin is necessary prior to scanning.

3 Methods

Although cytokine antibody microarrays allow for the analysis in parallel of thousands of cytokines, immunomodulators, and growth factors in multiple samples, the workflow to use them using labeled samples or in sandwich assays present some important differences.

Cytokine antibody microarrays are a version of high-density antibody microarrays specifically focused on the analysis of cytokines, immunomodulators, and growth factors in parallel. In the realm of cytokine antibody microarrays, the parallel analysis of multiple samples containing really scarce proteins and volumes is currently being utilized to detect changes in the abundance of hundreds to thousands of these proteins. This approach aids in

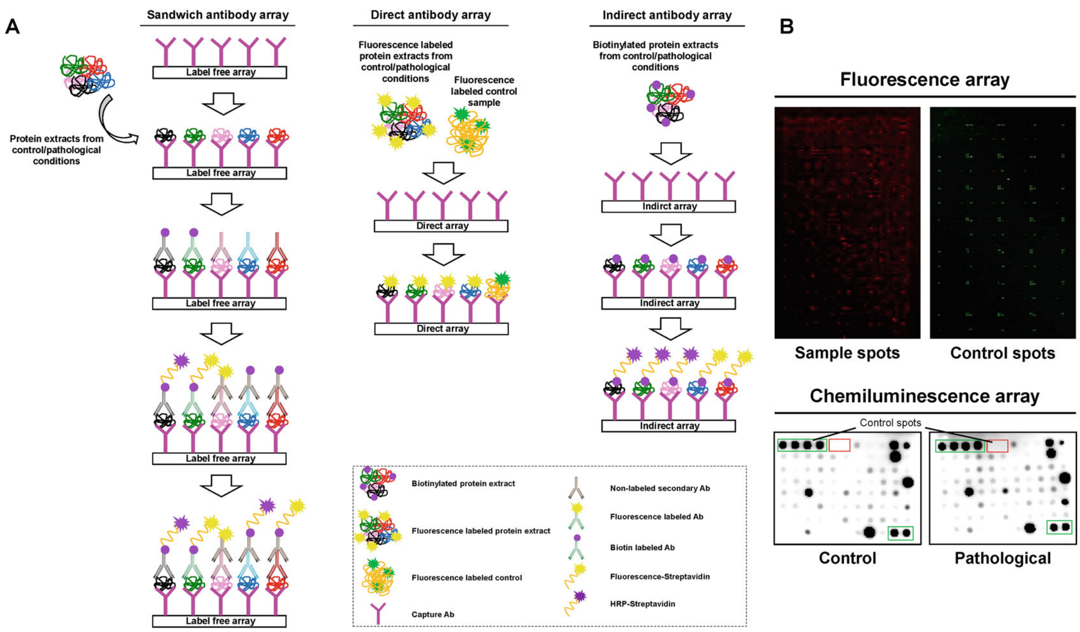

Fig. 1 Label and label-free cytokine antibody microarray approaches for characterizing the dysregulation of disease-specific cytokines. **a** Schematic representation of the three existing different cytokine antibody microarray approaches are depicted. On the left, a representative label-free sandwich antibody microarray using control and disease protein samples. Signal development occurs with HRP- or fluorescence-labeled antibodies. On the middle, a representative antibody cytokine microarray depicts a direct assay using protein samples (control or pathological) labeled with different fluorescence dyes. Direct labeling does not require any additional incubation for signal development. On the right, another representative cytokine microarray shows an indirect assay with one sample labeled with biotin per subarray. Indirect labeling involves sample biotinylation followed by incubation with the microarray, with signal development facilitated by HRP- or dye-labeled streptavidin. **b** Representative images of cytokine antibodies microarrays revealed with fluorescence (top) or chemiluminescence (bottom)

the discovery of potential biomarker candidates in biological specimens. The methodology involves protein extraction from cells or tissues and protein quantification of these samples, or those from the biological fluid of interest to be analyzed. Subsequent surveys for protein alterations are conducted through capture approaches using labeled samples or sandwich assays using label-free approaches (Fig. 1a). Commercially available cytokine antibody microarrays for capture or sandwich analysis are now readily available (*see* **Note 1**). Any of these analyses, using alternatively labeled samples or sandwich assays, might potentially lead to the identification of biomarker candidates.

3.1 Protein Extraction and Quantification

For the analysis of protein alterations in biological fluids or secretome samples, it is only necessary to determine the protein concentration (*see* **Note 2**). However, for determining protein changes in cells or tissues, a prior step of protein extraction is required.

3.1.1 Cells

A different methodology is needed depending on whether the cells are adherent or not in cell culture.

1. To extract proteins from adherent cultured cells, 1 mL of RIPA buffer supplemented with protease and phosphatase inhibitor cocktails is used to scrape off the cells. For cells growing in suspension, the same buffer is used to resuspend the pelleted cells.
2. The cells are then mechanically disaggregated using 16- and 18-gauge syringes and centrifuged for 10 min at 4 °C and 10,000 g for clarification.
3. The supernatant, which contains the protein extract, is then transferred to a new tube, and stored at −80 °C until use (*see* **Note 2**).

3.1.2 Tissue

For protein extraction from tissue samples, the samples are individually cut into small pieces on dry ice using a scalpel and lysed in 1 mL of tissue lysis buffer (RIPA buffer supplemented with protease inhibitor and phosphatase inhibitor cocktails). The tissues are then mechanically disaggregated using a TissueLyser, and the supernatant, which contains the tissue protein extract, is obtained as described above (*see* **Note 2**).

3.2 Labeling of Protein Samples for Screening of Antibody Microarrays: Capture Assays

For labeled approaches to determine cytokine alterations through capture assays, either indirect assays using biotin-labeled samples or direct assays using samples labeled with a compatible pair of fluorophores can be performed (Fig. 2).

3.2.1 Indirect Assays Using Biotin Labeled Samples

Before biotinylating the protein samples, it is essential to dialyze them to remove any residual amine and/or azide, as they can interfere with the biotinylation process (*see* **Notes 4** and **5**).

1. To biotinylate the protein samples, add 100–200 μL of protein samples at a concentration of 0.1–0.2 mg/mL into separate dialysis tubes. Place the tubes into a floating dialysis rack inside a beaker containing approximately 1000 times the volume of PBS at pH 8 compared to the sample volume. Ensure the buffer is stirring and dialyze the samples three times for 2 h each at 4 °C, with the exception of the third dialysis, which can be left overnight (O/N) (*see* **Note 4**).
2. Transfer the dialyzed samples into microfuge tubes and centrifuge them for 5 min at 1000 rpm. Collect the supernatant and store at −20 °C.
3. Just before labeling, prepare the labeling reagent by adding 100 μL of sterile filtered PBS to the vial containing the label

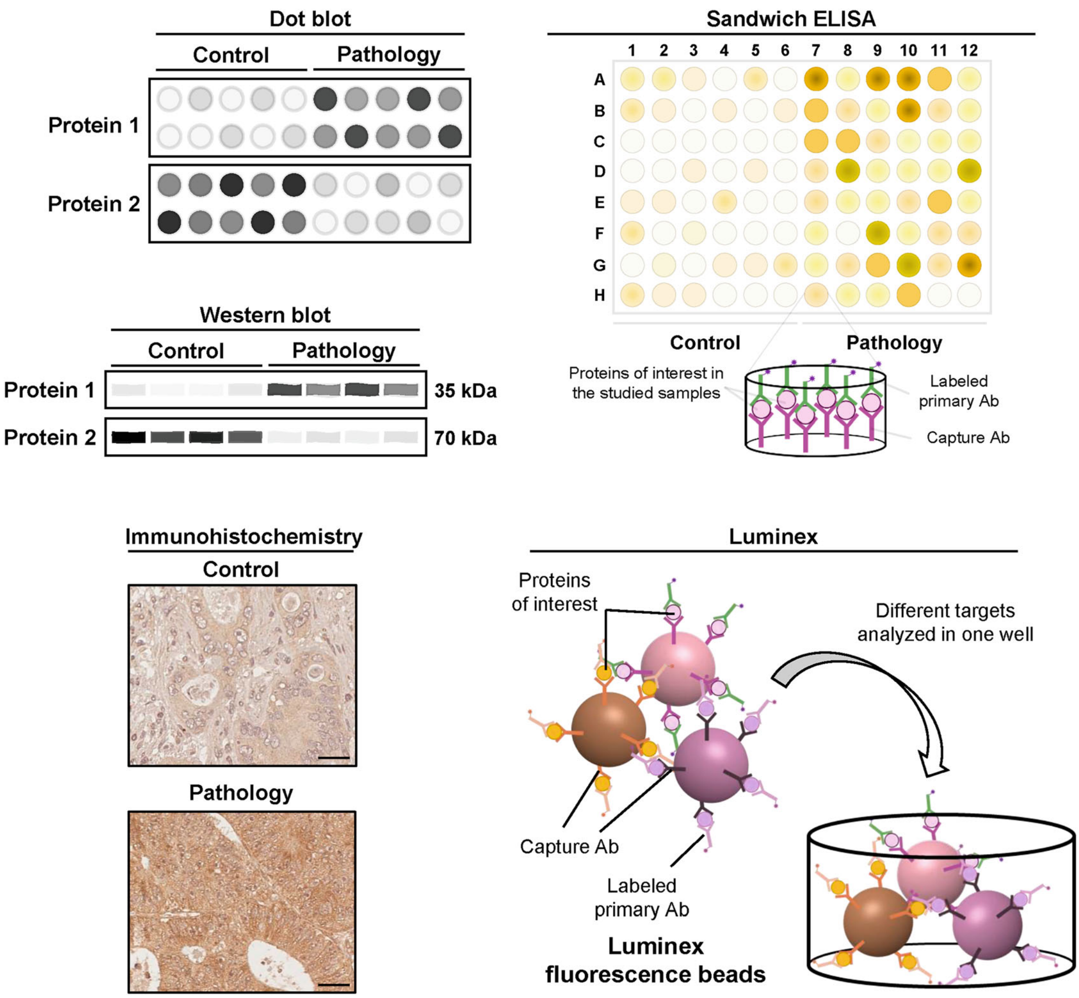

Fig. 2 Strategies for validation of antibody microarrays results. Validation is crucial due to potential false-positive protein identifications. At least three to four orthogonal validation approaches are recommended: dot blot, western blot (WB), immunohistochemistry (IHC), enzyme-linked immunosorbent assay (ELISA), and/or Luminex. Ab, antibody

reagent resuspended in DMSO (e.g., EZ-Link NHS-Biotin). Pipette up and down to dissolve the reagent.

4. For sample preparation, mix 30 μg of each protein sample with the appropriate volume of labeling reagent to achieve a ≥20-fold molar excess of biotin in a final volume of 300 μL, and shake the mixture for 30 min.
5. Stop the reaction by adding 3 μL of stop solution to the mixture.
6. Dialyze the samples as previously described. Then, centrifuge the samples for 5 min at 10,000 rpm to remove any floating particles and precipitates. Record the final volume and measure

the new protein concentration for subsequent use in the cytokine microarray hybridization (*see* **Note 4**). Confirm the quality of the biotin labeling by WB using an HRP- or phopho-streptavidin antibody (*see* **Notes 5** and **6**).

3.2.2 Direct Assays Using Labeled Samples with Fluorophores

For fluorophore labeling of samples, it would be recommended to use either Bodipy, Cyanine, or AlexaFluor dyes.

1. To prepare the labeling reagent, add 50 μL of labeling solution that does not contain azide or primary amine groups to the lyophilized fluorophore.
2. Add 6.25 μL of the reagent to each sample tube (20 μL of fluorophore for every 400 μg of protein). Incubate the labeling reaction for 30 min at 600 rpm. Then, add 50 μL of labeling reagent (*see* **Note 7**).
3. Prepare one SigmaSpin column to remove unbound dye for each labeling reaction.
4. Centrifuge for 2 min at 4 °C and 750 g to remove excess resin. Then, add the total volume of each labeling reaction to the center of independent SigmaSpin columns and centrifuge for 4 min at 4 °C and 750 g. Collect the eluate and store 130 μL at −20 °C and the remainder at 4 °C.
5. Calculate the dye/protein ratio and protein concentration using the eluate at 4 °C (*see* **Note 8**) and, in case of evaluating a phosphoactivated protein, confirm the quality of the labeling by WB using an anti-phospho residues (Ser, Thr, and/or Tyr) antibody (*see* **Notes 9** and **10**).

Alternatively, protein samples containing known proteins (i.e. BSA, IgG, etc.) can be fluorescence labeled as a positive control and printed onto the arrays. In this case, a Cy3/Cy5 or Alexa Fluor 555/Alexa Fluor 647 dye pair should be used for the labeling of control and problem protein samples, respectively.

3.2.3 Cytokine Microarray Hybridization Using Alternatively Biotin or Fluorophore Labeled Samples

Proteomics studies using individual samples enable the identification of specifically dysregulated cytokines, immunomodulators, and growth factors associated to any particular disease. Conversely, the use of pooled samples allows for the identification of dysregulated cytokines with common alterations across the tested samples, thus mitigating concerns regarding intra-individual variability. In both scenarios, it is advisable to hybridize cytokine microarrays using samples from individuals within a similar age range and to maintain a consistent female-to-male ratio.

The protocol involves the incubation of arrays coated with antibodies against the protein of interest with label-free protein samples followed by incubation with a cocktail of biotinylated or phosphorylated antibodies against the proteins printed onto the array. Alternatively, direct or indirect arrays can be performed by incubating antibody arrays with biotin or fluorescence labeled protein samples, respectively (Fig. 2).

Before microarray hybridization, it is important to incubate them at 4 °C for 15 min followed by an additional 15 min at room temperature (RT) to acclimate the microarrays to the working temperature. Subsequently, it is recommended to equilibrate the microarrays in the blocking buffer used for diluting the samples for 1 min at 30 rpm in the dark (*see* **Notes 6** and **10**).

For cytokine microarray hybridization with biotinylated samples involving indirect assays the protocol requires the next steps:

1. Incubate each microarray with 2–4 mL of blocking buffer in a quadriPERM Cell Culture Vessel for 1 h at RT with gentle shaking. Ensure no bubbles are present on the microarray surface (*see* **Note 6**).
2. After incubation, remove the blocking buffer and place the microarrays O/N at 4 °C with gentle shaking in a hybridization chamber containing 1.6 mL of each protein sample at a concentration of 5 μg/mL in blocking buffer (*see* **Note 9**).
3. Following incubation, remove the samples and wash the microarray six times (5 min per wash) with 4 mL of washing buffer while gently rocking in a quadriPERM Cell Culture Vessel.
4. Discard the washing buffer and incubate each microarray with a 1:1000 dilution of HRP or fluorescence-streptavidin in blocking buffer for 2 h at RT in the dark.
5. After incubation, remove the labeled-streptavidin solution and wash each array six times with washing buffer for 5 min per wash while gently rocking.
6. Subsequently, wash each array twice with PBS, followed by a wash with MilliQ water for 5 min each with gentle rocking.
7. If microarrays were incubated with HRP-streptavidin, immediately proceed to incubate them with the ECL reagent for revealing.
8. If microarrays were incubated with fluorescence-streptavidin, centrifuge each microarray at 1200 rpm for 10 min to dry them. Store them in the dark and proceed to scan them the next day to remove any interfering humidity during the scanning of the microarrays.

Alternatively, for cytokine microarray hybridization with fluorophore labeled samples involving direct assays the protocol requires the following steps:

1. Block the microarrays with blocking solution for 1 h at RT at 30 rpm in the dark.

2. Remove the PBS and add 5 mL of the blocking solution containing 20 μg of the fluorescence labeled sample or 20 μg of the fluorescence positive samples if required (*see* **Notes 11** and **12**).
3. Wash the arrays six times for 5 min each with washing buffer, followed by a wash in PBS containing no detergent for 5 min with gentle rocking. Submerge the microarrays in MilliQ water for 1 min.
4. Dry the microarrays by centrifugation at 1200 rpm for 10 min (*see* **Note 13**). Store in the dark and scan within the next 48 h.

3.3 Label Free Antibody Microarrays: Sandwich Assays

Whether utilizing commercial, custom, or homemade nitrocellulose membrane- or glass-based cytokine microarrays for sandwich assays, it is crucial to prepare all reagents immediately before starting the protocol.

Prior to commencing, it's recommended to equilibrate the microarray at RT and then in wash buffer for 1 min.

1. Add 2 mL of blocking buffer into each well and incubate at RT for 2 h in a quadriPERM Cell Culture Vessel.
2. After 2 h, aspirate the blocking buffer and add 1.6 mL of the protein sample in a hybridization chamber, then incubate O/N at 4 °C (*see* **Note 14**).
3. Carefully remove the samples from each well and wash the wells with 2 mL of wash buffer. Incubate for 5 min at RT. Repeat the washing step six times.
4. Remove the wash buffer from each well and incubate with 1.6 mL of either biotinylated or non-biotinylated antibody cocktail for 2 h at RT in a hybridization chamber.
5. After incubation, remove the antibody from each well and wash six times as described above.
6. Pipette 2 mL of HRP-Streptavidin into each well, or an HRP-labeled anti-mouse IgG antibody, or the same components conjugated to a desired fluorophore, depending on the antibody used in the sandwich assay. Incubate for 2 h at RT with gentle shaking in a quadriPERM Cell Culture Vessel.
7. Repeat the washing steps as described above (*see* **step 5**).
8. Remove any excess of wash buffer, prepare the detection buffer for each microarray to cover the slide completely, and incubate for 2 min at RT in the dark.
9. Transfer the membrane- or glass-based microarray to an appropriate scanner system (*see* **Note 15**).

3.4 Scanning and Data Analysis

For fluorescence microarray scanning, a microarray scanner equipped with at least two lasers (532 nm and 635 nm solid-state lasers) compatible with all microarray surfaces (i.e., glass, plastic, hydrogels, …, besides nitrocellulose microarrays), is required, such as the GenePix 4000B (Axon). The 635 nm laser detects signals from Cy5, Alexa Fluor 647, or similar dyes (red channel), while the 532 nm laser detects signals from Cy3, Alexa Fluor 555, or similar dyes (green channel). We recommend scanning the arrays at a minimum of two different laser intensities, such as 100% and 10%, to enhance the analysis of low and high reactive spots, respectively. For chemiluminescence microarrays, a chemiluminescence image analyzer is required, such as the Amersham ImageQuant 800 (Cytiva). We recommend scanning the arrays at different exposure times in a cumulative exposition mode to enhance the analysis of low and high reactive spots. Scan times longer than 30 min are not recommended due to signal lost in time.

The obtained images are quantified using microarray acquisition software like GenePix Pro 7, which allows intra-array normalization using the background signal around each spot. Subsequently, the data for each spot is inter-array normalized and processed using desired statistical tests with specific bioinformatics tools. Examples include free web-based programs like T-Rex from the GEPAS web-tool database included in the babelomics suite for functional profiling analysis, Pomelo II, or MultiExperiment Viewer [40–42], which allow performing both normalization and processing of the microarray data. Additionally, commercially available protein microarrays have developed their own bioinformatic tools, such as the ProtoArray Prospector Software [43], facilitating the identification of statistically significant up-regulated and down-regulated proteins associated with the disease.

3.5 Validation

Given the potential for the identification of false-positive identifications in proteomics (as occurs in other omics approaches), high-throughput cytokine antibody microarray technologies must be validated using alternative orthogonal approaches to ensure result accuracy. A common challenge in microarray approaches is the lack of validation of datasets, leading to potential misidentifications and hindering the translation of results into clinical applications.

Hence, it is imperative to further validate and functionally analyze data obtained from protein microarrays using alternative techniques, such as dot blot, Western blot (WB), enzyme-linked immunosorbent assay (ELISA), and/or Luminex. However, these approaches may be constrained by the availability and cost of antibodies recognizing the cytokines, immunomodulators, or growth factors of interest or the need for tagging novel proteins. Another validation method involves targeted mass spectrometry for low- or extremely low-abundant cytokines, immunomodulators, or growth factors in biological fluids or complex protein extracts [44].

Validation assay acceptance criteria should consider the study objectives, methodological nature, and biological variability of the biomarker. Ultimately, these validation assays should yield high-quality validated bioanalytical data.

4 Notes

1. When using commercial antibody microarrays, either for capture or sandwich assays, the manufacturer typically provides the optimal solutions for effective sample screening. Whether you're using commercial, custom-made, or lab-made microarrays, it's crucial to test the optimal concentration to detect changes in the protein abundance being analyzed.
2. It is important to determine both the protein concentration and the quality of the obtained protein extracts. While protein concentration can be determined by various conventional methods, we recommend the Tryptophan method, which is widely used in proteomics analyses, or BCA quantification [7, 15, 45–47]. Assess the quality of the sample by Coomassie blue staining and WB after SDS-PAGE, using antibodies against tubulin, β-Actin, or GAPDH as loading controls. If the protein bands are not well defined in the Coomassie blue staining or if multiple bands appear in the WB analysis, protein degradation may have occurred. In this case, repeat the protocol before labeling and hybridizing the samples on the microarrays.
3. If biotin-labeled antibodies are used, HRP-conjugated or fluorescence labeled (Bodipy, Cyanine, or Alexa Fluor dyes) streptavidin is required. Alternatively, HRP-conjugated streptavidin can be used if the protein samples are labeled with biotin. In all cases, SuperSignal West Pico PLUS Chemiluminescent Substrate (Thermo Fisher Scientific) ECL Detection Buffer is required to develop the chemiluminescent signal.
4. Sample volumes may change after dialysis. If the protein samples were equally concentrated before dialysis, the volumes should be the same after it. However, it's always recommended to measure the collected volume and determine the protein concentration for subsequent steps.
5. To verify the integrity of the protein sample after dialysis, perform a Coomassie Blue staining and a WB analysis after SDS-PAGE separation using antibodies against tubulin, β-Actin, or GAPDH as loading controls with the new protein samples. Compare the signal to the one obtained from the samples before dialysis. To check for a correct labeling, perform a WB analysis using HRP-streptavidin or fluorescence-streptavidin for the detection of biotinylated proteins. If the

labeling was done correctly, many bands along the whole lane of the WB should be observed.

6. Do not block the membrane with skimmed milk for the biotinylated samples. Use BSA instead of milk as a blocking agent, as the presence of biotin in milk could decrease or abrogate the specific signal of biotin-labeled proteins.
7. When handling samples with any fluorophore (Bodipy, Cyanine, or Alexa Fluor), use aluminum foil during the experiments (labeling, screening, etc.) to keep the fluorophores away from light.
8. Obtain a spectrum of the labeled protein from 750 nm to 220 nm using a NanoDrop System to determine the dye-to-protein ratio. Absorbance at 280 nm will allow you to obtain the protein concentration, and absorbance at 555 nm and 635 nm the dye concentration.
9. To check for a correct fluorescence labeling of phosphoactivated proteins within the protein extracts, perform a WB analysis using an antibody against phosphorylated residues for the detection of phosphorylated proteins. If the labeling was done correctly, many bands along the whole lane of the WB should be observed.
10. Do not block the membrane with skimmed milk for the phosphorylated samples. Use BSA instead of milk as a blocking agent, as milk contains phosphoproteins that might interfere with the detection of phosphorylated proteins, which might lead to false positive results or mask the signal of the protein of interest.
11. If possible, titrate the optimal protein concentration to achieve the best results in the microarrays, which should be between 2 and 10 μg/mL.
12. If arrays are incubated with two differently labeled samples, we recommend co-incubating a control sample with a sample of the disease to be studied per array. Additionally, it's highly recommended to perform a dye-swapping experiment for at least two of the samples. This helps to determine the absence of bias due to labeling with different dyes, avoid specific antibody-antigen interactions that may occur due to labeling, and allows for data normalization of the microarrays.
13. It's recommended to further dry the microarrays with dry air prior to scanning. It's even suggested to scan them the day after hybridization to remove any potential humidity in the microarray that could interfere during scanning.
14. Optimal sample concentrations depend on the nature of the sample. Conditioned media from cell culture does not require any dilution. For serum or plasma, a five-fold dilution in

blocking buffer is recommended, while 250 μg of protein from cell or tissue lysates are the recommended amount of protein extract.

15. For fluorescence-based arrays, scan on a compatible array scanner. For chemiluminescence-based arrays, the exposure should be made within 5–10 min after adding the detection reagents, as the signals will fade over time.

Acknowledgments

This work was supported by the financial support of PI20CIII/00019 and PI23CIII/00027 grants from the AES-ISCIII program cofounded by FEDER funds to R.B, and the financial support of grant PID2022-140307OB-I00 funded by MCIN/AEI/10.13039/501100011033 and by "ERDF A way of making Europe." P.SS-A. was supported by a Sara Borrell Postdoctoral contract of the AESIRRHH 2021–2022 call.

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Chapter 16

Unravelling Immunogenic Cell Death Mechanisms by Functional Proteomics Arrays

Ana Nuño-Soriano, Carlota Arias-Hidalgo, Pablo Juanes-Velasco, Rafael Gongora, Angela-Patricia Hernández, and Manuel Fuentes

Abstract

Immunogenic Cell Death (ICD) is a type of cell death regulated by an activation of endoplasmic reticulum stress. This signal results in an activation of the adaptative immune response against the affected cell. This characteristic has been used for the design of novel oncological drugs to complement antitumor therapy with this immunogenic response. However, ICD is complex and still needs a deep and massive study to unravel the key to its mechanism. Protein microarrays are a useful tool for large-scale protein detection and analysis, and thus may be the suitable platform for this study. In this study, antibody arrays have been designed and constructed as a method to detect different proteins involved in programmed cell death pathways in order to evaluate differential intracellular signalling profiles and correlate them with ICD.

Key words Antibody microarrays, Functional proteomics, Cell death, Antibodies, Cancer

1 Introduction

Cell death is a crucial process in the diagnosis, treatment, and biomarker search for all types of diseases. It is a complex and multi-stage process and, especially in cancer treatment, cell death is a key step in the success of treatments [1]. A primary mechanism of cell death is R*egulated Cell Death* (RCD), which includes several types: apoptosis, necroptosis, pyroptosis, ferroptosis, and immunogenic cell death (ICD), along with other forms [2]. ICD is a type of stress-mediated cell death that can activate an adaptative immune response, leading to a pro-inflammatory effect in immunocompetent hosts. During ICD, dying cells release damage associated molecular patters (DAMPs) that stimulate T cells and promote long-term immune memory, determining immunogenicity of the cell death [1].

Currently, different ICD activations have been described based on the "danger" signals released by dying cells due to various

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_16,

stimuli, such as infectious pathogens, chemotherapy, radiotherapy, necroptosis or physical hints, such as high hydrostatic pressure or photodynamic therapy. These processes lead to the release of immunostimulatory DAMPs molecules secreted as calreticulin, chaperones (HSP70, HSP90), chemokines (IFN, HMGB1 and Annexin A1), ATP, TNF, RIPK3, MLKL, HMGB, or pathogen associated molecular patterns (MAMPs) [1–4].

The ability of ICD to influence in immunosuppression and restore antitumor immunity makes it a potent immunotherapeutic strategy in oncology, addressing treatment resistance and enabling new therapies [1, 3]. Tumour reduction and long-term effects have been observed, but systematic investigation of ICD is needed to enhance existing chemotherapies and to understand multifactorial stress scenarios which involve various biomolecules that collectively define the cellular state at a specific time and/or moment [3, 5]. Indeed, proteins are the main molecular effectors of the cell and provide information about its specific state [6].

Bearing this in mind, high-throughput and high-content techniques are required. Protein microarrays are a valuable tool based on the specificity and selectivity of antibodies, enabling simultaneous identification of multiple individual proteins from a complex protein mixture. In turn, they allow for both quantitative and qualitative comparisons between different conditions [6]. This microarray technology enables large-scale sample analysis in parallel within a single miniaturized assay, using a minimal amount of sample and demonstrating minimal inter-assay variation, while providing high detection accuracy across a broad dynamic range [7].

Depending on the nature of the analytes immobilized on the surface, protein microarrays can be classified into: (1) Recombinant or functional protein arrays, where proteins are directly deposited onto a chemically activated surface and their interaction with other molecules (proteins, lipids, DNA, or antibodies) is analyzed [8]; (2) Antibody or analytical arrays, which are constructed from antibodies immobilized on the surface that detect and quantify specific proteins from a sample [9–11] (Fig. 1). (3) Reverse phase protein arrays (RPPA), which are based on the immobilization of cell lysates directly on the array surface to detect specific proteins via fluorescence [12].

Antibody arrays monitor protein expression and enable the simultaneous study of differential protein profiles and the selective screening of each protein [13]. This makes them an excellent tool for the molecular study of complex intracellular signalling pathways, as they allow for comprehensive comparisons of the proteome between different individuals. In oncology research, they validate biomarkers for diagnosis, prognosis, and treatments effectiveness (e.g., used in colon cancer) [8]. Additionally, they identify proteins involved in cellular processes such as extracellular matrix

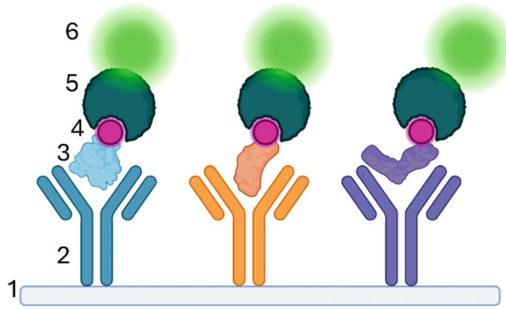

Fig. 1 Nanometric scheme of an antibody microarray assay. Each number points to a part of the assay: **1**. Functionalized microarray surface. **2**. Printed antibody. **3**. Capture protein. **4**. Biotin. **5**. Streptavidin. **6.** Fluorophore, for example, Cy3. (Created with BioRender.com)

regeneration, inflammation or tumorigenesis, and study activation and signalling cascades. The primary advantage of antibody microarray assays lies in their ability to examine multiple interactions simultaneously due to antibodies specific binding affinity [6] (Fig. 1).

This chapter details the critical aspects of the operational procedure for analytical protein microarrays designed to study ICD intracellular processes. These microarrays are especially valuable for clarifying intracellular signalling pathways in tumoral pathologies.

2 Materials

2.1 Cell Culture

1. Jurkat cell line (ATCC TIB-152).
2. HT-29 cell line (ATCC HTB-38).
3. Dulbecco's Modified Eagle Medium (DMEM) (1×) high glucose, pyruvate (Gibco ™, ThermoFisher Scientific, Waltham, Massachusetts, USA).
4. Roswell Park Memorial Institute Medium (RPMI) 1640 (1×) with L-glutamine (Gibco ™).
5. Fetal Bovine Serum (FBS) (Gibco ™).
6. Penicillin-streptomycin (10,000 U/mL) (Gibco ™).
7. 0.25% Trypsin-EDTA, Phenol Red (Gibco ™)
8. Culture plates, pipettes, etc.

2.2 Preparation of Samples

2.2.1 Anti-tumoral Drugs Incubation

1. 96 Well Plate.
2. 1× Phosphate-buffered saline (PBS) Na^+/K^+.

2.2.2 Protein Extraction and Quantification

1. 1× PBS Na^+/K^+.
2. Protease and phosphatase inhibitor cocktail Halt EDTA-free (100×) Halt™ (ThermoFisher Scientific, Cod:78441).
3. Lysis buffer: 5 mM HEPES pH 8.0, 10 mM $MgCl_2$, 140 mM NaCl, 0.1 (v/v) Tween20.
   (a) *Prepare Lysis buffer*: Dilute 0,119 g of HEPES in 50 mL of MiliQ water and add 0,20 mL of (5 M) $MgCl_2$, 2,8 mL of de (5 M) NaCl, 0,1 mL of 10% Tween20 and 1 mL of protease and phosphatase inhibitor cocktail. Finally, make up to volume of 100 mL with MiliQ water.
4. Coomassie brilliant blue R-250.
5. Ultrasound sonicator.
6. Nitrocellulose membrane.
7. Primary antibodies (anti-tag or anti-protein of interests).
8. HRP-labeled Anti-Mouse IgG secondary antibody.
9. Pierce BCA Protein Assay Kit (ThermoFisher Scientific).

2.2.3 Biotin Labeling

1. EZ-Link NHS-PEG4 Biotin.
2. 1× PBS Na^+/K^+.
3. Tris–HCl pH 8.0.

2.3 Preparation of Antibody Microarrays

2.3.1 Source Plate Preparation

This microarray consists of 59 human monoclonal antibodies whose targets are involved in 8 intracellular signaling death pathways (Cell Signalling, Massachusetts, EEUU): Apoptosis Antibody, Ferroptosis, Necroptosis, Pyroptosis, Human Exhausted CD8+ T Cell IHC, Human Exhausted T Cell, Pro-Apoptosis Bcl-2 Family, Pro-Survival Bcl-2 Family.

1. Thermoblock (ThermoFisher Scientific).
2. Thermal mixer (ThermoFisher Scientific).
3. JetStar™ Microarray-Specific 384-well Microplates (ArrayJet, Roslin, Scotland, UK).
4. JetStar™ Optimum Microarray Protein Printing Buffer C (ArrayJet).
5. 47% (v/v) Glycerol: Prepared from 99% glycerol (Sigma-Aldrich, St. Louis, MO, USA).

For the 384-plate preparation, the 59 commercial antibodies were diluted 1:1 (v/v) with Printing Buffer C, resulting in a 1:200 (v/v) dilution on the microarray.

2.3.2 Chemical Functionalization of Microarray Silica Surface

1. Acetone.
2. 3-(2-aminoethylamino)-propyldimethoxymethylsilane (ThermoFisher Scientific).

3. DEPC-treated water.
4. Epredia X50 Slides ground 45° (ThermoFisher Scientific).
5. Orbital shaker.

2.3.3 Non-contact Printing

1. Slides with chemically activated glass surfaces.
2. Nanoinjection printer Marathon ArrayJet (ArrayJet).
3. JetSpyder™ 12 samples (ArrayJet).
4. Sonicator.

2.3.4 Blocking the Microarray Surface

1. Super Block™ Blocking Buffer (ThermoFisher Scientific, Cod: 37535) are diluted in 1× PBS Na^+/K^+. For one microarray we need 6 mL of blocking buffer: 0.6 mL of Super Block™ Blocking Buffer and 5.4 mL of 1× PBS Na^+/K^+ (*see* **Note 1**).
2. Microarray washing chamber.
3. Orbital shaker.

2.3.5 Wash Microarray

1. Distilled water.
2. 1× Phosphate-Buffered Saline Tween20 (PBST): 0.05% (v/v) Tween-20 in PBS.

2.3.6 Microarray System Incubation

1. Incubation chamber.
2. Clips for microarrays systems.

2.4 Incubation of Samples in Microarrays of Antibodies

2.4.1 Preparation of Fluorescence Agent

1. Blocker™ BSA (Cod:37525) (ThermoFisher Scientific).

   Each sample was diluted in BSA 1× blocking solution (Cod: 37525) (1:100 v/v).
2. Cytiva Amersham Streptavidin-Fluor Cy3 (GE Healthcare, Little Chalfont, Buckinghamshire, UK).
3. MiliQ water.

   Cy3- Streptavidin was diluted with MiliQ water (1:200 v/v).

2.4.2 Incubation in Microarray

1. Transparent film.
2. Rotatory mixer rotate mild stirring.

2.4.3 Wash and Drying Microarray

1. 1× Phosphate-Buffered Saline Tween20 (PBST): 0.05% Tween20 in PBS.
2. Distilled water.

2.5 Scan of Microarrays and Data Analysis

1. Scanner GenePix 4000B (Axon, Molecular Devices, San Jose, CA, USA).
2. GenePix Pro 4.0 software (Axon, Molecular Devices).

3 Methods

The following protocol describe the procedures for incubation of protein solution by direct dye in microarrays of antibodies (Fig. 2).

3.1 Cell Culture

3.1.1 Cell Culture Conditions

1. Grow Jurkat cells in a 100 × 20 mm culture plate in RPMI with 10% FBS and 1% Penicillin-streptomycin at 37 °C in a 5% CO_2 atmosphere.
2. Grow HT29 cells in a 100 × 20 mm culture plate in DMEM with 10% FBS and 1% Penicillin-streptomycin at 37 °C in a 5% CO_2 atmosphere.

3.1.2 Anti-tumoral Drugs Incubation

1. Seed around 500,000 Jurkat cells and 250,000 HT-29 cells in 6-well plates.
2. Allow cells to attach overnight, adding the 12 different compounds.
3. After 48 h incubation at 37 °C in a 5% CO_2 atmosphere, trypsinize, collect, and wash the cells with 1× PBS Na^+/K^+ by centrifugation (x3 1200 rpm). Store the pellets at −80 °C until lysis.

3.2 Preparation of Samples

3.2.1 Extraction of Protein

1. Re-suspend the pellets in 100 μL of lysis buffer (5 mM HEPES pH 8.0, 10 mM $MgCl_2$, 140 mM NaCl, 0.1 (v/v) Tween20) and 1% (v/v) protease and phosphatase inhibitor cocktail (*see* **Note 2**).

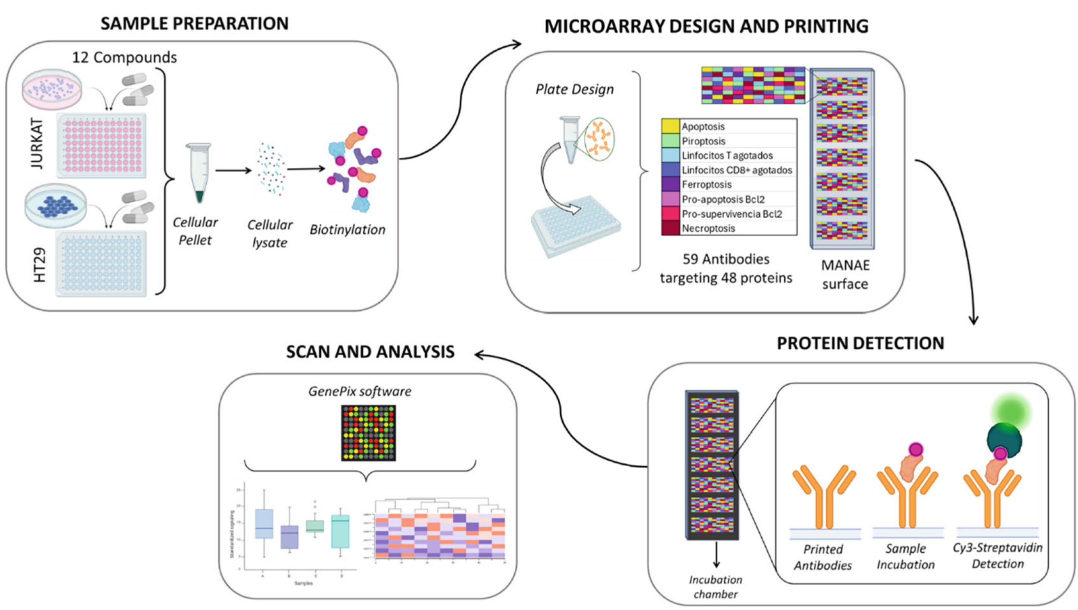

Fig. 2 Schematic Workflow for the antibody arrays. (Created with BioRender.com)

2. Sonicate the lysed cells 3 times with 10-s bursts with 1-min intervals on ice.
3. Centrifuge the samples at 12000 *g* at 4 °C.
4. Collect the supernatant in a new tube and store at −20 °C.
5. Additionally, the quality of protein lysis and samples should be verified using Coomassie blue staining and western blot (WB).

3.2.2 Biotin-Labelling

1. After sample quantification by Pierce BCA Protein Assay Kit, biotinylated the samples using 100 μg of each cell lysate and a solution of 6.2 mg/mL of biotin previously diluted with PBS (*see* **Note 3**).
2. Leave samples on ice for 2 h. A vortex after 1 h of incubation is recommended. Stop the reaction, adding 40 μL of Tris HCl pH 8.0.
3. Samples can be stored at −20 °C until use.

3.3 Preparation of Antibody Microarrays

The slides are printed as previously reported by Juanes-Velasco et al. [14].

3.3.1 Blocking the Microarray Surface

1. Add 6 mL of blocking buffer to each microarray washing chamber (*see* **Note 4**).
2. Immerse the microarray in the blocking buffer and incubate 1 h at room temperature (RT) with agitation on an orbital shaker (*see* **Note 5**).

3.3.2 Wash Microarray

1. Remove the blocking buffer.
2. Wash microarrays with distilled water and gently shake for 5 min (*see* **Note 6**).
3. Remove the distilled water.
4. Repeat **step 2** twice.
5. Remove the distilled water and keep the microarrays in clean distilled water until incubation.

3.3.3 Microarray System Incubation

1. Remove the microarray from the washing chamber.
2. Carefully place the microarray corner on absorbent paper without touching its surface to remove excess liquid (*see* **Note 7**).
3. Place the incubation chamber on the surface of the microarray (*see* **Note 8**).
4. Put fastening clips for microarrays systems.

3.4 Incubation of Samples in Microarrays of Antibodies

3.4.1 Incubation in Microarray

1. Dilute samples 1:100 in BSA Blocking Solution 1× (Cod: 37525) to a final volume of 100 μL.
2. For epitope retrieval process, keep 50 μL at RT and expose 50 μL to a 56 °C heat shock for 30 min.
3. After this time, combine both sample volumes into a single common Eppendorf tube.
4. Add samples onto each subarray (*see* **Note 9**).
5. Wrap the microarray with transparent film and put it on a rotatory mixer.
6. Incubate overnight at 4 °C in the rotatory mixer, with mild stirring (*see* **Note 10**).

3.4.2 Washing and Drying the Microarray

1. Remove the microarray incubation system (*see* **Note 11**).
2. Add 1× PBS Na^+/K^+- 0.05% Tween20 and put the microarray in a washing chamber.
3. Shake strongly in an orbital shaker for 5 min.
4. Remove the 0.05% PBSTween and add new.
5. Repeat **steps 2** and **3** two more times.
6. Perform the last wash by adding distilled water (*see* **Note 12**).
7. Dry microarrays by putting each microarray in a 50 mL falcon tube with a hole in the bottom, and centrifuge at room temperature for 3 min at 1000 *g* at RT.

3.5 Scan of Microarrays and Data Analysis

1. Microarrays are scanned with Scanner GenePix 4000B (Axon, USA), using Cy3-like settings (*see* **Note 10**).
2. Analyze the images with GenePix Pro 4.0 software (Axon, USA). The GenePix Array List file (.gal) generated after printing of array is used as reference to locate and identify each spot on the microarray surface.
3. Background correction for each spot is locally made by applying standardized settings of GenePix software.
4. The normalization of each slide is made according to Hernández et al. [15].

3.6 Method Validation

To assess the design of this new ICD microarray, 59 antibodies that target 48 different proteins and various controls were included to ensure the reliability of the study. Positive controls biotin (to verify streptavidin-Cy3 binding), and Cy3 and Cy3-IgG (positive printing controls) were included. As negative controls, various solutions used in the assay were deposited, such as blocking solution, printing solution, washing solution, PBS, 0.05% PBSTween, and Anti-AlexaFluor647.

After verifying that the printing was successfully conducted (Fig. 3), quantitative analysis of the antibody arrays was performed.

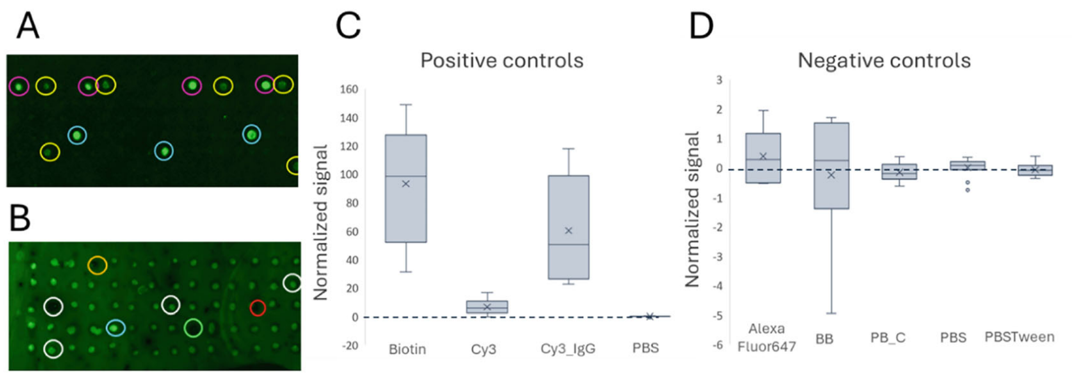

Fig. 3 ICD Antibody microarray image: **a** Image from GenePix Pro 6.0 analysis of subarray incubated with PBS. Examples selected from positive controls: Biotin (pink), Cy3 (yellow), Cy4_IgG (blue). **b** Image from GenePix Pro 6.0 analysis of subarray incubated with a sample. Examples selected from protein detection (white) 3and negative controls: PB_C (red), PBSTween (orange), AlexaFluor647 (green). **c** and **d**. Global quantitative analysis of positive controls compared to PBS **c** and negative controls **d** based on normalized signal (Z-score) in subarrays incubated with PBS

This involved evaluating the positive and negative controls on subarrays incubated with PBS, and general protein detection on subarrays incubated with 26 samples (controls from each untreated cell lines Jurkat and HT-29, and each cell line treated with 12 different compounds (A-L).

As observed in Fig. 3, the controls demonstrate qualitatively and quantitatively that the printing and detection of the array function correctly. Subsequently, differential proteomic profiles associated with ICD are determined by studying alterations in relative protein abundance under various conditions.

Significant variations in protein profiles are observed across six groups of compounds and five groups of proteins of interest (Fig. 4), regardless of cell line type [16].

The employed methodology has demonstrated to be able of decipher differential protein profiles between cell lines (based on relative abundance), between compounds, and between the same compound in different cell lines, as well as changes in the expression and activation of specific proteins. One of the opportunities offered by this protein microarray are the study of the activation and regulation status of the cell death pathways and the evaluation of the effect of the same compound on different tumor cells.

In conclusion, thanks to a novel functional protein microarray platform, different patterns of protein profiles (based of cellular immunogenic cell death) were observed at different conditions and tested cell lines. Therefore, it could be a powerful tool for studying the influence on immunosuppression and restoring anti-tumor immunity, as well as for developing new drugs or repositioning existing ones.

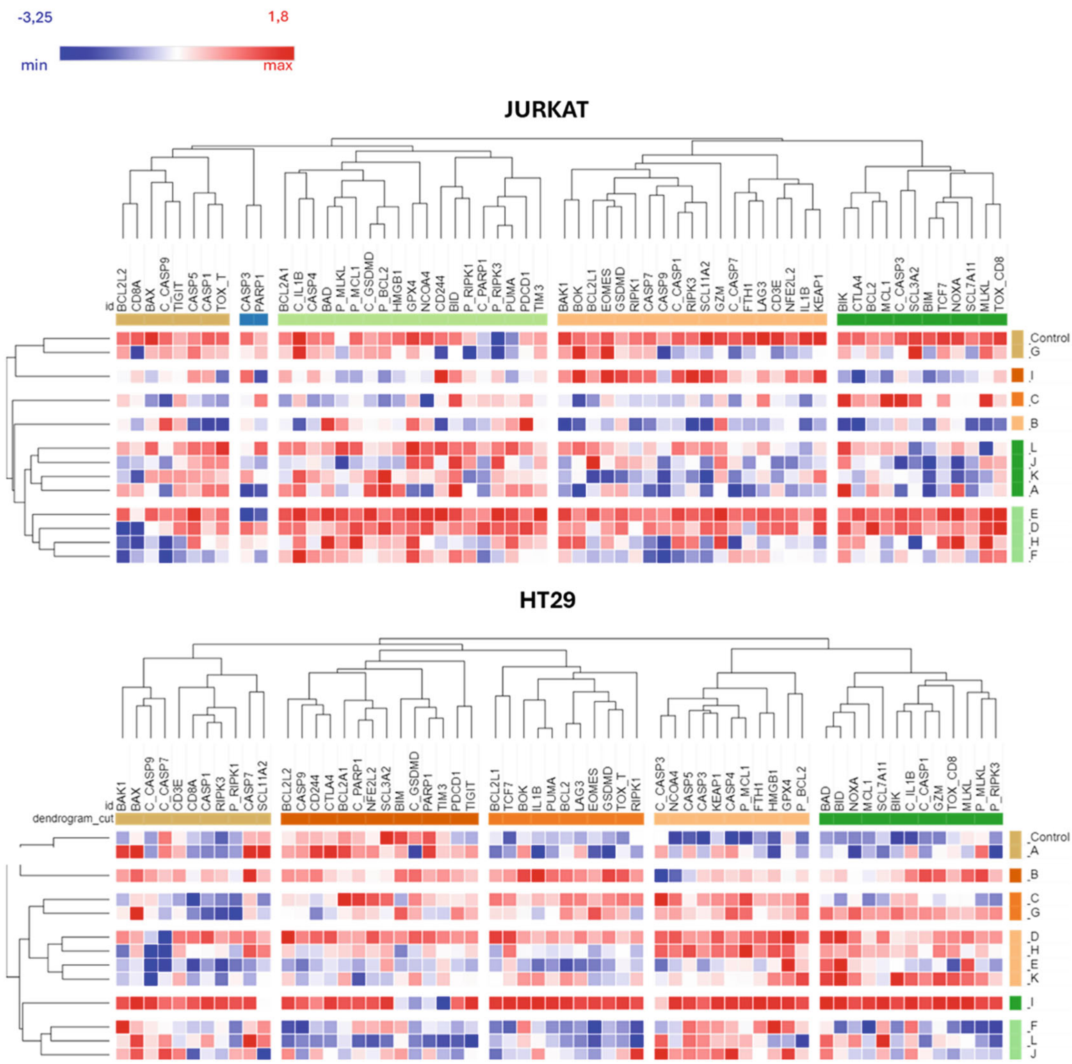

Fig. 4 Heatmaps of protein expression profiles in the presence of each compound (A-L) in each cell line of interest (Jurkat and HT-29). Each protein represents the normalized signal (z-score). Dendrogram for all proteins and compounds. (Created with Broadinstitute.org/Morpheus)

4 Notes

1. Blocking buffer must be thoroughly mixed by gently vortex, ensuring that no lumps remain.
2. Use different protein extraction protocols, depending on the needs of the study. In this chapter, a total protein extraction protocol is described.
3. Detergents, surfactants, and other reagents present in the lysis buffer may interfere with the quantification. For this reason, according to protocol extraction and needs of the study, other

quantification methods can be used, such as Bradford or NanoDrop.

4. The microarray must be completely covered by the blocking buffer.
5. We must be careful when introducing the microarray. Surface must never be touched to avoid spoil spots printing. Pay special attention to which side the spots are printed.
6. Tilt the washing chamber of microarrays to remove solution. Do not add on top of the microarray the different solutions necessary for the washing process, but always put it onto areas of the washing chamber that are not occupied by the microarray.
7. Lift the microarray out of the washing chamber very carefully with a pair of tweezers. Never touch the printed area of the slide with the tweezers or any other object or hand. Hold the microarray by the side edges and place it vertically. Once the microarray has been secured, gently tap on several sheets of absorbent paper to remove any excess solution.
8. Try to place the incubation chamber as centered as possible with the microarray the first time. Once the incubation chamber has been placed on the array, do not move it to try to square it with the edges of the array, even if they are not square, because the spots can be dragged and lose part of the printing.
9. Be very careful not to generate bubbles in the hatchery when the sample is added. If they have appeared, move the chamber from left to right and up and down, carefully and always keep it on a surface.
10. Ensure that the sample is distributed evenly throughout the chamber. Modify, if necessary, the inclination of the orbital mixer for gently stirring.
11. Remove the incubation chamber in one single step by lifting upwards, without sliding the chamber over the surface and holding the array tightly at the sides.
12. This last wash ensures that no traces of Tween20 reagent are left on the surface, making it difficult to image acquisition. Ensure a good rinsing without ever touching the surface of the array.

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Chapter 17

Combination of Antibody Arrays to Characterize the Effect of Neuropathological Stimulus in Human Olfactory Neuroepithelial Cells

Mercedes Lachén-Montes, Elena Anaya-Cubero, Paz Cartas-Cejudo, Leire Extramiana, Joaquín Fernández-Irigoyen, and Enrique Santamaría

Abstract

It has been established that smell impairment is a common early feature of neurodegenerative diseases (NDs), being most prevalent in Alzheimer's disease patients. As well as in other brain regions, the deposition of pathological substrates such as amyloid peptides (Aβ) or the hyperphosphorylated form of tau has been suggested as a potential origin of olfactory deficits. Nevertheless, other trends of thought are now suggesting the detriment in the soluble forms of these proteins as potentially responsible of the neurodegenerative process. Either way, the molecular mechanisms triggered by the presence and aggregation of these pathological substrates are not fully understood. In this chapter, we describe a workflow combining the use of neuropathological substrates depicting a neurodegenerative environment in an olfactory-related context with the employment of protein arrays targeting both extracellular and intracellular proteins aiming to characterize the molecular mechanisms underlying the established role of neuropathological hallmarks.

Key words Olfactory cells, Antibody arrays, Neurodegeneration, Aβ

1 Introduction

Olfactory dysfunction is a common and early feature in several neurodegenerative diseases (NDs) including Alzheimer's disease (AD) and others [1]. Importantly, this impairment is even considered as a premature sign of neurodegeneration and consequently, a reliable marker [2]. This dysfunction has commonly been related to the deposition and aggregation of pathological proteins, such as the beta amyloid (Aβ) peptide or the hyperphosphorylated form of Tau. Conversely, an alternative hypothesis concerning the loss of function of the soluble monomeric form of these proteins during the process of aggregation into amyloids is gaining currency. In this context, it is worth to note that the molecular mechanisms

Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929, https://doi.org/10.1007/978-1-0716-4595-6_17,

regarding the harmful effects of these deposits leading to neuronal death are still not fully understood.

The olfactory epithelium (OE)-bulb (OB)-tract (OT) axis constitutes the central site for the processing of olfactory information in the brain [3–5]. Interestingly, at OE level, Aβ deposition can predict mild cognitive impairment and is associated with dementia [6] and AD pathology correlates with brain pathology [7]. On the other hand, a massive analysis has revealed that the presence and severity of hyperphosphorylated Tau and Aβ pathology in OB-OT axis reflects the presence and severity of AD in other brain regions [7] whereas a significant degeneration of axons has been detected in the OT from AD subjects [8]. In pre-AD subjects, the loss of fiber OT integrity corresponds to a loss of gray matter density in parallel with a reduced glucose metabolism [9, 10]. Interestingly, OTs undergo early and sequential morphological alterations that correlate with the development of dementia [11]. All these data together with experimental evidences obtained in human AD and different animal models [12, 13] point out that the nasal cavity may be a useful site for early AD diagnosis [14] as well as for innovative intranasal therapies [15].

In this chapter, aiming to increase the knowledge concerning Aβ detrimental effects and its potential implication in the neurodegenerative process occurring within the olfactory system, we have built an exhaustive workflow combining the treatment of human olfactory epithelial cells and a wide molecular analysis of intracellular and extracellular potential alterations combining cytokine and phospho kinase antibody arrays.

The array technology refers to the miniaturization of thousands of assays on one small plate. Regarding protein arrays, the most representative model of analytical protein arrays are the antibody arrays and due to its high sensitivity, the sandwich assay format is now widely used [16, 17]. This format uses two different antibodies to detect the targeted proteins [16]. Capture antibodies are placed on the solid phase where they immobilize the targeted proteins or analytes, whereas reporter or detection antibodies, generate a signal for the detection system. Thus, the use of two antibodies significantly increases the specificity and sensitivity of this "sandwich" format [18]. The great versatility of this technology has led to the development of commercially available protein arrays in many different formats and researchers are able to analyze not only the extracellular presence of cytokines and growth factors, but also to monitor intracellular cascades related to autophagy, apoptosis, stress biomarkers, and phosphorylation events. This technology offers other advantages including its versatility in terms of the target biological samples (from fluids to cell or tissue protein extracts), to test and the small amount of sample. Required. Additionally, results can be obtained within 8–9 h and importantly,

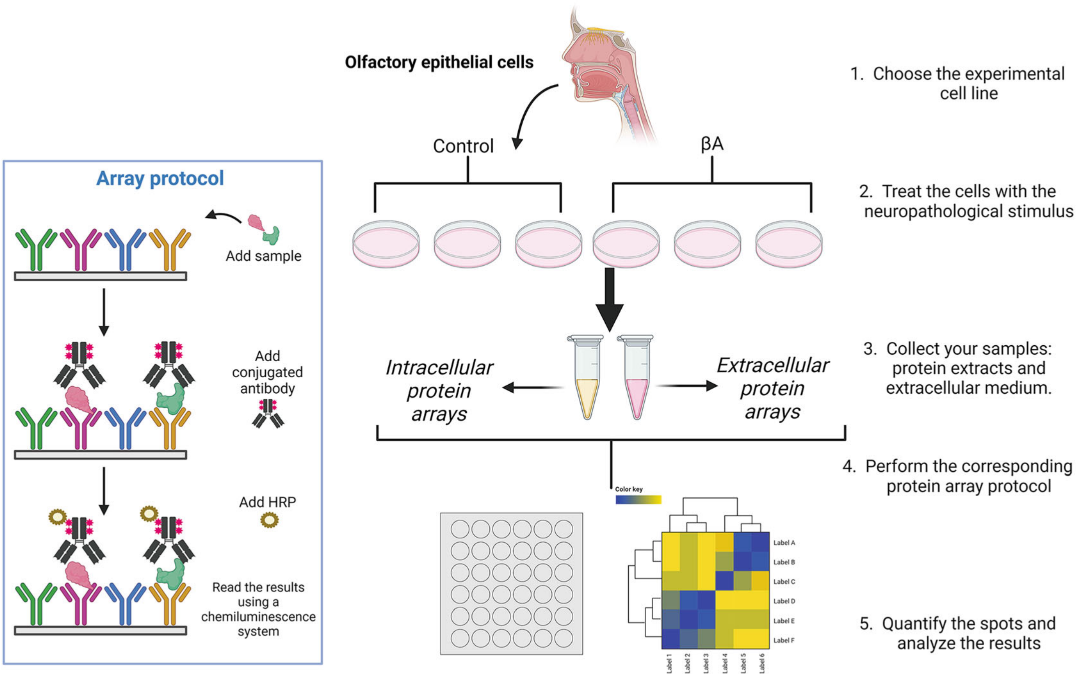

Fig. 1 Technical workflow based on antibody arrays used to characterize functional features of Aβ peptide in human olfactory epithelial cells

the possibility of creating a custom protein array is available in some commercial houses.

This chapter focuses on the utility of antibody arrays to establish a straightforward methodology with the aim to characterize the molecular mechanisms associated to the neuropathological effect of established pathological proteins such as Aβ peptides. This workflow can be adapted to monitor any neuro-related disorders that are accompanied by the presence and aggregation of pathological proteins, even considering the potential presence of co-pathologies [19, 20] (Fig. 1).

2 Materials

2.1 Reagents

2.1.1 Cell Culture

1. DMEM High glucose, pyruvate (Gibco).
2. Fetal Bovine Serum (Gibco).
3. Antibiotic-Antimytotic (Gibco).
4. Trypsin–EDTA (0.25%), Phenol Red (Gibco).
5. PBS.
6. Flasks, plates, pipette, etc.

7. Amyloid β-Protein Fragment 1–42 (Bachem).
8. Lysis buffer.

2.1.2 Cytokine and Phospho Kinase Arrays

1. Cytokine or phopho kinases Antibody Array Membranes (Abcam & R&Dsystems).
2. 8-well plastic tray.
3. Plastic sheets.
4. 1× blocking buffer.
5. 20×wash buffer I.
6. 20×wash buffer II.
7. 2000× Biotin-conjugated anti-cytokines.
8. 1000× HRP-conjugated streptavidin.
9. Detection buffer C.
10. Detection buffer D.
11. Additional required materials: tweezers, small plastic boxes/containers/centrifuge tubes (50 mL), tissue paper/blotting paper/chromatography paper.

2.1.3 Equipment

1. Centrifuge 5427 R (Eppendorf).
2. Orbital shaker/oscillating rocker/roller.
3. Chemidoc MP Imaging System (BioRad).
4. ImageLab Software (BioRad).
5. Perseus software.

3 Methods

3.1 Harvest, Isolation, Culture and Maintenance of Human Olfactory Neuroepithelial Cells

1. Sterilize forceps 2 min at 200 °C (or autoclave them).
2. Prepare the necessary eppendorfs to collect the samples inside a cabin with 500 μL of complete medium. Put them on ice.
3. Also prepare an autoclavated pipette tip.
4. Keep complete medium at 37 °C.
5. Nasal biopsies are then obtained using a giraffe forceps and immediately transferred to culture medium for transport to the laboratory.
6. Clean the eppendorfs with ethanol before putting them in the cabin.
7. Hold the brush with tweezers or forceps and pass it through the cut tip on the Eppendorf tube so that the sample stays on the Eppendorf tube and not on the brush.

8. With a P1000 pipette, add 250 ul of medium inside the tip to lower what may be left of the sample.
9. With a P200 pipette, pipet up and down with the cut tip of the Eppendorf tube. to clean it and make the whole sample stay in the middle.
10. Disintegrate with P1000 pipette (avoid bubbles) Disaggregate with P200 and P10 pipettes.
11. Cultivate in P24 wells. Cell growth should be monitored by microscope during the next days.
12. Change medium after a week. Carefully aspirate the old medium from the side of the well and add 500 μL of new complete medium.
13. Incubate for 3 weeks or until confluence and perform first passage (*see* **Note 1**).

3.2 Neuropathological Treatment

1. Culture cells in P10 plates (*see* **Note 2**).
2. The next day, remove medium in order to remove non-attached cells.
3. In new medium, add the Aβ peptide 1–42 reaching a final concentration of 10 μM (*see* **Note 3**).
4. Cultivate cells for 24 h.

3.3 Protein Extraction (see Note 4)

1. Remove medium and do not discard it (*see* **Note 5**).
2. Wash cells with PBS (*see* **Note 6**).
3. Trypsinize cells for 3–5 min at 37 °C and gently collect detached cells into a centrifuge tube.
4. Centrifuge cells during 5 min at 500 g to remove trypsin solution and discard supernatant.
5. Resuspend pellet with lysis buffer containing 7 M urea and 50 mM DTT.
6. Incubate samples on ice for 30 min, vortexing and spinning every 10 min (*see* **Note 7**).
7. Centrifuge samples for 1 h at 20000 g at 15 °C.
8. Quantify proteins using a protein quantification protocol such as Bradford, Lowry, etc.

3.4 Sample Incubation on Cytokine/Phospho-Kinase Antibody Membranes

1. Grasp membranes by the edges using tweezers and place each membrane into the 8-well tray printed side up (*see* **Note 8**).
2. Block membranes incubating with 2 mL 1×blocking buffer in a shaker at room temperature for 30 min (*see* **Note 9**).
3. Prepare samples. Diluted them if it is necessary with blocking buffer (*see* **Note 10**).

4. Aspirate 1× blocking buffer from each well (*see* **Note 11**).
5. Pipette 1 mL of sample into each well and incubate for 1.5–2 h at RT or o/n at 4 °C using a shaker or oscillating rocker (*see* **Note 12**).
6. Aspirate samples from each well.
7. Wash membranes using the 1× blocking buffer I (*see* **Note 13**).
   (a) After o/n incubation: wash membranes in separate containers with 20–30 mL of 1× wash buffer I per membrane for 30–45 min at RT on a shaker. Then proceed to 7b.
   (b) After 1.5–2 h incubation, pipette 2 mL of 1× wash buffer I into each well and incubate for 5 min at RT on a shaker. Repeat this step twice for a total of 3 washes.
8. Wash membranes using 1× wash buffer II. *See* **Note 13**. Pipette 2 mL of 1× wash buffer II into each well and incubate for 5 min at RT on a shaker. Repeat this step for a total of 2 washes.
9. Pipette 1 mL of 1× Biotin-conjugated anti-cytokines into each well and incubate 1.5–2 h at RT or o/n at 4 °C using a shaker (*see* **Note 14**).
10. Aspirate the 1× biotin-conjugated anti-cytokines from each well.
11. Wash membranes as directed in **steps 7b** and **8**.
12. Pipette 2 mL of 1×HRP-conjugated streptavidin into each well and incubate on a shaker for 2 h at RT or o/n at 4 °C on a shaker (*see* **Note 15**).
13. Aspirate the 1× HRP-conjugated streptavidin from each well.
14. Wash membranes as directed in **steps 7b** and **8**.

3.5 Cytokines and Phospho Kinase Detection (see Note 16)

1. Transfer each membrane printed side up into a sheet/chromatography/blotting paper (absorbent paper) on a flat surface. Remove any excess of washing buffer.
2. Transfer each membrane printed side up onto one of the plastic sheets provided with the kit on a flat surface.
3. Prepare equal volumes of detection buffer C and detection buffer D (*see* **Note 17**).
4. Pipette 500 ul of the detection mixture onto each membrane and incubate 2 min at RT (*see* **Note 18**).
5. Carefully place another plastic sheet on top of your membrane starting from one end and gently rolling the sheet across the surface to the opposite end to prevent bubble for forming.
6. Using forceps transfer the sandwiched membrane to the imaging system (*see* **Note 19**).

7. Let the exposure begin with the following parameters (*see* **Note 20**):

   Using ImageLab software (BioRad):

   (a) New protocol.

   (b) High resolution.

   (c) Select the size of the image (*see* **Note 21**).

   (d) Membrane exposure: automatic (*see* **Note 22**).

   (e) Save the image.

8. After exposure, remove the upper plastic sheet and place the membrane in the corresponding well of the tray. Clean both plastic sheets with water and dry them. Repeat the process from **step 4** for all of the membranes (*see* **Note 23**).

9. It is possible to store membranes. Sandwich them between the plastic sheets provided with the kit. It is possible to store 6 membranes in each sandwich. Afterwards, tape the sheets together or wrap in plastic wrap to secure them and store them at ≤ -20 °C (*see* **Note 24**).

3.6 Cytokines and Phospho-Kinase Quantification (ImageLab Software)

1. Use the Volume tools option to measure the density of each protein spot (*see* **Note 25**).

2. Click on Analysis table to get the quantitative density values of each spot.

3. Export table as Excel-table.

3.7 Statistical Analysis of the Differential Proteins

1. Name each spot with the corresponding protein.

2. Use the Adjust volume data.

3. Data normalization (Excel).

   (a) Calculate the average of the positive values of each membrane: Posn.

   (b) Calculate the correction factor for each membrane based on the reference membrane: (*see* **Note 26**).

      (i) Posn/Posref = Correction factorn (CFn).

   (c) Correct values of each membrane based on its correction factor: Prot.Spotn/FCn.

   (d) Save the normalized protein matrix as txt (*see* **Note 27**).

3.8 Data Statistics with Perseus Software (see Note 28) [21]

1. Import the txt matrix to Perseus and place data in its corresponding category.

   (a) Protein spot values in the "Main category."

   (b) Label (Protein names) in the "Categorical category."

2. Trasform the standardized values to Log2.

3. Using the "Categorical annotation row" option, grouped your samples as desired, e.g., Control and Treated.
4. Check how your samples are distributed visualizing on graphs such as Histograms, dendonrams, or boxplot.
5. Perform a t-test: Two-samples test.
6. If desired, rename columns –Log Students's T test p value as –Log p value and Student's T-test difference as Fold Change (FC).
7. If desired, filter rows applying a desired cut off for FC and pvalue values. Example:
   (a) For –log p value; $x \geq 1.3$ to filter proteins with a p value less than 0.05. (*see* **Note 29**). Select Add categorical column.
   (b) For FC; $x \geq 0.38$ and $x \leq -0.38$ to filter proteins with a differential expression of 30%. Select Add categorical column.
8. Select the differential protein based on p value or/and FC (*see* **Note 30**).
9. Visualize the data using the scatter plot (Volcano plot).
   (a) X axis.
   (b) Y axis: -log p value.
10. You can select the differential protein according to the previously established criteria (p value & FC) selecting those with the "Keep_Keep" label. Once selected, you can export them in a matrix clicking on the tab "points" and choosing "Export selection (reduced matrix)."
11. (Optional). You can add annotations related to the proteins using defined information packages that Perseus software offers (*see* **Note 31**)
12. To save any matrix, click on "Generic matrix export."
13. Save the session (*see* **Note 32**).

4 Notes

1. Perform as many cell passages as desired although it is highly recommended to perform the experiment during short cell passages. Moreover, you can choose an immortalized commercial cell line instead such as T9243 from ABM.
2. Cells should be about 75% confluent.
3. Different aggregation status of the stimulus might be assessed aiming to decipher the effect of both monomeric and oligomeric forms of Aβ. Likewise, other Aβ species might be

evaluated, as well as other neuropathological stimulus such as a the hyperphosphorylated form of Tau, α-synuclein aggregates, and other neurodegenerative-related proteins.

4. At this point, you need to check the toxicity of the stimulus in order to decide the accurate time-point to check the potential and consequent molecular alterations. Different approaches might be followed such as viability assays (MTT) or cytotoxicity assays (lactate dehydrogenase assays).
5. Changes in the proteostasis of your media may be happening after your treatment, so, further analysis of this sample may be of interest. Cytokine-based protein arrays might be a the suitable for choice to analyze this sample.
6. Add the PBS carefully, cells may get detached.
7. You might need to sonicate your samples due to a high presence of DNA.
8. The printed side of each membrane is denoted by a dash mark (−) in the upper left corner. It is recommended to use a permanent marker to write on the lid and on the side of the tray, the name of the sample that will be placed in each of the wells. At the end of the assay, you can clean them with ethanol. Use tweezers every time you handle the membranes.
9. After usage, membranes and 1× blocking buffer should be store at ≤20 °C. The rest of the kit reagents can be stored at 4 °C up to 3 months.
10. If samples have a consistent protein concentration among them (≤20%), you can dilute the samples using equal volumes of 1× Blocking buffer for each sample. In the case that samples present wider protein concentration variation among them, dilute each sample with 1× Blocking buffer to equal the concentrations.
11. To avoid damaging the membrane, tilt the tray to pick up the different reagents with the help of a pipette tip, from a corner.
12. Avoid spreading the reagents directly on the membrane, in order to avoid damaging it.
13. Both Wash buffers I and II have to be 20-fold diluted with distilled or deionized water. These 1× washing buffers can be store at 4 °C up to 1 month.
14. This antibody cocktail is concentrated (2000×) as small liquid bead, so using one vial you can test two membranes. Briefly centrifuge each vial prior to reconstruction as the concentrate can adhere to the inside walls and cap during transit. Reconstitute by pipetting 1 mL of 1× Blocking buffer into the vial, and mix the cocktail by softly pipetting up and down. Then pipette

500 μL of Blocking buffer and 500 μL of 1× antibody cocktail, into each well (=1 mL).

15. The HRP-Conjugated Streptavidin has to be 1000-fold diluted with 1× Blocking buffer.
16. Array kits will retain full activity although you store them up to 24 h at room temperature.
17. 500 μL of detection buffer are needed for each membrane.
18. The exposure should start within 5 min after adding the detection mixture and should last as much as 20 min. Otherwise chemiluminescent signals will fade over time. For this reason, it is strongly recommended to do this step, and the following ones, in each membrane separately.
19. Do not allow membranes to dry out during detection and avoid sliding the plastic sheet along the membranes printed surface.
20. Exposure the membranes one by one to making sure that all the cytokine spots will be visible.
21. The size of the imaging area must always be the same, so all the images have the same size and can be easily compared.
22. Typical exposure times take between few seconds to 2 min, but, if the imaging system allows it, expose the membranes in auto-expose mode. This ensures the detection of most of the cytokine spots, because it detects little abundant proteins without saturating those with the most abundant expression.
23. If the image has low quality, repeat the ECL incubation.
24. Membranes can be stored up to 5 days in 1× Wash buffer II or 1× Blocking buffer at 4 °C.
25. Use the same area (circle) on all the membranes you want to compare (use: copy and paste). If the size of the spots is too different, select three different sizes (small, medium, and large), that fit better the size of your spots.
26. The reference membrane can be any of them, although it is recommended to use one of the controls.
27. The excel matrix should contain the header: Label, each protein name and the sample names.
28. The Perseus software can be used for the quantification, visualization, and statistical analysis in a user-friendly manner. However, other statistical workflows might be used.
29. The pVal is represented in –Log10, so –Log pVal; $x >= 1.3$ means pVal ≤ 0.05 The selected FC chosen here is just an example, and you should use the one that best fits your experiment data or criteria. Consider that the FC is represented in

Log2. In the case of a 30% FC, x > = 0.38 means a FC ≥ 1.3 and x < = −0.38 means a FC ≤ 0.7.

30. If you choose both criteria to select the differential proteins, you should follow the option Rearrange—Combine categorical columns.
31. Go to Annotation columns and choose Add annotation. First of all, you must have downloaded the database with the information you want to have, and save it in the annotations folder where the program is. Select the annotations that you want to add.
32. Save the Perseus session and carefully annotate in which program version you have worked. Perseus session should always be opened in the same program version as the one in which they were created.

Acknowledgments

The Clinical Neuroproteomics Unit is member of the Spanish Olfactory Network—ROE (grant RED2022-134081-T from the Spanish Ministry of Science & Innovation) and is supported by grants PID2023-152593OB-I00 funded by MCIU/AEI/10.13039/501100011033 / FEDER, UE to ES and JFI and 0011-1411-2023-000028 (from Government of Navarra- Department of Economic and Business Development-S4) to ES.

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Cristina Ruiz-Romero et al. (eds.), *Protein Arrays: Methods and Protocols*, Methods in Molecular Biology, vol. 2929,
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